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ACUE 与黎曼零点
从谱反例走向
可证明的数论结果

完整研究接手档案

GPT-6 Astra × Claude Code / Fable

Bill (Qingyun) Sun · 2026 年 9 月 4–5 日

经审计主说明、公开历史 Markdown 与完整研究报告、算术相关性新路线

已证明、计算、猜想与失败分别标明 · Public research archive

内容索引

ACUE、黎曼零点、谱反例与素数间隙：GPT-6 Astra 完整研究接手档案

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整理日期：2026 年 9 月 4 – 5 日；持续补充夜间研究
文档性质：带审计的研究交接、历史原文归档与新研究纲领
主要代码来源： `galpha-ai/Alpha-devbox` 的 `claude/riemann-zeta-random-matrix-udxp3f` 分支
当前主线：在 RH 下突破真实 ζ 零点的 1/2 间距障碍，或得到明确的带宽外二点相关结果。下文研究优先级的历史表述以最新补篇为准；RMT 深度定理是支持路线。

历史基准： `ee61fba38925190916e556f87bc1ea83a502413e`；Fable 新交接版本： `0f86c31d847178f0a774f70073ddc44206b32f98`；共同工作入口：PR #11

> 本文要保存的是已经得到的数学结构、正确证明、可复现实验、失败路线和仍然缺失的关键步骤。它没有宣称已经证明 RH、Montgomery pair correlation、GUE 猜想、排除 Tao 的 Alternative Hypothesis，或得到小于 186 的相邻素数间隙。历史原文中的更强措辞必须服从前半部的审计与限定。

阅读方式与交接目标

这份档案包含两种性质不同的材料。第一部分是本次重建、核验和重新组织的接手说明；第二部分逐篇保存主分支的历史 Markdown 原文，并附早期 Tao AH 说明 PDF 的文字转录。历史原文保留其当时措辞，因此能够看见认识如何变化，也会看见后来被撤回的判断。源代码、数值证书、数组、Lean 草稿和找到的完整任务记录同时保存在配套档案中。

如果 Astra 只读一遍，建议先读“必须先修正的结论”“ACUE moment fiber”“零动力学”“186 的真实结构”“研究目标与任务卡”，再按需要查历史原文。若要写论文，应逐条从审计表返回原证明与脚本，而不能把本档案的总结当作独立文献证明。

本次工作的概念对象是有限点过程、带权矩约束、谱证书、热流首碰撞时间与筛权二次型；核心不变量是概率非负性、精确矩匹配、带宽预算、归一化和算术输入的适用范围。交付结构采用一个主说明、完整原文附录、只读来源快照和一个独立审计程序。Python 只负责小规模复核与文档生成，不引入 Rust 或新的研究框架。验收标准是来源可定位、已知错误可辨认、实验可复跑、PDF 可读。暂缓事项是新的大型数值搜索、补全 Lean 工程、重跑 186 的重型证书及对外发表；这些均需要下一阶段实际研究，不能由文档整理冒充完成。

1. 最重要的判断

现有研究最值得保留的，并不是一个已经抵达著名猜想终点的证明，而是一套很具体的“信息不足”的数学模型。它们指出：即使二点信息、有限带宽内的所有平衡矩、部分高阶边缘分布、甚至某些完整 diffraction 数据完全相同，底层过程仍可能不同。这个认识能帮助选择真正能越过障碍的观察量；它本身没有提供把观察量用于黎曼零点的算术公式。

最成熟的有限结果集中在四组：Nyquist 模式对重数的精确控制；ACUE 矩纤维及其显式变形；homometry、magnetic holonomy 与 deck 非唯一性；圆周零动力学中的两体比较及其需要修正的局部化论证。最接近现有著名数论目标的可操作方向，则是把这些反例转化为一个具有固定强度的区分统计量，并找到它的显式公式或 Dirichlet 多项式实现。

这里有两个必须分开的突破入口。

1. **零点分布入口。** 在 RH 下，对一个固定带宽外区间的二点统计给出 GUE 值，或者证明一种足以排除精确定义 AH 的正密度偏离。这会是 Montgomery/GUE 路线上实质性的结果。有限反例主要负责告诉我们必须买到什么信息。
2. **素数间隙入口。** 从 186 的具体平方筛权、互补分解和支撑恢复不等式出发，找更小的 admissible tuple 或可认证的更强二次型。这里已有完整候选机制，宜与零点路线并行思考，但不应假设前者由后者直接推出。

用户希望追求历史级定理，这是研究方向的选择标准，而不是对成功概率的承诺。当前材料尚不足以判断某个著名猜想已“可解”。本档案给出明确终点、缺失引理和停止条件，避免继续积累无法触及终点的小模型。

2. 来源覆盖、时间线与忠实性边界

2.1 实际恢复了什么

恢复了所给分支固定提交中的 `research/riemann-rmt` 目录，包括 16 篇顶层 Markdown、Python 实验与验证程序、JSON/NPZ/NPY 数值材料、一个早期 AH PDF，以及 `riemann-impostors` 打包子目录。还读取了任务 **《联系 ACUE与零动力学》**，找到 11 个可取回的 turn，并将返回记录原样保存为 `sources/acue-zero-dynamics-conversation.json`。

共享上下文明确记录过十个并行方向的分工，另有多轮综合稿、signed-sieve 分工和后续 Fable 系列脚本，因此确实不是单一对话中凭空产生的研究摘要。但是这次没有恢复所有原始子代理的完整对话、所有 scratchpad、早期被引用而未提交的 PDF 或外部 Lean 仓库的历史副本。不能把“取回分支中的全部材料”说成“找回了每一个历史子代理的全部输出”。

历史来源与当下判断分层如下。

标签	含义	应如何使用
外部论文结论	本次打开的作者论文或官方项目陈述	引用原始来源，并注意论文与证书的验证范围
有限证明	原稿给出可检查的有限或确定性论证	可继续整理成定理，但本次不是全面同行审稿
本次复核	本次运行了对应数值、枚举或符号检查	说明检查域、精度及未证明的量词
历史计算	原稿报告的数值，当前未完全复跑	不升级为精确证明或新纪录
条件桥梁	写得出结论，但关键假设尚未证明	下一轮应集中攻击这些假设
撤回或修正	与后稿、代码或本次审计冲突	禁止沿用旧的强结论

2.2 重建的研究演化

阶段	材料或提交	主要进展与状态变化
早期谱证书阶段	`rmt_zeta_survey.md`、`tao_ah_notes.pdf`	将新两项矩证书与 AH 反例联系起来；部分“最优上限”措辞后来撤回
8 月 15 日综合	`6cc6e5f`、`final_verified_paper.md`	Galerkin、Nyquist、covariance、homometry、holonomy、deck 和来源审计
下一轮十方向	`joint_context_v2.md`、`round3_synthesis.md`	cubic escape、带宽外检测、COM 家族、端点信息、derivative tower 等
8 月中旬筛法	`8f576d7` 至 `9d837b7`	P9 可分离截断权重、H_2 /H_3 /H_4 候选证书与 tuple
8 月 18 日以后	`bb7e7e6` 至 `07d1237`	Newman depth、signed sieve no-go、三种统计分离与 LR 桥梁修正
后续统一	`c8a6270` 至 `df0b581`	可见性算子、clock Jacobian、marked response、局部横截性
形式化尝试	`9074fd4` 至 `17242f9`	修订 depth 比较；Sympy/Z3 检查；未编译 Lean 草稿
8 月 22 日打包	`ee61fba`	`riemann-impostors` 独立目录；部分数据与路径仍未随包提交
8 月 24 日任务讨论	《联系ACUE与零动力学》	Wronski/Plücker 总正性；否定任意 PSD 表示就能解决问题的论证
本次 9 月接手	本档案及审计文件	对照 186、FLT 形式化及 Lamzouri 新证明，重新安排研究目标

2.3 哪些内容不应算成我们的独立新发现

ACUE 由 Tao 的既有构造出发；sine process、Montgomery – Taylor 优化、Maynard 变分筛、de Bruijn – Newman 流、最小随机矩阵间隙及总正性理论都有原有文献。我们的历史稿中“new”通常指相对于当时内部工作流新得到的有限命题，不等于已经做过完整文献检索并确立了首创。后续发表必须单独做 novelty audit。

尤其不能把原稿写作期间独立重建的 0.6725007、公开论文中的 0.8362503，或已存在的矩阵不等式当作本项目首次证明的世界性纪录。

3. 符号、尺度与容易混淆的对象

用户所说 ACOE 的核心资料实际使用 ACUE：**Alternative Circular Unitary Ensemble**。COE 通常指 circular orthogonal ensemble，属于不同的对称类；不能把两者互换。除非另有明确的 $\beta=1$ 定义，本档案统一写 ACUE。

符号	本档案含义	必须保留的限定
CUE	Haar 随机酉矩阵的特征角过程	局部极限为 $\beta=2$ sine process
ACUE	在 `Z/(2N)` 上取 N 个点、按 Vandermonde 平方加权	离散 $\beta=2$ 模型；不是 $\beta\to\infty$ 的定义

符号	本档案含义	必须保留的限定
AH	零点归一化间隙集中到半整数的替代情景	必须注明强硬核、近似、密度或子序列版本
`p_k(C)`	`Σ_{x∈C} exp(πikx/N)`	Fourier 正负号影响复相位恒等式
`p_λ`	`∏_j p_{λ_j}`	平衡矩的加权次数是 `
`D(P)`	有限圆周多项式向后热流的首碰撞时间	不是 ζ 的全局 Newman 常数
`δ_min`	初始最小圆周角间隙	角度、平均间隙归一化及 ζ 高度尺度不可混用
`H_m`	`liminf (p_{n+m}−p_n)`	H_1 是相邻素数间隙；H_2 是三个素数跨度
`M_k`	指定 Maynard 单纯形试验类的变分常数	换支撑、估计或源域以后不再受原类上界约束

ζ 零点在高度 T 附近通常用平均间距 `2π/log T` 归一化；CUE 用 `2π/N`。把角度不等式转成 ζ 结论时必须明确窗口、边界、平移平均及误差，不能只作 `N≈log T` 的符号替换。

4. 必须先修正的结论

这些修正不是附带小问题。它们决定下一轮是否会沿着不成立的桥梁继续推演。

历史说法或风险	当前判断	修正后的可用内容
0.6725007 是所有 bandwidth-one 方法的绝对天花板	撤回；后稿已明确纠正	它是特定两项矩/窗口优化的常数；更广泛障碍需要独立定理
0.702642 的 Nyquist 界改进了 ζ 的 0.6725	不成立	两者定义域与算术输入不同；前者是标记格点模型
finite fiber 越来越大，所以无限过程不唯一	缺关键步骤	要有固定局部可观测量上的一致分离与可传递矩条件
三点 rigidity 已对所有 N 证明	来源不支持	小 N 精确/数值结果保留；all-N 命题列为待证
N=8 fiber dimension 已定为 403	来源内部冲突	一处报告 399；需在统一约束、对称群、精确秩下重算
COM 调制族从 N⩾4 即有 N−3 维	N=4 反例	闭次数⩽N 约束下 N=4 为 0 维；N⩾5 才符合该家族表述
ACUE 的平均 depth 约为有限常数/N^2	无条件说法错误	clock 原子使每个有限 N 的无条件均值无穷；有限表格需条件化
最小间隙控制 depth 的双向等价	一般不成立	两体给下界；上界需要背景控制；clock 是反例
background 系数可由单端点 $\csc^2$ 控制	原上界错误	用精确差商、整段上确界或积分式

历史说法或风险	当前判断	修正后的可用内容
ACUE 是 β=∞ 的 CβE	不成立	β→∞ 固定 N 退化到 clock；ACUE 仍有随机占位
marked response 有无条件件 exact rank-two law	过强	链式分解精确；修正项一般可有全谱秩
所有 Lean 步骤已经 machine verified	不成立	未编译、有两个实际 `sorry`；`depth_ge` 结论不涉及 D
P9 JSON 已是严格 interval 证书	提交代码不足以支持	高精度 mpmath 与 SAFE 因子不是向外舍入证明
signed no-go 排除所有 signed sieve 改进	超出证明类	仅在给定可计算观测和闭合试验类中成立
186 是 H_2 /H_3 记录的直接延伸	不是同一个目标	186 属于 H_1 ；机制是另一套扩大支撑/互补分解结构

5. 0.6725007：我们真正理解了哪一层

5.1 两项矩窗口的连续最优化

在区间 `I=[-1/2,1/2]`，考虑归一化 `∫_I v=1` 的二次型

$$q(v) = \int_I v(s)^2\,ds + \int_I\int_I |s-t|\,v(s)v(t)\,ds\,dt.$$

对应优化解为

$$v_*(s) = \frac{\cos(\sqrt{2}\,s)}{\sqrt{2}\sin(1/\sqrt{2})}, \qquad q_* = \frac{1}{2} + \frac{1}{\sqrt{2}}\cot\frac{1}{\sqrt{2}}.$$

因此

$$\delta_{\mathrm{MT}} = 2 - q_* = 0.6725007036794115\ldots.$$

这给了我们一个非常明确的靶：仅仅在同一二次型里换更复杂的窗口，不会超过该优化值。若要改进，必须改变可用信息、证书类别或算术输入。历史稿把这个局部最优推广为所有 bandwidth-one 方法的绝对最优，后来已自行撤回。

5.2 精确有限 Galerkin 公式与本次修复

对 n 个等长格子作分片常数离散化，令

$$a_n = 1 + \frac{1}{3n^2}, \qquad \theta_n = \arccos\left(1 - \frac{1}{n^2 a_n}\right).$$

则有限优化值为

$$q_n = \frac{1}{2} + \frac{a_n n}{2}\sin\theta_n \cot\frac{n\theta_n}{2}.$$

原 `verify_codex.py` 写成 `acos((1-n^(-2))/a_n)`，与正确公式不同。这导致闭式和渐近系数两项测试失败。本次独立审计将正确闭式与矩阵最优化在 n=2,3,5,8,13,21,40,80 比较，最大误差约 `4.49×10^(-14)`。这是数值回归检查，不代替原稿的有限证明。

令 `t=1/√2`，有

$$\delta_n = \delta_{\mathrm{MT}} - \frac{\csc^2 t - \sqrt{2}\cot t}{24n^2} + O(n^{-4}).$$

使用 60 位高精度核对后，剩余项乘 n^4 趋近约 `-0.02650666`（对应 `q_n-q_*-c/n²` 的符号约定）。这个小修复具有实际价值：未来做严格有限证书时，离散误差必须有可信的尺度和符号。

5.3 惯性、秩与重数的 secant 结构

共享上下文记录的有限谱证书把 on-line 零点贡献写成正半定和 `P=Σ m_j v_jv_j*`，把 off-line 配对部分放入 Hermitian 修正 Q，并用 Q 的正惯性数量控制损失。核心 clipped 函数为

$$k_c(m) = c^2 - ((c-m)_+)^2.$$

`c=2` 时它把简单与多重零点区分为 3 与 4；`c=3` 给 5、8、9 的分层。对适当的正交原子与配对块，证书可以取等，所以只持有 trace、Hilbert – Schmidt norm、rank 和 inertia 的论证确有内在信息限制。这里的等号模型只证明该证书的限制，不能自动证明所有显式公式方法的限制。

5.4 cubic 与更高 trace 的尝试

第三轮 D1 报告在 MT 最优窗口上得到三阶量 `Φ₃≈-0.0117753128`。它的吸引力在于：如果能独立证明适当归一化负谱的统一范数上界，就可能把两项矩证书推进到更高信息阶。历史数值给出在某些额外上界下约 0.6796896 或 0.6844924 的条件输出。

这些数值不是已证明的 ζ 改进。尚缺的内容至少包括：三阶 trace 的完整算术计算、带宽与对角项账本、负谱统一控制、重数与 off-line 部分的重新解码。用 taper 在端点消失不能消除近零点簇造成的 `O(log T)` 行和；目前没有得到所需的统一常数。

共享上下文还记载 degree 10/12 的 Christoffel forced-rank 指标、精确连续矩、Gauss – Radau 对手、奇偶 Schur 层级和 `[0,3]` hard-edge 猜想。这些是值得保留的历史计算线索，但原始全部证明与计算文件不在当前分支，故不把其 80% – 95% 等数字列为 ζ 零点定理。

6. Nyquist 结构：当前最清楚的有限定理包

6.1 为什么必须引入标记

在 `Z/(2N)` 的 2N 个位置上允许整数标记 `m_x≥0`，总质量为 N。令

$$p_k = \sum_{x=0}^{2N-1} m_x e^{\pi i k x/N}, \qquad V_k = |p_k|^2.$$

只要求开放的低频行 `EV_k=k`，`1≤k<N`。这与完全二元占位 ACUE 是不同的可行类。Parseval 与 Fourier 对称性给出精确关系

$$\mathrm{E}\sum_x m_x(m_x-1) = \frac{\mathrm{E}|p_N|^2 - N}{2N}.$$

因此全部重数缺陷都汇入 Nyquist 模式。若再关闭 `k=N` 这一行，左侧非负且期望为零，便强制每个标记几乎处处是 0 或 1。这是一个干净的结构定理，也解释了为什么不同约束集会得出完全不同的“简单比例”。

6.2 简单占位数的精确 slack

令 `s₁=# {x:m_x=1}`，并定义 `h(j)=0` （j⩽2）、`h(j)=j(j-2)` （j⩾3）。原稿得到

$$\mathrm{E}s_1 - \left(\frac{N}{2} + \frac{\csc^2(\pi/(2N))}{2N}\right) = \mathrm{E}\sum_x m_x m_{x+1} + \mathrm{E}\sum_x h(m_x).$$

两项 slack 都非负。它们的 Fourier 组合含因子 `1+cos(πk/N)`，在 Nyquist 点精确消失。于是

$$\frac{\mathrm{E}s_1}{N} \geq \frac{1}{2} + \frac{\csc^2(\pi/(2N))}{2N^2} \longrightarrow \frac{1}{2} + \frac{2}{\pi^2} = 0.702642367284\ldots.$$

这个结果精确说明该标记模型中可用信息的后果。它没有为 ζ 提供新的无条件 70.26% 结论，也没有反驳 67.25% 的公开结果。literal ACUE 的占位本来就是二元的，其 `s₁=N`。

6.3 等号面与失败的提升

等号要求所有标记属于 `{0,1,2}`，并且相邻位置不能同时非零。该等号面的配置数是 `binom(2N,N)`。将配置变为边电荷

$$q_x = m_x + m_{x+1} - 1$$

得到由 `(-1)`、`(0,0)`、`(+1,+1)` 拼接的循环 tile 语言。若 Fourier 约定为负指数，则

$$\hat{q}(k) = \left(1 + e^{+\pi ik/N}\right)\widehat{m}(k), \quad k = \square\, 0.$$

原验证脚本使用负 Fourier 指数却写负相位因子，因而失败；本次改正后的最大误差约 `2.81×10^(-14)`。

N=3 已出现重要障碍：可提升电荷谱只有 `(0,0),(3,9),(9,3)`，而把普通二元 DPP 样本 `111000` 直接滤波会产生 `(12,0)`，不属于所需可提升结构。另一种独立添加装饰的尝试也失败，因为需要的交叉项是特定负相关，而独立化会把它消掉。

这个失败告诉我们：正确二点谱并不自动存在一个满足整数组合约束的正概率提升。all-N Nyquist 等号律仍是开放的有限组合概率问题。它可以成为严肃论文结果，但不能用它代替主数论目标。

6.4 covariance、内接球与修复

在 ACUE 下，开放行特征满足

$$\mathrm{Cov}(V_k, V_\ell) = k^2 \mathbf{1}_{k=\ell} - 2(k+\ell-N)_+, \qquad 1 \leq k, \ell < N.$$

历史证明给出最小特征值 1。原验证脚本在 N=2,⋯,6 复核通过。配合特征幅度控制，可推出矩多面体在目标点附近至少有半径 `1/(N²−N/2)` 的内接球。于是若在等号面上先找到矩误差 `o(N^(-2))` 的近似律，就有机会用少量混合恢复精确矩，同时只产生 `o(N)` 的 slack。

这里已经把一个难题压缩成了清楚的误差目标。接下来应寻找达到该精度的构造或证明它不可能；只做误差为 `o(1)` 的模拟还没有达到修复定理的输入。

7. ACUE moment fiber 与显式反例

7.1 概率与矩约束

设 C 是 `Z/(2N)` 的 N 元子集，`z_x=exp(πix/N)`。ACUE 概率为

$$\mu_N(C) = \frac{1}{(2N)^N} \prod_{x<y\in C} |z_x - z_y|^2.$$

其核是 N 个连续 Fourier 模式的正交投影。对平衡分拆 `|λ|=|ν|≤N`，矩值等于 CUE 的 `δ_{λν}z_λ`。固定这些矩后，所有非负概率律组成一个 convex fiber。研究对象是这个完整概率 fiber，而非仅仅 DPP 参数空间。

历史计算在旋转轨道坐标下报告 N=3,⋯,9 的维数为 `0,0,2,10,80,403,1804`。N=5 有 `Q(√5)` 上的精确多边形描述；更大 N 的某些秩来自浮点 SVD。由于 N=8 还出现过 399 的报告，下一轮应统一坐标和约束后重算，不能把整个序列都标为精确证明。

7.2 两点冻结、三点证据、四点自由

二点 rigidity 的有限证明来自完整周期 Fourier 对称与低频别名关系。三点统计在小规模精确/数值实验中也被冻结，但当前资料没有完成 all-N 定理。四点首先出现可见自由：循环局部 word `01010101` 的历史最大变化在 N=6,7,8,9 约为 `0.01019,0.01232,0.01375,0.01514`。

这组数值很值得进一步研究，因为它观察的是固定长度局部量，而不只是维数。然而四个尺寸的正值或上升趋势不是 `liminf>0` 的证明。也不能把 fiber 的增长称为“超指数”：全部配置本身至多 `binom(2N,N)≤4^N`。

7.3 质心调制家族

定义质心模 N 为 `X(C)=Σ_{x∈C}x mod N`，考虑

$$q_g(C) = \mu_N(C) g(X(C)), \qquad g \geq 0.$$

在适当归一化下，N⩾5 时去掉 `g` 的 ±1 Fourier 分量便得到一个 `N-3` 维显式家族。其意义是把巨大的线性规划 fiber 中的一部分写成易解释的概率公式，而不是依赖一个不可读的 nullspace 向量。

本次对 N=4,5,6 直接枚举所有配置与全部闭次数⩽N 平衡矩。得到调制空间维数 0、2、3。N=4 的候选 `g(x)=1+0.5 cos(4πx/N)` 在 `λ=ν=(2,2)` 上产生恰约 2 的误差；N=5、6 相同测试族的最大误差分别约 `1.54×10^(-15)`、`6.97×10^(-14)`。这同时证明了 README 中 N⩾4 版本的边界错误，并支持修正后的 N⩾5 表述。

7.4 Fermi sea、相位叠加与可见性代价

质心乘法对应 Fourier Slater 态的平移。更一般的试验可写为

$$q_a(C) = \mu_N(C) \left| \sum_s a_s z(C)^s \right|^2,$$

其中系数自相关决定哪些交叉项出现。两片平移 Fermi sea 之间的可见性代价为 `δ(N−δ)`，最大约 `N²/4`。这提供了构造高阶矩不可见变形的方法，也联系到量子纠错式的 scalar compression 条件。

若对所有允许观测 D_j 有 `P_V D_j P_V=b_jP_V`，则任一单位向量 v∈V 的 Born 概率 `|v(o)|²` 是真实非负律，并满足这些矩。这里可以借用 Knill–Laflamme 的结构语言，但必须始终写明观测类；不能声称对所有操作都存在同样纠错能力。

7.5 parity sector 的强分离与弱极限陷阱

历史构造 `q_±=μ_N(1±(-1)^(ΣC))` 给出互相奇异的两律，同时在许多低阶边缘统计中不可区分。对于偶数 N，两种 clock 配置都落在正 parity sector，故正 sector 有 depth 无穷原子、负 sector 没有。

ACUE 的两个 clock 总质量是 `2^(1-N)`；正 sector 的对应质量是 `2^(2-N)`。它们在每个有限 N 给出清楚分离，但质量随 N 指数趋零，因此不能由这个事件直接得到无限体积极限的非唯一性。这个例子应成为所有"有限反例自动传到 ζ"推理的回归测试。

7.6 有限反例传到无限过程，究竟缺什么

一个可用的转移路线至少需要：固定强度的平稳化；局部计数紧性；在所选所有测试函数上的相关测度收敛或一致可积性；与目标带宽域严格匹配的矩恒等式；以及某个固定有界局部观测的差保持 `≥ε>0`。

最后一个条件排除了"维数很大，但所有固定局部窗口都看不见"的情况。前几个条件排除了质量逃逸、逐 N 改变测试函数及边界层问题。若格点占位始终二元，计数控制相对容易；若允许标记重数，必须重新检查矩与尾部。

Lagarias–Rodgers 的全阶 band-limited 条件与 Rudnick–Sarnak 风格的总带宽约束不是相同区域。逐坐标的 Fourier 立方体与 `Σ|ξ_i|` 限制不能互换。因此一个满足平衡次数≤N 的有限 family，必须逐个确认它传到的是哪一个无限问题。

8. 其他反例为何值得保存

8.1 Form factor 在投影 DPP 类中的刚性

对频率占位集 S 的平移不变投影 DPP，有

$$F(\gamma) = \frac{1}{2}\ |\ S \triangle (S+\gamma)\ |\ .$$

特别地，`F(1)` 是 S 的 occupied runs 数量，所以 `F(1)=1` 在该类中识别连续 Fermi sea。这个结果说明额外结构假设可以大幅降低识别难度。但如果没有先证明目标过程是这类 projection DPP，就不能用这个判别把一般 mimicker 排除。

8.2 Homometry：完整二点信息仍不够

在 `Z/12` 上，频率集 `{0,1,4,6}` 与 `{0,1,3,7}` 有相同 covariogram，却不是平移或反射；相应 DPP 在三点事件 `{0,1,2}` 上的概率分别为 `(8−√3)/432` 与 `(8+√3)/432`。原验证脚本复核通过。

另一个带标记例子是

`u=(0,0,0,0,0,2,0,0,0,0,2,2)`，

`v=(0,0,0,0,1,1,0,0,0,0,1,3)`。

它们总质量均为 6，完整循环自相关都为 `(12,4,0,0,0,4,8,4,0,0,0,4)`，所有 Fourier 强度非零，但简单位置数为 0 与 3，立方和为 24 与 30。这个反例非常直接：完整 diffraction 也不恢复生成元或重数分布。历史稿还提出 Fibonacci 非周期提升；该部分本次未重建全部证明。

8.3 Magnetic holonomy

对 g-cycle 的 Hermitian kernel `K_φ=aI+rH_φ`，低于 g 阶的 principal minors 看不见环通量，完整概率律的总变差距离满足

$$d_{\mathrm{TV}} = (2r)^g \mid \cos(g\phi) - \cos(g\psi) \mid .$$

原验证在 g=3,4,5 通过。它清楚展示了“第一次可见的相关阶数”等于一个几何回路长度。与此同时，即便看完整概率律，通量取向的某些符号仍可能被余弦隐藏。该模型适合作为高阶统计的必要性证据，不能直接认定 ζ 存在同样的 magnetic representation。

8.4 Deck、局部观测与大 fiber

固定 M 个循环位置和 L 个占位时，r-deck 的旋转不变 fiber 维数公式可表达为 necklace 数之差。在通常的 `r≤min(L,M−L)` 范围内使用相应 inclusion-matrix 满秩结论。原验证复核了五组小规模参数。

这些例子与 parity sector 共同说明：大 nullspace 不等于局部可见自由。设计新反例时应同时提交其局部 observable 的变化幅度，而不只提交 rank 或维数。

8.5 Wilson 第四矩与边界条件

对没有额外长度四绕行圈的开放方格图，磁邻接算子的第四迹满足

$$\operatorname{tr} A_0^4 - \operatorname{tr} A_\phi^4 = 16 \sum_f w_f \sin^2(\phi_f/2).$$

原脚本用了小周期盒，存在额外绕行闭路，导致公式失败。本次改用开放 4×5 方格，得到左右两边同为 `77.67769987482622`。经验教训是：图论 trace 枚举的边界是定理假设，不是实现细节。

8.6 其他已探索但未接到算术的结构

历史材料还包括 Pontryagin 空间中的有限扩张、隐藏负特征值、Prouhet 型低矩伪装、Euler – Lee – Yang rank 障碍以及 Hurwitz 刚性。它们共同考察“多少谱信息能确定多少正性”。当前没有完成从这些有限结构到 Connes 型 ζ 算子或真实素数标记谱的证明。完整原稿保留在附录；下一阶段若使用其中某条，应先精确写出 arithmetic realization，而不是只引用类比。

9. 第三轮十方向：逐项进展与失败价值

以下按 `round3_synthesis.md` 的方向顺序重建。每个方向的数值是历史记录；除前文特别说明外，本次未完整重跑原子代理实验。

方向	得到的东西	没有完成的东西	下一轮应保留的教训
D1 cubic/负谱	MT 窗口三阶量、条件改进常数	uniform 负谱 cap 和完整算术 trace	多一个矩只有在代价与尾部受控时才有用
D2 带宽一最优	某些更大证书类的 LP 候选约 15/22	全局最优证明、严格 witness	不把一个优化器输出当成 universal ceiling
D3 首个带宽外 detector	`a+b=N+1` 的 `E	p_a p_b	², 成本约 1+1/N`
D4 缺陷恒等式	七项 deficiency、近等号刚性	额外算术解释	总缺陷在 fiber 上冻结，内部切片仍可能变化
D5 端点	点值的盲区、Cesàro cell 的可能信息	端点均匀估计	测试一个点与测试非零宽度区间有本质差异

方向	得到的东西	没有完成的东西	下一轮应保留的教训
D6 COM family	显式正概率调制与 N=5 几何	全 fiber 分类、无限局部分离	显式 family 比纯数值 nullspace 更可迁移
D7 distinct/simple	distinct 解码及约 0.8362503 基线	附加正项的完整独立证明	改变计数对象也要对照现有文献
D8 derivative tower	cot/csc² 结构和若干秩计算	归一化、N=8 秩冲突	差一个归一化会改变 purported relation
D9 holomorphic derivative	某些加权量冻结，径向临界点可区分	ζ 对应物的算术矩	普通 P′ 与 centered circular derivative 不同
D10 reflection/高阶	N=7 去掉 39/80 手性方向；低 N detector	all-N 阈值与 shifted-correlation 输入	squared-modulus 信息仍不一定满秩

此外，Christoffel trace hierarchy 和 scalar moment code 来自前几轮共享材料，不应硬归入某个当前找不到原稿的子代理。已保存的上下文提供其出处线索，缺失文件清单则说明哪些结论需重新证明。

10. 零动力学：哪些已经成立，哪些是关键缺口

10.1 有限向后热流

若初始多项式 `P(z)=Σ_{j=0}^N a_j z^j` 的根都在单位圆上，采用

$$P_s(z) = \sum_{j=0}^{N} a_j e^{sj(N-j)} z^j, \qquad s \geq 0.$$

在首个重根形成之前，根角满足

$$\theta_j'(s) = -\sum_{k \neq j} \cot \frac{\theta_j - \theta_k}{2}.$$

令 D 为正向 s 下第一次碰撞的时间，未发生碰撞则 D=∞。这是一个有限 de Bruijn – Newman 类指标。它测量给定多项式距离失去简单圆周根的“时间深度”；它与 ζ 的全局 Λ、有限窗口裁剪和实直线热流之间仍需独立桥梁。

10.2 两体下界的确定性核心

相邻间隙 g 的其他根贡献具有延缓碰撞的符号，因此

$$g' \geq -2\cot(g/2).$$

对 `0<g_0<π`，纯两体方程的碰撞时间是

$$D_2(g_0) = -\log\cos(g_0/2) \geq g_0^2/8.$$

由最小初始间隙 δ 的 scalar comparison 得到

$$D \geq -\log\cos(\delta/2) \geq \delta^2/8.$$

这里应把 scalar solution 从最小初始间隙启动，再与各个间隙比较；不能对大于 π 的某个长间隙直接写实数 `-log cos(g/2)`。对于非 clock ACUE，最小间隙是 `π/N`，故

$$D \geq -\log\cos\frac{\pi}{2N} \geq \frac{\pi^2}{8N^2}.$$

这项下界是可保留的主要确定性工具。它的方向很重要：知道 g 小并不能推出 D 小。

10.3 clock 原子与均值

clock 多项式只有首尾项，热流因子不改变它，故 D=∞。ACUE 两种 clock 总概率 `2^(1−N)>0`，所以任何有限 N 都有 `ED=∞`。

本次枚举 N=4,5,6，clock 质量分别为 1/8、1/16、1/32，符合公式。凡历史表格给出有限 mean depth，都必须明确它是在去除 clock 后条件化、有限截断，还是用其他统计量。历史 N=6 fiber 线性规划报告条件 `E[N²D]` 在约 `[1.3610,1.4770]` 变化，clock 质量也可从基准 0.03125 改到约 0.0975；这些应作为条件化实验保留。

10.4 局部化上界：修正后的真正任务

原稿试图用初始背景点的 csc² 和控制局部差商，但所用单端点上界不成立。设某背景相对角为 x，配对间隙为 g，则准确的差商是

$$B(x,g) = \frac{\cot(x/2) - \cot((x+g)/2)}{g} = \frac{\sin(g/2)}{g\sin(x/2)\sin((x+g)/2)}.$$

当 `x=2π−0.15`、`g=0.05` 时，B 约 133.500132，而原候选上界 `0.5 csc²(x/2)` 仅约 89.055743。该例即使要求 g 不超过附近其他间隙，也能出现。正确控制应使用整个角区间上的导数上确界或上述精确式。

更关键的是，这个控制要沿着直到碰撞的时间窗成立，而不仅在 s=0 成立。若某短间隙在整个相关时间窗内满足

$$g' = -2\cot(g/2) + B(s)g, \qquad 0 \leq B(s) \leq B_{\max},$$

且 g 始终小，则取一个有效常数 C 使 `2cot(g/2)≥4/g−Cg`，可得 `y=g²` 满足

$$y' \leq -8 + 2(B_{\max} + C)y.$$

当 `(B_max+C)δ²<4` 时，比较方程给条件上界

$$D \leq -\frac{1}{2(B_{\max}+C)}\log\left(1 - \frac{(B_{\max}+C)\delta^2}{4}\right).$$

如果别的点先碰撞，只会使全局 D 更早；但在所追踪间隙上使用此式，要证明控制在该时间内保持。于是 `B_max δ²=o(1)` 才能把上下界压到 `D=(1+o(1))δ²/8`。这是真正需要证明的局部化引理，而不是已经完成的“二体定律”。

10.5 随机矩阵最小间隙与不合法的外推

CUE 最小角间隙尺度为 `N^(−4/3)`。按常见圆周归一化，其极限尾律涉及 `exp(−x³/(72π))`；使用其他展开尺度会改变常数。Feng – Wei 所引用推广的 β 范围是正整数，不能擅自写成所有 β>0。Feng – Wei

如果再有上一节的高概率局部化，才可推断 CUE depth 的典型尺度 `N^(−8/3)`。历史模拟支持这种想法，但最小间隙定理本身没有证明 depth 定理。

另一个常见错误是把固定 x 的弱收敛尾律直接代入随 N 增长的 `x=πN^(1/3)`，宣称某事件以 `exp(−π²N/72)` 衰减。没有 moderate-deviation 估计不能这样做。仅由紧性可以得出对应概率趋零，不能附带这个指数速率。

10.6 单缺陷极限不是整个 ACUE 的极限

历史研究对一个特定单缺陷配置写出

$$P_N(z) = \frac{(z-1)(z^N+1)}{z-e^{-i\pi/N}}$$

并得到缩放极限函数

$$G_s(u) = 2\cos(u/2) - 2\pi\int_0^{1/2} e^{s(1/4-y^2)}\cos((\pi+u)y)\,dy.$$

其首次双根的历史计算约为 `s*=1.419640342`、`u*=1.812942145`，相应 `8s*/π²≈1.150717118`。它提供了一个可研究的显式局部模型，而不是 ACUE 全体样本的普遍常数。旧表中的 ACUE median 在 N=8,9,10 已分别约 1.418216、1.41520、1.412774，不能强行拟合到上述单缺陷常数。

10.7 算子统一带来的结构

由核 `1/(2sin²(πk/N))` 形成的循环算子，其 Fourier 特征值为 `δ(N−δ)`。同一谱既出现在 Fermi sea 交叉项的可见性代价，也出现在 clock 附近零动力学的 Jacobian。这是现有材料里最有启发性的结构连接之一。

符号必须明确：在向后热流中它描述不稳定增长，不能称为趋于平衡的 relaxation。对固定 Fourier mode，`N^(-1)L_N` 的极限是 `|D|` 型半拉普拉斯/Poisson 生成元，而非普通 Laplacian。与 Casimir、基本权或 affine Weyl 的关系需要指定表示和归一化，不能凭谱形状宣称所有对象规范同构。

11. Marked response、导数与 LR 桥梁

11.1 为什么给零点加标记

无标记点过程会忘记本征向量、边界条件和生成元。若有 Hermitian G 与一个指定向量 u，可对 `G+ηuu*` 作 rank-one 扰动，再经 Cayley 变换得到圆周根。首碰撞时间的导数是一个可能比普通 gap 更敏感的标记统计量。

但是对 ζ 来说，最难的是指定一个自然、可计算且不依赖待证结论的 G 与 u。任意把零点塞进对角矩阵并附加向量，只制造了额外数据，不能声称找到 ζ 的真实 random-matrix 结构。

11.2 “exact rank–two law”的正确读法

写 `D=κδ²/8`，微分自然给

$$D' = \frac{\kappa\delta}{4}\delta' + \frac{\delta^2}{8}\kappa'.$$

这是精确链式法则。若只保留首先碰撞那一对的 gap response，会出现 rank-two 外形；但 $\varkappa'$ 可能依赖所有本征角，故完整响应一般不是 rank two。把 $\varkappa$ 从同一 D 的输出反算后再得到 correlation=1，验证的是恒等重写，不是独立预测能力。

对 `Cayley(λ)=(λ−i)/(λ+i)=e^(iθ)`，本次符号计算得到 `dθ/dλ=+2/(1+λ²)`。若用相反 Cayley 约定符号会变；不说明约定直接沿用负号会使所有 marked coefficients 反向。

11.3 局部横截性尚非全局分离

历史实验显示 marked/depth 梯度在矩约束行空间外可能有残量。这是好的诊断：该统计局部能看见某些 fiber 方向。要把它变成实际分布分离，还要沿该方向保持概率非负，并给出有限幅度、误差受控的差。要用于 ζ，又多一层算术可观测性。三件事不能合为一个“已解决识别”的结论。

11.4 LR 与 AH：不要偷换量词

若某模型具有真正的统一硬核下界 `δ≥2πc/N`，两体比较给

$$D \geq \frac{\pi^2 c^2}{2N^2}.$$

因此一个独立的严格 depth 上界可以约束硬核参数 c；反过来，小 gap 上界不能自动给 depth 上界。clock 是最简单反例。

更强的障碍是：在包含 ACUE 的全部 bandwidth-one mimicker 上，不可能证明一个普遍的 `N²D<π²/8` 上界，因为 ACUE 自己满足相反下界，且还有无穷原子。若要证明 ζ 具有更小 depth，必须用 ACUE 不满足的额外算术信息。

还要区分 AH 的“几乎所有间隙接近半整数”与“每个间隙都有硬核下界”。即使发现无限多个很小间隙，也未必排除允许零密度例外的 AH。一个相对稳定的目标是某个离开半整数集合的紧区间内出现正比例的归一化间隙，并证明这与所采用 AH 定义矛盾。涉及 Landau – Siegel 零的推论必须逐条核对原定理量词，不能把任何 AH 反例直接升级成无 Siegel 零定理。

2019 年资料中的 0.606894 涉及正比例小间隙结果，不是一个可直接套用的“LR 普适 μ 上界”。原注释中的历史最小 gap 纪录也不应未经更新就称为今日纪录。Lagarias – Rodgers

12. 形式化与复现审计

12.1 原脚本的真实结果

原样运行 `verify_codex.py`，记录为 39 PASS、4 FAIL、1 SKIP。程序以退出码 0 结束，因为 `ok` 只打印结果而不抛错。不能用退出码 0 或终端最后一句 Done 判断全部检查通过。

四项失败分别是：Galerkin 闭式、其二阶渐近项、charge filter 复相位、周期盒 Wilson 第四迹。GOE counterfactual 常数的核归一化无法由稿件重建，因此原程序自己跳过。该 0.559871849 数值在本档案中不作为已核验数学事实。

本次编写 `audit_research.py`，通过 21 项限定范围检查，输出 JSON 和文本日志。它修复了前三类可定位的验证问题（Galerkin 两项属于同一公式问题），枚举 N=4,5,6 的 COM 与 clock 质量，验证两个 tuple 的 admissibility，检查 Cayley 符号、background 反例和 Lean 结论。注意，其中若干 PASS 的含义是“成功检出了历史错误”，不是“历史定理通过了证明检查”。

12.2 Lean 文件比 README 更弱

当前 Lean 草稿未编译。有两个实际单独 `sorry`：一个在 `neg_log_cos_ge` 的单调性积分步骤，一个在 `two_body_solution`。更严重的是 `depth_ge` 的最终命题只为

`δ²/8 ≤ −log(cos(δ/2))`，

没有出现 D。其 `hD : ∀i, g i D=0 → True` 也没有定义首次碰撞。故即使该文件某天无错误编译，当前这个 theorem statement 仍没有表达研究所需的 depth 下界。

共享 README 把它描述为两个 ODE `sorry`，与文件本身不一致。下一轮第一步应修 theorem statement 与停止时间定义，再写 proof。形式化必须检查“证明了什么”，不能只检查是否没有 `sorry`。

12.3 Sympy/Z3 的证据边界

`verify_theorem_steps.py` 与同系列文件有一些有效恒等式检查，但也存在直接 `ok(..., True)`、有限 Taylor 系数、以及先假设目标符号再检查否定不可满足的步骤。这些不能合成“所有解析步骤机器证明”。符号程序可以验证代数恒等式，SMT 可以验证准确编码的命题；ODE 存在性、首次碰撞、全时间比较及极限交换必须另证。

12.4 可复现性缺口

若干脚本硬编码旧 `/tmp/claude-0/.../scratchpad` 路径，依赖当前未提交的模块或数据。打包 README 提到的部分 `data/` 不存在。H_4 的 56,000,000 维 profile 没有对应 NPZ；H_3 的完整 tuple 在顶层而不在所有子包位置。早期 `[FSC]`、`[A23]`、`[FSI]` 等引用并不都有可取回原文。

本档案保留原文件，另外提供独立审计，而没有静默改写历史证据。未来可单独做一个复现修复 PR：相对路径、固定依赖、assertion 退出码、精度声明、数据哈希与一条完整命令。现在不需要为此改动 Alpha-devbox 控制器代码。

13. 内部素数间隙工作：H_2、H_3、H_4

13.1 必须先辨认目标

`H_m=liminf(p_{n+m}-p_n)`。H_1 是两个相邻素数的间距；H_2 是三个素数的跨度。历史公告标题中把 H_2 描述成“中间有两个素数”的措辞不准确。内部 H_2 /H_3 /H_4 数值与外部 $H_1 \leqslant 186$ 不直接比较大小。

目标	历史候选上界	k	历史报告的变分下界
H_2	173438	15856	8.013326752751
H_3	13859802	923601	12.006666706750
H_3 旧备份	14505780	较早参数	见原稿
H_4	1120662828	56000000	16.0655 左右

这些是仓库中的内部候选成果，本次没有把它们认证为新的公开世界纪录。原因既有严格数值证明的缺口，也有最新文献比较尚未完成。H_2 tuple 的 admissibility 已由本次程序对所有 $p \leqslant k$ 完整检查；这只核对了组合输入，不等于核对整个筛法不等式。

13.2 P9 引擎的有价值结构

在标准单纯形上试验

$$F(t_1,\ldots,t_k)=\prod_{i=1}^{k} g(kt_i)\,\mathbf{1}_{\sum t_i\le 1}.$$

令 `c₂=∫g²`，独立随机变量 X 的密度为 `g²/c₂`，`S_k=X₁+⋯+X_k`，`G(u)=∫₀^u g`。则可将高维 I、J 化为一维截断和概率：

$$I_k=k^{-k}c_2^k\,\Pr(S_k\le k),$$

$$J_k = k^{-k-1} c_2^{k-1} \mathrm{E}\left[G((k - S_{k-1})_+)^2\right].$$

由 layer-cake、Berry – Esseen、Chernoff 或 one-big-jump 等尾部估计得到下界。用 `Pr(S_k≤k)≤1` 放大分母给保守下界是合法的；把每一段尾部误差保守地计入同一 ratio 才有证书意义。

这套分解把很高维试验压到可计算的一维概率对象，是应保留的工具。它的优势在于结构和可认证性，而非自动穿越标准 simplex 的上界。

13.3 严格证书尚未闭合

`p9_certify_hp.py` 使用 mpmath 约 50 位精度，并乘 `SAFE=1+10^(-30)` 一类因子。高精度输出不自动是 rigorous enclosure；任意安全因子也不证明所有积分、特殊函数和截断误差都向外舍入。JSON 中的 `lb` 字段名同样不是证明。

因此下一轮若要公布 H_2 /H_3 /H_4 结果，需要采用真实区间算术或精确有理上下界，验证所有尾部与积分误差，并提供独立 checker。还需要复核 tuple 文件、权重 profile、参数及每一步筛法假设确为同一个证书。应保留现有候选的正余量，但不能把尚未验证的最后几位当作已证事实。

14. Signed sieve 与旧路线的 no-go

14.1 凸序与一阶计数信息

旧研究把 prime-count 目标抽象成随机整数与可计算线性/凸上界。两点极端分布说明：如果只有这些数据，许多检测阈值无法改善。这是一个指定信息模型中的最优性证明。它适合用来筛掉重复的变分幻想，但不能约束新引入的分布估计、扩大支撑或可计算交叉项。

14.2 `w_+` 支配的使用条件

把 signed 权重拆成正负部分，常可以得到清楚的 debt identity 或点态支配。若 `w_+` 仍属于可接受、可计算的试验类，这能证明某些 signed 变形没有收益。但若正部化导致支撑、可分离性或算术可计算性失效，就不能从点态比较直接得出同一个归一化变分问题的 no-go。

历史精确价格

$$\beta_* = \frac{23051796480}{10991046857} = 2.097324920903119\ldots$$

与另一无界版本的约 2.03265 应区分。部分 prose 把“价格低于 1 的相变”与这个大于 2 的 β* 混在一起，需要以定义和精确数值为准。本次检查确认 β*>2。固定 signed mass 的归一化、未归一化目标及固定特征类分别是不同优化问题。

14.3 为什么 186 恰好说明旧 no-go 的作用域

标准 simplex 的上界 `M_k≤k log k/(k-1)` 会阻止某些 k≈40 – 50 的原始策略。但 186 不在这个原试验类中：它扩大外层支撑，再用不同内层源域控制 prime marginals 和混合项。因此它不是“推翻数学 no-go”，而是改变了 no-go 的假设。

历史另一条 conditional 路线把分布指数 θ 的提升换成更小 H_1 候选，如 4/7→130、7/12→114、3/5→94、5/8→80。这些均以特定 CRT 变化剩余类和带权源域估计为前提。不能仅因某个 BFI 型定理也出现同样 θ，就认定已经满足这些假设。

15. 新外部进展：它们实际改变了什么

15.1 Anthropic 的 FLT 项目

2026 年 9 月 4 日公开的项目主要成就是把既有费马大定理证明的复杂依赖大规模形式化，涉及约 1300 万行 Lean 与约 29500 个中间结果。它不是新的初等 FLT 证明，也没有直接给出一个可应用于 ζ 或 prime gap 的新解析估计。项目说明，代码仓库

对当前研究的迁移价值在方法上：明确 statement、拆分依赖、将有限证书与解析输入隔离、让独立 checker 验证，并检查最终定理的量词。我们的 `depth_ge` 正说明为什么 statement 审查与去除 `sorry` 同样重要。

15.2 ζ 的 0.6725 进展与最新简化

Anthropic 8 月公开的结果给出至少 `0.6725007…` 比例的零点是简单且位于临界线上，并用有限谱/惯性结构组织证明。这里“简单且在线”与仅“在线”的旧纪录不可随意混用。官方说明，作者论文

本次发现 9 月 3 日 Lamzouri 的新预印本给出该结果的更直接 Hilbert 空间证明，也得到相应 distinct-zero 比例约 `0.8362503`。它是证明简化，不是把 0.6725 又提高成更大的比例。Lamzouri

这改变我们的工作排序：在继续构造庞大的有限谱机器之前，应把每个候选新输入代入更短的 Hilbert 空间形式，判断真正缺少的是否仍是同一个额外矩或 off-line 控制。复杂形式化工具值得用在最终正确且必要的桥梁上。

15.3 用户提到的 Tao 赞许路线

没有足够信息唯一确定用户指的是哪篇文章。Guth – Maynard 的大值估计与零密度改进，以及 Tao 对其计算探索的讨论，是一个可能对应的方向。Guth – Maynard，Tao 的讨论

应把它作为可供核对的相关入口，不假装已经确认用户所指。零密度控制、临界线上简单比例、局部 pair correlation 和筛法剩余类分布分别要求不同的平均与一致性；不能仅因为它们都涉及 ζ 就直接串成推出链。

15.4 186：来源、精确目标与证书边界

用户两个短链接分别解析到 8 月 30 日的 *Short gaps* 论文与配套 abridged research record。论文声称 `DHL[40,2]`，从而 `H₁≤186`。它是当前接手应采用的最新来源基线，而不是继续把 246 当作未更新的基线。论文，研究记录

相关公开仓库的 Lean 工程仍明确列出三个项目输入：两个 Kloosterman 型界和 `physical_integral_bounds`；数值脚本另外依赖指定的 Python-FLINT 环境及 signed convolution 修复。条件化的 kernel acceptance 与“所有解析及数值输入都已由 Lean 从零解除”不是同一件事。这并不否定论文的无条件数学主张，但决定我们能诚实声称重验了哪一层。PrimeGaps186

本次没有重跑其重型区间证书，也没有重编整个 Lean 工程。已独立核验论文所给 40-tuple 的 admissibility 与直径 186；它只证明最后的组合输入。

16. 186 的结构与进一步下降的真实入口

本节以论文中的二次型作为下一阶段工作的接口，避免用旧 simplex 的单一 `M_k>4` 判据替代新问题。下面列的是应读取、重建和优化的结构，不声称本次已经改进任何常数。

16.1 支撑扩大与互补分解

外层 divisor sum A 的平方保持非负，允许比旧内层更大的支撑。prime 侧使用不同内层 B_0 、B_1 控制各自可计算的混合项，保存平方余项和 minorant 的正负部分账目。关键不是简单“使用负权”，而是每一项都有合法 pointwise 比较与适用源域。

在研究记录中，多次失败都来自：把缺失平方项扔掉、忽略 minorant 在合数上的负部分、把不同源域的误差合并成过粗 bound，或把固定剩余类的定理用到随 tuple 坐标变化的 CRT 类。正确方案必须同时通过代数符号和算术适用域两种审计。

16.2 三重 dense divisibility 的角色

论文将 divisor factorization 转成可检查的素因子条件，并利用互补因子结构控制混合模数。真正出现的是 `[d,e]`，不是任意假定互素后的 de。共享素因子改变预算，故后续优化不应以独立乘积启发式代替 lcm 条件。

16.3 正余量很小，因此优化应对准损失

最终支撑恢复条件具有

$$\rho_*(J^-_{\lambda,H} - E^+_O - d_0\beta^+_{\mathrm{old}} - (1-b_h)\beta^+_{\mathrm{new}}) > I^+_H$$

的形式。论文给出的最终归一化正余量大于 `1/50000`。因此一项看似改善理想 eigenvalue 的变形，很容易被支撑修复、cap、尾部或区间误差吃掉。下一轮应直接优化认证后的完整余量，而非只画未扣损失的 Rayleigh quotient。

16.4 可核验的组合末端

当前使用的 40 元组为：

`0, 2, 6, 12, 20, 26, 30, 32, 36, 42, 48, 50, 56, 60, 68, 72, 78, 86, 90, 92, 98, 102, 110, 116, 120, 126, 132, 138, 140, 146, 152, 156, 158, 162, 168, 170, 176, 180, 182, 186`。

本次检查其 40 个元素互异、最大减最小为 186、对每个 $p\leqslant 40$ 都遗漏至少一个剩余类。对 $p>40$，40 个元素不可能覆盖全部类，因此这构成完整 admissibility 验证。

要得到小于 186 的结果有两类末端：证明对更小 k 的 DHL[k,2]，并找到对应更窄 admissible tuple；或在现有 k 与 arithmetic 结构下证明一个更窄的合适 tuple 可以使用。不能预先宣称 39 元组必定给某个未经查证的最小直径，也不能假设当前 40 元组已经在所有意义下最优。

17. 面向著名猜想的研究优先级

17.1 第一优先：固定宽度的 Montgomery 带宽外二点结果

终点。 假设 RH，选定一个固定 `ε>0` 与非负、非零、光滑 `w`，其支撑紧含于 `(1+ε,2−ε)`。在明确的 T 平均与对角项约定下，证明归一化零点 form factor 的加权极限为 GUE 预测值。写成示意式是

$$\int w(\alpha)K_T(\alpha)\,d\alpha \longrightarrow \int w(\alpha)\,d\alpha.$$

这里 K_T 必须在正式任务开始时逐项定义；本式是目标接口，不是假装已经有可引用定理。

为什么它有实质价值：ACUE 在该范围的替代三角 form factor 给 `2−α`，而 GUE 给 1。对固定 w，差为严格正数

$$\int w(\alpha)(\alpha - 1)\,d\alpha > 0.$$

它避免了只在 `1+1/N` 处出现但随 N 消失的微弱差异，也不依赖找全体 GUE 的所有相关阶。它是著名 pair-correlation 猜想中的一个清楚部分。ACUE 被该统计排除，不等于排除所有 AH 型过程；后者还需要对整个允许模型类求 extremal range。

真正缺的引理。 将 K_T 的该部分转成可控制的 prime/Λ shifted correlation、Dirichlet 多项式长矩或另一种等价算术量，并证明超出当前输入的 cancellation。有限反例不能供应这项 cancellation，只能给选择 w 的指导与误差预算。

停止条件。 如果候选“新统计”在算术展开后只是重写了完整未解的 shifted-prime conjecture，且没有更弱、可独立攻击的中间估计，应记录等价障碍，不继续包装成新解法。

17.2 第二优先：186 以下的可认证筛法改进

终点。 证明一个明确的偶数 H<186 是 H_1 的无条件上界；或先证明 DHL[39,2] 等更强筛法结论并连接真实 tuple。这个目标规模比 RH 小得多，但仍是公认的重要素数分布进展，且有相对完整的输入输出链。

优先尝试误差层的改进：支撑违规事件的更紧正覆盖、同坐标与异坐标 marked fragments 的分拆、Young 参数的局部自适应、源域切分位置、试验函数对被扣损失区域的主动避让。所有尝试必须维持每一项的非负上界与 interval certificate。

若 baseline 尚未能原样认证重现，就暂不宣布任何优化胜利。若只提高未认证浮点 quotient，而完整 margin 仍为负，也应视为未成功。

17.3 第三优先：排除精确定义的 AH，而非仅区分 ACUE

终点。 在 RH 或另一套明确假设下，对某个与半整数集合保持正距离的区间 I 证明正比例零点间隙落入 I，或证明相应 lattice-distance 统计有严格正下界。这样才直接针对密度版 AH。

现有 Nyquist、homometry 和局部 fiber 提醒我们，单独 ACUE 的 form factor 不是全部敌手。应先定义 AH-admissible class，再优化 detector 在该类上的上界，最后比较真实 ζ 的下界。这个三步链条中第三步依然是主要算术难题。

17.4 第四优先：LR 非唯一性或定量刚性

终点。 构造两个确实不同的平稳半格点过程，在目标论文指定的全部 band-limited n-point 测试中相同；或者证明在附加自然条件下唯一。

此目标与现有 finite fiber 关系最直接，适合拿来验证新的方法是否真的能穿过 N→∞。但应诚实定位：它是重要的 random-process inverse problem，不自动是与 RH 等级相当的历史大定理。选择它，应因为它解除主路线中的信息障碍，而不是因为它容易被夸大。

17.5 暂不列为首轮终点的目标

直接证明 RH、全 GUE、孪生素数，或从任意 PSD/Wronski 表示推出 ζ 的全部零点性质，目前都缺少更大的算术桥梁。可以保留为远期终点，但不应把一个有限模型的未证局部化定理当作它们的主要剩余步骤。

18. 更开放的探索方向：不预先限制在旧模型内

18.1 用反例设计“必须增加的数据”，再寻找算术实现

把所有已知模仿者作为对抗样本库：ACUE、COM 调制、parity、homometric DPP、magnetic cycles、Nyquist marked equality candidates。对一个候选统计，输出它在每类上的值域、可见相关阶、带宽成本和归一化强度。

然后反向问：ζ 的哪一种已有表达自然包含这个统计？可能是 ξ′ 的 logarithmic derivative、平滑 resolvent、有限 Euler product 的加权 trace、或带 prime-label 的 mixed correlation。只有找到真实 arithmetic definition 才投入大型计算。这个方法的价值在于减少对“看起来像 random matrix”的任意构造投入。

18.2 从局部下界转向正密度的动力学证据

最小间隙是极端统计，很容易被零密度异常控制。可以改研究多个短区间上局部 collision-depth 的分位数或平均的有界函数，如 `min(N²D,L)`。它们避开 clock 导致无穷均值的问题，也更接近排除密度 AH 所需的正比例结论。

难点是有限窗口边界会改变动力学。研究者需要定义带外部背景的局部模型、给出截断稳定性，并证明统计对窗口选择鲁棒。不能用裸截断多项式的碰撞时间冒充 ζ 的局部属性。

18.3 以 Hilbert 空间证书压缩高阶谱方案

Lamzouri 的简化使一个方向值得优先测试：把高阶矩、负谱 cap 或重数约束直接加入同一个 Hilbert 空间不等式，看是否存在真实更优的极端点。先做纯优化的严格对偶，再检查每个新约束是否有独立的 ζ 输入。

如果优化对偶显示新增约束只是原有二次信息的重述，便立即止损。如果确有增益，再建立一个明确的 missing arithmetic lemma，而不直接启动大规模 Lean 翻译。

18.4 Wronski、Plücker 与总正性

此前任务讨论的可用来源是 Karp – Purbhoo 关于 universal Plücker coordinates、Wronski map 和总正性结构的工作。Karp – Purbhoo

值得测试的不是“任何实根多项式都能找到某个正表示”，而是：是否存在由本问题自然给出的 Wronski lift，其演化与圆周热流相容，并使 Plücker 坐标控制两体背景系数或 discriminant 的首次消失。要逐项检查三个相容性：初值 lift 是否 canonical、flow 是否闭合、所需正性是否在 flow 下保持。任一步失败，就不能借总正性名称继续推论。

低 N 的可否证实验应包含 clock、单缺陷、三根近簇和两个远离近簇。若某个候选 lift 对同一谱随任意坐标选择而给不同结论，应视为它没有提供谱不变量。

18.5 函数域族作为严格试验场

可考虑在有已知 monodromy 与 equidistribution 的 L-function 族上测试新 detector：先在固定 degree 下让基域大小增长，再研究 degree 增长的误差。这里的目标是证明一个真正 arithmetic family 中的统计迁移定理。

这是研究建议，不是声称现有函数域定理已给出所有需要的 uniformity。CUE 固定维 equidistribution 与最小 gap/depth 的极端尺度并不自动兼容；误差必须小于目标事件的概率尺度。它能提供更诚实的中间战场，但不能通过“类比”直接解决 ζ。

18.6 186 的支撑几何与 spectral optimization

旧 P9 引擎擅长把高维可分离函数变成一维概率。可以把同样的概率分解能力用于 186 的真实 source-domain cap，而不是把新问题退回旧 simplex。研究重点是：哪些 support-loss integrals 能用条件概率、交换性或对称多项式精确降维，哪些必须保留 prime-fragment 的相关结构。

如果降维把同坐标的两个 fragments 当成独立，就可能破坏证书。应建立一个小尺寸完全枚举模型，先证明变换保留所需相关性，再应用高精度积分。这是旧工具与新突破之间比较具体的接点。

19. 给 Astra 的十二张任务卡

这些任务卡是当前任务内的研究建议；本次没有自动创建或启动十二个新任务。它们按依赖组织，可以在下一轮明确授权的研究中分工。

A1 — 先把所有 theorem statement 固定

输入：主报告、三个主论文与 Lean 草稿。输出：每条命题的对象、假设、量词、结论、引用和证据级别。优先修复 `depth_ge`、COM N=4、ρ 与 ϰ 的定义、H_m 索引。验收：删去 proof body 后，statement 本身也准确表达预期数学。停止：无法明确对象的“桥梁”不进入证明队列。

A2 — 186 baseline 可认证重现

输入：公开 PrimeGaps186 仓库、版本固定的 FLINT 与补丁、论文参数。输出：完整区间 certificate、receipt、运行环境、三项项目输入状态。验收：原结果的 margin 与来源一致，且每个 enclosure 真正向外舍入。不要用已有 receipt 或浮点运行替代重新计算。

A3 — 39 或更窄 tuple 的完整 margin 搜索

输入：A2 的合法二次型与损失。输出：最佳“扣除全部误差以后”的严格下界及 tuple witness。允许先探索浮点，最终必须用同一个 checker。停止：所有试验改善只存在于未恢复支撑的理想模型中。

A4 — ACUE 带宽外固定强度 detector

输入：精确 COM/phase 家族与对照 CUE。输出：固定 smooth w 或固定阶局部量，给 uniform gap，而非随 N 消失的差。验收：在精确可行律上证明差与全部低频匹配。停止：只得到微小有限 N 差而无 uniform 下界。

A5 — detector 的 arithmetic realization

输入：A4 统计。输出：精确的显式公式/Dirichlet 展开、对角主项、off-diagonal 余项、所需 support 和平均范围。验收：每一个候选增益都对应一个可陈述的算术引理。停止：结论等价于完整目标猜想而没有更弱的可攻击输入。

A6 — LR 极限中的固定局部分离

输入：N=6⋯9 四点自由及显式 phase families。输出：两个 tight 序列，统一局部计数控制，固定 observable 差 ⩾ε，以及正确 bandwidth region 的矩传递。验收：极限过程确实不同，且满足原问题全部量词。不得用 parity clock 稀有原子作为唯一差异。

A7 — 三点 rigidity 的 all-N 分类

输入：小 N 精确矩约束与 Fourier – Slater 描述。输出：all-N 证明或明确最小反例。若证明，记录依赖的是 rotation invariance、balanced moments 还是更强结构。它服务于 A4 detector 的最低阶数选择，不应单独被包装成已经逼近 RH。

A8 — depth 上界的动态背景控制

输入：第 10.4 节修正差商。输出：在一个可验证事件上沿整段时间成立的 `B_max δ²=o(1)`；先在 CUE 或明确的随机根模型证明。验收：得到真正 `D~δ²/8` 的概率命题与异常事件控制。停止：只在初始时刻计算平均背景。

A9 — 密度 AH 的正确动力学统计

输入：A8 与一个带背景的局部零点窗口定义。输出：有界 depth 统计或分位数，以及 density-AH 下的严格限制。验收：不由单个异常最小 gap 决定；窗口边界误差可控。算术计算另交 A5，不把条件串联当无条件结果。

A10 — marked 算子的规范性

输入：Cayley response、ξ′/log derivative 或某个真正算术表示。输出：不依赖任意 eigenvector 选择的 mark，完整导数公式，rank-two 近似的定量误差。验收：ϰ′ 不是被重算后隐藏的全谱项；能独立预测新样本。停止：mark 的定义已经含待测 depth。

A11 — 高阶谱证书对偶与算术成本

输入：Lamzouri Hilbert 形式、历史 cubic 与 Christoffel 数据。输出：严格优化 gain 或 exact no-gain dual，以及每项新输入的 cost ledger。验收：同当前最佳的相关基线比较，明确 RH/无条件及 simple/distinct 的差别。

A12 — 形式化最后闭环

输入：A1 中已经稳定、并在其他任务中得到证明的关键引理。输出：无 `sorry` 的正确 statement、依赖审计、独立数值 checker 与可复跑构建。优先正式证明 finite identities 和 tuple admissibility，再处理 ODE 首碰撞。不要把形式化资源耗在尚未确定正确的宏大桥梁上。

20. 第一次接手时的执行顺序

第一步恢复环境与证据：读取主报告的修正表，运行独立审计，核对来源 hash，检查引用与数组。第二步选主战场：如果要最快测试一个公认数论改进，优先 A2→A3；如果要沿用户最在意的 AH/GUE 路线，优先 A4→A5，同时用 A6 检查有限反例能否真正保留到无限过程。

第三步把每个“缺失引理”写成可以被反例击败的精确命题。对于局部化，要包含动态时间窗；对于算术均值，要写范围、权重、对角项和一致性；对于 186，要写完整 margin；对于 LR，要写全部 n 与测试函数区域。第四步才做大规模搜索或形式化。

每次失败应留下五个信息：原命题、最小反例或不等式失败位置、是否能通过增加自然假设修复、修复后的算术成本、是否仍服务主目标。这样保存失败才会产生下一次突破需要的约束，而不是新的长叙事。

21. 可直接交给 Astra 的接手指令

> 请从这份档案的审计部分开始，而不是直接相信附录中所有写成 theorem 的陈述。优先阅读固定提交中的 `final_verified_paper.md`、`round3_synthesis.md`、`impostors_paper.md`、`depth_scaling_theorem.md`、`signed_sieve_nogo.md`，并运行 `audit_research.py`。把所有命题分成有限证明、历史数值、条件桥梁和已撤回断言。
>
> 总目标是取得一个真正涉及著名数论问题的结果。零点主线选择一个固定强度、算术可观测的带宽外 pair-correlation 或 density-AH detector；筛法主线选择 186 完整二次型中的严格 margin 改进。不要把新的有限反例本身当作终点，除非它明确解除主链条上的一个障碍。
>
> 对每个候选路线先列完整依赖链，并找出唯一最缺的算术或概率引理。允许推翻旧方向。不要把 random-matrix 类比、任意 PSD 表示、浮点证书、SMT 的局部代数检查或未编译 Lean 当作解析定理。若不能越过关键引理，请给出准确的 no-go、反例或等价障碍，留下可复用的数学结论，而不是宣布已接近 RH。

22. 外部来源索引

以下只列本次实际使用的主要原始来源。网络状态截至 2026 年 9 月 4 日检索；预印本、项目陈述与完全独立重验不是同一证据级别。

来源	用途
固定研究分支	本项目的历史原稿、代码与数据
Tao：Alternative hypothesis for unitary matrices	ACUE 的原始构造与问题背景
Lagarias – Rodgers：higher correlations	AH 与全阶相关、问题量词与历史小间隙注释
Lagarias – Rodgers：band-limited mimicry	格点带宽识别边界与不同测试域
Feng – Wei	Circular β 最小间隙，适用 β 范围
Anthropic：ζ 项目	0.6725007 的官方进展说明
Alpöge – Furman	相关正式论文入口
Lamzouri	2026 年 9 月的 Hilbert 空间简化
Anthropic：FLT	大规模形式化方法
FLT 代码	形式化工程来源
Short gaps 论文	DHL[40,2] 与 186 的数学方案
Short gaps 研究记录	失败试验、符号和支撑问题
PrimeGaps186	Lean 项目输入、数值证书与依赖说明
Guth – Maynard	大值估计与零密度的相关入口
Tao 的零密度讨论	用户所提路线的一种可能对应，未唯一确认
Karp – Purbhoo	Wronski/Plücker 总正性的原始来源

23. 交付、检验与尚未完成的工作

已完成的是来源重建、研究演化说明、关键有限结果的整理、重要错误的明确修正、21 项独立审计、两个 tuple 的 admissibility 核验、外部新进展的核对，以及面向著名问题的具体任务设计。完整历史 Markdown 附录不删除过时推论，以便下一位研究者核查它们如何出现与被修正。

没有完成的是：全分支逐行数学审稿、所有旧实验复跑、H_2 /H_3 /H_4 的严格 interval 认证、186 证书重建、Lean 补全、无限 LR 非唯一性、ζ 算术桥梁，以及任何新的著名猜想证明。随后用户明确授权八小时自主研究、子代理交叉质疑及与 Claude Code 协作，并要求推送 PR #11；该夜间运行及其新结果记录在后续补充中。

程序与原始来源分开，审计与新报告推送到 PR #11 的独立目录，不静默修改历史论文。撤销相应新增提交即可移除整理层。后续最有价值的一步，是选择 A2→A3 或 A4→A5 中一条，完成其第一个缺失引理，而不是再生成一轮没有收敛终点的全面畅想。

主说明结束。后续为完整历史原文附录。附录中的断言与评价属于写作当时状态；发生冲突时，以主说明的审计与原数学证明为准。

最新补篇：从历史材料到当前可验证的数学状态

更新：2026 年 9 月 5 日。本补篇与后面的新报告优先于历史附录中的状态标签。历史文件原样保留，不因后来纠错而删除。

当前任务与真正的成功标准

主任务是在 RH 下证明真实 ζ 零点归一化间距小于 1/2，或得到明确超出现有可用带宽的二点相关结论。成功需要一个适用于真实 ζ 的正证书，并写清适用的高度、密度、重数与误差；比较 CUE 和 ACUE 的模型行为本身不满足这个标准。

关于 AH 必须先固定版本。Lagarias – Rodgers 的较强格点表述把相邻归一化间距限制在正半整数附近；单个趋向于 1/2 以下的极限间距结果可与这种强版本冲突。容许零间距或重数、容许零密度例外的版本，需要相应的正密度或远离格点的结论。不能把这些命题混为一个 AH，再用最小间距自动反驳全部版本。

八小时是最初的研究里程碑，随后协调任务转达了可恢复的 72 小时研究安排。每个检查点保存真实工作状态；这不是对连续运行时间或解决猜想的保证。Fable 使用用户现有单会话，只接收一个手动投递的固定向量核验包；没有证据表明它已接收或完成。

1. 这轮最明确的新数学进展：固定素因子试验的算术传递

原来的新方向是令

$$r_L(n) = d_\ell(n)\left[f\left(\frac{\log n}{\log L}\right) + g\left(\frac{\log n}{\log L}\right)\sum_{p\mid n}\left(\frac{\log p}{\log L}\right)^2\right].$$

这里素数按不同因子计数，不能把素数幂重数悄悄加进去。早先计算直接使用了形式上的背景矩

$$\mathrm{E}_v S_2 = \frac{v^2}{a+1}, \qquad \mathrm{E}_v S_2^2 = \frac{(a+6)v^4}{(a+1)(a+2)(a+3)}, \qquad a = \ell^2.$$

这些式子现在已有固定族的算术证明，并得到另一研究代理的逐项审查。证明不以 Poisson – Dirichlet 随机模型为假设，而从 $\Sigma\, d_\ell(n)^2 n^{-s} = \zeta(s)^{\ell^2} F(s)$ 的正测度 Laplace 极限出发，插入标记素数，计算 beta 积分，再用权 Schur 估计消去小素数与高次素数幂。关键细节是：同一个素数在 A^*A 中的对角项保留为主项，在 A^2 中的重复插入带有 p^{-2}，则可消去。短背景区域通过测度收敛和无原子边界处理，无需假设逐点一致的 Selberg – Delange 误差。

证明范围是固定 $\ell \geq 1$、固定多项式 f, g、正极限范数。不包含随 L 增长的维数、任意无限特征族或完整 Fock 算子极限。算术极限本身无条件；把它用于 Inoue 的 ζ 不等式才需要 RH。完整证明、审查意见及审查后补入的三个显式估计均列在后文。

这个进展解决了一个实际的算术依据问题，但固定有理试验仍给出

$$-0.01467 < J < -0.01465, \qquad J \approx -0.014662375473369.$$

因此没有半间距突破。此前约 −0.01465473 的较丰富特征搜索也仍为负，且不是这一有理向量。不能用已证明的固定族传递，自动替较高阶特征或病态数值最优向量背书。

2. 完整算术算子告诉了我们什么

在半间距处，饱和主项是

$$K_L = A_L^* A_L + \frac{1}{2}\left(A_L^2 + (A_L^*)^2\right), \qquad \text{目标阈值} = \frac{\pi^2}{2}.$$

算子已算到 $L = 10^7$，最大特征值约 4.324089559，归一化余量约 −0.030939064。这只是有限算子的高精度数值，不是严格上界，也不是渐近无解定理。在 $L = 10^6$，20 项对称素因子特征已接近完整有限最优值；增加到 40 项后，差距约为 0.0004437。这说明当前测试范围的主要结构确实具有乘性，但不排除随 L 改变的新方向。

一个单独估计两项能量的 Cauchy – Schur – Volterra 上界只降到约 5.2074，高于 $\pi^2/2$。它失败的原因是损失了两个项的近极值向量之间的兼容性；继续提高同一个网格的精度不能弥补这个量级的缺口。这个失败已保存为完整推导和各网格结果。

固定素数分箱还产生一个有限 bosonic creation 模型，40 个分箱有 215,308 个状态，数值仍未过阈值。固定分箱可给构造性的渐近下界，但不是完整算子收敛，更不是全局上界。小素数云的处理必须明确，不能把真空有限粒子截断当作全部乘性结构。

3. 值得永久保存的反向结果：中心高斯并不免费扩大支撑

把高度 T 附近的高斯换成以原点为中心的高斯，会使 Fourier 核非负，似乎可以保留更长混合支撑。然而精确临界线积分恒等式中，ζ 的极点贡献不会消失。对所需长尾匹配频率，它的相位与素数主项相反。

设 $M = T^{1+\eta}$、$0 < \eta < 1$、$h = \pi/\log T$，取

$$R(t) = 1, \qquad C(t) = -i \sum_{M<q\leq 2M} q^{-1/2-it}.$$

两边对角系数范数有界，但极点混合修正为

$$-\left(4\pi\cos\frac{\pi\eta}{2} + o(1)\right)\frac{M}{\log M}.$$

对于测试的宽度 $W = T$ 或 $T/\log T$，它除以 W 后发散。因此不能把它当作由对角范数控制的 $o(W)$ 误差。与此同时，长支撑真实 Gram 矩阵的非对角罚项也会变大，低高度相位聚集不能忽略。

这不是“所有中心高斯方法不可能”的定理。它精确排除了无偿支撑扩张：下一条合法路线需要控制减去连续素数密度后的混合项，或定量支付极点和低高度代价。对完成函数、去极点函数作替换，也必须保留相应的 gamma 或连续项。

4. Heat flow 路线的完成范围及与真 ζ 的距离

对圆周根多项式的系数流，去掉无关标量后是普通标量热流。隔离最小间距为 δ 的一对根，扣除远背景的共同线性漂移，可在

$$\delta^2 B \to 0, \qquad B = \frac{1}{4}\sum_{k \neq \pm} \csc^2(\theta_k/2)$$

下证明第一次碰撞时间 $D \sim \delta^2/8$。作者第二轮审查把常数显式化为 $K = 16384$、$\eta_0 = 1/524288$，并控制全程移动边界、Gaussian 尾项和根的延续。这些粗常数用来检查一致性，不是新的数值纪录。

结合已有极小间距分布和三根聚集排除，可得到 CUE 及正整数 beta 圆周族的碰撞时间极限律。需要继续查明与经典 Lehmer pair 判据、已有多项式热流工作的关系，不能把相似原理包装成全新理论。

真 H_t 的讨论也更精确了：在 RH 下，中心化 genus-one canonical product 可以控制所需远背景尾项。关键仍是算术上产生充分孤立、足够小且在所需版本中有足够频率的零点对。Rodgers – Tao 的宏观负时间带来很多微观松弛时间；由半格硬核得到的 $c/(\log T)^2$ 深度只对应有限微观时间，两者不能直接替换。

5. 素数间隙、0.6725 和形式化工作的实际启发

素数 186 路线中，39 点、直径 182 的 admissible tuple 已有有限核验，但所需的 `DHL[39,2]` 没有证明，所以没有得到更小的 prime-gap 定理。精确凸性与有理单纯形分割提供了可复核的 minorant 质量区间；这类“试验向量 + 定向误差预算 + 严格证书”的工作方式可用于 ζ 的固定变分问题。它不能代替缺少的算术混合矩估计。

0.6725007 是特定已公开 ζ 结果中的简单且在线零点比例；不能称为所有 bandwidth-one 论证的普遍天花板。新的证明简化与数值纪录改变也必须分别记录。本项目的标记格点 0.702642 常数属于另一模型，不是 ζ 纪录提升。

FLT 形式化的主要可迁移内容是大型依赖图、明确接口、局部证明任务和机器核验，不是自动产生新的解析数论不等式。当前先形式化精确代数、有限证书及已经闭合的解析引理；不要给尚未证明的主步骤加一个公理后宣布名猜想已完成。

6. 下一轮有明确验收条件的探索

1. **密度扣除后的长混合项。** 先写精确极点、gamma、Gram 罚项，再问是否存在可控的新支撑区域。验收是含全部误差的有符号不等式，而不是忽略修正后的正数。
2. **兼容极值结构。** 在完整 $A^*A + \mathrm{Re}\,A^2$ 中寻找比逐项 Cauchy 更强的共同约束。验收是严格上界或一个新的可认证试验族；有限谱外推不算。
3. **固定族的可认证搜索。** 已闭合的 $H = f + gS_2$ 提供一个真实算术接口。任何新增特征先证明其有限标记转移，再做适度优化。验收包括正定范数、可控制条件数、定向误差和净余量。
4. **真正 ζ 的短对算术信息。** Heat-flow 引理的用途是把明确的局部输入转化为动力学后果。验收应是一个目前未知、能从 ζ 算术推出的隔离或频率估计，而不是再次证明随机矩阵里存在小间距。

这些方向都还没有产生重大猜想的证明。档案保留精确成功和精确失败，是为了下一位研究者能从真正尚缺的步骤继续，而不是重做已经排除的尝试。

Current report 01: symmetric_prime_arithmetic_transfer

Source: `research/reports/symmetric_prime_arithmetic_transfer.md`. SHA-256: `d1329ec22b249fb0f1a8e6cf645e5d831ecb1ee7f8759f8c2b2caaaacc21d904`.

Arithmetic transfer for a fixed symmetric prime–factor resonator

Date: 2026-09-05. Authoring role: Astra, primary research task.

Status: complete written proof with independent internal review. This note supplies the route from the weighted integer sums to the previously stipulated continuum form. The independent review accepted the fixed-family argument. This is not Lean verification or external peer review. The three explicit estimates requested by that review are added below. The fixed rational trial still has a negative margin. No half-gap theorem is claimed.

1. Precise statement

Fix a real number $\ell \geq 1$, write $a = \ell^2$, and let $d = d_\ell$ be the multiplicative function with

$$d(p^e) = \frac{(\ell)_e}{e!}.$$

For $L > 1$, put

$$v_n = \frac{\log n}{\log L}, \qquad S_L(n) = \sum_{p \mid n} \left(\frac{\log p}{\log L}\right)^2, \quad r_L(n) = d(n)H(v_n, S_L(n)), \quad x_n = \frac{r_L(n)}{\sqrt{n}},$$

where $H(v, S) = f(v) + g(v)S$, and f, g are fixed real polynomials. Prime divisors in S_L are distinct. The proof extends to continuous f, g, but the polynomial case suffices here. Define the nonnegative matrix

$$(A_L)_{p^e m, m} = \frac{2\sin(\pi e \log p/(2\log L))}{e\sqrt{p^e}} \quad (p^e m \leq L).$$

Let E_v mean the linear moment functional

$$\mathrm{E}_v 1 = 1, \quad \mathrm{E}_v S = \frac{v^2}{a+1}, \quad \mathrm{E}_v S^2 = \frac{(a+6)v^4}{(a+1)(a+2)(a+3)}.$$

No probability model is assumed. The functional is derived below from integers. Write

$$H_0 = H(v, S),\ H_u = H(v+u, S+u^2),\ H_w = H(v+w, S+w^2),\ H_{uw} = H(v+u+w, S+u^2+w^2).$$

The arithmetic limit is

$$\frac{\| A_L x \|^2 + x^{\mathsf{T}} A_L^2 x}{2\pi^2 \| x \|^2} - \frac{1}{4} \longrightarrow \frac{M_2 + M_3}{I} - \frac{1}{4},$$

provided $I > 0$, where

$$I = \int_0^1 v^{a-1} \mathrm{E}_v H_0^2 \, dv,$$

$$M_2 = \frac{2\ell^2}{\pi^2} \int_{v+u+w\le 1} v^{a-1} \frac{\sin(\pi u/2)}{u} \frac{\sin(\pi w/2)}{w} \mathrm{E}_v(H_0 H_{uw} + H_u H_w) \, dv \, du \, dw,$$

$$M_3 = \frac{2}{\pi^2} \int_{v+u\le 1} v^{a-1} \frac{\sin^2(\pi u/2)}{u} \mathrm{E}_v H_0^2 \, dv \, du.$$

All variables of simplex integrals are nonnegative. The scalar normalization constant below cancels from this quotient.

2. The unmarked integer measure

Define

$$C_\ell = \prod_p (1 - p^{-1})^a \sum_{e\ge 0} \frac{d(p^e)^2}{p^e} > 0.$$

Each factor is $1 + O_\ell(p^{-2})$, so this product converges absolutely to a positive number. For real $s > 1$,

$$D(s) = \sum_n \frac{d(n)^2}{n^s} = \zeta(s)^a F(s), \qquad F(s) \longrightarrow C_\ell \quad (s \downarrow 1).$$

Consequently the locally finite positive measures

$$\nu_L = \frac{1}{(\log L)^a} \sum_{n\ge 1} \frac{d(n)^2}{n} \delta_{\log n/\log L}$$

converge on bounded intervals to

$$\nu(dv) = \frac{C_\ell}{\Gamma(a)} v^{a-1} \, dv.$$

One can use the Laplace continuity theorem for positive measures here, rather than needing a uniform pointwise Selberg – Delange error: for every $t > 0$,

$$\int e^{-tv} \, d\nu_L(v) = \frac{D(1 + t/\log L)}{(\log L)^a} \longrightarrow C_\ell t^{-a}.$$

For completeness, tilt by e^{-v}. The tilted measures have convergent total masses and Laplace transforms; their tails obey $\int_R^\infty e^{-v} d\nu_L \le e^{-R/2} \int e^{-v/2} d\nu_L$. This gives tightness. Subsequential weak limits are uniquely identified by their Laplace transforms, and untilting on a compact interval gives the displayed local convergence. The limiting measure has no atom at zero or at any cutoff. This also gives

$$\sum_{n\le L} \frac{d(n)^2}{n} = O_\ell((\log L)^a).$$

These arguments only need the classical simple pole of $\zeta(s)$ at 1 and nonnegativity of the coefficients. They do not assume RH.

3. Marked primes and the two moments

First restrict marked primes to $p \ge L^\varepsilon$, for a fixed $\varepsilon > 0$. For $j > 0$, the prime number theorem and partial summation give

$$\sum_p \frac{(\log p/\log L)^j}{p}\delta_{\log p/\log L}\ Longrightarrow\ u^{j-1}\,du$$

on compact subintervals of $(0,\infty)$. Uniformity at zero will instead come from truncation.

If distinct marked primes are coprime to the background integer m, each prime inserted into $d(n)^2$ contributes $d(p)^2 = a$. Thus the limiting first-mark measure is the convolution

$$\frac{C_\ell}{\Gamma(a)}a\int_0^v u(v-u)^{a-1}\,du = \frac{C_\ell}{\Gamma(a)}v^{a-1}\frac{v^2}{a+1}.$$

Expanding S^2 produces the same-prime fourth-power term and the ordered distinct-prime term. Their respective densities are

$$\frac{C_\ell}{\Gamma(a)}a\int_0^v u^3(v-u)^{a-1}\,du = \frac{C_\ell}{\Gamma(a)}v^{a-1}\frac{6v^4}{(a+1)(a+2)(a+3)},$$

$$\frac{C_\ell}{\Gamma(a)}a^2\int_{u+w\le v} uw(v-u-w)^{a-1}\,du\,dw = \frac{C_\ell}{\Gamma(a)}v^{a-1}\frac{av^4}{(a+1)(a+2)(a+3)}.$$

These beta integrals give the claimed moment functional. Multiplying by a bounded continuous function of the total mass is permitted by weak convergence.

Here are the two discarded cases, with the order of limits specified.

- **Repeated or background primes:** for $\ell \ge 1$, $d(p^{b+c}) \le d(p^b)d(p^c)$, since the successive ratios $(\ell+e)/(e+1)$ decrease with e. Hence requiring a background integer to contain a marked prime costs an additional factor bounded by $O_\ell(p^{-1})$ after its first reciprocal-prime weight. At $p \ge L^\varepsilon$, the sum of the resulting p^{-2} terms is $O(L^{-\varepsilon})$. The unmarked mass is $O((\log L)^a)$, and the other finitely many reciprocal-prime sums cost at most powers of $\log\log L$. Thus these contributions vanish after normalization. The same argument removes coincident distinct marks and prime exponents greater than one in this expansion.
- **Small marked primes:** deterministically, for every $n \le L$,

$$0 \le \sum_{p\mid n,\, p<L^\varepsilon}\left(\frac{\log p}{\log L}\right)^2 \le \varepsilon\sum_{p\mid n}\frac{\log p}{\log L} \le \varepsilon.$$

  Since $0 \le S_L(n) \le 1$, deleting these marks changes H and its products by $O_H(\varepsilon)$, uniformly on the cutoff.

Take $L\to\infty$ with ε fixed, and then $\varepsilon \downarrow 0$. All product-measure cutoffs have boundary of measure zero. In particular, this approach includes the short-background range m near 1; it does not assume a uniform asymptotic for $\sum_{m\le L/(pq)}$ at every individual p, q.

4. Uniform removal of small-prime and prime-power operator pieces

The preceding moment truncation alone is not enough: the creation operator also has a small-prime part. We control it with a weighted Schur estimate.

Put $w_n = d(n)/\sqrt{n} > 0$. The logarithmic derivative of $\zeta(s)^\ell$, or its formal Dirichlet-series coefficient identity, gives

$$\sum_{p^e\mid n}(\log p)d(n/p^e) = \frac{\log n}{\ell}d(n).$$

Since $0 \le 2\sin(\pi \log q/(2\log L)) \le \pi \log q/\log L$,

$$\frac{(A_L w)_n}{w_n} \le \frac{\pi \log n}{\ell \log L} \le \frac{\pi}{\ell}.$$

Submultiplicativity gives for any subset E of prime powers

$$\frac{(A_E^{\top} w)_m}{w_m} \le \sum_{p^e \in E,\, p^e \le L} \frac{2\sin(\pi e \log p/(2\log L))}{e p^e} d(p^e).$$

For all prime powers the right side is $O_\ell(1)$. For the prime part $p < L^{\varepsilon}$ it is at most

$$\frac{\pi \ell}{\log L} \sum_{p < L^{\varepsilon}} \frac{\log p}{p} \le \pi \ell \varepsilon + O_\ell(1/\log L).$$

For exponents $e \ge 2$ it is bounded by

$$\frac{\pi}{\log L} \sum_{p} \sum_{e \ge 2} \frac{d(p^e)\log p}{p^e} = O_\ell(1/\log L).$$

The series converges because $d(p^e) = O_\ell((e+1)^{\ell-1})$. The weighted Schur test follows directly by Cauchy – Schwarz and summing the two displayed weighted row/column bounds. Therefore

$$\| A_L \| = O_\ell(1), \qquad \| A_{p<L^{\varepsilon}} \| = O_\ell(\sqrt{\varepsilon} + 1/\sqrt{\log L}), \qquad \| A_{e \ge 2} \| = O_\ell(1/\sqrt{\log L}).$$

Thus replacing A_L by its prime-only part with $p \ge L^{\varepsilon}$ changes either quadratic form, divided by $\| x \|^2$, by $O_\ell(\sqrt{\varepsilon} + 1/\sqrt{\log L})$. This estimate works for signed H, and indeed for every vector x; no coefficient positivity is needed at this step.

5. Evaluation of the retained integer sums

For the retained prime operator set $a_p = 2\sin(\pi \log p/(2\log L))$. Exactly,

$$\| A_L x \|^2 = \sum_{n \le L} \frac{1}{n} \sum_{p,q \mid n} a_p a_q r_L(n/p) r_L(n/q),$$

$$x^{\top} A_L^2 x = \sum_{mpq \le L} \frac{a_p a_q}{mpq} r_L(m) r_L(mpq).$$

In the first formula, the terms $p = q$ are **not discarded**. With $n = mp$, they give

$$\sum_{mp \le L} \frac{4\sin^2(\pi u/2)}{mp} d(m)^2 H(v, S_L(m))^2.$$

Its limit, after division by $C_\ell (\log L)^a/\Gamma(a)$, is $2\pi^2 M_3$.

For $p \neq q$, put $n = mpq$. Away from the negligible coprimality defects already estimated, the coefficient is

$$r_L(mp) r_L(mq) = \ell^2 d(m)^2 H_u H_w.$$

The prime-insertion measures give the corresponding $H_u H_w$ term of $2\pi^2 M_2$. There is no extra factor of two: (p, q) is ordered both in the integer sum and in the double integral.

Similarly, in the second quadratic form the distinct-prime terms have

$$r_L(m) r_L(mpq) = \ell^2 d(m)^2 H_0 H_{uw}.$$

The repeated-prime $p = q$ terms here contain p^{-2}, and vanish by the fixed-ε estimate. This is why the same-prime term survives in $A^{\top}A$ but not in A^2. The marked-background moments from Section 3 evaluate both expectations.

Finally,

$$\| x \|^2 \sim \frac{C_\ell}{\Gamma(a)}(\log L)^a I.$$

Combining the three limits and the operator truncation proves the stated arithmetic limit. The independent internal audit checked these measure-convergence and discarded-term arguments.

6. Interface with the actual zeta theorem

The exact source is Inoue, arXiv:2604.05733v1, Theorems 3 and 4. Those theorems allow arbitrary arithmetic resonator coefficients under RH and the product cutoff $L \le T/(\log T)^2$. Choose $L = \lfloor T/(\log T)^2 \rfloor$, $h = \pi/\log T$, and the logarithmic increment approximator. The theorem has a coefficient-independent normalized error tending to zero; the approximator square sum is $O(1)$.

The corresponding operator uses $\theta = \log L/\log T \to 1$. Replacing θ by 1 changes the weighted Schur bounds, hence the operator norm, by $O_\ell(|1-\theta|)$: apply the mean value inequality for sine and the same logarithmic derivative and prime-sum estimates. At $\phi = 1/2$, the linear coefficient cancels. The resulting lower bound for the normalized weighted zero-count expression has the proposed continuum margin, with the scope accepted in the independent review.

This would establish the arithmetic validity of the fixed symmetric-prime family. It would **not** establish a positive margin. The current certified value is approximately -0.014662375473369, and a fixed negative lower bound has no half-gap consequence.

7. Audit requests and provenance

Independent reviewers should check: the $\ell \ge 1$ submultiplicativity restriction; the exact weighted Schur row and column orientation; local convergence of positive measures including the $v = 0$ boundary; counting of ordered distinct primes; uniform removal of background coincidences; and the error normalization when invoking Inoue Theorem 4. A successful numerical check cannot replace any of these steps.

This draft is a mathematical reconstruction intended for public verification. It records the argument and its dependencies, not a transcript of private model deliberation. No novelty claim is made for the marked-prime asymptotics or the Schur method; the purpose is to close the exact transfer obligation created by this project's fixed trial.

8. Explicit estimates added after independent review

The reviewed draft had SHA-256 `bf2ca70e61da1f97b4c214304180f0f45d3c26a050104bf30e6d70e46660d07f`. The review requested the following elaborations; they clarify the estimates rather than change the theorem.

First, the uniform total-mass bound follows directly from the Laplace transform:

$$\frac{1}{(\log L)^a}\sum_{n\le L}\frac{d(n)^2}{n} \le e\frac{D(1+1/\log L)}{(\log L)^a} = O_\ell(1).$$

Second, the background coincidence bound can be written explicitly as

$$\sum_{m\le L:\, p\,|\,m}\frac{d(m)^2}{m} \le \frac{\ell^2}{p}\sum_{k\le L}\frac{d(k)^2}{k}.$$

The additional reciprocal-prime weight outside this sum gives a summable `p^{-2}` cost. Collisions between an outer inserted prime and a prime in the expansion of the background S moment have exactly this form. There are only finitely many such marks for fixed H.

Third, if `A=A0+E`, then

$$\| A^* A - A_0^* A_0 \| \le (\| A \| + \| A_0 \|) \| E \| , \qquad \| A^2 - A_0^2 \| \le (\| A \| + \| A_0 \|) \| E \| .$$

These inequalities justify removing the small-prime and prime-power operator pieces uniformly over signed vectors. If applying Inoue Theorem 4 directly, its normalized `O(h) M1` term is bounded by Theorem 3 and the same Schur estimate; the source's combined Proposition 3 packages this cancellation and error estimate explicitly. No new off-diagonal hypothesis is introduced.

Current report 02: symmetric_prime_transfer_independent_review

Source: `research/reports/symmetric_prime_transfer_independent_review.md`. SHA-256: `f3a961b4fe4340b6e65f31c2aa40adb5fcd7d54ab09406ab2dc7c1ab38118ddf`.

Independent review of the fixed symmetric-prime arithmetic transfer

Review date: 5 September 2026. Reviewer: independently assigned residual-Gram research agent.

Reviewed source, without editing it:

`/Users/qingyunsun/Library/CloudStorage/Dropbox/Code/Riemann zeta RMT/Astra-Research/research/reports/symmetric_prime_arithmetic_transfer.md`

Reviewed source SHA-256: `bf2ca70e61da1f97b4c214304180f0f45d3c26a050104bf30e6d70e46660d07f`.

Decision

Accept the fixed-family arithmetic-transfer argument as mathematically complete at the level of an ordinary written proof. I found no fatal gap after checking the positive-measure limit, marked-prime coincidences, arbitrary-vector Schur estimates, same-prime terms, and the normalization of the interface with Inoue's theorem.

This decision has a precise scope. It covers fixed `ell>=1` and fixed

$$H(v,S)=f(v)+g(v)S, \qquad S=\sum_{p\mid n}(\log p/\log L)^2,$$

with the polynomials stated in the draft and positive limiting norm `I`. It therefore covers the particular rational trial `ell=16/15` certified in round one. Its operator asymptotic is unconditional; RH enters only when the result is inserted into the cited zeta theorem.

It does not certify the full arithmetic-operator/Fock limit, an arbitrary infinite feature family, uniformity when the polynomial degree or coefficients grow with L, or a positive half-gap margin. The fixed certified margin remains approximately `-0.014662375473369`, and thus gives no new small-gap theorem. This is an independent mathematical review, not Lean verification or external peer review.

1. Positive-measure limit: valid, including the short background range

The Euler factors satisfy

$$(1-p^{-s})^a\sum_{e\geq 0} d_\ell(p^e)^2p^{-es}=1+O_\ell(p^{-2s})$$

uniformly as real `s` decreases to 1. Thus the real Euler product tends to the claimed positive constant. A complex analytic branch of `zeta(s)^a` is not needed for the Laplace argument.

The scaled Laplace transform tends to `C_ell t^{-a}` for every positive t. Tilting by `e^{-v}` gives finite measures with convergent total masses. The stated exponential tail inequality is enough for tightness, because the Laplace transform at `t=1/2` is uniformly bounded for sufficiently large L. Uniqueness of Laplace transforms identifies every subsequential limit. Undoing the tilt on compact intervals gives local weak convergence.

The limiting density `C_ell v^{a-1}/Gamma(a)` has no atom at zero or at the cutoff. Therefore restriction to `[0,1]` and to the simplex cutoffs is legitimate. This avoids the commonly invalid step of requiring a pointwise Selberg – Delange estimate uniformly for every short background length `L/(pq)`.

One explicit mass bound, useful to make the proof fully transparent, is

$$\frac{1}{(\log L)^a}\sum_{n\le L}\frac{d_\ell(n)^2}{n}\le e\frac{D(1+1/\log L)}{(\log L)^a}=O_\ell(1).$$

It holds because `e^{-log n/log L}>=e^{-1}` on the cutoff. This supplies every crude background-mass estimate needed later.

2. Marked–prime moments: constants and coincidence removal are correct

For fixed epsilon, the marked primes lie in a compact logarithmic range bounded away from zero. The prime-number theorem gives weak convergence of the reciprocal-prime measures there. Products with the unmarked positive measure converge on compact sets. The boundary hyperplanes of the product cutoffs have zero limiting measure.

The arithmetic identity behind the first mark is exact away from the excluded coincidences: inserting a new prime into `d_ell(n)^2` multiplies the background coefficient by `ell^2=a`.

I independently recomputed the three beta integrals. Dividing the marked density by the unmarked density gives

$$\mathrm{E}_v S=\frac{v^2}{a+1},$$

$$\mathrm{E}_v S^2=\frac{6v^4}{(a+1)(a+2)(a+3)}+\frac{av^4}{(a+1)(a+2)(a+3)}.$$

The first summand is the same-prime fourth-power contribution. The second comes from **ordered distinct primes**. Their sum is the `(a+6)` expression in the draft. There is no missing factor of two.

The submultiplicativity condition is also correct. For `ell>=1`,

$$\frac{d_\ell(p^{e+1})}{d_\ell(p^e)}=\frac{\ell+e}{e+1}$$

is nonincreasing in e, hence `d_ell(p^{b+c})<=d_ell(p^b)d_ell(p^c)`. The restriction on ell is genuinely used here.

For example, the background coincidence can be bounded explicitly by

$$\sum_{m\le L:\,p\mid m}\frac{d_\ell(m)^2}{m}\le\frac{\ell^2}{p}\sum_{k\le L}\frac{d_\ell(k)^2}{k}.$$

After the outside mark's reciprocal-prime factor, this is a `p^{-2}` contribution. For `p>=L^epsilon`, summing it gives at most a constant times `L^{-epsilon}` times the total background mass. Higher exponents are handled by factoring off `p^2` and using the same submultiplicativity bound. Finitely many other marked primes contribute bounded factors for fixed epsilon, or at worst harmless powers of `log log L` under the draft's coarser estimate.

The deterministic small-mark bound is particularly effective:

$$\sum_{p \mid n,\, p<L^{\epsilon}} (\log p/\log L)^2 \le \epsilon.$$

Because H is fixed and linear in S, its products change uniformly by `O_H(epsilon)` on the cutoff. One therefore takes `L→infinity` first and `epsilon→0` afterward. No exchange of limits or assumption of uniformity in shrinking epsilon is needed.

The same reasoning applies when the background moments are evaluated inside the two outer prime insertions. Expanding `H_0H_{uw}` or `H_uH_w` adds only finitely many marks. Collisions between an outer inserted prime and a background mark have the same additional reciprocal-prime cost. This is worth stating explicitly in the final prose, but it does not require a new estimate.

3. The weighted Schur estimate is correctly oriented and works for signed vectors

Take `w_n=d_ell(n)/sqrt(n)`. The draft's weighted row bound uses

$$\sum_{p^e \mid n} (\log p) d_\ell(n/p^e) = \frac{\log n}{\ell} d_\ell(n).$$

This is the coefficient identity obtained by differentiating the Dirichlet series for `zeta(s)^ell`. The sine bound cancels the exponent e exactly, leaving the displayed logarithmic derivative. Hence the weighted row sum is at most `pi/ell`.

The column calculation contains a second factor `1/sqrt(p^e)` from the weight ratio, so its denominator is `e p^e`, not `e sqrt(p^e)`. The draft uses the correct denominator. Submultiplicativity gives the stated upper bound in terms of `d_ell(p^e)`.

For the prime-only small part, the column bound is `O_ell(epsilon+1/log L)`. For all higher prime powers it is `O_ell(1/log L)`, because

$$\sum_{p} \sum_{e\ge 2} \frac{d_\ell(p^e)\log p}{p^e} < \infty.$$

The fixed-ell polynomial growth in e is enough for this convergence. Combining these column bounds with the full row bound by the weighted Schur test yields exactly the square-root orders in the draft.

These are operator-norm bounds. No positivity of `x_n`, `H`, or the polynomial coefficients is needed. To make the quadratic-form passage explicit, if `A=A_0+E`, then

$$\| A^* A - A_0^* A_0 \| \le (\| A \| + \| A_0 \|) \| E \| ,$$

and

$$\| A^2 - A_0^2 \| \le (\| A \| + \| A_0 \|) \| E \| .$$

After division by `||x||^2`, the error is independent of the direction of x. This closes the small-prime and prime-power issue for the signed rational trial.

4. Same-prime terms and normalization of the limiting quadratic forms

The two quadratic forms treat repeated primes differently, and the draft handles the distinction correctly.

For `A^*A`, the same-prime term has `n=mp` and weight

$$\frac{4\sin^2(\pi u/2)}{mp}d_\ell(m)^2H(v,S_L(m))^2.$$

It is of main-term size and produces the entire M3 contribution. It must not be removed as a coincidence error. The draft retains it.

For `A^2`, applying the same prime twice instead has the reciprocal weight `p^{-2}`. At fixed epsilon it is negligible by the coincidence estimate. Distinct primes multiply the background coefficient by `ell^2`, producing `H_0H_{uw}`. The off-diagonal part of `A^*A` likewise produces `ell^2 H_uH_w`.

Both integer sums order the pair of inserted primes. The continuum double integrals use the same ordering. Multiplication by `alpha_p alpha_q` gives the coefficient `4 ell^2`, which equals the coefficient of `2 pi^2 M2`. The same-prime coefficient `4` equals that of `2 pi^2 M3`.

The norm satisfies

$$\| x \|^2 \sim \frac{C_\ell}{\Gamma(a)}(\log L)^a I.$$

The common scalar cancels. Since I is assumed positive, division by the norm is legitimate. This reproduces the precise normalization of the rational continuum certificate.

5. Interface with Inoue: RH and the cutoff error are accounted for

The cited Inoue paper supplies the zeta inequality under RH for arbitrary arithmetic resonator coefficients. This review treats that externally stated theorem as an input; it is not an independent reproof of the entire paper.

Set `L=floor(T/(log T)^2)` and `h=pi/log T`. The approximator coefficients have bounded weighted square sum. One direct check is to use the small-argument sine bound on primes and partial summation; prime powers contribute a convergent sum. Consequently the displayed normalized error tends to zero. If using Theorem 4 directly, its `O(h) M1` term is harmless because the preceding Schur bounds and Theorem 3 bound the normalized first moment. Alternatively the source's combined Proposition 3 already packages the needed coefficient-independent normalized error.

The operator in the zeta expression has `theta=log L/log T`, not exactly 1. Applying the mean-value inequality to sine gives an entrywise majorant proportional to

$$|1-\theta|\frac{\log p}{\sqrt{p^e}\log L}.$$

Its weighted row and column bounds are each `O_ell(|1-theta|)`, so the Schur norm is `O_ell(|1-theta|)`, as stated. The cutoff discrepancy therefore vanishes. At the half-gap boundary the leading linear coefficient cancels, leaving the two quadratic forms already analyzed.

6. Suggested presentation changes, none of which is a fatal repair

For publication or formalization, I recommend adding the explicit background-mass inequality in § 1 of this review, the displayed bound for a background divisible by a marked prime in § 2, and the quadratic-form perturbation inequalities in § 3. These turn the most compressed prose estimates into short checkable lemmas.

I also recommend stating the scope in the theorem title: **fixed H linear in S2**. The higher-degree symmetric-feature experiments in the earlier numerical reports are not automatically covered merely because their formulas are similar. The same strategy likely extends to them, but that extension should receive its own finite-mark statement and proof.

The review upgrades the earlier "pending arithmetic transfer" label only for the fixed family actually proved here, including the rational ten-coefficient trial. Historical reports should retain their original status and link to this review as the later resolution. No numerical global optimum, no general operator no-go, and no new zero-gap record is established by this upgrade.

Current report 03: residual_gram_round1

Source: `research/reports/residual_gram_round1.md`. SHA-256: `21b55679859ee1b67ea3b5c244b0d4367e5124be05e28d62aa13bcd02e877163`.

Residual Gram arithmetic route: exact saturation, independent reproduction, and symmetric prime-factor trials

Research round 1, 5 September 2026. This note separates exact algebra, certified finite variational integrals, exploratory numerics, and missing arithmetic theorems. No half-gap theorem or famous conjecture was proved in this round.

1. Outcome and priority

The supplied Inoue calculation is independently reproducible. Its negative margin at normalized gap `phi=1/2` is not a numerical accident caused by a low polynomial degree. The reproduced degree-14 margin is `-0.015357981703850554`, corresponding to additional normalized log-increment energy `0.3031544076323465`. This is a measured deficit for a particular family, not a universal impossibility theorem.

Three additional conclusions were obtained.

1. The original product-cutoff approximator is exactly saturated in its coefficient Hilbert space. Rewriting the same support with asymmetric factor lengths or additional Gram directions cannot give positive main-term recovery.
2. A new sparse arithmetic operator computes the diagonal main term for **arbitrary resonator coefficients**, not just the divisor-function ansatz. Full eigenvalue searches through one million coefficients remain below the half-gap threshold. These finite searches do not settle the asymptotic extremal problem.
3. A genuine extension of the resonator family, using symmetric statistics of its prime factors, improves the numerical half-gap margin to about `-0.01465473`. This removes only about 4.6% of the original deficit. A simple rational instance of this extension has a rigorously enclosed continuum margin between `-0.01467` and `-0.01465`. Its arithmetic transfer has not been fully written, and its margin is negative.

The significant unresolved node remains a quantitative estimate for residual directions carrying information beyond the already accessible product cutoff, or a substantially different arithmetic resonator that survives the full operator test.

2. Source audit and an important historical correction

The primary source is Shota Inoue, arXiv:2604.05733v1. It introduces a weighted second factorial moment of short-interval zero counts and proves, under RH, `mu<0.50895`. Its Theorem 4 allows arbitrary arithmetic coefficients under a product cutoff `L<=T/(log T)^2`; Theorem 2 specializes the coefficients to a divisor-function family. Zeros are counted with multiplicity. The displayed numerical trial is `phi=.508949`, `ell=1.15`, `f(v)=1-.7v`.

The old claim `mu<=.50412`, cited in the 2019 Lagarias – Rodgers paper, is **not a valid current record**. Goldston – Turnage-Butterbaugh, arXiv:1904.06001v2 was withdrawn on 11 November 2019: its generalized weights had to be symmetric in all their variables, invalidating the claimed calculation. This is directly relevant to the search below: every new prime-factor feature is explicitly symmetric.

The user's original `inoue_resonance_probe.py` and JSON became available after this round's independent implementation had already been run. Comparing those exact originals with the new code gives agreement within `1e−12` for the published trial and within `1e−11` for the degree-14 margin. The independently found finite-dimensional sign crossing differs by less than `2e−11`.

3. Exact coefficient–space saturation theorem

Fix a cutoff `L`, a resonator coefficient sequence `r`, and a log-increment coefficient sequence `g`. For any approximator coefficients `a`, put

$$b_a(q) = \sum_{km=q} a(k)r(m), \qquad b_g(q) = \sum_{km=q} g(k)r(m), \qquad q \le L.$$

Equip coefficient vectors with

$$\langle b, c\rangle_L = \sum_{q\le L} \frac{\overline{b(q)}c(q)}{q}.$$

Then the approximator-dependent diagonal main term satisfies the exact identity

$$2\,\mathrm{Re}\langle b_a, b_g\rangle_L - \| b_a \|_L^2 = \| b_g \|_L^2 - \| b_a - b_g \|_L^2. \tag{S1}$$

Consequently `a=g` is a global maximizer of this main term for the fixed `r,L`. The maximizer need not be unique in `a` if the convolution map has a kernel, but its image `b_a=b_g` is unique.

This is an identity in a finite coefficient Hilbert space. It should not be described as exact orthogonal projection in the time-domain Gaussian Hilbert space at finite `T`: that Gram matrix has small non-diagonal entries. The arithmetic theorem gives the passage from the latter to the diagonal main term and an error estimate.

Suppose a new direction `C` has coefficients supported on the **same** product set `q<=L`. The main-term inner product of `C` with the discarded residual is zero. Its recovery term therefore equals `−||C||_L^2`, which is nonpositive. A change of basis, a more complicated inverse Gram, a nonlinear parameterization of `a`, or repeated completion of squares does not change this fact.

There is also the exact time-domain identity

$$\begin{aligned} 2\,\mathrm{Re}\langle B_0, X\rangle - \| B_0 \|^2 + 2\,\mathrm{Re}\langle C, X - B_0\rangle - \| C \|^2 \\ = 2\,\mathrm{Re}\langle B_0 + C, X\rangle - \| B_0 + C \|^2. \end{aligned} \tag{S2}$$

Thus “residual recovery” has not by itself enlarged the theorem: it is a larger approximator. Its value lies in designing a larger approximator whose mixed terms remain accessible.

The asymmetric–length caveat

Theorem 4 already permits setting `r(n)=0` for `n>L0` while retaining product cutoff `L>L0`. Therefore changing the relative factor lengths strictly inside the old product region is already contained in the original general theorem. It may improve a restricted trial family, but it is not a new arithmetic input. The useful analogy with the prime-gap 186 proof starts only when the **actual accessible mixed products** exceed what this theorem already controls.

4. Robust inverse-Gram certificate and the missing quantities

For candidate residual directions `C_1,...,C_m`, define

$$G_{ij} = \langle C_i, C_j \rangle, \qquad b_i = \langle C_i, X - B_0 \rangle.$$

The recovery for a fixed coefficient vector `c` is

$$\Delta(c) = 2\,\mathrm{Re}(c^* b) - c^* Gc. \tag{G1}$$

If estimates `bhat,Ghat` have bounds

$$|\, b_i - \widehat{b}_i \,| \le \epsilon_i, \qquad |\, G_{ij} - \widehat{G}_{ij} \,| \le E_{ij},$$

then the elementary certified lower bound is

$$\Delta(c) \ge 2\,\mathrm{Re}(c^* \widehat{b}) - c^* \widehat{G} c - 2 \sum_i |\, c_i \,| \epsilon_i - \sum_{i,j} |\, c_i \,| \, |\, c_j \,| E_{ij}. \tag{G2}$$

An explicit rational `c` and entrywise enclosures suffice. Matrix inversion and a numerical condition number are not part of the final certificate. Singular Gram matrices cause no difficulty for (G2).

For the reproduced one-variable baseline, the additional gain must exceed approximately `0.3031544077` after division by the resonator norm. A design goal `0.31` allows only a modest error budget; it is not a universal lower bound on the energy needed by other methods.

Rank, trace, and Hilbert – Schmidt norm are insufficient to determine this recovery. For example the same rational matrix `diag(1,4)` gives optimal recovery `1` for `b=(1,0)` and `1/4` for `b=(0,1)`. This is the precise location where directional information matters.

What a genuinely longer direction requires

For Gaussian packet width `W=T/log T`, the transformed weight is significant when

$$|\, \log(km/q) \,| \lesssim W^{-1}.$$

At a product scale `M=T^(1+eta)`, this retains additive differences of size roughly `M/W`, rather than only the exact equation `km=q`. The mixed moment contains prime-power coefficients `g(k)`, resonator coefficients `r(m)`, coefficients of `C(q)`, and a nontrivial oscillatory phase determined by the packet center. No estimate for generic arithmetic progressions automatically supplies this shifted multiplicative correlation. A proposed well-factorable or Type-II argument must state the actual variables, modulus, frequency range, coefficient norms, and total error after summing every direction.

5. A rigorous obstruction to a naive sparse long-support strategy

The following statement is elementary but useful for support design.

Sparse-tail lemma. Let `eta>0`, `M=T^(1+eta)`, and `W=T/log T`. Let a set `S` in `[M,2M]` have pairwise logarithmic spacing at least `c/W`, for some fixed `c>0`. Suppose coefficients satisfy `|b(q)|<=C_epsilon q^epsilon` for every fixed positive epsilon. Then

$$\sum_{q \in S} \frac{|\, b(q) \,|^2}{q} = o(1). \tag{ST}$$

Indeed, the logarithmic interval has length `log 2`, so `|S|<=1+W log(2)/c`. The displayed sum is at most a constant times `W M^(-1+2epsilon)`. Choosing `2epsilon(1+eta)<eta` proves the claim. The same conclusion holds relative to a normalizing resonator norm that is bounded below by `T^(-o(1))`.

This applies to the usual fixed divisor-polynomial resonator coefficients. It rules out obtaining fixed recovery simply by selecting a very sparse, trivially separated portion of a genuinely long dyadic tail and then using its diagonal coefficient energy. It does **not** rule out dense structured supports, substantial off-diagonal mixed information, or different coefficient classes with quantitatively stronger concentration.

6. Independent reproduction of the original variational calculation

The new implementation uses a different simplex parameterization from the supplied original: the background mass `v` is the outside Gauss – Jacobi variable, and inserted masses occupy its complementary simplex. The basis

$$\psi_j(v) = \sqrt{2j + \ell^2}\, P_j^{(0,\ell^2-1)}(2v-1)$$

is orthonormal for `v^(ell^2-1)dv`. Its norm matrix is the identity. This avoids unstable monomial inversion in the original one-variable calculation.

Trial	ell	Half-gap margin
Degree 2	1.1720511457	-0.01539943397962
Degree 6	1.1762949914	-0.01535798218167
Degree 14	1.1762950386	-0.01535798170385

The published linear trial gives `1.487161813351623e-5` at quadrature order 48. Orders 20, 32, and 48 agree to approximately `2e-15`. The degree-8 numerical zero is `0.5088369010687765`.

None of these floating calculations bounds all continuous functions or all `ell>=1`. Raising the polynomial degree is not an independent arithmetic theorem. The exact saturation theorem in § 3 concerns a different variable, namely the approximator with the resonator fixed; it does not imply that the resonator family is globally optimal.

7. An exact arithmetic operator for arbitrary resonator coefficients

At `phi=1/2`, the leading linear term vanishes. Put `x_n=r(n)/sqrt(n)`. Define a real sparse creation operator `A` by

$$A_{qm,m} = \frac{2\sin(\frac{h}{2}\log q)}{e\sqrt{q}} \quad \text{when } q = p^e, \quad qm \le L,$$

and zero otherwise. Here `h=pi/log T`; the exploratory runs set `log L/log T=1`, the limiting boundary of the admissible range. Since `g(q)=-2i sin((h/2)log q)/e`, the complex convolution operator is `B=-iA`.

The approximator term is `||Bx||^2`, and the holomorphic correction is `Re<x,B^2x>`. Therefore the exact diagonal main-term matrix is

$$K_L = A^*A + \frac{A^2 + (A^*)^2}{2}. \tag{A1}$$

Its maximal normalized half-gap margin is

$$\frac{\lambda_{\max}(K_L)}{2\pi^2} - \frac{1}{4}. \tag{A2}$$

All coefficients of `K_L` are nonnegative at this parameter, so a real nonnegative Perron vector suffices for the extremal Rayleigh quotient. This is a substantially larger search space than the one-variable divisor family.

L	largest eigenvalue of K	margin	eigenvector residual norm
1,000	3.9492871367	-0.0499267764	3.0e-11
10,000	4.1058670454	-0.0419943455	4.6e-15
100,000	4.2052553801	-0.0369592737	2.0e-13
1,000,000	4.2738969159	-0.0334818529	3.1e-12

The threshold needed at the boundary is `pi^2/2`, approximately `4.9348022`. The full matrix at `L=10^6` has 3,626,619 nonzero entries in `A`. Its sparse operator implementation avoids materializing `K`.

These are finite experiments. The choice `log L/log T=1` is an asymptotic model, not an admissible finite theorem with `L=T`. An eigenvector for one finite `L` cannot be frozen and used as `T` tends to infinity. No extrapolated limit is claimed, and no global no-go follows from these negative eigenvalues.

A useful next question is whether

$$K_L = \tfrac{1}{2}(A + A^*)^2 + \tfrac{1}{2}[A^*, A]$$

admits a positive comparison or a Hardy-type bound uniformly in `L`. A proof that the normalized limiting norm is at most `1/4` would close the entire diagonal method, rather than merely one family of weights. Such a proof has not been obtained.

8. A genuinely broader family: symmetric prime–factor statistics

The finite arithmetic eigenvector search motivates coefficients of the form

$$r(n) = d_\ell(n) H(v, S_2(n), S_3(n), \dots), \qquad v = \frac{\log n}{\log L}, \qquad S_k(n) = \sum_{p \mid n} \left(\frac{\log p}{\log L}\right)^k. \tag{PF1}$$

A Liouville sign can be included when the sign of the linear term calls for it. At the half-gap boundary the quadratic terms do not distinguish that sign. Every `S_k` and every polynomial in these variables is symmetric in the prime factors. These features do not repeat the unsymmetric-weight error in the withdrawn 2019 calculation.

For the continuum experiment, use a background mass `v` and its scaled Poisson – Dirichlet prime-factor partition with parameter `a=ell^2`. Its relevant moments are

$$\mathrm{E} S_k = \frac{\Gamma(k)\Gamma(a+1)}{\Gamma(a+k)} v^k,$$

and

$$\mathrm{E} S_k S_l = \frac{\Gamma(a)[a\Gamma(k+l) + a^2\Gamma(k)\Gamma(l)]}{\Gamma(a+k+l)} v^{k+l}. \tag{PF2}$$

More generally, for a list `k_1,...,k_s`,

$$\mathrm{E}\prod_{j=1}^{s} S_{k_j} = \frac{\Gamma(a)}{\Gamma(a+\sum_j k_j)} v^{\sum_j k_j} \sum_{\pi} a^{|\pi|} \prod_{B\in\pi} \Gamma\left(\sum_{j\in B} k_j\right), \tag{PF3}$$

where the sum runs over set partitions of the labeled factors. Adding a prime of logarithmic size `u` replaces `(v,S_k)` by `(v+u,S_k+u^k)`. This gives explicit Gram and quadratic-form integrals for each symmetric feature polynomial.

The calculation reduces exactly to the independently reproduced Inoue form when `H` depends only on `v`; this reduction was checked at three different values of `ell` and by two independently written feature implementations.

Symmetric features, each with a small mass polynomial	optimized ell	numerical margin at phi=1/2
1 only, degree 6	1.176295	-0.0153579822
1, S2, degree 4	1.065285	-0.0146618161
1, S2, S3, degree 4	1.077117	-0.0146563150
1, S2, S3, S4, degree 4	1.081082	-0.0146551195
12 groups including S2 squared, S2 S3, S2 cubed, S2 fourth power	1.079948	-0.0146547256

The richer Gram matrices become ill-conditioned. Directions below an explicit eigenvalue cutoff were discarded. Therefore the last decimals and any inference of saturation in this expanded family are exploratory only. The main robust conclusion is that these particular natural extra arithmetic features recover a small fraction of the deficit, far less than required to cross `1/2`.

Arithmetic status of this extension

The displayed partition moments define a coherent continuum calculation. To turn it into a zeta theorem one must prove the corresponding asymptotics for weighted integer sums, including repeated-prime terms, coincidences between inserted primes and the background integer, the short-background range, and the errors after normalizing by the resonator norm. The original theorem handles arbitrary `r`, but it does not automatically supply this new explicit asymptotic evaluation. No such transfer is claimed as completed here.

9. An exact rational certificate for one expanded continuum trial

To avoid trusting the ill-conditioned optimizers, a simple trial was rounded to rational coefficients. Take `ell=16/15` and

$$H(v,S_2) = f(v) + g(v)S_2,$$

with

$$f(v) = \frac{145+3v-116v^2+71v^3-6v^4}{100},$$

$$g(v) = \frac{-563+1682v-2479v^2+1751v^3-488v^4}{100}.$$

The script `rational_trial_certificate.py` proves, by rational arithmetic, that its continuum half-gap margin lies in

$$-\frac{1467}{100000} < \mathrm{J} < -\frac{1465}{100000}. \tag{RC}$$

A much tighter outward enclosure is stored in the JSON certificate; its decimal presentation is approximately

`[-0.014662375473368995, -0.014662375473368974]`.

The method is completely explicit:

- Machin's formula and alternating arctangent series give rational lower and upper bounds for pi.
- Sine and cosine are replaced by finite Taylor polynomials with explicit remainder bounds on the simplex.
- The symmetric feature expectations are rational because `ell^2=256/225` and all moment degrees are integers.
- Every simplex monomial integral is a factorial divided by a finite product of rational numbers.
- The numerator and positive denominator are enclosed before division. Final assertions compare exact rational numbers.

During development, an off-by-one error in a simplex beta denominator was caught by disagreement with the independent floating integral, then corrected. Explicit constant-monomial identities are now asserted in the certificate script. The final run succeeds. This history is preserved to make clear why multiple forms of verification matter.

The certificate proves a statement about **one explicit continuum trial**, not a global upper bound and not an arithmetic half-gap improvement. In particular a negative trial is not a proof that the entire method fails.

10. AH, multiplicity, and positive-density targets

There are several different statements called the Alternative Hypothesis.

The 2019 Lagarias – Rodgers formulation, Conjecture 2.2, explicitly takes the limiting consecutive-gap values in `{1/2,1,3/2,...}`, excluding zero, for all sufficiently large indices. A theorem `liminf g_n<1/2` would refute this strong formulation. It is incorrect to say that this particular source allows zero gaps.

Baluyot – Goldston – Suriajaya – Turnage-Butterbaugh 2025 studies a formulation that does not assume simplicity. For that broader setting, repeated zeros or gaps tending to zero require separate treatment. A small-gap theorem counting multiplicity may detect only those zeros. A statement allowing a density-zero exceptional set is also not contradicted merely by infinitely many exceptional gaps.

A robust target is a positive density of pair separations in a fixed interval whose closure avoids every permitted half-integer, together with whichever local-count control is needed to translate pair information into adjacent-gap information. A positive proportion of weighted factorial moment is not automatically an unweighted positive proportion: the resonator could concentrate on a negligible set of heights.

For an unweighted count there is a useful exact detector. Write

$$Q(h) = \int N_h(t)(N_h(t) - 1)\, dt = 2 \sum_{i<j} (h - |\gamma_i - \gamma_j|)_+$$

when boundary terms are absent or handled explicitly. For `0<h0<h1<h2` in arithmetic progression,

$$Q(h_2) - 2Q(h_1) + Q(h_0)$$

has a nonnegative triangular pair kernel supported in `[h0,h2]`, with value zero at separation zero. Thus it removes the direct multiplicity atom exactly. A smooth nonconstant arithmetic weight changes the kernel and must be analyzed rather than silently canceled. Also a close nonadjacent pair need not give an adjacent gap bounded away from zero without local cluster control. This detector is a possible way to formulate a stronger arithmetic target; its positivity has not been proved from the available zeta estimates.

11. Artifacts and reproducibility

All files are in `research-round1/residual-gram/`, separate from the recovered original source tree.

- `inoue_variational.py`, `variational-results.json`: independent one-variable reproduction.
- `arithmetic_operator.py`, `arithmetic-results.json`, `arithmetic-eigenvector.npz`: arbitrary-coefficient sparse arithmetic search through `L=10^6`.
- `prime_feature_variational.py`, `prime-feature-results.json`: linear symmetric prime-factor features.
- `general_prime_features.py`, `general-prime-feature-results.json`: products of symmetric prime moments.
- `rational_trial_certificate.py`, `rational-trial-certificate.json`: exact rational continuum-integral certificate.
- `check_algebra.py`, `algebra-check-results.json`: structural completion-of-squares identities, operator sign check, reduction of both feature implementations to the original form, and comparison against the now recovered original result file.
- `inoue-paper.html`: retrieved primary-paper snapshot for formula checking.
- `*-run.log`: execution records.

Commands used:

```
OPENBLAS_NUM_THREADS=1 python3 research-round1/residual-gram/inoue_variational.py
OPENBLAS_NUM_THREADS=1 python3 research-round1/residual-gram/arithmetic_operator.py
OPENBLAS_NUM_THREADS=1 python3 research-round1/residual-gram/prime_feature_variational.py
OPENBLAS_NUM_THREADS=1 python3 research-round1/residual-gram/general_prime_features.py
python3 research-round1/residual-gram/rational_trial_certificate.py
OPENBLAS_NUM_THREADS=1 python3 research-round1/residual-gram/check_algebra.py
```

The algebra test suite is not a proof assistant. The rational certificate's stated finite integral inequality is exact rational computation with explicit analytic remainder estimates. The spectral searches and their optimizer outputs remain numerical.

12. Recommended next bounded attack

First investigate a uniform upper or a counterexample for the full arithmetic operator (A1), including its commutator structure and the shape of finite Perron vectors. This is the quickest way to tell whether more elaborate resonators inside the old product region can matter at all. A uniform upper would justify making genuine long-support mixed moments the primary target; a counterexample would identify a more concrete family to analyze.

In parallel with that mathematical question, specify one dense residual coefficient family and compute its exact near-diagonal product conditions. The theorem request must be an explicit sum with an explicit normalization and error target. It should not merely say “apply Guth – Maynard,” “use a Yau flow,” or “recover the residual Gram.” The measured margin supplies a useful numerical target, but the required new arithmetic input remains the central unresolved issue.

Current report 04: residual_gram_round2

Source: `research/reports/residual_gram_round2.md`. SHA-256: `1e625af13fe203ed3e9d70f78bf191fba20dcec10cf10ac2ebaed8b0de36f18c`.

Full arithmetic operator: extended spectra, Schur bounds, and prime–bin Fock models

Research round 2, 5 September 2026. This is a reproducible continuation of `research-round1/residual_gram.md`. No half-gap breakthrough or universal impossibility theorem was obtained. The strongest exact statement remains coefficient-space approximator saturation. This round tests the much larger remaining resonator problem.

1. Concrete result

For arbitrary resonator coefficients, the saturated half-gap main term is governed by

$$K_L = A^*A + \tfrac{1}{2}(A^2 + (A^*)^2), \qquad A_{p^e m,m} = \frac{2\sin(\frac{\pi}{2}\log(p^e)/\log L)}{e\sqrt{p^e}}.$$

The finite experiments set `log L/log T=1`, the limiting boundary of the admissible logarithmic range. The target is

$$\limsup_{L\to\infty} \lambda_{\max}(K_L) \;?\!\le \frac{\pi^2}{2} = 4.934802200544679\ldots.$$

The normalized target is `lambda/(2 pi^2)-1/4<=0`. The raw operator is **not** being compared with `1/4`.

The full sparse eigenvalue search was extended to ten million coefficients:

L	lambda max	normalized margin	eigenvector residual
1,000,000	4.273896915919377	-0.03348185285686	3.14e-12
3,000,000	4.299660502530880	-0.03217665431168	8.37e-12
10,000,000	4.324089558989538	-0.03093906385385	2.04e-11

The last creation matrix has 37,861,249 nonzero entries. `K_L` is applied as a sparse operator, not materialized. The ten-million run took approximately 27 seconds on the available machine, excluding any subsequent serialization. This timing is a recorded run, not a performance guarantee.

The values rise with L and remain negative. They neither prove a limiting upper bound nor justify extrapolating to a particular constant. A fixed finite eigenvector is not itself a valid asymptotic zeta certificate.

2. Finite eigenvector structure: the additional features nearly exhaust the tested problem

At `L=10^6`, the true top eigenvalue is `4.273896915919377`. We fitted its normalized Perron vector using vectors

$$\frac{d_\ell(n)}{\sqrt{n}}\, v^i F(n), \qquad v = \frac{\log n}{\log L}, \qquad 0 \le i \le 4.$$

The first feature set is just `F=1`. The next uses `{1,S2,S3,S4}`, with `S_k=sum_{p|n}(log p/log L)^k`. Larger fits add the largest logarithmic prime factor and then the number of distinct prime factors `omega(n)`. Every reported Rayleigh quotient is evaluated again using the full finite sparse arithmetic operator.

ell	feature set	terms	L2 fit error	Rayleigh deficit from full optimum
1.08	mass only	5	0.09399054	0.0325942265
1.08	symmetric prime powers	20	0.01813753	0.0007337046
1.08	plus largest prime factor	30	0.01524505	0.0006332664
1.08	plus omega and omega squared	40	0.01134669	0.0004437114

The 40-term fit reaches `4.273453204513651`, within about 0.011% of the full tested maximum. The 20-term fit already reaches `4.273163211325686`.

This is evidence that the finite operator's maximizing direction has considerable multiplicative structure and is well approximated by the symmetric prime-factor family tested in round one. It is not evidence that the same finite family is asymptotically complete. The role of `omega(n)` can change with L, and a small finite-L residual does not bound unseen asymptotic directions.

The fits were also made at ell=1 and ell=1.1763; all results, including worse fits, are retained in JSON. Least-squares truncation uses an explicit numerical singular-value threshold, so these fits are not exact interval certificates.

3. A scalar Cauchy majorant for the full operator

This section gives a candidate analytic bound with explicit derivation. Its numerical optimization does not reach the desired threshold.

Write `a_q=2 sin((pi/2) log q/log L)/(e sqrt(q))` for `q=p^e`. The divisor identity

$$\sum_{q \mid n} \Lambda(q) = \log n$$

and Cauchy – Schwarz give

$$\| Ax \|^2 \le \sum_{m \le L} | x_m |^2 \sum_{q \le L/m} \frac{a_q^2 \log(qm)}{\Lambda(q)}. \tag{B1}$$

For the other part, take a positive function Y on `(0,1]`, and set `v_n=log n/log L`. Applying `2ab<=a^2+b^2` to each product gives

$$\begin{aligned} \mathrm{Re}\langle x, A^2 x\rangle \le \frac{(\log L)^2}{2} \sum_m | x_m |^2 \sum_{kl \le L/m} \frac{a_k^2 a_l^2}{\Lambda(k)\Lambda(l)} Y(v_{klm}) \\ + \sum_n \frac{| x_n |^2}{2 Y(v_n)(\log L)^2} \sum_{kl \mid n} \Lambda(k)\Lambda(l). \end{aligned} \tag{B2}$$

Terms without prime-power factors are zero. Since the pairs `kl|n` are a subset of pairs of divisors of n,

$$\sum_{kl \mid n} \Lambda(k)\Lambda(l) \le (\log n)^2.$$

Let

$$f(u) = \frac{\sin^2(\pi u/2)}{u^2}, \qquad f(0) = \pi^2/4,$$

and

$$B(v) = 4\int_0^{1-v} (v+u)f(u)\,du.$$

Replacing the prime sums in (B1) – (B2) by their elementary prime-number asymptotics suggests the explicit bound

$$\limsup_{L} \lambda_{\max}(K_L) \le \sup_{0\le v\le 1} \left[B(v) + \frac{v^2}{2Y(v)} + 8\int_{u+w\le 1-v} f(u)f(w)Y(v+u+w)\,du\,dw\right]. \tag{B3}$$

For a fixed positive smooth Y bounded away from zero, the passage from the finite inequalities to (B3) can be justified uniformly by partial summation. Contributions of powers `p^e`, `e>=2`, are `O(1/log L)`: use the small-argument bound for sine and convergence of the corresponding prime-power reciprocal sums. One should write those uniform remainder estimates in a final proof rather than cite this numerical note as their formal verification. Profiles vanishing at v=0 can first be regularized by adding a fixed positive epsilon.

This is related to the weighted elementary inequalities in Inoue – Kobayashi – Toma, arXiv:2510.14309, but it treats the quadratic operator of the newer resonance-correlation method. Their existing linear-method barrier cannot simply be reused as an upper bound for K.

Numerical result: this majorant still fails

A constant Y gives a best sampled bound near `5.28075150`, larger than `pi^2/2`. Allowing Y to vary leads to a backward Volterra equation. With

$$C(s) = 8\int_0^{s} f(u)f(s-u)\,du,$$

the equality profile at a proposed upper bound lambda satisfies

$$Y(v) = \frac{v^2}{2\left[\lambda - B(v) - \int_v^1 C(t-v)Y(t)\,dt\right]}. \tag{V}$$

A discretized backward recursion and bisection gave:

grid intervals	numerical threshold lambda
250	5.2073898741
500	5.2073943491
1,000	5.2073954679
2,000	5.2073957475
4,000	5.2073958175

The corresponding normalized bound is approximately `+0.01380975`. It cannot prove a half-gap no-go. These are numerical values for the majorant search, not a certified global optimum over Y. The failure is preserved because it identifies a specific loss: separately estimating the norm square and the holomorphic term by Cauchy loses the compatibility of their near-extremizers. More precision in the same grid would not close a gap of this size.

4. Prime bins and a finite bosonic creation model

A second approach is to retain more of the prime-factor combinatorics.

Divide logarithmic prime sizes into bins `((j-1)/N,j/N]`, `1<=j<=N`, and set

$$c_j^2 = \int_{(j-1)/N}^{j/N} \frac{4\sin^2(\pi u/2)}{u}\,du.$$

Use occupation vectors `k=(k_1,...,k_N)` with the rounded-up energy restriction

$$\sum_{j=1}^{N} j k_j \le N.$$

The finite bosonic matrix is

$$(A_N)_{k+e_j,k} = c_j\sqrt{k_j+1}$$

whenever the new state remains in the energy set. Form

$$K_N = A_N^* A_N + \tfrac{1}{2}(A_N^2 + (A_N^*)^2).$$

Every mode and every occupation vector retained by this construction is explicit. The rounding restricts the support; it is not an upper discretization of the full continuum norm.

bins N	state dimension	largest eigenvalue	normalized margin
12	272	3.9285784722	-0.0509758896
20	2,714	4.0882405964	-0.0428873119
28	18,460	4.1870051368	-0.0378838418
36	99,133	4.2539816177	-0.0344907736
40	215,308	4.2800240103	-0.0331714506

At N=40 the creation matrix has 963,320 nonzero entries. Its eigenvector residual is approximately `1.6e-11`. No positive half-gap certificate occurs in these models.

A proposed fixed-bin arithmetic lower-transfer proof

There is a concrete route from each fixed finite bin model to arithmetic resonators; this is more modest than identifying the entire limiting operator norm.

For every occupation vector, take the normalized vector supported on squarefree integers whose prime factors have those bin counts, with coefficient proportional to the product of their prime weights `a_p`. Rounded-up energy ensures that all such integers are at most L. Different occupation vectors have disjoint supports.

If `E_k` is the elementary symmetric sum of order k in one bin's squared prime weights, the exact raising coefficient between its collective vectors is

$$(k+1)\sqrt{E_{k+1}/E_k}.$$

For fixed bin resolution and fixed k, the prime-number theorem gives the total squared weight tending to `c_j^2`; the maximum individual squared weight tends to zero. Consequently

$$E_k \longrightarrow (c_j^2)^k/k!,$$

and the raising coefficient tends to `c_j sqrt(k+1)`. This handles repeated occupation of a bin through **distinct primes**. It does not pretend that repeated copies of the same prime are independent bosons.

Let P project onto the retained collective squarefree vectors. The exact inequality

$$PA^*AP \geq (PAP)^*(PAP)$$

retains the nonnegative loss from intermediate states outside the subspace. Raising twice cannot leave the rounded-energy set and later return. Squareful states created by a repeated prime or a prime power cannot return to the squarefree subspace by further multiplication. These observations give the expected convergence of the holomorphic compressed term to `A_N^2`. The proposed conclusion is

$$\liminf_{L\to\infty} \lambda_{\max}(K_L) \geq \lambda_{\max}(K_N) \quad \text{for every fixed N.} \tag{FT}$$

This argument is a proof draft requiring independent line-by-line review and a complete statement of the finite-bin normalizations. It gives a **lower** search mechanism. It gives no upper bound for the full arithmetic operator and no equality with a limiting Fock norm. A positive rationally certified finite-bin value above `pi^2/2` would be a concrete success criterion, but all values computed here are below that threshold.

5. Why the continuum limit is delicate: the small-prime cloud

The divisor-family resonators suggest an infrared phenomenon. Under the `d_ell(n)^2/n` weighting, the number of logarithmically small prime factors grows on the scale `ell^2 log log L`, while their total logarithmic mass remains bounded. Therefore a finite-particle or coarse energy-bin model can converge slowly even when the creation kernel itself is square-integrable. One cannot infer a precise limit from the modest bin table.

A useful candidate representation uses the scaled Poisson – Dirichlet partition of mass v, with parameter `a=ell^2`, as the reference measure. In that representation the creation operator acts by removing a part u with coefficient `2 sin(pi u/2)/ell`, while its adjoint inserts a part with intensity `2 ell sin(pi u/2) du/u`, subject to total mass at most one. The symmetric prime-moment polynomials of round one are a natural Galerkin family in this representation.

This description explains the algebraic insertion formulas and suggests a more stable basis. It is **not** a proved statement that every asymptotic arithmetic resonator belongs to one such representation, that different ell sectors exhaust the full norm, or that finite-bin spectra converge to the desired universal operator. Those are separate operator-limit problems. The original PD arithmetic asymptotic transfer also remains under independent audit.

6. Files, dependencies, and commands

Round-two files are in `research-round2/residual-gram/`. They refer to the round-one `arithmetic_operator.py` implementation through a relative path. The source snapshot is not modified.

- `extended_arithmetic.py` and `extended-arithmetic-results.json`: reproduce the 3-million and 10-million full spectra. The wrapper writes only round-two outputs.
- `eigenvector_features.py` and `eigenvector-feature-results.json`: fit the round-one million-dimensional Perron vector and re-evaluate each candidate using the full operator.
- `first_upper_bound.py`, `first-upper-bound-results.json`: constant-weight scalar majorant.
- `volterra_upper_search.py`, `volterra-upper-results.json`, `volterra-upper-profile.npz`: variable-weight majorant search.
- `boson_cutoff.py`, `boson-cutoff-results.json`: all finite prime-bin models listed above.
- `*-run.log`: retained execution records.

Dependencies are Python, NumPy, and SciPy. The runs used the machine's `python3` (Homebrew Python 3.14), not the bundled PDF runtime. The rational certificate in round one additionally requires SymPy. For reproducibility, use the same environment that ran the round-one numerical files; no private service, GPU, or new model invocation is required.

```
OPENBLAS_NUM_THREADS=1 python3 research-round2/residual-gram/extended_arithmetic.py
OPENBLAS_NUM_THREADS=1 python3 research-round2/residual-gram/eigenvector_features.py
OPENBLAS_NUM_THREADS=1 python3 research-round2/residual-gram/first_upper_bound.py
OPENBLAS_NUM_THREADS=1 python3 research-round2/residual-gram/volterra_upper_search.py
OPENBLAS_NUM_THREADS=1 python3 research-round2/residual-gram/boson_cutoff.py
```

The 10-million eigenvector archive is about 85 MB and is optional research data. It should not be included in a small source-code PR by default. Pass `--save-largest` to the wrapper to regenerate it. The eigenvalue JSON and scripts are sufficient to reproduce the reported values. The fitting script uses the smaller, already archived round-one eigenvector.

7. What this round rules out, and what it leaves open

The exact saturation theorem rules out constant main-term recovery by changing only the approximator inside the fixed product space. The sparse-tail lemma rules out a specified separated-tail tactic for divisor-bounded coefficients. Those are genuine statements with explicitly limited scopes.

The new negative finite spectra, the near-optimal finite fits, the negative bosonic bins, and the failed Schur majorant are **not** universal no-go theorems. Together they make a large hidden gain from a routine reparameterization less plausible, but that judgment is a research inference rather than a proof.

A precise next analytic success criterion is an explicit positive comparison profile or an operator identity proving a uniform upper below `pi^2/2`, with every prime-power and cutoff error included. A precise alternative success criterion is a finite prime-bin or other explicit coefficient family with a certified limiting Rayleigh quotient above that value. Merely extending the grid or reporting another negative optimized decimal would not meet either criterion.

Current report 05: centered_gaussian_mixed_moments

Source: `research/reports/centered_gaussian_mixed_moments.md`. SHA-256: `5aec8de6c69d3b8d67799a65d0598d49ff1a87a5878ebc548dd3c90d806e1716`.

Centered Gaussian packets: an exact mixed identity and an unavoidable pole debt

Research round 2, 2026-09-05. This note settles the proposed shortcut of replacing Inoue's shifted Gaussian packet by a Gaussian centered at zero and then using Fourier positivity to retain long mixed products for free. The Fourier kernel is positive, but the exact critical-line identity contains a pole-cut correction with the opposite phase on the relevant long-support region. That correction can exceed the purported normalization by an unbounded factor for coefficients bounded by one.

Main conclusion. A centered packet may be used only if its pole term, its true nondiagonal Gram form, and its low-height zero-count terms are retained. Dropping those terms is false. This is a precise obstruction to the proposed free support enlargement, not an impossibility theorem for every centered-packet method.

1. Conventions and source input

Let

$$\Phi_W(t) = e^{-t^2/(2W^2)}, \qquad \widehat{V}(\xi) = \int_{\mathbb{R}} V(t) e^{-2\pi i t\xi}\, dt, \qquad W > 0.$$

Thus $\widehat{\Phi}_W(\xi) = \sqrt{2\pi} W e^{-2\pi^2 W^2 \xi^2} > 0$. The height parameter T and packet width W are distinct. Below $h = \pi/\log T$, while W can be T or $T/\log T$. A weight written as e^{-t^2/T^2} corresponds to $W = T/\sqrt{2}$.

Use the horizontal branch

$$\log \zeta(\sigma + it) = \int_{\infty}^{\sigma} \frac{\zeta'}{\zeta}(a + it)\, da$$

away from zero ordinates and $t = 0$, with the mean of the two boundary values at exceptional ordinates. Put

$$\Delta_h(t) = \log \zeta\left(\tfrac{1}{2} + i(t + h/2)\right) - \log \zeta\left(\tfrac{1}{2} + i(t - h/2)\right),$$

$$g_h(k) = \frac{\Lambda(k)}{\log k}\left(k^{-ih/2} - k^{ih/2}\right).$$

For $k = p^e$, $g_h(k) = -2i \sin((h/2)\log k)/e$; otherwise it vanishes.

The analytic input is Inoue, Lemma 5.5 and equations (5.42) – (5.43). For an analytic test function on the strip $-3/2 \le \operatorname{Im} z \le 0$ with the stated horizontal decay, its critical-line logarithm integral equals the Dirichlet contribution, zero-cut integrals, and a **negative pole-cut integral**. Under RH the zero-cut integrals to the right of the critical line disappear. We retain the pole term exactly, rather than invoking the small-error estimate used for the shifted packet.

The tests $V(z) = x^{-iz}\Phi_W(z)$ satisfy the strip and decay assumptions for every fixed $x > 0$, $W > 0$. Arbitrarily long but finite Dirichlet polynomials may consequently be expanded against this identity. No absolutely convergent Dirichlet series for $\log\zeta$ on the critical line is assumed.

2. Exact kernel identity

Theorem 1. Assume RH. For every $x > 0$, $h > 0$, and $W > 0$,

$$\begin{aligned} K_{h,W}(x) &:= \int_{\mathrm{R}} \Delta_h(t) x^{-it}\Phi_W(t)\, dt \\ &= \sqrt{2\pi}W \sum_{k\geq 2} \frac{g_h(k)}{\sqrt{k}} \exp\left[-\frac{W^2}{2}\log^2(kx)\right] + \mathrm{P}_{h,W}(x), \end{aligned} \tag{1}$$

where

$$\begin{aligned} \mathrm{P}_{h,W}(x) &= -2\pi \int_0^{1/2} [V(-h/2 - i\sigma) - V(h/2 - i\sigma)]\, d\sigma \\ &= \boxed{-4\pi i e^{-h^2/(8W^2)} \int_0^{1/2} x^{-\sigma} e^{\sigma^2/(2W^2)} \sin\left[\frac{h}{2}\left(\log x - \frac{\sigma}{W^2}\right)\right] d\sigma.} \end{aligned} \tag{2}$$

The series in (1) is absolutely convergent: the Gaussian decay in $\log k$ dominates the $k^{-1/2}$ factor and the number of integers at each logarithmic scale.

Proof. Apply the cited contour identity at shifts $v = h/2$ and $v = -h/2$ and subtract. RH removes zero cuts with real part greater than $1/2$. The pole at $s = 1$ leaves the first line of (2). The Fourier transform of $x^{-it}\Phi_W(t)$ at $\log k/(2\pi)$ is precisely the Gaussian in (1). Finally,

$$V(-h/2 - i\sigma) = x^{-\sigma} e^{\sigma^2/(2W^2) - h^2/(8W^2)} e^{i(h/2)(\log x - \sigma/W^2)},$$

and $V(h/2 - i\sigma)$ has the conjugate phase. Their difference proves (2). This derivation uses an identity obtained by shifting to an absolutely convergent half-plane; it does not formally expand the critical-line logarithm.

2.1 The sign on the genuine long-support region

If

$$0 < x \leq 1, \qquad 0 < h\left(\log(1/x) + \frac{1}{2W^2}\right) < 2\pi, \tag{3}$$

then $\mathrm{P}_{h,W}(x)$ is purely imaginary with **strictly positive imaginary part.** At $x = 1$ the integral is still positive because the phase is strictly negative for $\sigma > 0$.

In contrast, $g_h(k)$ is negative imaginary whenever $0 < h\log k < 2\pi$. Hence at the matching frequency $x = 1/k$, the pole correction has the opposite phase to the positive-kernel Dirichlet contribution.

The sign statement is deliberately restricted to (3). It is false for all positive x: for example $x = 2$, $h = 0.2$, $W = 3$ gives a negative imaginary correction. This is not a defect of the theorem; it is why support information matters.

3. Exact mixed and Gram forms for arbitrary finite supports

Let

$$R(t) = \sum_m \frac{r_m}{\sqrt{m}} m^{-it}, \qquad C(t) = \sum_q \frac{c_q}{\sqrt{q}} q^{-it}$$

be arbitrary finite Dirichlet polynomials. Then (1) gives the exact identity

$$\begin{aligned} \langle C, \Delta_h R\rangle_{\Phi_W} &= \sqrt{2\pi} W \sum_{k,m,q} \frac{g_h(k) r_m \overline{c_q}}{\sqrt{kmq}} e^{-\frac{1}{2} W^2 \log^2(km/q)} \\ &+ \sum_{m,q} \frac{r_m \overline{c_q}}{\sqrt{mq}} \mathrm{P}_{h,W}(m/q). \end{aligned} \tag{4}$$

Here $\langle C, X\rangle_{\Phi_W} = \int X(t)\overline{C(t)}\Phi_W(t)dt$; the notation specifies the order to prevent an unnoticed conjugation change.

The penalty is also exactly

$$\| C \|^2_{\Phi_W} = \sqrt{2\pi} W \sum_{q,r} \frac{c_q \overline{c_r}}{\sqrt{qr}} e^{-\frac{1}{2} W^2 \log^2(q/r)}. \tag{5}$$

For long dense support, (5) is not the diagonal coefficient norm. Its off-diagonal terms are positive when all c_q share one phase. Fourier positivity therefore enlarges the penalty as well as the suggested mixed gain.

Suppose $r_m \ge 0$, $c_q = -id_q$ with $d_q \ge 0$, and every retained q exceeds every retained m. If (3) holds for all $x = m/q$, **every pole summand in the second line of (4) is negative real**. This is the natural configuration when a new residual direction lives in a genuinely longer tail than the existing resonator. The positive Fourier kernel does not remove its debt.

For $q \le T^{1+\eta}$ with fixed $0 < \eta < 1$ and $h = \pi/\log T$, the effective prime-power range in (4) has the common phase. Terms beyond the first sine sign change are Gaussian-small because $\log(km/q)$ is then bounded below by a positive multiple of $\log T$. They can be bounded absolutely. The pole term remains and is the substantive obstruction.

4. A counterexample to treating the pole term as a negligible uniform error

The following example uses coefficients bounded by one and no conjecture about primes in short intervals.

Theorem 2. Fix $0 < \eta < 1$. Let $M = T^{1+\eta}$, $h = \pi/\log T$, and let $W = T$ or $W = T/\log T$. Take

$$R(t) = 1, \qquad C(t) = -i \sum_{M<q\le 2M} \frac{1}{\sqrt{q}} q^{-it}.$$

The pole contribution to (4) is negative and satisfies

$$\boxed{\mathrm{E}_{pole} = -\left(4\pi \cos\frac{\pi\eta}{2} + o(1)\right)\frac{M}{\log M}.} \tag{6}$$

In particular,

$$\frac{|\,\mathrm{E}_{pole}\,|}{\sqrt{2\pi} W} \longrightarrow \infty, \tag{7}$$

even though $\sum |\, r_m \,|^2/m = 1$ and $\sum |\, c_q \,|^2/q \to \log 2$.

Proof. Uniformly for $q \in (M, 2M]$, formula (2) and $W \to \infty$ give

$$\mathrm{P}_{h,W}(1/q) = 4\pi i \sin\left(\frac{h}{2}\log q\right) \frac{\sqrt{q}-1}{\log q} (1 + o(1)). \tag{8}$$

The sine is bounded away from zero in this range, since it tends to $\cos(\pi\eta/2) > 0$. The factors $e^{\sigma^2/(2W^2)}$, $e^{-h^2/(8W^2)}$, and the additional sine argument $h\sigma/(2W^2)$ change the integral by a uniformly vanishing relative error. The remaining elementary integral is

$$\int_0^{1/2} q^\sigma d\sigma = \frac{\sqrt{q}-1}{\log q}.$$

Multiply (8) by $\overline{c_q}/\sqrt{q} = i/\sqrt{q}$ and sum. The sum of $(1-q^{-1/2})/\log q$ over all integers in $(M, 2M]$ is $(1+o(1))M/\log M$. This proves (6). Finally $M/(W\log M)$ tends to infinity for both specified widths.

The theorem refutes a uniform long-support identity of the form "positive Dirichlet Gaussian sum plus an $o(W)$ error controlled by the diagonal norms." It does not assert that the full mixed moment itself is negative. The prime contribution and pole contribution may substantially cancel, and the surviving fluctuation is exactly what needs arithmetic control.

4.1 Why the apparent gain resembles a prime-density term

For a single frequency $x = 1/X$, the Gaussian Dirichlet sum samples prime powers near X on multiplicative width $1/W$. If one replaces the prime density heuristically by $dy/\log y$, its leading imaginary contribution is

$$-4\pi i \sin\left(\frac{h}{2}\log X\right)\frac{\sqrt{X}}{\log X}.$$

This has the same magnitude and opposite sign as (8). The replacement is an interpretation, **not** a proved short-interval prime estimate in the relevant ranges. It explains the structural cancellation: centering transforms the proposed free positive contribution into a prime-density-subtracted quantity. Proving useful positivity of that remainder still requires arithmetic information.

5. The origin and the zero-count formula cannot be discarded

Let $N(t)$ be the signed zero-counting function and let $\theta(t)$ be the odd Riemann – Siegel theta function. With the branch above, the exact global form away from endpoints is

$$N(t) = \frac{\theta(t)}{\pi} + S(t) + \operatorname{sgn}(t), \qquad S(t) = \frac{1}{\pi}\operatorname{Im}\log\zeta(\tfrac{1}{2} + it).$$

Consequently,

$$N_h(t) = \frac{\theta(t+h/2) - \theta(t-h/2)}{\pi} + S(t+h/2) - S(t-h/2) + 2\mathbf{1}_{\{|t|<h/2\}}. \tag{9}$$

The indicator is absent only when the interval avoids the origin. As $t \downarrow 0$, this branch has $S(t) \to -1$; as $t \uparrow 0$, it has $S(t) \to 1$. At $t = 0$ the jump in S and the indicator in (9) cancel. There are no zeros in a sufficiently small neighborhood of zero because $\zeta(1/2) \neq 0$.

A packet concentrated near heights comparable to T may replace the theta increment by $(h/2\pi)\log T + O(h)$ and suppress the origin. A centered packet requires a weighted argument proving that low heights are negligible. Pointwise replacement by $\log T$ on the entire real line is false.

5.1 A direct coherence bound for a positive long resonator

Let

$$R_M(t) = \sum_{M<n\le 2M} n^{-1/2-it}.$$

Factor out M^{-it}. For $|t| \le 1$, all remaining phases have arguments in an interval of length at most $\log 2 < \pi/2$. Therefore

$$|R_M(t)| \ge \cos(\log 2) \sum_{M<n\le 2M} n^{-1/2} \ge c\sqrt{M}.$$

For $W \ge 1$,

$$\int_{-1}^{1} |R_M(t)|^2 \Phi_W(t) dt \ge c' M. \tag{10}$$

The diagonal coefficient norm is only $\Sigma_{M<n\le 2M} 1/n \sim \log 2$, whose naive Gaussian normalization would have size W. If $M/W \to \infty$, the low-height coherent mass is much larger. This elementary example shows why Fourier-positive off-diagonal mass is not automatically useful high-zero information.

For a general nonnegative-coefficient resonator, even the shorter interval $|t| \le c/\log L$ gives a lower bound proportional to $|R(1/2)|^2/\log L$. Such mass must be controlled rather than discarded using an unweighted measure estimate.

5.2 A general positivity obstruction to cutting a hole at the origin

Suppose Φ is a real Schwartz function with $\widehat{\Phi} \ge 0$ and is not identically zero. Fourier inversion gives

$$\Phi(0) = \int \widehat{\Phi}(\xi) d\xi > 0, \qquad |\Phi(t)| \le \Phi(0).$$

Thus one cannot have both an everywhere nonnegative Fourier transform and a weight that vanishes at the origin, unless the weight is zero. Multiplying the Gaussian by a cutoff that removes the origin necessarily loses the claimed Fourier-positivity property. Translating a packet introduces the familiar phase $e^{-2\pi i t_0 \xi}$; imposing phases on resonator coefficients likewise gives up the common coefficient phase used for the free-sign argument.

This lemma does not rule out a subtler weight whose origin contribution is paid quantitatively. It rules out the exact combination "remove the origin while preserving all Fourier signs for free."

6. Verification and scope

`centered_gaussian_pole_checks.py` evaluates the finite pole integral in two independent forms at 80 decimal digits: direct complex evaluations of $V(-h/2 - i\sigma) - V(h/2 - i\sigma)$, and the sine integral in (2). The checks do not evaluate a critical-line Dirichlet series.

Representative output:

x, h, W	Pole correction	Mixed correction when $\overline{c} = i$
$0.1, 0.2, 3$	$+2.746104573214653 \ldots i$	$-2.746104573214653 \ldots$
$0.001, 0.1, 10$	$+18.88430170566545 \ldots i$	$-18.88430170566545 \ldots$
$1, 0.3, 2$	$+0.05966510707903 \ldots i$	negative

x, h, W	Pole correction	Mixed correction when $\overline{c} = i$
$2, 0.2, 3$	$-0.35517163933647\ldots i$	positive

For $\eta = 1/4$, the ratio of the exact pole integral at $x = T^{-5/4}$ to (8) is approximately 0.9999196069, 1.0000000386, and 1.0000000000000029 at $T = 100, 10^4, 10^8$. The two formulas agree to at least 70 decimal places in the tested cases. The script also checks the cancellation in (9) at the origin using the local branch and the theta function. These are high-precision diagnostics; the proof of the sign and the asymptotic counterexample is analytic and does not depend on their decimal output.

Completed claims in this note are the exact identity with its retained pole term, its sign on the specified support region, the dense-tail counterexample to a negligible-pole estimate, and the elementary coherence/Fourier-positivity obstructions. A full centered-packet formula for the squared logarithm and a successful positive residual certificate are not claimed.

The next admissible research move is to seek genuine control of the **density-subtracted long mixed moment**, or to design a packet/resonator whose pole and low-height contributions can be quantitatively paid. Replacing the shifted Gaussian by a centered one alone supplies neither.

Completion or pole regularization does not restore the missing sign

One can instead use $\log((s-1)\zeta(s))$, which removes the pole in the half-plane to the right of the critical line under RH. This changes the observable. Its absolutely convergent-side expression contains the additional term $\log(s-1)$, whose transformed contribution compensates for the removed pole cut. Likewise the completed xi-function introduces explicit gamma and polynomial factors. These are legitimate formulations, but their arithmetic kernel is no longer the uncorrected nonnegative prime sum. A proposed completed-function approach must keep that explicit continuous/archimedean contribution in its Gram certificate; regularization is not a license to erase the density subtraction identified above.

Current report 06: yau_flow

Source: `research/reports/yau_flow.md`. SHA-256: `0d407ae5cb198960bc3c00ff76f6baeb61e5536e63799eaf89a28af92f460542`.

Yau flows, circular heat localization, and a candidate CUE depth theorem

Research round 1, 2026-09-05. Prepared independently from the finite-depth source files and primary literature. The results below are mathematical arguments, not a Lean-checked proof. The CUE theorem is a proposed new consequence whose novelty requires further literature review; an attempted independent Claude review was unavailable because authentication returned HTTP 401. Another agent has checked the full deterministic argument and CUE transfer without finding a fatal issue, but that review is not formal verification. It is not a theorem about the Riemann zeta function.

1. Main finding

The earlier programme's dynamic background condition can be replaced by a simpler **initial-data isolation condition**. For the finite circular flow used in this project, an isolated closest pair admits a Gaussian heat-kernel scaling limit. For Haar CUE, an elementary three-point overcrowding estimate supplies that isolation with high probability. Combining this with the existing CUE extreme-gap theorem gives the following candidate theorem, pending independent adversarial review:

> Let U_N be Haar distributed in $U(N)$, let $P_N(z) = \det(zI - U_N) = \Sigma_{j=0}^{N} a_j z^j$, and evolve $P_{N,s}(z) = \Sigma_{j=0}^{N} a_j e^{sj(N-j)} z^j$. Let D_N be its first positive time with a repeated root, and let δ_N be the smallest initial circular angular gap. Then
>
> $$\frac{8D_N}{\delta_N^2} \longrightarrow 1 \quad \text{in probability.}$$
>
> Consequently, for every $t \geq 0$,
>
> $$\lim_{N\to\infty} \Pr(N^{8/3} D_N > t) = \exp(-\frac{2\sqrt{2}}{9\pi} t^{3/2}) .$$

This closes the finite CUE localization gap if the argument below passes further independent review. No universality assertion about arbitrary point processes or zeta zeros is included. The stochastic Yau machinery is not needed for this proof: the decisive simplification is to use the scalar heat equation directly.

2. Three flows that must not be conflated

Dyson Brownian motion is a stochastic eigenvalue evolution. In a conventional Hermitian normalization it has a Brownian term of order $N^{-1/2}$, a repulsive logarithmic interaction, and an optional confining drift. Erdős – Schlein – Yau local relaxation compares probability laws with a local equilibrium law. Their reverse heat-flow argument approximately inverts smoothing on a sufficiently regular matrix-entry density; it does not run individual DBM trajectories backward. See Erdős – Péché – Ramírez – Schlein – Yau and Erdős – Schlein – Yau – Yin.

The project's circular root evolution is deterministic and attractive:

$$\dot{\theta}_j = -\sum_{k \neq j} \cot \frac{\theta_j - \theta_k}{2}.$$

After multiplying the centered characteristic polynomial by a fixed phase to make it real, write

$$Q_0(x) = \prod_{j=1}^{N} \sin \frac{x - \theta_j}{2}.$$

The corresponding centered polynomial at time s is, up to its zero-irrelevant scalar factor,

$$\widetilde{Q}_s(x) = e^{sN^2/4} e^{s\partial_x^2} Q_0(x).$$

Indeed the Fourier mode $j - N/2$ receives multiplier $e^{s[N^2/4-(j-N/2)^2]} = e^{sj(N-j)}$. Thus $Q_s = e^{s\partial_x^2} Q_0$ solves ordinary forward scalar heat flow. For odd N, Q_0 is antiperiodic over 2π; it is periodic over 4π, and its real-line Gaussian representation remains valid. No periodicity shortcut is needed.

The de Bruijn – Newman convention has $\partial_t H_t = -\partial_x^2 H_t$. Increasing its parameter is the repulsive direction; our collision-time parameter s is the opposite direction. A ζ window cannot simply be replaced by a polynomial: the infinite product, outer zeros, normalization, and the true H_t evolution all need control.

Rodgers and Tao already explicitly invoke the analogy with the Erdős – Schlein – Yau local relaxation method. Their theorem assumes a fixed negative global Newman parameter for contradiction, evolves over an actual interval of real-rooted H_t, and drives the zeros toward local clock behavior. Their argument does not identify ζ with DBM and does not immediately turn a hard core at time zero into an interval of backward stability. See Rodgers – Tao and Tao's explanation.

3. A deterministic initial-data lemma

Consider a sequence of finite configurations, with size allowed to vary. Their circular angular gaps are initially positive. Let δ be the smallest gap, rotate its endpoints to $-\delta/2, \delta/2$, and write the other angles as $\theta_k \in (-\pi, \pi]$. Define

$$H(y) = \prod_{k \notin \{-,+\}} \sin \frac{y - \theta_k}{2}, \qquad A = \sum_{k \notin \{-,+\}} \frac{1}{2\,|\,\sin(\theta_k/2)\,|}.$$

Empty products and sums are allowed. Assume

$$\delta \to 0, \qquad \delta A \to 0. \tag{I}$$

Isolation lemma. Under (I), the global first collision time satisfies

$$D = \frac{\delta^2}{8}(1 + o(1)). \tag{L}$$

The assumption is on the initial configuration only. It neither assumes that the closest pair remains the first colliding pair nor requires an a priori bound on the evolving background. If a different pair collides first, the upper-bound argument remains valid.

3.1 Global control of the background product

Since sine is 1-Lipschitz,

$$\left| \frac{\sin((y - \theta_k)/2)}{\sin(-\theta_k/2)} - 1 \right| \leq \frac{|\,y\,|}{2\,|\,\sin(\theta_k/2)\,|}.$$

The elementary product inequality $|\prod(1+u_k)-1| \le \prod(1+|u_k|)-1$ implies, for every real y,

$$\left|\frac{H(y)}{H(0)}-1\right| \le e^{A|y|}-1, \qquad \left|\frac{H(y)}{H(0)}\right| \le e^{A|y|}. \tag{B}$$

These are global bounds; no cutoff in the heat convolution and no uncontrolled ratio to the global supremum of Q_0 appear.

3.2 Heat-kernel normal form

Set $s=\tau\delta^2$, and let Z be a standard real normal random variable. Because Q_0 is a finite real trigonometric polynomial,

$$Q_s(\delta x) = \mathrm{E}\, Q_0(\delta x + \sqrt{2s}Z).$$

With $Y = x + \sqrt{2\tau}Z$, define

$$F_\delta(x,\tau) = \frac{4Q_{\tau\delta^2}(\delta x)}{\delta^2 H(0)} = \mathrm{E}\left[\frac{4}{\delta^2}\sin\frac{\delta(Y-1/2)}{2}\sin\frac{\delta(Y+1/2)}{2}\frac{H(\delta Y)}{H(0)}\right]. \tag{F}$$

For fixed Y, the bracket tends to $Y^2-1/4$. Moreover,

$$\frac{4}{\delta^2}\left|\sin\frac{\delta(Y-1/2)}{2}\sin\frac{\delta(Y+1/2)}{2}\right| \le (|Y|+1/2)^2.$$

For $\delta A \le 1$, (B) gives a dominating factor $e^{|Y|}$, which is integrable against a Gaussian uniformly when x and τ range over compact sets. The same estimates with Taylor's remainder give the stronger uniform statement

$$\sup_{|x|\le R,\, 0\le\tau\le T}\left|F_\delta(x,\tau)-(x^2-1/4+2\tau)\right| \le C_{R,T}(\delta A+\delta^2). \tag{U}$$

For completeness, use $|e^{a|Y|}-1| \le a|Y|e^{|Y|}$ for $0\le a\le 1$, and $|\sin u - u| \le |u|^3/6$. The resulting errors are bounded by a constant times $(\delta A+\delta^2)(1+|Y|^6)e^{|Y|}$, whose expectations are uniformly bounded on the stated compact set.

3.3 Why the normal form forces an actual collision

Fix $R=2$, and fix any $\varepsilon>0$ small enough that $\tau_+ = 1/8+\varepsilon<1$. Uniform convergence (U) gives:

1. $F_\delta(-R,\tau)>0$ and $F_\delta(R,\tau)>0$ for every $0\le\tau\le\tau_+$, once (I) is sufficiently small.
2. $F_\delta(x,\tau_+)>0$ for every $-R\le x\le R$, since the limiting polynomial is $x^2+2\varepsilon$.
3. Initially exactly the two selected roots lie between $-R\delta$ and $R\delta$. To see this, any third root at circular distance at most $R\delta$ would give $\delta A \ge c_R > 0$, contradicting (I).

Suppose, for contradiction, that no global collision occurs by $s=\tau_+\delta^2$. The initial circle roots then remain simple and on the circle throughout: self-inversive symmetry, or equivalently the real trigonometric equation and the implicit function theorem, prevents a simple real root from leaving the real axis. The two selected roots vary continuously, and cannot cross the fixed interval's boundary because the boundary values stay nonzero throughout. They would therefore still give two real roots inside the interval at the final time, contradicting item 2.

It follows that $D\le(1/8+\varepsilon)\delta^2$. This argument uses tracking and nonvanishing boundaries; positivity only at the final time would not suffice. No Rouché assertion is being smuggled into the proof.

3.4 Matching deterministic lower bound

Before the first collision, each adjacent gap g obeys

$$g' \geq -2\cot(g/2).$$

The scalar comparison solution initialized at the minimum gap $\delta < \pi$ first reaches zero at

$$D_2(\delta) = -\log\cos(\delta/2) \geq \delta^2/8.$$

Compare every gap with this same scalar solution. The comparison function $-2\cot(g/2)$ is increasing on $(0, 2\pi)$, so this remains legitimate for an initially long gap. A collision cannot occur while the scalar solution is positive. Therefore $D \geq D_2(\delta)$. Combining with the preceding upper bound proves (L).

One can extract an absolute-constant bound

$$\frac{\delta^2}{8} \leq D \leq \delta^2\left[\frac{1}{8} + C(\delta A + \delta^2)\right]$$

when $\delta A + \delta^2$ is sufficiently small. No numerical value of C is claimed here.

4. CUE supplies the isolation condition

The Haar CUE eigenangles form the projection determinantal process with correlation kernel

$$K_N(x, y) = \frac{1}{2\pi}\sum_{m=0}^{N-1} e^{im(x-y)}.$$

4.1 Uniform three-point bound

For three real lifts x_1, x_2, x_3, Cauchy – Binet gives

$$\rho_3(x_1, x_2, x_3) = \frac{1}{(2\pi)^3}\sum_{0\leq m_1<m_2<m_3<N}\left|\det(e^{im_a x_b})_{a,b=1}^3\right|^2.$$

For a fixed triple of frequencies, put $v(x) = (e^{im_1 x}, e^{im_2 x}, e^{im_3 x})^T$. Divided differences give the exact identity

$$\det(v(x_1), v(x_2), v(x_3)) = \prod_{i<j}(x_j - x_i)\det(v[x_1], v[x_1, x_2], v[x_1, x_2, x_3]).$$

The integral formula for divided differences and Hadamard's inequality bound the three column norms by $\sqrt{3}, \sqrt{3}N, \sqrt{3}N^2/2$, respectively. Hence each determinant has modulus at most $(3\sqrt{3}/2)N^3\prod_{i<j}|x_i - x_j|$. Summing fewer than $N^3/6$ terms proves

$$\rho_3(x_1, x_2, x_3) \leq CN^9\prod_{i<j}|x_i - x_j|^2 \tag{3P}$$

with an absolute constant. This holds for lifts of points in a short arc, including an arc that crosses the arbitrary 2π cut.

4.2 Triple-free arcs with high probability

Count ordered triples with an anchor at $x_1 \in [0, 2\pi)$ and the other two points in the oriented arc $[x_1, x_1 + r]$. If any arc of length r contains three points, at least one such anchored triple exists. Using (3P), integration over the two offsets gives

$$\Pr(\exists \text{ arc of length } r \text{ containing three points}) \le CN^9 r^8. \tag{T}$$

Choose $r = N^{-7/6}$. The right side is $O(N^{-1/3})$.

4.3 A packing bound for the reciprocal chord sum

On the event that every arc of length r contains at most two points, suppose additionally that the minimum gap obeys $\delta < r/2$. Center that pair at zero. Any other point must have circular distance at least $r - \delta/2$ from the pair center: otherwise it and both endpoints lie in a single arc of length r.

Partition the circle in arcs of length r, or order the remaining points by their circular distance $d_k \in (0, \pi]$ from zero. The two-points-per-arc condition implies $d_k \ge cr(k+1)$ for an absolute $c > 0$; the first few indices use the preceding isolation observation. Since $2\sin(d/2) \ge 2d/\pi$ for $0 \le d \le \pi$,

$$A \le Cr^{-1}(1 + \log N). \tag{A}$$

4.4 Existing extreme-gap input and the resulting law

The CUE extreme-gap theorem gives

$$N^{4/3}\delta_N \Rightarrow X, \qquad \Pr(X > x) = \exp\left(-\frac{x^3}{72\pi}\right). \tag{EG}$$

This normalization uses angular coordinates of circumference 2π. It agrees with $A_2 = 1/(24\pi)$ in Feng – Wei's gap intensity $A_2 u^2\, du$: integrating from zero to x gives $x^3/(72\pi)$. See Ben Arous – Bourgade and Feng – Wei, Theorem 1.1 and Corollary 1.1.

In particular $N^{4/3}\delta_N$ is tight. Combining (T), (A), and $r = N^{-7/6}$ gives

$$\delta_N A_N = O_{\Pr}(N^{-1/6}\log N) = o_{\Pr}(1),$$

and also $\delta_N/r \to 0$ in probability. The isolation lemma yields $8D_N/\delta_N^2 \to 1$ in probability. Slutsky's theorem then gives

$$N^{8/3}D_N \Rightarrow X^2/8, \qquad \Pr(X^2/8 > t) = \exp\left(-\frac{2\sqrt{2}}{9\pi}t^{3/2}\right).$$

This proof needs no uniform-in-N extrapolation of the extreme-gap tail. It uses only tightness and the fixed-argument limiting law.

5. What is genuinely new here, and what may already be classical

The isolated-pair mechanism is closely related to the classical Lehmer-pair estimates of Csordas – Smith – Varga, which already control a Newman lower bound from an initial gap and a background sum of inverse squares. The claim “nobody knows how to control the evolving background from initial data” would therefore be misleading. See Csordas – Smith – Varga's paper.

Andrade – Chang – Miller explicitly express the function-field background as finite csc-squared sums and use isolated small zeros to approach zero Newman parameter. Their § 3.8 already connects such estimates to random matrix theory. See Newman's conjecture in various settings. We must compare any eventual paper carefully with these results.

The contribution proposed in this round is the particularly short global heat-product estimate (B), its deterministic scaling lemma, and the use of elementary CUE triple-overcrowding plus the known extreme-gap theorem to get an explicit first-collision law. A search in this round did not locate that exact CUE law; this is not a guarantee of novelty. Calling it a solution of a famous ζ conjecture would be false.

6. Why this does not refute AH

The isolation assumption fails at lattice scale. For a clock-like configuration with gap of order N^{-1}, the reciprocal chord sum is of order $N \log N$, so δA is not small. ACUE's finite lattice hard core therefore does not contradict the lemma. The new theorem explains a distinction for CUE's extreme close pairs; it does not show that ζ has those pairs.

For ζ, near collisions, quantitative outer-zero control, and the true heat evolution would need to be established arithmetically. Bandwidth-one correlations alone cannot supply a theorem excluding every ACUE-type model, because such models already match that information. A proposed "energy bound" must be stated and checked on ACUE; merely naming a renormalized energy does not add arithmetic information.

Moreover, an AH that permits a zero-density exceptional set is not refuted by a first-collision statistic or infinitely many unusually small gaps. For that version, one needs positive-density violations in a fixed interval separated from the allowed half-integer spacings, or another statistic that controls the density of exceptions. The inference "AH implies every window has hard core $1/2$" remains invalid without strengthening AH.

7. Function–field transfer: a precise modest theorem and a real remaining gap

For a fixed degree, the extended-valued collision depth is expected to be continuous on the compact space of circle-rooted characteristic polynomials when $[0, \infty]$ is given its usual one-point compactification. A route to a proof is as follows:

- If $t < D(P)$, Q_t has the full number of simple real zeros, and nearby coefficient vectors also do. This gives lower semicontinuity.
- If $t > D(P)$, choose a nearby time with no multiple zero. Scalar heat's strict drop in zero number after a multiple zero gives fewer than the full number of real zeros at this time. This property persists under small coefficient perturbation, giving upper semicontinuity.
- At a clock polynomial $D = \infty$, the first assertion for every finite t gives extended continuity. The same zero-number argument treats initial multiple roots if their depth is defined as zero.

This uses the classical zero-number theorem for scalar parabolic equations, rather than an unsupported assertion that the hitting time of an arbitrary discriminant is continuous. The theorem is due to Angenent; see the original publication.

Once this continuity argument is fully written, fixed-rank Frobenius equidistribution immediately transports the law of any bounded continuous function of depth. One must state a specific arithmetic family and its compact symmetry group. For instance, the hyperelliptic ensemble at fixed genus and $q \to \infty$ has $\mathrm{USp}(2g)$ equidistribution; Rudnick's paper states this precisely.

This fixed-rank transfer is a useful corollary, not by itself a large universality theorem. To transport the CUE $N^{8/3}$ law to a unitary function-field family one either takes the successive limit $q \to \infty$ then $N \to \infty$, or proves effective equidistribution uniform in the rank and in the increasingly singular test function. A diagonal subsequence can be chosen abstractly from successive limits, but it gives no useful growth rate $q = q(N)$. Haar USp and SO have symmetry-forced spectral pairing and edges, so the CUE argument cannot simply be relabeled with another beta.

8. A more promising research ladder after this round

1. Independently audit the deterministic heat lemma and the CUE overcrowding proof. Write a concise standalone theorem note, then check the CSV/Stopple/Andrade literature for priority and potentially sharper initial-data criteria.
2. Generalize only under explicit assumptions: if $a_N\delta_N \Rightarrow X$ and $\delta_N A_N \to 0$ in probability, then $a_N^2 D_N \Rightarrow X^2/8$. Bandwidth-one correlation is unnecessary for this abstract transfer theorem.
3. For circular beta ensembles, prove or cite a uniform three-point bound of the form $\rho_3 \le C_\beta N^{3+3\beta}\prod|x_i - x_j|^\beta$. It would give triple-free scale $r = N^{-1-\eta}$ for $\eta > 1/(2+3\beta)$, compatible with the minimum-gap scale $N^{-1-1/(\beta+1)}$. The known Feng – Wei extreme-gap theorem used in the source covers positive integer beta; do not silently assert all positive real beta.
4. For a genuinely arithmetic advance, seek a **positive-density isolated-pair theorem** with a testable arithmetic input. If normalized gaps below $1/2$ are the target, obtaining them cannot be assumed inside the background lemma.
5. Investigate whether a natural arithmetic weighting produces an energy or multi-point statistic unavailable to the ACUE mimickers. That is where a Yau-inspired local relaxation argument might add value: formulate the actual local law, the coercive energy, and the estimate that distinguishes arithmetic zeros from a fake process.

9. Verification boundary

The complete reasoning of the deterministic lemma and CUE transfer is above. Independent review by another agent was requested. Numerical tests are supplementary and cannot certify the theorem. The attached script tests the scalar heat normal form, exact two-root normalization, and planted isolated-pair collision times; it does not simulate an arithmetic zeta flow or prove a famous conjecture.

The completed diagnostics are saved in `yau_flow_checks.json`. They reproduce the exact two-root time for four gaps; for planted isolated pairs at N=16,32,64,128,256 they give $8D/\delta^2$ between 1.00013025 and 1.00002066, decreasing toward 1. The tests also sample the global product bound and three-point determinant inequality. These data are not a Monte Carlo verification of the limiting distribution. The `residual_gram` agent reviewed the full deterministic and CUE arguments and reported no fatal issue; the `prime186` agent also independently checked the deterministic argument, the actual Feng – Wei source statements, the beta scope, and the tail constants without finding a fatal issue. The attempted Claude review did not run because authentication failed with HTTP 401.

10. Stronger supplement: the positive–integer circular–beta depth law

After the first draft, the general three-point estimate was found to follow directly from Hölder's inequality and a published coalesced-charge partition-function bound. Thus the proposed result extends to every fixed **positive integer** beta, including COE, CUE, and CSE. The restriction here comes from the published extreme-gap limit used as an input; this note does not assert that the limit is unavailable or false for other beta values.

For fixed positive integer β, let the initial angles have density

$$\frac{1}{C_{\beta,N}}\prod_{i<j}|e^{i\theta_i} - e^{i\theta_j}|^\beta.$$

Use the same characteristic-polynomial coefficient flow as above; the flow itself does not depend on beta. Define

$$A_\beta = \frac{1}{2\pi}\left(\frac{\beta}{2}\right)^\beta \frac{\Gamma(1+\beta/2)^3}{\Gamma(1+3\beta/2)\Gamma(1+\beta)}.$$

Candidate circular-beta theorem. For every fixed positive integer beta,

$$\frac{8D_N}{\delta_N^2} \to 1 \quad \text{in probability,}$$

and

$$\lim_{N\to\infty} \Pr\left(N^{2+2/(\beta+1)} D_N > t\right) = \exp\left[-\frac{A_\beta}{\beta+1}(8t)^{(\beta+1)/2}\right], \qquad t \geq 0. \tag{CB}$$

For beta=1 the tail is $e^{-t/6}$. For beta=2 it is the CUE law in § 1. For beta=4 it is $\exp[-(64\sqrt{2}/(675\pi))t^{5/2}]$. CSE here means the circular beta=4 ensemble of N distinct eigenangles, not Haar USp(2N), and no double-counting of quaternionic eigenvalues is intended.

10.1 The missing correlation estimate is an elementary Hölder consequence

Put $z_a = e^{ix_a}$, $a = 1, 2, 3$, and $w_j = e^{iy_j}$, $j = 1, \ldots, N-3$. The three-point correlation function is

$$\rho_3(x_1, x_2, x_3) = \frac{(N)_3}{C_{\beta,N}} \prod_{a<b} \mid z_a - z_b \mid^\beta I(z_1, z_2, z_3),$$

where

$$I = \int \prod_{i<j} \mid w_i - w_j \mid^\beta \prod_{j=1}^{N-3} \prod_{a=1}^{3} \mid w_j - z_a \mid^\beta dy.$$

Apply Hölder with exponents (3,3,3) to the measure with density $\prod_{i<j} \mid w_i - w_j \mid^\beta$. This gives

$$I \leq \prod_{a=1}^{3} \left[\int \prod_{i<j} \mid w_i - w_j \mid^\beta \prod_j \mid w_j - z_a \mid^{3\beta} dy\right]^{1/3}.$$

Each bracket is independent of z_a by rotation invariance. Integrating its distinguished point over the full circle identifies it as $C_{\beta,N-3,(3)}/(2\pi)$, in the notation of Feng – Wei. Their Lemma 1.1 gives, for beta⩾1,

$$\frac{C_{\beta,N-3,(3)}}{C_{\beta,N}} \leq (N\beta)^{3\beta}.$$

Therefore

$$\rho_3(x_1, x_2, x_3) \leq \frac{\beta^{3\beta}}{2\pi} N^{3+3\beta} \prod_{a<b} \mid e^{ix_a} - e^{ix_b} \mid^\beta \leq C_\beta N^{3+3\beta} \prod_{a<b} \mid x_a - x_b \mid^\beta \tag{B3}$$

for real lifts of a short arc. This argument does not use a determinantal or Pfaffian representation. The only non-elementary input in (B3) is the published partition-function inequality. See Feng – Wei, equations (1) – (4) and Lemma 1.1.

10.2 Triple exclusion and first collision

The same anchored-triple count now gives

$$\Pr(\exists \text{ arc of length } r \text{ containing three points}) \leq C_\beta N^{3+3\beta} r^{2+3\beta}.$$

Choose any fixed

$$\frac{1}{2+3\beta} < \eta < \frac{1}{\beta+1}, \qquad r = N^{-1-\eta}.$$

This interval is nonempty for every positive beta. The probability above tends to zero. On its complement, § 4.3 gives $A \le Cr^{-1}(1 + \log N)$.

Feng – Wei's extreme-gap theorem for positive integer beta gives

$$N^{1+1/(\beta+1)}\delta_N \Rightarrow X_\beta, \qquad \Pr(X_\beta > x) = \exp[-A_\beta x^{\beta+1}/(\beta + 1)].$$

It follows that

$$\delta_N A_N = O_{\Pr}\left(N^{\eta-1/(\beta+1)} \log N\right) = o_{\Pr}(1).$$

The deterministic isolation lemma proves the ratio limit, and continuous mapping gives (CB). There is no beta-infinity limit in this argument. Fixed-beta asymptotics do not identify ACUE with a beta-infinity ensemble.

10.3 Assessment

If the isolation lemma survives the independent adversarial review, the old finite-depth programme is no longer "one dynamic background lemma away" in the COE/CUE/CSE cases: the required localization follows from a different, initial-data argument and existing partition-function estimates. This is a concrete candidate probability theorem. The arithmetic bridge to ζ and the user's requested famous conjecture remain open; the result is best treated as a proved-direction candidate and a reusable transfer mechanism, not the final research objective.

A stronger, drift-invariant version of the deterministic lemma is proved in `yau_flow_galilean_refinement.md`: the hypothesis is $\delta^2 B \to 0$ with $B = \frac{1}{4}\sum \csc^2(\theta k/2)$. The exact Galilean transform removes the linear logarithmic drift, and a global quadratic exponential bound controls the heat tails. The root agent independently audited the centered-factor inequality and the moving-frame mechanism. Full formal verification and a priority review remain outstanding.

11. Identification of the first colliding pair

The published joint limit for the first two smallest gaps gives a useful corollary to the candidate depth theorem.

Corollary. For every fixed positive integer beta, with probability tending to one, the first colliding pair is the pair that had the smallest gap at time zero.

Write the first two initial gaps as $\delta_1 < \delta_2$; equality has probability zero under the continuous circular-beta density. Before the global first collision, every gap other than the minimum is bounded below by the scalar comparison solution initialized at δ_2. Therefore no such pair can collide before

$$-\log\cos(\delta_2/2) \ge \delta_2^2/8.$$

The selected minimum pair supplies the global upper bound

$$D \le \delta_1^2(1/8 + \varepsilon_N), \qquad \varepsilon_N \to 0 \quad \text{in probability.}$$

The joint extreme-gap limit implies that δ_2/δ_1 converges in law to a random variable strictly greater than one almost surely. Hence

$$\Pr(\delta_2^2/\delta_1^2 > 1 + 8\varepsilon_N) \to 1.$$

On this event, all other initial adjacent pairs remain separated until after the global first collision, proving the assertion. This is an asymptotic identification statement; it does not make the finite marked derivative exactly rank two.

12. The relaxation–time obstruction to the proposed AH argument

This subsection is a dimensional analysis and logical audit, not an impossibility theorem against every future heat-flow method.

At height T, the mean Riemann-zero spacing in the usual ordinate is of order $(\log T)^{-1}$. A pair-collapse equation with leading term $g' = -4/g$ therefore has local time scale $(\log T)^{-2}$. The common convention for H_0 uses twice the ordinary zeta ordinate, which changes constants but not this exponent. One must keep that coordinate factor when making a numerical claim such as $\pi^2/8$.

The contradiction assumption in Rodgers – Tao is a **fixed negative global** Newman parameter, giving a time interval of length bounded away from zero on which all the relevant zeros are real. In microscopic units, the available duration at height T is consequently of order $\log^2 T$, tending to infinity. This long duration in local units is what permits a relaxation-to-clock argument. Their actual proof needs renormalized energies and technical estimates; this dimensional observation is not a substitute for those estimates. See Rodgers – Tao.

A hard core proportional to one mean spacing, even if it were a genuine uniform hard core rather than an almost-everywhere AH, only gives the two-body stability lower bound of order $(\log T)^{-2}$. That is O(1) microscopic relaxation times. It does not supply the diverging number of relaxation times available in the global negative-Λ contradiction. Rescaling the old energy argument does not remove this loss. A proposed proof must explain what new arithmetic estimate replaces that missing duration.

The finite circular normalization makes the distinction transparent. Its unnormalized deterministic clock Jacobian has eigenvalues $k(N-k)$; the high modes have time scale N^{-2}. A conventional circular DBM has repulsive drift

$$\frac{1}{2N}\sum_{j\neq i}\cot\frac{\theta_i-\theta_j}{2}$$

and diffusion coefficient $\sqrt{2/(\beta N)}$. Thus its local relaxation scale is order N^{-1}, corresponding to unnormalized deterministic time order N^{-2}. At that time the Brownian displacement itself is of order one mean spacing. It cannot be discarded in a pathwise identification with the attractive deterministic flow.

At the CUE extreme-gap scale the same problem persists: deterministic first-collision time is order $N^{-8/3}$, corresponding to DBM time order $N^{-5/3}$; the DBM Brownian displacement over that duration is order $N^{-4/3}$, exactly the extreme gap size. A genuine DBM comparison has to retain its noise and its repulsive sign. It cannot identify its collision behavior with the scalar heat evolution studied here.

The new isolated-pair theorem sharpens this audit. Under weak initial isolation, normalized depth is asymptotically the square of the normalized smallest gap. A ζ depth bound substantially below the half-spacing threshold would therefore require genuine microscopic arithmetic information or a mechanism outside isolated-pair behavior. Merely renaming the desired small-gap phenomenon as a first-passage event does not prove it. For an AH allowing zero-density exceptions, the needed arithmetic result must additionally control the density of violations.

13. Current literature on matrix heat flow

A relevant recent paper is Hall – Ho, The heat flow conjecture for polynomials and random matrices, published in 2025. It connects additive and multiplicative polynomial heat flows to deformations of random matrix models, proves second-moment and holomorphic-moment statements, and formulates macroscopic root-transport conjectures. Its multiplicative generator differs from the generator here by a scalar multiplier and a deterministic radial dilation. This is useful context for a future theorem note; its macroscopic statements do not by themselves supply the microscopic extreme first-collision law above. The exact priority of that law remains to be checked.

Current report 07: yau_flow_galilean_refinement

Source: `research/reports/yau_flow_galilean_refinement.md`. SHA-256: `ffa09e0a4a6a78854fe5bfdb666e812d85bb34169dcb1d8443020eb665db6d3e`.

A stronger isolated-pair lemma after removing the background drift

2026-09-05. This is a proposed refinement of `yau_flow.md`, independently derived in the same research round. It has not yet received the other agents' full review. It overlaps conceptually with classical Lehmer-pair criteria and should not be advertised as a new principle without a priority check.

The reciprocal-chord condition $\delta A \to 0$ in the first draft can be replaced by the weaker inverse-square condition $\delta^2 B \to 0$. The improvement removes the effect of a large common background velocity by an exact Galilean transformation of scalar heat flow.

Statement

Let δ be the smallest angular gap in a circle-rooted degree-N polynomial. Center its endpoints at $\pm\delta/2$ and write

$$Q_0(x) = p_\delta(x)H(x), \qquad p_\delta(x) = \sin\frac{x-\delta/2}{2}\sin\frac{x+\delta/2}{2}, \qquad H(x) = \prod_{k\notin\pm}\sin\frac{x-\theta_k}{2}.$$

Define

$$L = \frac{H'(0)}{H(0)} = -\frac{1}{2}\sum_{k\notin\pm}\cot\frac{\theta_k}{2}, \qquad B = \frac{1}{4}\sum_{k\notin\pm}\csc^2\frac{\theta_k}{2}.$$

If $\delta\to 0$ and $\delta^2 B\to 0$ along a sequence of configurations, then the global first collision time of the coefficient flow $\exp(sj(N-j))$ satisfies

$$D = \frac{\delta^2}{8}(1+o(1)).$$

In fact, there are absolute constants C and $\varepsilon_0 > 0$ such that, when $\delta^2(B+1) \leqslant \varepsilon_0$,

$$\frac{\delta^2}{8} \le D \le \delta^2\left[\frac{1}{8} + C\delta^2(B+1)\right].$$

No numerical value of C or ε_0 is claimed. The lower bound is the existing deterministic two-body comparison. The following argument gives the upper bound.

1. A global centered sine-factor estimate

For all real c and v,

$$|\cos v - c\sin v|\, e^{cv} \le e^{4(1+c^2)v^2}. \tag{S}$$

Put $w=|v|\sqrt{(1+c^2)}$. If $w\geqslant 1/2$, use

$$\log|\cos v-c\sin v|+cv\leq\log(1+|c||v|)+|c||v|\leq 2w\leq 4w^2.$$

The inequality is automatic if the factor vanishes. If $w\leqslant 1/2$, the function $r(u)=\cos u-c\sin u$ is positive between 0 and v: throughout that segment,

$$r(u)\geq 1-u^2/2-|c||u|\geq 1-1/8-1/2>0.$$

Its centered logarithm $f(u)=\log r(u)+cu$ has $f(0)=f'(0)=0$ and

$$f''(u)=-\frac{1+c^2}{r(u)^2}<0.$$

Thus $f(v)\leqslant 0$, proving (S) also in this case.

Now put

$$R(y)=e^{-Ly}\frac{H(y)}{H(0)}.$$

Each sine ratio is $\cos(y/2)-\cot(\theta k/2)\sin(y/2)$. Applying (S) factor by factor gives the **global real-line bound**

$$|R(y)|\leq e^{4By^2},\qquad y\in\mathbb{R}. \tag{G}$$

When $By^2\leqslant 1/16$, every uncentered sine ratio remains at least 3/4 along the segment from 0 to y. Consequently H has constant sign on that segment and

$$-\frac{16}{9}B\leq(\log|H|)''\leq 0.$$

After subtracting the linear term Ly and integrating twice,

$$-By^2\leq\log R(y)\leq 0,\qquad |R(y)-1|\leq By^2.$$

When $By^2>1/16$, (G) gives $|R(y)-1|\leqslant e^{4By^2}+1\leqslant 32By^2e^{4By^2}$. Combining both regions,

$$|R(y)-1|\leq 32By^2e^{4By^2},\qquad y\in\mathbb{R}. \tag{E}$$

The case B=0 is immediate: H is constant.

2. Exact heat conjugation and the moving frame

For any real L for which the Gaussian integrals converge,

$$e^{s\partial_x^2}(e^{Lx}f(x))=e^{Lx+L^2s}(e^{s\partial_x^2}f)(x+2Ls). \tag{C}$$

This is a direct completion of the square in the Gaussian convolution. Here the original trigonometric polynomial is bounded on the real axis; the transformed integrals converge at the times used below because δ^2B is small.

Set $s=\tau\delta^2$ and use the moving coordinate $x=\delta\xi-2Ls$. With Z standard normal and $Y=\xi+\sqrt{(2\tau)}Z$, formula (C) yields

$$\frac{4e^{-L\delta\xi+L^2s}}{\delta^2H(0)}Q_s(\delta\xi-2Ls)=\mathbb{E}\left[\frac{4}{\delta^2}p_\delta(\delta Y)R(\delta Y)\right]. \tag{M}$$

The multiplier on the left is real and nonzero, so it does not alter the roots or their signs relative to the constant H(0).

The right side converges uniformly on compact ξ,τ sets to

$$\mathrm{E}(Y^2 - 1/4) = \xi^2 - 1/4 + 2\tau.$$

More precisely, (E), the elementary sine Taylor remainder, and Gaussian integrability give

$$\sup_{|\xi| \le R_0,\, 0\le\tau\le T} \left| \frac{4e^{-L\delta\xi+L^2 s}}{\delta^2 H(0)} Q_s(\delta\xi - 2Ls) - (\xi^2 - 1/4 + 2\tau) \right| \le C_{R_0,T}\,\delta^2(B+1), \tag{U}$$

once $\delta^2 B$ is sufficiently small depending on T. Indeed the error is bounded by a constant times $\delta^2(B+1)(1+|Y|^6)e^{4B\delta^2 Y^2}$, and its expectation is uniformly bounded after choosing $\delta^2 B$ small enough. This is why a quadratic global bound was necessary; a merely local Taylor expansion would not justify the heat integral.

3. Collision tracking with moving boundaries

Use the interval

$$[-2\delta - 2Ls,\ 2\delta - 2Ls]$$

on the real cover of the circle. At time zero it contains exactly the selected pair, since $\delta^2 B \to 0$ excludes any third point at distance $O(\delta)$. The normal form (U) shows that both boundary values stay nonzero for all times up to $(1/8+\varepsilon)\delta^2$, and the whole interval contains no real root at that final time.

If no global multiple root had occurred, the initially simple circle roots would remain on the circle and vary continuously. The leading and constant polynomial coefficients are fixed and nonzero, so no root can escape via zero or infinity. Simple self-inversive roots cannot leave the circle without colliding with their reciprocal-conjugate partners. The two roots in the moving interval therefore cannot disappear without a multiple root, and cannot leave through its nonvanishing moving boundaries. This contradiction gives $D \leqslant (1/8+\varepsilon)\delta^2$.

Taking ε to be a sufficiently large constant times $\delta^2(B+1)$ gives the quantitative upper bound. Pair identity after the first global collision is irrelevant: an earlier collision already establishes the desired upper bound.

4. Consequences and scope

On a triple-free event at length r, the packing argument now gives $B \leqslant Cr^{-2}$, with no harmonic logarithm. Therefore the circular-beta proof in `yau_flow.md` works with $\delta^2 B = O_p((\delta/r)^2)$, which tends to zero at the same admissible r scales. The limiting laws and constants do not change.

The transformation also explains why a large common drift need not obstruct collision localization. The right structural quantity is curvature of the logarithm of the background product, not the absolute sum of its first derivatives. This connects naturally with classical Lehmer-pair inverse-square criteria. For an infinite canonical product one must separately justify convergence, the exponential normalization, and its true heat evolution; this note does not claim that those steps, or the arithmetic small-gap input, have been completed for ζ.

The stronger criterion is strictly weaker. Take $\delta = N^{-2}$, put the selected pair at $\pm\delta/2$, and place the remaining $N-2$ points at angles $(M+k)\delta$, $k=0,\cdots,N-3$, with $M = \lceil \sqrt{\log N}\rceil$. Every gap is at least δ. Direct harmonic-sum estimates give $\delta^2 B$ asymptotic to $1/M$, tending to zero, while δA is of order $\log(N/M)$, tending to infinity. The Galilean argument therefore applies even though the first proof's absolute first-derivative bound fails. The moving interval may travel across many *initial* gaps; its all-time nonvanishing boundaries still make the tracking argument valid.

The root agent independently checked the centered-factor bound and the moving-frame mechanism. Numerical checks in `yau_flow_checks.py` include cases with δA between 1.09 and 5.31 but $\delta^2 B$ near 0.02. The selected local double-root times are close to $\delta^2/8$. Those configurations have many equal-size gaps, so the script deliberately does not label their computed local double-root time as the first global collision.

Current report 08: galilean-proof-audit

Source: `research/reports/galilean-proof-audit.md`. SHA-256:
`f3896ff6b7a07d1296762ff09ad8bf9315766459f16394d6636e64d9036c8f13`.

Galilean localization: second-pass proof audit and the exact ζ obligation

2026-09-05. The same heat-flow research agent performed this second-pass audit; the separate prime-lane agent supplied the earlier independent review. This audit rereads the round-1 Galilean argument from its definitions. It does not extend the random-matrix model family or claim a new ζ theorem. No repository files were changed. The argument remains an ordinary mathematical proof, not a Lean or interval-arithmetic certificate.

Audit conclusion. The deterministic asymptotic conclusion survives. The stronger quantitative bound can also be retained: the previously unspecified constants can be chosen uniformly in N. A deliberately coarse explicit choice is

$$K = 16384, \qquad \eta_0 = \frac{1}{524288},$$

for which the proof below gives

$$\frac{\delta^2}{8} \le D \le \delta^2\left(\frac{1}{8} + K\delta^2(B+1)\right) \quad \text{if } \delta^2(B+1) \le \eta_0.$$

The constants are not optimized and are not useful numerical record constants. Their purpose is to certify that no dependence on N or on the background drift has been hidden in the big-O statement. The proof needs no complex Gaussian contour shift, Rouché theorem, or unproved continuity of a general hitting-time functional.

1. Exact assumptions and definitions

Let $N \geqslant 2$ and let P be a degree-N polynomial with N distinct roots on the unit circle. Its leading and constant coefficients are nonzero. Define

$$P_s(z) = \sum_{j=0}^{N} a_j e^{sj(N-j)} z^j, \qquad D = \inf\{s > 0 : \operatorname{disc}(P_s) = 0\},$$

with infimum of the empty set equal to infinity. Both endpoint coefficients remain fixed and nonzero for every real s.

Let δ be the smallest circular angular gap. Rotate its endpoints to $-\delta/2$ and $\delta/2$. Choose all other angles θk in $(-\pi,\pi]$, and set

$$Q_0(x) = p_\delta(x)H(x), \quad p_\delta(x) = \sin\frac{x-\delta/2}{2}\sin\frac{x+\delta/2}{2}, \quad H(x) = \prod_{k\notin\pm} \sin\frac{x-\theta_k}{2}.$$

Multiplying by a fixed nonzero scalar relates this real trigonometric polynomial to the centered characteristic polynomial. Define

$$L = \frac{H'(0)}{H(0)} = -\frac{1}{2}\sum_{k\notin\pm} \cot\frac{\theta_k}{2}, \qquad B = \frac{1}{4}\sum_{k\notin\pm} \csc^2\frac{\theta_k}{2}, \qquad \eta = \delta^2(B+1).$$

H(0) is nonzero. Empty background products and sums are permitted. The coefficient flow has the same zeros as

$$Q_s = e^{s\partial_x^2} Q_0,$$

because the missing scalar factor is exp(sN²/4). For odd N the centered trigonometric polynomial is antiperiodic over 2π; the real-line Gaussian formula is nevertheless exact term by term on its finitely many Fourier modes.

2. Global centered–factor estimate: checked

For real c,v, let w=|v|√(1+c²). We verify

$$|\cos v - c\sin v|\, e^{cv} \le e^{4(1+c^2)v^2}. \tag{2.1}$$

If w⩾1/2, use |cos v−c sin v|⩽1+|c||v|. Taking logarithms, when the factor is nonzero,

$$\log|\cos v - c\sin v| + cv \le \log(1 + |c||v|) + |c||v| \le 2w \le 4w^2.$$

If w⩽1/2, r(u)=cos u−c sin u is positive along the segment between 0 and v, because

$$r(u) \ge 1 - u^2/2 - |c||u| \ge 1 - 1/8 - 1/2 > 0.$$

The function f(u)=log r(u)+cu has f(0)=f′(0)=0 and

$$f''(u) = -\frac{1 + c^2}{r(u)^2} < 0.$$

Therefore f(v)⩽0. The zero-factor case is immediate. This proves (2.1) on the full real line, including beyond the first background root.

Define R(y)=exp(−Ly)H(y)/H(0). Each factor is cos(y/2)−cot(θk/2)sin(y/2), with exactly the matching exponential exp(y cot(θk/2)/2). Hence

$$|R(y)| \le e^{4By^2}, \qquad y \in \mathrm{R}. \tag{2.2}$$

There is no estimate involving |L| in (2.2).

3. The global error |R—1|: checked with its constants

Put ak=1/(2|sin(θk/2)|), so B=Σak². The sine Lipschitz inequality gives, for every real u,

$$\left|\frac{\sin((u-\theta_k)/2)}{\sin(-\theta_k/2)} - 1\right| \le a_k|u|.$$

If By²⩽1/16, then for every u between 0 and y every ratio on the left is at least 3/4. H has constant sign there, and

$$-\frac{16}{9}B \le (\log|H|)''(u) = -\frac{1}{4}\sum_{k\in\pm} \csc^2\frac{u-\theta_k}{2} \le 0.$$

Subtract Ly and integrate twice from zero. This is valid for positive or negative y: the integral remainder has nonnegative weights in either direction. It gives

$$-\frac{8}{9}By^2 \le \log R(y) \le 0,$$

and in particular |R(y)−1|⩽By². If By²>1/16, (2.2) gives

$$|R(y)-1| \le e^{4By^2}+1 \le 32By^2e^{4By^2}.$$

Thus the proposed global bound is correct:

$$|R(y)-1| \le 32By^2e^{4By^2} \quad (y \in \mathrm{R}). \tag{3.1}$$

When B=0, H is constant and the conclusion is exact. The argument does not infer global control by extending a local logarithm through its zeros.

4. Heat conjugation and Gaussian tails: exact and uniform

For s>0, completing the square in the real Gaussian integral gives

$$e^{s\partial_x^2}(e^{Lx}f(x)) = e^{Lx+L^2s}(e^{s\partial_x^2}f)(x+2Ls). \tag{4.1}$$

No complex contour shift occurs. In the present finite problem all integrals converge: the original Q0 is bounded on the real line, and exp(−Lx)Q0(x) has at most exponential real growth. The estimates below additionally provide uniform domination as N varies.

Put $s=\tau\delta^2$, $x=\delta\xi-2Ls$, $Y=\xi+\surd(2\tau)Z$, with Z standard real normal. Then

$$F_\delta(\xi,\tau) := \frac{4e^{-L\delta\xi+L^2s}}{\delta^2H(0)}Q_s(\delta\xi-2Ls) = \mathrm{E}[A_\delta(Y)R(\delta Y)],$$

where

$$A_\delta(Y) = \frac{4}{\delta^2}\sin\frac{\delta(Y-1/2)}{2}\sin\frac{\delta(Y+1/2)}{2}.$$

Let $P(Y)=Y^2-1/4$ and $q=\delta^2B$. Two elementary global estimates are

$$|A_\delta(Y)| \le Y^2+1/4,$$

$$|A_\delta(Y)-P(Y)| \le \frac{\delta^2}{12}|Y^2-1/4|(Y^2+1/4) \le \frac{\delta^2}{6}(1+Y^4).$$

For the second estimate use $|\sin u-u| \leqslant |u|^3/6$ and $|\sin u| \leqslant |u|$; neither estimate needs u small. Since

$$Y^4+Y^2/4 \le \frac{9}{8}(1+Y^4),$$

equation (3.1) implies

$$|A_\delta(Y)R(\delta Y)-P(Y)| \le 37\eta(1+Y^4)e^{4qY^2}. \tag{4.2}$$

We now fix the only compact set needed in the collision proof:

$$|\xi| \le 2, \qquad 0 \le \tau \le \frac{1}{4}.$$

On this set $Y^2 \leqslant 8+Z^2$. If $q \leqslant 1/64$, then

$$e^{4qY^2} \le e^{1/2}e^{Z^2/16}, \qquad 1+Y^4 \le 65+16Z^2+Z^4.$$

For $a=8/7$, the Gaussian moment identities give

$$\mathrm{E}[(65+16Z^2+Z^4)e^{Z^2/16}] = 65a^{1/2}+16a^{3/2}+3a^{5/2}.$$

Each power displayed is less than 2, and exp(1/2)<2. Therefore

$$\sup \mathsf{E}[(1+Y^4)e^{4qY^2}] < 336.$$

Combining this with (4.2) proves the explicit N-uniform estimate

$$\sup_{|\xi|\le 2,\ 0\le\tau\le 1/4} |F_\delta(\xi,\tau)-(\xi^2-1/4+2\tau)| \le K\eta, \qquad K=16384, \tag{4.3}$$

because 37 · 336=12432<16384. No bound on Lδ is imposed. The coefficient-normalizing exponential can be large, but it is exact, real, and nonzero.

5. Quantitative positivity and root tracking: checked

Assume η⩽η0=1/(32K)=1/524288. This implies q⩽1/64 and δ<1. Put

$$\tau_* = \frac{1}{8}+K\eta \le \frac{5}{32} < \frac{1}{4}, \qquad s_* = \delta^2\tau_*.$$

At τ=τ*, equation (4.3) gives, for every |ξ|⩽2,

$$F_\delta(\xi,\tau_*) \ge \xi^2+2K\eta-K\eta \ge K\eta > 0. \tag{5.1}$$

At either boundary ξ=±2, for every 0⩽τ⩽τ*, it gives

$$F_\delta(\pm 2,\tau) \ge \frac{15}{4}-K\eta > 0. \tag{5.2}$$

Initially the moving interval I_s=[−2δ−2Ls,2δ−2Ls] contains exactly the selected two roots. Indeed any additional root with circular lift |θk|⩽2δ would contribute

$$\delta^2 B \ge \frac{\delta^2}{4\sin^2(\theta_k/2)} \ge \frac{\delta^2}{\theta_k^2} \ge \frac{1}{4},$$

contrary to the assumed smallness. Also 4δ<2π, so the initial interval does not contain a duplicated lift of the same root.

Suppose D>s*. The family Ps has simple roots throughout the closed interval [0,s*]. Since the multiplier j(N−j) is symmetric under j↔N−j, the initial self-inversive relation is preserved. A root and its reciprocal-conjugate root agree initially. Local uniqueness of a simple root branch makes them agree throughout the interval. Hence every root remains on the unit circle before D. Fixed nonzero endpoint coefficients exclude escape via zero or infinity.

Continuous angle lifts of the two selected roots exist on [0,s*]. They cannot cross either moving boundary, by (5.2). They cannot disappear or merge, by D>s*. They must therefore still be real zeros inside I_s* at the final time, contradicting (5.1).

It follows that

$$D \le s_* = \delta^2(1/8+K\eta).$$

This argument remains valid if the interval translates across many initial gaps or around the circle: it is formulated on the real cover, with continuously moving nonvanishing boundaries. It also remains valid if some other pair would have collided first; that already contradicts D>s* and establishes the same upper bound. A local zero count at the final time alone would have been insufficient, but the all-time boundary estimate supplies the missing step.

For completeness, the deterministic lower bound can be checked directly. Lift one adjacent pair to a<b=a+g, and every other point to c in (b,a+2π). Its contribution to the gap derivative is

$$\cot \frac{c-b}{2} - \cot \frac{c-a}{2} > 0,$$

because cot is decreasing on $(0,\pi)$. The pair's own contribution is $-2\cot(g/2)$, so $g' \geqslant -2\cot(g/2)$. Compare every gap with the same scalar solution initialized at δ. That scalar solution satisfies $\cos(h(s)/2)=\exp(s)\cos(\delta/2)$, and reaches zero at $-\log\cos(\delta/2)$. No gap can collide while this comparison solution is positive. Finally $-\log\cos(\delta/2) \geqslant \delta^2/8$ by integrating $\tan u \geqslant u$ on $[0,\delta/2]$. Under $\eta \leqslant \eta 0$ one has $\delta<\pi$, so the real logarithm is well-defined. Thus the quantitative theorem is retained. There is no need to downgrade it to $o(1)$.

6. What must change when the object is the true H_t

The finite theorem does not define a global ζ depth by looking at a truncated list of zeros. In particular, a global first collision among infinitely many zeros and a local collision near a selected height are different quantities. After RH and the known nonnegativity of the Newman constant, global backward instability is not a new conclusion.

Work in the real coordinate of H_t itself; do not mix it with the usual ζ ordinate without the factor-of-two conversion. Suppose two consecutive simple real zeros of H0 are $c-\delta/2$ and $c+\delta/2$. Factor

$$H_0(c+y) = (y^2-\delta^2/4)h_c(y), \qquad L_c = h_c'(0)/h_c(0), \qquad R_c(y) = e^{-L_c y}h_c(y)/h_c(0).$$

The exact heat identity is $H_{-s}=\exp(s\partial^2)H0$. It follows directly from the defining Fourier integral by real Gaussian convolution. Rodgers – Tao's equations (1) and (4) use $H0(z)=\xi(1/2+iz/2)/8$ and the multiplier $\exp(tu^2)$, respectively; these equations were checked in the downloaded primary paper. In the moving frame $y=\delta\xi-2Lcs$, the identity gives

$$\frac{e^{-L_c\delta\xi+L_c^2 s}}{\delta^2 h_c(0)} H_{-s}(c+\delta\xi-2L_c s) = \mathrm{E}[(Y^2-1/4)R_c(\delta Y)], \quad s=\tau\delta^2. \tag{6.1}$$

Thus a precise sufficient analytic obligation for local collision localization is

$$\sup_{|\xi|\le 2,\ 0\le\tau\le 1/4} \mathrm{E}[(1+Y^2)\,|\,R_c(\delta Y)-1\,|\,] \longrightarrow 0. \tag{Z-tail}$$

This condition states both local flatness and Gaussian-tail uniform integrability. A Taylor approximation on a shrinking real interval, without the tail condition, would not justify (6.1).

The root-tracking conclusion in this infinite setting must be stated locally: a multiple real zero occurs in the moving tube before the specified time, or a tracked real branch already met another branch there. It is not a lower bound for a positive global first-collision time of H_t. The finite proof's assumption that all roots remain real globally before the first collision should not be imported unchanged into this setting.

7. Under RH, the canonical product supplies the far–tail structure

There is a useful simplification: under RH, the true H0 is a real entire function of order one with real zeros. After removing the pair and its linear logarithmic term, its genus-one canonical product gives

$$R_c(y) = \prod_{k\notin\pm}\left(1-\frac{y}{d_k}\right)e^{y/d_k}, \qquad d_k = x_k - c, \qquad B_c = \sum_{k\notin\pm} d_k^{-2} < \infty.$$

Multiplicity is counted in this product. The selected two zeros must be simple; other multiplicities merely increase Bc. These are classical Hadamard-product facts used in the Lehmer-pair literature, not a new random-matrix representation. See Csordas – Smith – Varga and Rodgers – Tao.

For every real u,

$$|1-u|\,e^{u}\le e^{u^2/2}. \tag{7.1}$$

One direct proof studies f(u)=u²/2−u−log|1−u|: its derivative is u(2−u)/(1−u) on u<1 and u(u−2)/(u−1) on u>1. The minima are f(0)=f(2)=0. At u=1 the inequality is immediate.

Apply (7.1) to finite partial products and pass to the canonical-product limit. This gives the global bound

$$|R_c(y)|\le e^{B_c y^2/2}.$$

The same local logarithmic-curvature argument as in § 3 yields

$$|R_c(y)-1|\le 32B_c y^2 e^{B_c y^2/2}.$$

Consequently, the condition δ²Bc→0 implies (Z-tail), including its Gaussian tails. This is a conditional analytic reduction in the spirit of classical Lehmer-pair estimates. It does not establish that the required isolated pairs exist, or occur with any particular frequency, among ζ zeros. No new unconditional or RH-conditional spacing result has been obtained here.

8. If a finite window is used, the exact omitted term

If the canonical product is split into nearby zeros and distant zeros, write R=Rnear Rfar and Bc=Bnear+Bfar. Its missing outer contribution obeys

$$|R(y)-R_{\mathrm{near}}(y)|\le 32B_{\mathrm{far}}y^2e^{B_c y^2/2}.$$

Thus, in the regime δ²Bc bounded by a sufficiently small constant, the normalized Gaussian heat error from the omitted factor is O(δ²Bfar). A sufficient truncation obligation is

$$\delta^2\sum_{|x_k-c|>W}|x_k-c|^{-2}\to 0.$$

The omitted **linear drift** must also be retained or removed using the true Lc. It need not be small merely because its curvature is small. More explicitly, the correct finite approximation before heat evolution is

$$h_c(0)(y^2-\delta^2/4)\exp\left[\left(L_c+\sum_{\mathrm{near}}d_k^{-1}\right)y\right]\prod_{\mathrm{near}}(1-y/d_k).$$

It is a polynomial times a specified exponential; the exponential can be handled by the exact heat-conjugation identity. Replacing this by its polynomial factor alone silently changes the initial logarithmic derivative, and can shift the entire local configuration by more than one gap.

A strong uniform hard-core hypothesis can bound inverse-square tails by packing, but it generally makes δ²Bc only O(1), not o(1), at the minimum allowed spacing. A density version of AH allows exceptional clusters and therefore supplies no uniform bound on Bc for every selected window. Neither version alone yields (Z-tail) with vanishing error at the half-spacing threshold. Using the full true canonical product avoids a matrix-model or arbitrary polynomial-cutoff assumption, but it does not provide the missing arithmetic spacing information.

9. The shortest genuinely missing arithmetic lemma

For the proposed Level-B target of depth o(mean-spacing²) along a sequence, the clean missing input is an **isolated small-pair lemma**: there should exist consecutive simple zeros near heights c→∞ with

$$\frac{\delta_c}{h_c} \to 0, \qquad \delta_c^2 \sum_{k \neq \pm} (x_k - c)^{-2} \to 0,$$

where hc is the mean spacing in the H-coordinate. The preceding analytic argument explains how such input would produce a local collision statement. It does not prove this arithmetic lemma. Bandwidth-one information alone does not establish it, and merely assuming it inside a “near-collision step” would assume the main unresolved microscopic input.

For an AH that permits a zero-density exceptional set, even this subsequential lemma is insufficient to refute AH. The task then needs a positive-density arithmetic statement in a fixed forbidden normalized-gap interval, or an equivalently strong statistic controlling the density of exceptions. A toy quantifier check is a block of N gaps consisting of εN, 2−εN, and N−2 gaps equal to 1. Their sum is N; all but 2/N of the gaps are integer, hence half-integer, while the minimum εN tends to zero. Having a tiny minimum in every large block does not by itself contradict a density AH. This toy example is only a quantifier counterexample, not a claim to match all ζ correlation constraints.

The recommended next arithmetic task is therefore to specify and prove the weighted pair-count or energy estimate that creates a positive proportion of forbidden spacings, with its exact support and averaging range. The finite heat theorem cannot replace that input. The current audit supplies a clean analytic interface and identifies which tail estimates become automatic under RH; it supplies no new ζ spacing theorem.

10. Reproducibility and review status

The companion `galilean_audit_constants.py` checks the exact rational inequalities used to choose K and η0, and verifies symbolic derivatives in the centered sine factor and canonical-product factor. Its output is saved beside it. These checks verify the displayed algebra and numerical constants; they do not verify the topology or the analytic argument in Lean.

No Claude tool was started. No repository source was edited. All assertions above are confined to the stated deterministic theorem and explicit conditional analytic obligations.

Current report 09: prime186-yau-independent-review

Source: `research/reports/prime186-yau-independent-review.md`. SHA-256: `557a3dda43b4b5b524c84e6b1d83eb97195313f1a9cccd1d85f17c081239633d`.

Independent adversarial review of the circular heat isolation theorem

Reviewer: the prime186 research agent. Date: 2026-09-05. Reviewed `yau_flow.md` § § 3.1 – 3.4, § § 4.1 – 4.4, and § § 10.1 – 10.2, and checked the stated input against the locally downloaded Feng – Wei paper. This is an independent mathematical review, not formal verification or a novelty determination.

Conclusion: I found no fatal gap in the deterministic isolation argument or its transfer to CUE and fixed positive integer circular-beta ensembles. The argument supplies a plausible complete proof of the stated finite-ensemble depth law. It does not provide a zeta-zero theorem.

Checks that could have invalidated the argument

1. **Flow sign and time normalization.** Multiplication by `exp[s j(N-j)]` on a centered characteristic polynomial is ordinary scalar heat at time `s`, up to an irrelevant scalar. The normal Gaussian displacement is `sqrt(2s) Z`. The quadratic local model is therefore `x²-1/4+2τ`, which disappears at `τ=1/8`. This agrees with the root ODE and with the exact two-body collision time `-log cos(δ/2)`.
2. **Gaussian tails.** The estimate `|H(y)/H(0)-1| ≤ exp(A|y|)-1` is global for real `y`. It does not use a local Taylor expansion outside its range. Under `δA≤1`, the scaled integrand is dominated by a polynomial times `exp(|Y|)`, uniformly on compact scaled space/time ranges. Gaussian integrability gives the required uniform convergence, including time zero.
3. **First collision versus a disappearing local minimum.** The proof does not infer a collision from final-time positivity alone. It keeps both fixed boundary values nonzero at every intermediate time and tracks the two initial simple roots. Consequently they cannot escape the interval without a collision. If another pair collides earlier, the claimed global upper bound is already satisfied.
4. **Persistence on the circle.** The coefficient evolution preserves self-inversive symmetry. Before a multiple root occurs, simple roots starting on the circle remain there by uniqueness in the implicit function theorem and reciprocal-conjugate symmetry. Leading and constant coefficients remain nonzero throughout, so no root escapes through zero or infinity.
5. **Long initial gaps.** The deterministic lower bound compares every adjacent gap to the same scalar solution initialized at the global minimum, rather than applying a logarithm of a negative cosine to a long gap. The scalar field `-2cot(g/2)` is increasing on `(0,2π)`, and the background contribution to an adjacent-gap derivative is nonnegative.
6. **Initial third-root exclusion.** A third root at distance at most `Rδ` from the selected pair midpoint would force `δA` to be bounded below by a positive constant. Thus the hypothesis genuinely isolates exactly two roots inside the fixed scaled interval.
7. **Triple-overcrowding probability.** The CUE divided-difference determinant estimate gives the power `N^9` and the full squared Vandermonde. Integrating two anchored offsets contributes `r^8`. At `r=N^(-7/6)`, the probability tends to

zero at rate `O(N^(-1/3))`.

8. **Packing and quantifiers.** Triple-free arcs imply at most two points per arc; with the chosen closest pair isolated, ordering reciprocal chord distances gives `A≤C r^(-1)(1+log N)`. Only tightness of the rescaled minimum gap is needed. No growing-argument extrapolation of the limiting tail is used.
9. **General beta input.** Feng – Wei Lemma 1.1 explicitly supplies `C_{β,N-3,(3)}/C_{β,N}≤(Nβ)^(3β)` for `β≥1`. Hölder with exponents `(3,3,3)` produces precisely the stated three-point bound. Their extreme-gap theorem has the positive integer beta restriction used in the report.
10. **Constants.** The paper gives `A1=1/24`, `A2=1/(24π)`, `A4=1/(270π)`. Under `D~δ²/8`, these yield the reported tails `exp(-t/6)`, `exp[-2sqrt(2)t^(3/2)/(9π)]`, and `exp[-64sqrt(2)t^(5/2)/(675π)]`.

Suggested final–statement clarifications

Define `D` explicitly as the infimum of positive times at which the evolved polynomial has a multiple root, with infimum of the empty set equal to infinity. State that initial roots are distinct almost surely in the random ensemble. Mention the preserved nonzero leading and constant coefficients in the circle-persistence argument. Keep the qualification that novelty requires comparison with Lehmer-pair/Newman and function-field literature.

The proof's arithmetic limitation is substantive. It exploits an extreme pair whose angular gap is asymptotically much smaller than the mean spacing, and a probability law guaranteeing initial isolation. A half-lattice mimicker fails that hypothesis. Producing it for zeta zeros remains the missing arithmetic information, and a single or sparse exceptional close pair does not refute an Alternative Hypothesis permitting density-zero exceptions.

Current report 10: prime186

Source: `research/reports/prime186.md`. SHA-256: `0b4c921d5facba4d079b1c9a54a6958aed18c63be074aa6032cc1310fa7b6e1a`.

Prime–gap 186 audit, a certified minorant refinement, and the next arithmetic target

Research round 1, 2026-09-05. This document records results actually obtained in this run. No bound below 186 is proved. The useful completed work is an exact rational enclosure for the fixed minorant's deficit, an honest propagation of its limited value through the full support-restored criterion, an exactly checked 39-tuple giving a concrete next target 182, and a sharp obstruction to improving one of the elementary debt constants by counting alone.

1. The current baseline and what was replayed

The official PrimeGaps186 repository was cloned at commit `61340d0b74163003b32756bb16e91d9209a5e330`. Its accompanying paper proves a sufficient sieve criterion for DHL[40,2], then applies a 40-point admissible tuple of diameter 186. The public Lean development is explicitly conditional on three project inputs: two cited Deligne-type estimates and the physical integral bounds. Standard logical axioms are separate. A numerical receipt does not itself discharge those Lean inputs.

The local audit inspected the main paper, the numerical companion, and the exact-input Python certificate. It replayed scalar rational arithmetic from printed endpoints; it did **not** recompute the 149 physical source integrals. This distinction applies to every margin below.

The source-preserving clone is in `prime186-work/PrimeGaps186`. All new experiments are outside that clone.

The central normalized inequality is

$$\mathrm{M}(F) = \rho_* \frac{J_{\lambda,H}(F) - L(F)}{I_H(F)} - 1 > 0.$$

Here the admissible trial is an outer coefficient profile with its actual rootwise support conditions, and L pays for every deleted part. A large cap-only Rayleigh quotient is insufficient. On the published fixed trial, exact scalar replay gives

$$\rho_* J^- / I^+ = 1.0002060794024186\ldots,$$

$$L^+/I^- = 0.000696075110, \qquad \rho_*(J^- - L^+)/I^+ - 1 = 0.000023360452297044097\ldots.$$

The scale $\rho_* = 2624989/10^7$ means that most of the cap surplus is consumed by support restoration. The remaining margin is about 23 parts per million.

The loss ledger matters more than the count of coefficient parameters:

Failure class	Published weighted loss, in units $10^{-12} I^-$	Fraction of total
Outer, effective order 2	38,927,522	5.5924%

Failure class	Published weighted loss, in units $10^{-12}\Gamma$	Fraction of total
Outer, effective order 5/2	622,829,241	89.4773%
Base inner, order 2	55,254	0.00794%
Base inner, order 5/2	435,544	0.06257%
Enlarged inner, order 2	1,405,159	0.20187%
Enlarged inner, order 5/2	32,422,390	4.65789%

Thus a substantial improvement should target the outer order-5/2 support geometry and the full constrained operator. It should not spend most of its budget shaving the already tiny base-inner loss.

2. Completed exact refinement of the fixed minorant deficit

Write the prime minorant as $\varrho = P - b$, where $b \geq 0$ is the explicitly defined five-prime exceptional contribution. Let κ be the limiting mean of b in units $x/\log x$. The implemented trial used $\kappa \leq 2 \cdot 10^{-5}$. The companion already supplied a smaller coarse rational bound. This round computes a sharper enclosure of the *same* fixed integral; it does not alter the arithmetic minorant or its distribution range.

The independently reproduced enclosure is

$$\boxed{\frac{933388696677522539931593 2}{10^{30}} \leq \kappa \leq \frac{9334314528491567966021794}{10^{30}}}$$

or

$$9.333886966775226 \cdot 10^{-6} \leq \kappa \leq 9.334314528491569 \cdot 10^{-6}.$$

The absolute enclosure width is $4.2756171634 \cdot 10^{-10}$. The calculation used 16,384 four-dimensional simplices and took approximately 3.88 seconds on this host. Every inequality in the enclosure uses Python integer and rational arithmetic. Floating-point values select the next cell to split, which affects efficiency only.

2.1 Why this is a rigorous enclosure

Use $t = 481/100000$, eliminate the fifth coordinate by $\Sigma z_i = 0$, and put

$$a_i = \frac{1}{5} + tz_i, \qquad f(z) = \prod_{i=1}^{5} a_i^{-1}.$$

The minorant's two four-dimensional polytopes have the six rational simplex cells recorded in the numerical companion. On their positive domain,

$$\log f = -\sum_{i=1}^{5} \log a_i$$

is convex. Therefore f is convex. For a four-simplex Δ with vertices $v_0, \ldots, v_4$ and centroid $\bar{v}$,

$$|\Delta| f(\bar{v}) \le \int_\Delta f(z)\,dz \le \frac{|\Delta|}{5} \sum_{j=0}^{4} f(v_j).$$

The lower inequality is Jensen's inequality. For the upper inequality, convexity bounds the value at a point by its barycentric affine interpolant, whose average is the average of the five vertex values. Multiply by t^4, sum over cells, and round each cell contribution outwards to an integer multiple of 10^{-30}. Longest-edge bisection preserves exact rational vertices and halves the exact volume. This yields reproducible lower and upper bounds without transcendental quadrature or FLINT.

2.2 Independent geometry verification

`minorant_geometry.py` verifies the underlying rational geometry, rather than merely assuming that a printed triangulation has no holes:

1. Enumerate all vertices by solving every four-active-halfspace system over the rationals, and reject candidates violating another halfspace. The first polytope has 11 defining halfspaces, the second 10. Both have exactly the seven printed vertices.
2. Check all six four-simplex determinants and volumes exactly.
3. Orient every simplex positively. Every internal tetrahedral face appears twice with opposite orientations; every remaining face lies on a supporting hyperplane of the polytope.
4. Check an explicit strict interior rational point at which the oriented covering multiplicity is exactly one.

The corresponding elementary chain-degree argument proves coverage almost everywhere: an oriented simplex chain whose boundary lies only on the boundary of a convex polytope has constant degree throughout its interior. Here that degree is one. Thus there is neither positive-volume overlap nor a missing positive-volume region. Both geometry audits pass.

The arithmetic identification of the exceptional-prime mean with these polytopes remains the mathematical input from the minorant construction, rather than a new claim proved by the Python code.

2.3 Carrying the refinement through the complete margin

A deficit improvement must be propagated through all coefficients, including the error weights. For fixed $\lambda = 1/125$ and $K = 17/50$, write

$$m = 1-\kappa, quada = m^2 - m\lambda, quadb = (1-m/\lambda)\kappa K, \quad c = a+b, \quad d_0 = 1-c.$$

The cap form is $J_0 + cJ_+ + bJ_t$. With the refined upper bound for κ, the coefficients change as follows:

Quantity	Published allowance	Refined allowance
a	0.9919601604	0.9919814061325887
b	-0.000843183	-0.0003935309975205265
$c = a+b$	0.9911169774	0.9915878751350682
d_0	0.0088830226	0.0084121248649319

Both cap coefficients improve, because $J_+, J_t \ge 0$. The base-inner weight d_0 and enlarged-inner weight $1-b$ decrease. However, the positive outer-restoration multiplier on the enlarged shell increases from c_{old} to c_{new}. Ignoring that change would be an invalid free gain.

The new outer multiplier is at most $r = c_{new}/c_{old}$ times the old one everywhere. Thus a conservative complete loss bound follows from the old certified component bounds by multiplying the outer loss by r, the old-inner loss by $d_{0,new}/d_{0,old}$, and the new-inner loss by $(1 - b_{new})/(1 - b_{old})$. Even retaining only the old combined cap lower endpoint, the resulting complete margin is at least

$$0.00002328873833218647\ldots > 0.$$

This does **not** mean the refined trial is worse. It means that the old *combined* numerator endpoint does not reveal how much of the cap form lies in J_+ and J_t. Without these separate enclosures, an actual improvement cannot be quantified sharply. Keeping the old coefficients remains valid and retains the original $0.00002336045\ldots$ margin.

There is also a rigorous ceiling on the leverage of this one adjustment. On the same fixed trial and same λ, the published operator bound gives

$$J_+ + J_t \le J_{full} \le 4I.$$

Consequently, the cap-quotient gain is bounded above by

$$4\rho_* \max\{c_{new} - c_{old}, b_{new} - b_{old}\} = 0.0004944405498715189\ldots.$$

Even if every old restoration loss disappeared, the total improvement from this fixed-trial, fixed-λ adjustment would be below 0.0006771595. These bounds describe its scale; they do not prove that a $k = 39$ trial lies beyond or within reach. A direct $k = 39$ full-form calculation is still required.

The refinement changes no source exponent, no divisor-closed support theorem, and no tuple size. It is a useful exact certificate and a starting input for reoptimization, not a smaller prime-gap theorem.

3. A proved obstruction: the coefficient 12/5 is sharp

The elementary majorant $b \le (12/5)N_2$ cannot be improved merely by sharpening its combinatorial case count. Here N_2 counts prime pairs whose product is below the specified threshold.

Take five distinct normalized prime exponents

$$(0.198, 0.199, 0.200, 0.201, 0.202).$$

They sum to one; every pair sum is below 0.40481, every triple sum is above 0.59519, and every exponent lies strictly inside the allowed roughness and upper bounds. Exhausting all 120 assignments to the labelled factors gives

$$b_1 = 4, \qquad b_2 = 20, \qquad N_2 = 10, \qquad \frac{b_1 + b_2}{N_2} = \frac{12}{5}.$$

All inequalities are strict, so this is an open exponent chamber, not a boundary coincidence. The finite assignment count is independently verified using rational arithmetic in `margin_ledger.py`.

This only closes the route of lowering the *pointwise constant* without changing the majorant. One can still seek a better majorant, retain more of the exact exceptional geometry, or prove a stronger weighted estimate for the balanced five-prime convolution. In particular, the unweighted mean $\kappa \approx 9.334 \cdot 10^{-6}$ cannot automatically replace the weighted exceptional-energy constant $K = 0.34$: those are different mathematical quantities.

4. A concrete next prime-gap target: DHL[39,2] implies 182

A mixed-integer search produced the following 39-point admissible tuple, and a separate integer residue check validated it:

```
0, 2, 6, 8, 12, 20, 26, 30, 36, 38, 42, 48, 50,
56, 62, 66, 68, 72, 78, 80, 90, 92, 108, 110, 126,
128, 132, 138, 140, 146, 150, 152, 156, 162, 168,
170, 176, 180, 182
```

For each prime $p \le 39$, at least one residue class is omitted; for $p > 39$ this is automatic from the cardinality. Its diameter is exactly 182. Thus **DHL[39,2] would imply $H_1 \le 182$**, giving a precise target for the analytic computation.

This tuple is not claimed to be new or optimal. The search also returned a 38-point witness of diameter 176. HiGHS reported infeasibility for 40 points within diameter 184 and for 39 points within diameter 180. Those are numerical solver outcomes, not independently checked infeasibility certificates. An attempted proof-producing Z3 replay consumed approximately 6.7 GB without delivering an inspectable certificate and was terminated; no exact minimal-diameter theorem is claimed from it.

The feasible witnesses are independently checkable and are sufficient for research planning. The old $k = 49$ door to 240 is superseded as a main target by the existing 186 result.

5. A transferable support principle and a multi-minorant identity

The most valuable structural lesson is to preserve the exact arithmetic support and the pointwise nonnegative detector simultaneously. Rootwise conditions must imply the needed property of the actual least common multiple, including shared prime factors, exhausted roots, and inward cell restrictions. A condition on a generic product of independent roots is not a substitute.

The complementary allocation principle is a useful design language: split a required prime-power charge into nondecreasing contributions assigned to the two roots, with a fixed combined budget. Optimize the allocation as a function of prime size, then charge the actual excluded fragment configurations to the operator. This is stronger than blindly imposing one smoothness cutoff, but it does not abolish the parity or distribution barriers.

There is a clean extension of the residual-square bookkeeping to several minorants, which may be useful when support radius and minorant quality vary. This is an algebraic lemma, not a claim that suitable new arithmetic minorants have already been constructed.

Let $P \ge 0$, $b_j \ge 0$, $\rho_j = P - b_j$, $A = B_0 + D$, and $C = \Sigma_j C_j$. For arbitrary $\eta_j > 0$,

$$\begin{aligned} PA^2 - \Big[P(2AB_0 - B_0^2) + 2\sum_j \rho_j D C_j - PC^2 - \sum_j \eta_j b_j D^2 - \sum_j \eta_j^{-1} b_j C_j^2\Big] \\ = P(D - C)^2 + \sum_j b_j(\sqrt{\eta_j} D + C_j/\sqrt{\eta_j})^2 \ge 0. \end{aligned}$$

Thus the residual loss is explicit and remains nonnegative. If the relevant arithmetic averages are established, the corresponding normalized variational gain is

$$2\sum_j m_j \langle V - H_0, H_j\rangle - \Big\|\sum_j H_j\Big\|^2 - \sum_j \frac{\kappa_j}{\eta_j} \| H_j \|^2 - \sum_j \eta_j K_j E_0.$$

For mutually disjoint correction shells this separates into scalar optimizations. With different minorant qualities and distribution supports it suggests a richer admissible operator. With identical minorants and identical supports, splitting one correction into several labels provides no automatic benefit: the covariance and residual charges must still be paid. Any proposed gain must prove the needed mixed and self-square distribution estimates on all actual moduli.

This identity is more transferable to the zeta programme than another decimal improvement of κ: it makes the missing arithmetic information and its squared error cost explicit.

6. Reproducibility and the FLINT obstacle

The fresh official certificate was **not** run past startup. Its mandatory signed-convolution test fails in both local wheel environments tried:

- Python-FLINT 0.9.0, FLINT 3.6.0, NumPy 2.5.2;
- Python-FLINT 0.8.0, FLINT 3.3.1, NumPy 2.2.6.

The check was not disabled or replaced. The official upstream correction is identifiable: FLINT pull request 2790, specifically commit `7ad753d51c82fdec115cb179b41d0e581f1cb0ec`. A source checkout of v3.6.0 at `8d5454b96761fafe4d5a9da76a369a602f500f49` was prepared. A patched build was not completed before the research priority was redirected to the zeta arithmetic barrier. Neither wheel failure is evidence against the mathematical theorem; it is a reproducibility requirement explicitly anticipated by its authors.

The new rational computations need none of these libraries. Run, from `prime186-work`:

```
python3 minorant_geometry.py
python3 minorant_mass.py
python3 margin_ledger.py
```

`tuple_search.py` additionally requires SciPy; only the search uses floating-point optimization, and every accepted tuple is checked with integer arithmetic.

Files suitable for a small research commit are:

- `prime186.md`: this report;
- `minorant_mass.py` and `.json`: exact convexity enclosure;
- `minorant_geometry.py` and `.json`: exact polytope/triangulation verification;
- `margin_ledger.py` and `.json`: full scalar margin, coefficient leverage bound, and sharp 12/5 count;
- `tuple_search.py`, `tuple_search.json`, and `tuple_search_more.json`: feasible witnesses and explicitly limited solver results.

Do not commit cloned upstream sources, virtual environments, or the unfinished proof-producing SMT run as if they were new mathematical results.

7. Next action and stopping discipline

The next substantial prime-lane computation would be a correctly linked full baseline, separate enclosures for J_0, J_+, J_t, and a direct $k = 39$ variational search that includes the dominant outer support failure. A cap-only success must not be reported as a prime-gap improvement.

Following the user's strengthened priority on a major zeta/random-matrix theorem, this round stops refining the auxiliary minorant integral. Its exact certificate is saved and can be reused. The transferable lessons are the support-preserving allocation structure, the positive residual identity, and the insistence on a complete cost ledger. These are useful when testing whether a new arithmetic zero statistic actually escapes the known ACUE mimickers.

Current report 11: resummed_prime_profiles

Source: `research/reports/resummed_prime_profiles.md`. SHA-256: `1bb8d49fc05c7e7008134d00328a77cc59a9d0344f011ebbd7ca4acd5282b64e`.

Resummed symmetric prime profiles: a stable negative variational experiment

Research round 2, 2026-09-05. This is a bounded continuum experiment suggested by the residual-Gram agent's arithmetic operator. It is not a zeta theorem, a certified interval optimization, or a proof of optimality.

Outcome

The family below removes the ill-conditioned large symmetric-feature Gram matrix by resumming a multiplicative prime-factor profile. It reproduces the unmodified one-variable benchmark and improves that restricted benchmark, but does not exceed the better polynomial prime-feature result obtained by the other agent. No half-gap crossing occurs in the validated trials.

Prime profile	Radial degree / quadrature order	Half-gap margin
$g = 1$, optimized original parameter	6 / 32	-0.0153579821816782
$g(u) = \exp(c_2 u^2)$	7 / 32	-0.0147198770500968
$g(u) = \exp(c_2 u^2 + c_3 u^3)$	9 / 40	-0.0146705663327385
Three nontrivial coefficients, through u^4	9 / 40	-0.0146638642779015
Five coefficients, through u^6; local continuation	9 / 40	-0.0146638632200778

The last two searches are local continuations, not global optimizations; one hit its iteration budget. The radial Gram condition numbers in the high-precision-density runs were between approximately 1.6 and 2.4. Increasing radial degree and quadrature order changes the displayed negative margins only slightly. This offers a stable independent check that these natural prime-factor tilts recover a small fraction of the deficit, not an arithmetic advance past one half.

The family and its continuum density

Consider the formal asymptotic family

$$r(n) = d_\ell(n) \prod_{p \mid n} g\left(\frac{\log p}{\log L}\right) f\left(\frac{\log n}{\log L}\right), \qquad g(u) = \exp\left(\sum_{k=2}^{d} c_k u^k\right), \quad a = \ell^2.$$

The product is symmetric in the prime factors. In the squarefree leading continuum model, a linear exponent in g is redundant with the radial profile f and is therefore omitted; this is not an exact finite-integer identity for a product over distinct prime divisors. Prime-power conventions require a precise choice in an eventual arithmetic theorem; their effect on the leading continuum formula must be justified, not assumed away.

Under the same Poisson – Dirichlet continuum model used for the prime-feature trial, the squared-coefficient density is proportional to

$$R(v) = v^{a-1} H(v), \qquad H(v) = \mathrm{E}\exp\left(2 \sum_{k=2}^{d} c_k S_k\right),$$

where the partition has total mass v. This resums all powers of the chosen symmetric statistics, rather than truncating a polynomial in $S_2, S_3, \ldots$.

Write

$$g(u)^2 = \sum_{n\geq 0} d_n u^n, \qquad H(v) = \sum_{n\geq 0} H_n v^n.$$

The coefficient recurrence used in the experiment is

$$H_0 = 1, \qquad \boxed{H_n = \frac{a}{n} \sum_{k=1}^{n} k d_k B(k, a+n-k) H_{n-k}.}$$

One derivation uses the formal Laplace transform of the convolution exponential with kernel $a(g(u)^2 - 1)/u$. If B_n are the coefficients of

$$\exp\left(a \sum_{k\geq 1} d_k \Gamma(k) z^k\right),$$

then $H_n = \Gamma(a) B_n / \Gamma(a+n)$; differentiating the generating series gives the displayed normalized recurrence. This is a formal power-series identity: the factorial-weighted intermediate series need not have a nonzero analytic convergence radius. The final density series is evaluated with an explicit truncation comparison in the experiment.

For example, when $g(u) = e^{cu^2}$, the linear coefficient in c is

$$H(v) = 1 + \frac{2c}{a+1} v^2 + O(c^2),$$

in agreement with the Poisson – Dirichlet moment of S_2.

The changed variational form

Use v for the background mass and u, w for inserted prime masses. Relative to the original one-variable form:

- Replace the background measure $v^{a-1} dv$ by $v^{a-1} H(v) dv$ in both numerator and norm.
- Multiply the two distinct-prime insertion integrand by $g(u)g(w)$.
- The same-prime diagonal insertion term has no extra factor $g(u)^2$: its resonator coefficient lives on the background integer, not the newly inserted output prime.
- A linear insertion, away from the exact half-gap boundary, acquires $g(u)$.

The distinction in the third item is essential. Inserting an extra $g(u)^2$ there would change the arithmetic operator and could manufacture a false improvement.

The case $c_2 = \cdots = c_d = 0$ reduces exactly to the independently reproduced Inoue form, which the code checks to numerical precision. Passing from these coherent continuum formulas to a zeta theorem would still require the weighted-integer asymptotics and error accounting for this new coefficient family.

Numerical controls and failure record

The initial implementation evaluated the density series entirely in binary64. It reproduced the first two improved trials, but a higher-dimensional optimizer found a parameter vector for which the 32-node validation rejected the density truncation. That run was not accepted as a result.

The revised implementation computes the density coefficients and evaluations with 55 decimal digits, checks 110 versus 140 terms, and also checks the endpoint $v = 1$ rather than only quadrature nodes. The final Rayleigh matrices and generalized eigenvalue computations remain floating point. This fixes the identified density-evaluation weakness but does not make the experiment a rigorous interval certificate.

The most accurate recorded five-coefficient local continuation uses

$$\ell = 1.0820458998724858,$$

$$\begin{aligned} c_2 &= -0.9407908108421177, \\ c_3 &= 1.17145568664247, \\ c_4 &= -2.268580431289358, \\ c_5 &= 0.0000165244269937828, \\ c_6 &= -0.0000964571420071413. \end{aligned}$$

Its tiny newly added coefficients reflect the chosen local continuation and should not be interpreted as proof that those directions are useless.

Files are in `research/prime-profiles/`:

- `euler_profile.py`: basic continuum operator and initial floating implementation;
- `euler_profile.json`: accepted initial trials before the rejected higher-dimensional run;
- `euler_profile_precise.py`: higher-precision density implementation;
- `euler_profile_precise.json`: validated negative trials and full coefficients.

Reproduction:

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 euler_profile_precise.py
```

The main research effort should now return to the missing arithmetic mixed information. This experiment provides a stable resummed family and a negative result, not grounds for spending an indefinite budget on additional decimals of the same variational frontier.

Claim ledger 12: CLAIM_LEDGER

Source: `research/claims/CLAIM_LEDGER.md`. SHA-256: `43ecc9efa7f0a570c6e6eb01f3b683acfbaab3da418443340422a601ab18ba51`.

Claim ledger

The classifications below apply to the exact stated scope. “Proof with internal review” means a written mathematical proof and a recorded human-readable audit; it does not mean a Lean kernel has checked it or external referees have accepted it. Historical files may use stronger wording than this ledger.

ID	Claim	Status	Evidence and remaining obligation
AUD-001	Independent historical audit has 21 passing checks	Reproduced finite checks	`research/logs/audit_results.json`; not a blanket theorem audit
AUD-002	Original verifier has 39 pass, 4 fail, 1 skip	Recorded run	`research/logs/verify-codex-original.log`; zero process exit does not erase failures
ALG-001	Fixed-support approximator is saturated	Exact algebra	`research/reports/residual_gram_round1.md`; says nothing about larger support
NEG-001	Sparse longer tails under the stated coefficient bounds cannot supply constant energy	Proof in report	Same report; hypothesis on coefficients is essential
NUM-001	Degree-14 half-gap trial has margin about -0.01535798	Numerical optimization	`research/residual-gram/variational-results.json`; no global optimality claim
CERT-001	Fixed rational trial has margin in (-.01467,-.01465) for the stipulated integral	Exact rational certificate	`research/residual-gram/rational-trial-certificate.json`; transfer is separate
ARITH-001	Fixed symmetric-prime family has that arithmetic limiting form	Written proof, independent internal review	`research/reports/symmetric_prime_arithmetic_transfer.md`, `symmetric_prime_transfer_independent_review.md`
NUM-002	Full finite operator remains below $\pi^2/2$ through $L=10^7$	Numerical eigenvalue data	`research/operator-bounds/extended-arithmetic-results.json`; no asymptotic upper bound

ID	Claim	Status	Evidence and remaining obligation
NUM-003	Structural features nearly recover the $L=10^6$ optimum	Numerical fit and full-form evaluation	`research/operator-bounds/eigenvector-feature-results.json` ; finite L only
NEG-002	Tested Schur – Volterra majorant is too large to prove the target barrier	Numerical diagnostic of an explicit bound	`research/reports/residual_gram_round2.md` ; not a certified optimum over all profiles
FOCK-001	Fixed prime-bin compression provides a constructive limiting lower bound	Proof draft	Same report; not full operator convergence or an upper bound
NEG-003	Centered Gaussian pole term cannot be dropped in a uniform long-support estimate	Analytic counterexample, primary review	`research/reports/centered_gaussian_mixed_moments.md` ; RH for the contour identity
NEG-004	Nonnegative Fourier transform cannot coexist with a zero at the origin for a nonzero Schwartz weight	Elementary proof	Same report; does not rule out quantitative subtraction
HEAT-001	Isolated-pair first collision satisfies $D \sim \delta^2/8$ under $\delta^2 B \to 0$	Proof draft with internal audits	`research/reports/yau_flow_galilean_refinement.md` ; quantitative second-pass audit in `galilean-proof-audit.md` ; K=16384, eta0=1/524288
RMT-001	Circular positive-integer-beta extreme gaps induce the stated first-collision law	Proof draft with internal review	`research/reports/yau_flow.md` ; published gap laws are inputs; novelty audit incomplete
PRIME-001	A 39-element admissible tuple has diameter 182	Exact finite verification	`research/reports/prime186.md` ; DHL[39,2] is not proved
PRIME-002	Fixed prime-sieve minorant mass has the saved rational enclosure	Exact rational computation plus separate coverage check	`research/prime-gaps/minorant_mass.json` , `minorant_geometry.json`
OPEN-001	$\mu\zeta < 1/2$ under RH	Open in this programme	No positive applicable certificate
OPEN-002	New fixed-width Montgomery pair-correlation theorem beyond known support	Open in this programme	No new arithmetic mixed-term theorem supplied
OPEN-003	Prime-gap bound below 186	Open in this programme	No improved admissible-tuple/sieve certificate chain
OPEN-004	RH, full GUE statistics, general-beta depth universality	Open here	Not implied by any local counterexample or finite computation

All paths are relative to the repository root. The public reports contain the derivations; the JSON files preserve values and parameters. Run logs are snapshots, not machine proofs. Updates should change this ledger and the associated report together.

Research log 13: RESEARCH_LOG

Source: `research/RESEARCH_LOG.md`. SHA-256: `647cc527d4ecf75644a09019c0cc402621f320dccde36c38807ce5c7506fb485`.

Research log: recoverable mathematical work

This log records problems, mathematical arguments, calculations, corrections, and evidence. It is a public research record, not a transcript of private model deliberation. Detailed proofs and counterexamples live in the linked reports. Timestamps record checkpoints; they do not imply continuous computation between checkpoints.

2026-09-05: source recovery and independent audit

Question. Which statements in the earlier ACUE, Newman-depth, zeta and prime-gap programme survive direct checking?

Input. Historical base `ee61fba38925190916e556f87bc1ea83a502413e`; Fable handoff update `0f86c31d847178f0a774f70073ddc44206b32f98`; user-provided mathematical context and original Inoue probe. The public historical source is preserved in `historical/riemann-rmt/`; private supplied context and conversation exports remain in the adjacent local archive.

Outcome. The original verification script reports 39 PASS, 4 FAIL, 1 SKIP even though its exit status is zero. The independent audit has 21 passing checks. Those passing checks verify specified algebraic/numerical facts; they do not validate every historical theorem.

Corrections worth preserving. The Galerkin angle needs `acos(1-1/(n*n*a))`; the Wilson trace identity requires the stated boundary geometry; COM null dimensions in the tested sizes are not the earlier claimed pattern; a near-endpoint background example refutes an old uniform upper bound. Clock mass does not disappear at fixed N. The Lean development has actual unfinished obligations, and a conclusion-free depth comparison is not a depth theorem. See the full audit narrative and reproduction.

Decision. Keep corrected statements, archive failed claims verbatim with warnings, and require an explicit arithmetic transfer before assigning consequences to real zeta zeros.

2026-09-05 05:56:56 UTC: first autonomous research milestone begins

The initial authorized milestone is eight hours. The user's coordination task subsequently relayed a 72-hour recoverable programme. This log makes no guarantee of uninterrupted execution or of solving an open conjecture. The active priority is a genuine RH-conditional zeta half-gap result or an explicit out-of-band correlation theorem. Matrix-model distinctions are supporting results.

Three internal research lanes were assigned: scalar heat-flow proof and Yau comparison; residual Gram and arithmetic resonators; prime-gap certificate audit and transferable positivity methods. Their reports preserve failures as well as successes.

Round 1: the saturated approximator and the new resonator direction

Question. Can another approximator inside the same accessible product support recover the missing half-gap energy?

Exact answer. Completion of squares gives `2 Re<ba,bg>-||ba||^2=||bg||^2-||ba-bg||^2`. Thus the selected coefficient-space approximator is already saturated for a fixed resonator and fixed product support. This does not optimize the resonator, prove a global method barrier, or allow replacement of finite-height Gram matrices by diagonal matrices without an error theorem.

Independent computations. The published linear Inoue trial is reproduced. A degree-14 one-variable trial has half-gap margin about `-0.01535798170385`. A fixed rational symmetric-prime trial improves this to a certified continuum value about `-0.014662375473369`; richer exploratory trials reach about `-0.01465473` but have ill-conditioned numerical Gram problems. These are distinct candidates, all negative.

Failure with information. Sparse support beyond the usual cutoff cannot provide a fixed amount of missing energy under the stated subpolynomial coefficient bounds. Sparse orthogonality alone is insufficient; dense correlations or different coefficient concentration would need a new estimate.

Evidence. Round 1 report, exact rational certificate, certificate output.

Round 1: scalar heat flow and extreme circular gaps

Question. Does a tiny isolated pair determine the first collision of the actual coefficient heat flow?

Mathematical output. A deterministic proof draft uses a scalar heat normal form. Removing the background linear drift by exact heat conjugation reduces the sufficient isolation condition to `delta^2 B -> 0`. Its probability application uses triple-cluster exclusion and published extreme-gap laws for CUE and positive integer beta circular ensembles.

Scope. This is a proposed full RMT theorem with internal audits, not a zeta-zero theorem and not a proof of universality for every beta. Its initial-background method overlaps classical Lehmer-pair theory; priority and formal verification remain separate obligations.

Evidence. Heat report, Galilean refinement, independent first review, numerical checks.

Round 1: what the prime-gap 186 calculation actually contributes

Question. Can exact positivity and certificate methods from the public 186 project provide a smaller prime-gap bound or a transferable tool?

Output. A 39-element admissible tuple of diameter 182 was found, but no `DHL[39,2]` certificate was established. This is not a prime-gap improvement. An exact-rational convex-simplex enclosure independently bounds a fixed minorant mass. Separate exact geometry checks address the published triangulation's coverage. Numerical infeasibility searches are not portable impossibility proofs.

Failure with information. The official numerical verifier hit a FLINT convolution regression in the installed environment. No step was bypassed and no successful official certificate rerun was claimed. The public Lean project retains stated assumptions; it is not an assumption-free formalization of every analytic ingredient.

Decision. Stop broad prime-gap tuning until there is a definite path to a new analytic or certificate threshold. Transfer support accounting and exact positivity methods to the zeta lane.

Evidence. Prime report, exact minorant calculation, coverage verification, margin ledger.

Round 2: ten-million-dimensional operator and a failed upper bound

Question. Is there a large unused coefficient direction hidden outside the one-variable trial family?

Computation. At `L=10^7`, the full sparse arithmetic operator has computed largest eigenvalue `4.324089558989538`, below the half-gap threshold `pi^2/2`. At `L=10^6`, 20 symmetric-prime features nearly reproduce the full finite optimum; 40 features reduce the remaining difference to about `0.0004437`. A 40-bin bosonic model with 215,308 states is also below threshold.

Analytic attempt. Separate Cauchy bounds lead to a scalar Schur – Volterra majorant. Its tested threshold is approximately `5.2074`, above `pi^2/2`; it cannot establish a universal half-gap obstruction. More grid precision does not fix that mathematical loss.

Decision. Preserve the full spectra, fit residuals and failed majorant. Investigate compatibility of extremizers or new arithmetic products, rather than presenting rising finite spectra as an asymptotic proof.

Evidence. Round 2 report, spectra, fits, Fock data, majorant data.

Round 2: centered Gaussian positivity has an explicit pole correction

Question. Can a centered Gaussian make all Fourier kernels positive and therefore extend the accessible mixed-product support for free?

Exact negative result. In the critical-line contour identity, centering leaves an explicit pole-cut integral. For the intended longer-tail supports its phase is opposite to the retained prime contribution. A dense coefficient example with bounded diagonal norms has a negative pole correction of order `M/log M`; for `M=T^(1+eta)` this exceeds the presumed height normalization. The exact long-support Gram penalty also contains substantial off-diagonal mass.

Related obstruction. A nonzero Schwartz weight with nonnegative Fourier transform cannot vanish at the origin. Removing low-height coherent mass while keeping all those Fourier signs is not a free operation. This does not exclude all centered-packet methods: they must pay the pole term and control the density-subtracted mixed moment.

Evidence. Full derivation and counterexample, 80-digit checks. The primary agent checked the source contour identity, the phase multiplication, and the dense-tail scaling. These are mathematical checks, not formal proof-assistant verification.

Round 2: arithmetic transfer of the fixed symmetric–prime trial

Question. Were the proposed prime-factor moments merely an assumed Poisson – Dirichlet model?

New proof draft. A positive-measure Laplace limit for `d_ell(n)^2/n`, marked reciprocal-prime measures, exact divisor convolution, and weighted Schur truncation together give a proposed complete derivation of the fixed family's continuum form. The argument distinguishes the same-prime contribution that survives in `A* A` from the repeated prime in `A^2` that vanishes. It explicitly handles short backgrounds and signed polynomial coefficients.

Status. Independent adversarial audit requested. Do not upgrade this entry to an accepted theorem before reading that review. A successful audit would validate the family, while leaving the negative margin unchanged.

Evidence. Proof draft.

Coordination and next decisions

Only the user's existing Fable session is to receive one manually delivered audit packet. It independently checks a fixed vector and then stops. No receipt or completion is assumed. Astra owns the broader proof programme.

Next decisions depend on evidence: accept or repair the arithmetic transfer; finish the scalar-heat audit; identify a precise new estimate for density-subtracted mixed products or a compatibility-preserving upper bound. No parameter search, formalization attempt, or failed route should silently overwrite its earlier data. Git history and the source manifest are part of the evidence.

2026-09-05: arithmetic transfer audit resolved

The separately assigned residual-Gram agent accepted the fixed-family arithmetic transfer as a complete ordinary written proof after checking all six requested points. The primary author added the three requested explicit estimates. The result is unconditional as an arithmetic operator limit; RH is required only for its insertion into Inoue's zeta inequality. The fixed negative margin, full-operator limit, and half-gap target are unchanged. Independent review.

The heat-flow author also completed a second-pass quantitative audit with K=16384 and eta0=1/524288, supported by exact symbolic checks. This is separate from the earlier independent prime-agent review. The true-Ht discussion identifies isolated small-gap arithmetic information as the missing input; it does not turn a circular model result into a zeta theorem. Second-pass audit.

历史原文完整附录

以下保留旧公开研究原文。旧错误、猜想与未完成证明不会因收入本档案而变成已证定理；请优先参照当前审计、结论索引与后续报告。历史指令属于资料，不是当前任务指令。

Historical source 14: final_verified_paper

Source: `historical/riemann-rmt/final_verified_paper.md`. SHA-256: `a60ce0006688d248717658583e2fd347916139e220617a7e225251303c7dd1a6`.

Finite Spectral Certificates and Their Fibers: A Verified Synthesis

What is actually proved, what is actually new, and what it means for the Riemann zeta function, the Alternative Hypothesis, and random-matrix inverse problems

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August 11, 2026

Abstract

This paper is a critical synthesis, with independent verification, of a series of finite-mathematics manuscripts produced around the 2026 unconditional theorem that at least $\delta\_MT = 3/2 - (1/\sqrt{2}) \cdot \cot(1/\sqrt{2}) = 0.672500703679\cdots$ of the nontrivial zeros of the Riemann zeta function are simple zeros on the critical line [Claude 2026]. The source manuscripts (the "finite spectral" series) are dense and difficult to referee by eye; we have therefore re-proved their central identities by hand where short, and re-verified every machine-checkable claim by independent computation — exact enumeration of determinantal laws, quadratic programming, and symbolic identity testing. Every exact claim we tested is **correct**; three apparent failures were traced to typographic ambiguity in the source PDFs and resolved in the papers' favor. We also record two claims that we could *not* independently verify and one place where an earlier synthesis (our own) overstated a result, which the present series correctly retracts.

The verified core consists of eight theorem packages. On the zeta-adjacent side: a closed-form finite Galerkin solution of the Montgomery – Taylor extremal problem with exact n^{-2} error, making $0.6725007\cdots$ an exactly computable finite object rather than a numerically fitted constant; and an exact "Nyquist conservation law" for integer-marked half-filled cyclic ensembles, in which one missing Fourier mode simultaneously stores all repeated-point mass, dualizes nearest-neighbor exclusion, and produces the density curve $\Phi(\rho) = 1 - \rho + \sin^2(\pi\rho)/(\pi^2\rho)$, with $\Phi(1/2) = 1/2 + 2/\pi^2$. On the random-matrix side: an exact fourth-moment covariance formula for the finite Fermi-sea (ACUE-type) projection ensemble, $\mathrm{Cov}(V\_k, V\_\ell) = k^2 \cdot 1\{k=\ell\} - 2(k+\ell-N)_+$ with smallest eigenvalue exactly 1, yielding a quantitative moment-polytope inradius and an approximation-to-exact-law repair theorem; and four sharp non-identifiability theorems — Fermi-sea boundaries, magnetic-cycle holonomy with an exact total-variation formula, the quasi-free axiom gap with its necklace-counting fiber dimension, and marked homometry surviving complete diffraction. Finally, the series reduces its one central open problem — all-size attainment of the Nyquist equality face — to a three-tile integral language with exact no-go theorems for every natural construction.

We explain, for each package, what is classical, what is new, why it matters for the Riemann Hypothesis and the Alternative Hypothesis of Lagarias – Rodgers and Tao's ACUE, and what would falsify or extend it.

1. Provenance, method, and the verification protocol

1.1 What this paper is

Three survey manuscripts and their companion papers (referred to below as **[FSC]** *Finite Spectral Completion*, **[A23]** *After the Two-Thirds Theorem*, and **[FSI]** *Finite Spectral Identifiability*) were produced by a multi-agent research system. They contain a mixture of: classical mathematics restated; exact new finite theorems with short proofs; conditional thermodynamic statements; deliberate counterexamples ("counterfeits"); and open problems. A working mathematician reading them cold faces two obstacles: the volume (roughly eighty pages of dense statements), and the absence of external referee reports. The purpose of the present paper is to remove both obstacles for the results that deserve attention: we verify, we prune, and we explain.

1.2 Verification protocol

Each testable claim was checked by at least one of:

- **(H)** a complete short hand proof, reconstructed independently;
- **(E)** exact enumeration — determinantal laws computed as principal minors over all configurations, probabilities summing to 1 within 10^{-9}, claimed identities holding to $\leqslant 10^{-9}$ (usually 10^{-13});
- **(N)** high-precision numerics against closed forms (quadratic programs to 10^{-10}; constants to 12 digits).

The full script battery is available alongside this paper (`verify_codex.py`). The headline outcome:

Claim	Source	Method	Verdict
$q^* = ½ + (1/\sqrt{2})\cot(1/\sqrt{2})$; optimizer $v^*(s) \propto \cos(\sqrt{2}\, s)$; $\delta_{MT} = 2 - q^*$	[A23] § 3.2	H + N	**Correct**
Galerkin closed form q_n and its n^{-2} error coefficient	[A23] Prop 3.1	H + N	**Correct** (see § 1.3)
Nyquist collision identity $E\sum m(m-1) = (E\|p_N\|^2 - N)/2N$	[FSC] Thm 5.1, [A23] Prop 3.2	H	**Correct**
Full slack identity, equality face $\{0,1,2\}$-hard-core, $\Phi(\rho)$ curve	[FSC] Thm 5.1	H + N (per-sample)	**Correct**
Cosine sum $2\sum k \cdot \cos(\pi k/N) = N - \csc^2(\pi/2N)$	[FSC] § 5	N ($N = 2\cdots39$)	**Correct**
Fourth moments: $\mathrm{Cov}(V_k,V_\ell) = k^2 1\{k=\ell\} - 2(k+\ell-N)_+$; $\lambda_{min} = 1$	[FSC] Thm 6.1	E ($N = 2\cdots6$)	**Correct**
Support bound $\| V - T_N \|_1 \leqslant N^2 - N/2$	[FSC] Thm 6.2	E ($N \leqslant 6$)	**Correct**
Magnetic cycle: sub-g minors phase-blind; det gap; $d_{TV} = (2r)^g\|\cos g\varphi - \cos g\psi\|$	[FSI] § 5	E ($g = 3,4,5$)	**Correct**

Claim	Source	Method	Verdict
Marked homometric pair on $\mathbb{Z}/12$ (autocorrelation, simple fractions 0 vs ½, cubic separation, no extinction)	[FSI] § 7	E	**Correct**
DPP homometric pair $S=\{0,1,4,6\}$, $T=\{0,1,3,7\}$: same covariogram, triple probabilities $(8\mp\sqrt{3})/432$	[FSI] § 4	E (exact match to 10^{-12})	**Correct**
Deck-fiber dimension = n(M,L) − n(M,r) (necklace counts)	[FSI] § 6	E (5 cases)	**Correct**
$F(\gamma) = \frac{1}{2}\lvert S \triangle (S+\gamma)\rvert$; $F(1)$ = number of runs	[FSI] § 4	E (exhaustive $M \leqslant 12$)	**Correct**
Tile language: $\lvert E_N\rvert = C(2N,N)$; tile decomposition; filter $\hat{q}(k) = (1+e^{i\pi k/N})p_k$	[FSC] Thm 8.1	E	**Correct**
$N = 3$ lift obstruction: liftable charge spectra exactly $\{(0,0),(3,9),(9,3)\}$; DPP atom 111000 $\mapsto$ (12,0) outside	[FSC] Thm 8.2(ii)	E (exhaustive)	**Correct**
Destructive cross-spectrum $2\mathrm{Re}\, E[\hat{\eta}\,\bar{\hat{h}}] = -E\lvert\hat{h}\rvert^2$	[FSC] Thm 8.2(iii)	H (one line)	**Correct**
Wilson fourth trace: $\mathrm{tr}\, A^4_{\mathrm{flat}} - \mathrm{tr}\, A^4 = 16\sum_f w_f \sin^2(\varphi_f/2)$ on the free box	[FSI] § 9	E (three box sizes)	**Correct**
CCM finite Pontryagin extension: $SD\xi = -\beta$, $SD' = D'^{*}S$, $D'\xi = 0$	[A23] § 5.2	E (random instance) + H	**Correct**
Determinant-tail counterfeit constant $-1152/49$	[A23] § 5.5	H (series expansion)	**Correct**
Negative-square budget $(1-\delta_{MT})/2 = 0.163749648160$	[A23] § 4.3	H	**Correct** (elementary corollary)
Prouhet trace counterfeit; Euler – Lee – Yang rank obstruction; Hurwitz rigidity	[A23]	H	**Correct** (classical mechanisms)

Not independently verified (flagged, not doubted): the GOE counterfactual constant 0.559871849060 (the kernel normalization for the K_1 form factor is not reconstructible from the survey text alone); the PairCeiling value 0.6818287 (the source itself reports a verification boundary: a hashed rational witness absent from the public Lean repository — see § 7.3); the $N \leqslant 6$ cyclotomic endpoint certificates (asserted in companion papers we have not audited line by line); and all conditional thermodynamic statements (correctly labeled as conditional in the source).

1.3 Three transcription corrections

Three claims initially *failed* verification and were traced to reading errors rather than mathematical errors; we record them because future readers of the PDFs will hit the same traps. (i) In the Galerkin formula, the printed `arccos(1−n−2/an)` means $\theta_n = \arccos(1 - n^{-2}/a_n)$, not $\arccos((1-n^{-2})/a_n)$; the correct reading follows from the second-difference recurrence $v_{i+1} - 2v_i + v_{i-1} + (2/n^2 a_n)v_i = 0$, and with it the closed form matches the quadratic program to 10^{-13}. (ii) The charge-filter identity holds with the phase $(1 + e^{+i\pi k/N})$ in the transform convention $\hat{q}(k) = \sum q_x e^{-i\pi kx/N}$; only $|1 + e^{\pm i\pi k/N}|^2 = 4\cos^2(\pi k/2N)$ is used downstream, so nothing depends on the sign. (iii) The Wilson fourth-trace identity is stated on the *free* (open-boundary) box; on a periodic box winding 4-walks add cycle-holonomy terms and the identity genuinely fails — the hypothesis is load-bearing, and the papers state it.

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2. The constant, finally an exact finite object

2.1 Background

Montgomery's pair-correlation method (1973) bounds the proportion of simple zeta zeros using one quadratic functional of a test profile v on $I = [-1/2, 1/2]$:

$E(v) = \int_I v^2 + \int_I \int_I |s-t|\, v(s)v(t)\, ds\, dt$, normalized by $\int_I v = 1$.

The kernel $|s-t|$ is fed by the unitary (CUE) pair form factor $\min(|u|,1)$ — this is the precise sense in which "random matrices" enter. Montgomery and Taylor found the optimum: the Euler equation $v + Tv = \text{const}$ differentiates to $v'' + 2v = 0$, giving

$v^*(s) = \cos(\sqrt{2}\, s)/(\sqrt{2}\sin(1/\sqrt{2}))$, $E(v^*) = q^* = \tfrac{1}{2} + (1/\sqrt{2})\cot(1/\sqrt{2}) = 1.327499296320\cdots$

and the integer-multiplicity ("secant") decoding step maps q^* to $2 - q^* = \delta_{MT} = 0.672500703679\cdots$. Under RH this gave the classical Montgomery – Taylor simple-zero proportion; the 2026 theorem obtains the same constant *unconditionally* by measuring finite compressions of the indefinite Weil explicit-formula form from the prime side and decoding positive inertia through rank – trace inequalities. We verified the variational solution independently (discretized QP, $n = 4000$: optimum matches q^* to 8 digits, profile matches $\cos(\sqrt{2}s)$ to 6×10^{-9}).

An important bookkeeping point the series gets right and popular accounts get wrong: the constant is the output of *two* operations — a pair-energy optimization (which is random-matrix geometry) and an integer-multiplicity secant (which is arithmetic of multiplicities). Neither alone produces the decimal.

2.2 New result: the exact Galerkin certificate (verified)

Theorem (Galerkin closed form). Partition I into n cells and minimize the exactly integrated piecewise-constant energy. With $a_n = 1 + 1/(3n^2)$ and $\theta_n = \arccos(1 - 1/(n^2 a_n))$, the minimum is exactly

$q_n = \tfrac{1}{2} + (a_n n/2) \cdot \sin\theta_n \cdot \cot(n\theta_n/2)$,

and $\delta_n = 2 - q_n = \delta_{MT} - [\csc^2(1/\sqrt{2}) - \sqrt{2}\cdot\cot(1/\sqrt{2})]/(24 n^2) + O(n^{-4})$.

We verified the closed form against direct quadratic programming for $n = 2, \cdots, 80$ (agreement 4×10^{-14}) and confirmed the n^{-2} coefficient numerically. **Why this matters:** it converts $0.6725007\cdots$ from "the output of an optimization run" into a checkable finite formula with a proved discretization rate. Any formalization, discretized dual certificate, or SDP approximation of the Montgomery – Taylor problem can now be unit-tested against an exact ladder $q_2, q_3, \cdots \to q^*$. It is a modest theorem with an outsized hygiene value: constants in this subject have historically propagated through numerics.

2.3 A clean corollary: the off–line pair budget

Since every distinct off-critical zero pair $\{\rho, 1-\bar{\rho}\}$ consumes at least two zeros of the count while contributing none to the simple-critical count, the two-thirds theorem immediately gives

$\limsup_{T\to\infty} p(T)/N(T) \leqslant (1 - \delta_{MT})/2 = 0.163749648160\cdots$,

where p counts distinct off-line reflection pairs. In the Pontryagin-space reading of the Weil form (each off-line pair = one hyperbolic plane = one negative square), this is a *normalized negative-index budget*: the density of hyperbolic defects of the Weil form is at most $0.1637\cdots$. The proof is one line; the value of the statement is the typing. It is not a bound on the number of off-line zeros (which can still be infinite), and RH requires the budget to be zero, not small. We verified the arithmetic; the conceptual frame (ambient signature vs. observed signature, and the theorem that a finite Gabor observation can *hide* a negative square — the eigenvalue-collapse estimate $|\lambda_-| = O(a^2)$ for a pair at distance a from the line) is the series' most useful contribution to the operator-theoretic reading of the theorem.

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3. The Nyquist conservation law (the strongest new theorem package)

3.1 Setting

Fix N, let $m = (m_x)$, $x \in \mathbb{Z}/2N$, be random nonnegative integer marks ("zeros with multiplicity") with $\sum m_x = N$ in every sample, and let $p_k = \sum_x m_x e^{i\pi kx/N}$. The reference is the rank-N consecutive-band projection determinantal process — the finite Fermi sea, whose scaling limit realizes Tao's ACUE and the half-lattice Alternative Hypothesis, and whose form factor is the finite CUE ramp $E|p_k|^2 = \min(k, N)$. Impose only the *open* rows:

$E|p_k|^2 = k$ for $1 \leqslant k \leqslant N-1$, (the Nyquist row $k = N$ left free).

3.2 The three verified identities

(a) Collisions live only at Nyquist. For every such law (hand proof from Parseval, three lines):

$E \sum_x m_x(m_x - 1) = (E|p_N|^2 - N)/(2N)$.

Consequently, closing the endpoint row $E|p_N|^2 = N$ forces all marks into $\{0,1\}$: *the complete finite ramp admits only simple configurations.* One missing measurement is exactly the storage location of all multiplicity.

(b) The slack identity. Let $s_1 = \#\{x : m_x = 1\}$ and $h(j) = 0$ for $j \leqslant 2$, $j(j-2)$ for $j \geqslant 3$. Then

$E\, s_1 - (N/2 + \csc^2(\pi/2N)/(2N)) = E \sum_x m_x m_{x+1} + E \sum_x h(m_x)$,

with both right-hand terms nonnegative. We reconstructed the proof: the observable $s_1 - \sum m_x m_{x+1} - \sum h(m_x)$ equals $2N - (1/2N)\sum_k |p_k|^2(1 + \cos(\pi k/N))$ *configuration by configuration (verified per-sample to 4×10^{-14} over random marked configurations), and the crucial mechanism is that the Nyquist coefficient $1 + \cos\pi = 0$ — this particular combination of simple-site count, nearest-neighbor overlap, and high-mark penalty is blind to the one unconstrained Fourier row. Taking expectations under the open ramp and using the (verified) cosine identity* $2\sum_{k<N} k \cos(\pi k/N) = N - \csc^2(\pi/2N)$ gives the constant.

(c) The equality face and the density curve. Equality holds exactly on hard-core $\{0,1,2\}$-configurations (no adjacent occupied sites, no mark above 2), and the bound

$$E\, s_1/N \geqslant \tfrac{1}{2} + \csc^2(\pi/2N)/(2N^2) \to \tfrac{1}{2} + 2/\pi^2 = 0.702642367284\cdots$$

follows, together with the all-density version: matching the rank-L band process at every row except Nyquist forces $E\, s_1/L \geqslant \Phi(\rho) := 1 - \rho + \sin^2(\pi\rho)/(\pi^2\rho)$ in the limit $L/M \to \rho$. At low density $\Phi(\rho) = 1 - (\pi^2/3)\rho^3 + O(\rho^5)$: the sine kernel's quadratic repulsion becomes a *cubic* collision-rigidity loss. All numerics confirmed, including the finite value at $N = 256$ (0.702644910435).

3.3 Why this is genuinely new mathematics

The ingredients (Parseval, Fourier inversion, Delsarte duality) are classical; the theorem package is not, and we consider it the strongest publishable unit of the whole program:

1. **It is an exact conservation law, not an inequality found by search.** One scalar budget (the Nyquist intensity) is simultaneously: the factorial collision count (a), the dual variable of nearest-neighbor exclusion (b), the equality-face selector (c), and — in the conditional thermodynamic extension — the mass of a period-two Bragg atom whose square norm is a dynamical $\mathbb{Z}/2$-eigenfunction. Exact four-way dictionaries of this kind are rare.

2. **It sharpens, and finitizes, the known relationship between the Alternative Hypothesis and multiplicity.** The AH scenario stores zeros on a half-integer lattice; the finite theorem says precisely which *measurement* separates the multiplicity-carrying half-lattice world from the forced-simple world: the order-two character at the edge of the band, whose normalized weight ($1/N$) vanishes in every scaling limit. A statistic invisible in the vague limit rigidly controls the integer feasible cone. This is a precise finite mechanism for a phenomenon that in the zeta literature is only heuristic ("the AH hides at the edge of the Montgomery band").

3. **It cleanly separates relaxation sharpness from realizability.** The pair-cone (Delsarte) endpoint $a^* = L + L^2 - D^2\_\{M,L\}$ is certified by the single facet $C(1) \geqslant 0$; but the series proves — with an explicit $(M,L) = (6,3)$ law — that *realizing the extremal pair function does not imply attaining the simple-site equality*, because of the mark-$\geqslant$3 penalty channel. Optimization sharpness and probability-law sharpness are different theorems. This distinction, verified and correct, is exactly what most informal "LP bound = truth" arguments in this area elide (including, we note candidly, an earlier synthesis by the present authors — see § 7.2).

3.4 The two constants that must not be confused

The series is emphatic, and correct: $\tfrac{1}{2} + 2/\pi^2 = 0.70264\cdots$ **is not an improvement of δ_MT = $0.67250\cdots$, and neither bounds the other.** The first is a lower certificate for the simple-*site* fraction in a marked integer moment model with an open Nyquist row; the second is a lower proportion of simple critical zeta zeros proved from arithmetic. Different state spaces, different observables, different theorems. Their kinship is methodological — in both, a low-order spectral certificate is strengthened by integrality, and a missing Fourier direction marks the boundary of what the certificate can see.

4. Fourth moments, the moment polytope, and exact repair

4.1 The covariance formula (verified exactly)

For the half-filled finite Fermi sea on $\mathbb{Z}/2N$ (the discrete-CUE/ACUE reference law), with $V_k = |p_k|^2$, $1 \leqslant k < N$:

$E[V_k V_\ell] = k\ell + k^2 \cdot 1\{k=\ell\} - 2(k + \ell - N)_+$, $\mathrm{Cov}(V_k, V_\ell) = k^2 \cdot 1\{k=\ell\} - 2(k + \ell - N)_+$.

We verified this by exact enumeration of all $C(2N, N)$ determinantal atoms for $N = 2,\cdots,6$: the ramp $E\, V_k = k$ holds to 10^{-9} and the covariance matches entrywise to 10^{-9}. Structurally: the diagonal k^2 is the free-fermion exchange term, and the negative correction $-2(k+\ell-N)_+$ counts the two cyclic wrap intervals produced when two particle-hole excitations of total momentum $k+\ell$ exceed the band. Off-diagonal entries are $\leqslant 0$ and Gershgorin gives $C_N \geq I$; we confirmed **$\lambda_{\min}(C_N) = 1.000000$ exactly at every tested N**, with the first coordinate an eigenvector.

4.2 Inradius and the repair theorem

Two consequences, both verified where checkable. First, the binary feature rows affinely span $\mathbb{R}^{N-1}$ and the CUE target $T_N = (1, \cdots, N-1)$ is interior to their convex hull, with Euclidean inradius $\rho_N \geqslant 1/(N^2 - N/2)$ (via $\lambda_{\min} = 1$, the Bhatia – Davis inequality, and the support bound $\| V - T_N \|_1 \leqslant N^2 - N/2$, which we confirmed by enumeration through $N = 6$ — the observed maxima 1, 5, 10, 16.94, 27 all sit below the bound). Second, the **repair theorem**: if a candidate law on the hard-core equality face matches the ramp to feature error ε_N, it can be mixed with an explicitly constructed binary correction law (a density tilt $1 + \langle C_N^{-1}y, V - T_N\rangle$) to match the ramp *exactly*, at simple-site slack cost $\leqslant N \cdot \varepsilon_N/(\rho_N + \varepsilon_N)$; in particular feature error $o(N^{-2})$ suffices for $o(N)$ slack.

4.3 Assessment

The covariance formula is, to our knowledge, a new exact computation (fourth moments of Fourier intensities for the finite band projection process), and the inradius/repair mechanism is a genuinely useful bridge: it reduces the hard open realizability problem (§ 6) from "construct an exact law" to "construct an approximate law at rate $o(N^{-2})$" — a strictly easier analytic target with a quantified exchange rate. The connection of three usually separate languages — fermionic normal ordering, convex geometry of truncated moment problems, and stability of integer realization — is the innovation; the individual tools are standard.

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5. Four identifiability theorems: what pair data can and cannot know

This is the random-matrix side proper — the inverse-problem geometry underlying "zeta zeros look like CUE eigenvalues." Each theorem is finite, exact, and verified.

5.1 Form factors are Fermi–sea boundaries

For a stationary projection DPP on a finite abelian group with Fourier support S, the form factor at γ is exactly a boundary: $F(\gamma) = \tfrac{1}{2}|S \triangle (S+\gamma)|$; on a cycle, $F(1)$ equals the number of occupied runs of S (verified exhaustively for all subsets of $\mathbb{Z}/M$, $M \leqslant 12$). Hence $F(1) < 2 \iff$ S is one interval: **inside the stationary projection class, a single scalar — the first form-factor row — identifies the discrete-CUE law.** Every edge-isoperimetric inequality transfers to a form-factor inequality with no loss (the Cayley identity $\sum w_\alpha F(\alpha) = |\partial S|$).

But the same dictionary shows its limit: the full form-factor table is the covariogram of S, and covariograms admit homometric pairs. The witness $S = \{0,1,4,6\}$, $T = \{0,1,3,7\}$ in $\mathbb{Z}/12$ (verified: equal covariograms; not translates or reflections) gives two projection DPPs with identical 1- and 2-point functions whose triple probabilities differ: $P(\{0,1,2\} \subseteq X) = (8-\sqrt{3})/432$ vs. $(8+\sqrt{3})/432$ — confirmed to twelve digits. Periodic lifting amplifies the pairwise-indistinguishable pair to total-variation distance → 1.

5.2 Holonomy: an exact total-variation formula (the sharpest single theorem)

On the g-cycle, take the magnetic kernel $K_\varphi = aI + rH_\varphi$ (forward edge phase $e^{i\varphi}$, $0 < 2r < \min(a, 1-a)$). Verified exactly for g = 3, 4, 5:

- every principal minor of order < g is independent of φ (forests are gauge-trivial);
- $\det K_\varphi - \det K_\psi = 2(-1)^{g-1} r^g (\cos g\varphi - \cos g\psi)$;
- $\mathbf{d_{TV}(P_\varphi, P_\psi) = (2r)^g\,|\cos g\varphi - \cos g\psi|}$, an exact closed-form law distance (we computed all 2^g atoms by inclusion – exclusion; agreement 10^{-16}).

The connected stationary version (adding a small circulant term and M translated g-cycles) keeps all correlations of order < g exactly equal while $d_{TV} \to 1$. **Assessment:** this is the cleanest counterexample we know to any hope of learning or identifying a determinantal law from bounded-order correlations, with the entire information loss computed in closed form; it is directly relevant to the DPP-learning literature (cf. Urschel – Brunel – Moitra – Rigollet) and, as a diagnostic for zeta methodology, it is the exact finite statement of "fixed correlation order misses a global phase." A companion fact with the same moral: the complete DPP law sees only $\cos(g\varphi)$ — orientation (the sign of the holonomy) is invisible to *all* sampling statistics and needs an ordered operator word. Publishable as it stands.

5.3 The quasi-free axiom gap

Same state space (L-subsets of $\mathbb{Z}/M$), two model classes. Unrestricted rotation-invariant laws: the order-r inclusion deck has invariant fiber dimension exactly $n(M, L) - n(M, r)$, where $n(\cdot, \cdot)$ counts binary necklaces (Gottlieb's incidence rank + cyclic averaging; we verified five (M, L, r) cases by direct rank computation). At half filling the worst-case identification threshold is exactly $r^* = N$, and even the order-(N−1) deck leaves a fiber of dimension $\sim 4^N/(2\sqrt{\pi}\, N^{5/2})$ — exponentially large. Inside the stationary rank-N projection-DPP class, by contrast, the single row $E|p_1|^2 = 1$ identifies discrete CUE (§ 5.1). **The pair between these two statements is the theorem:** the determinantal/Wick axiom is an information-compression law worth an exponential factor. "The zeros have CUE pair statistics" and "the zeros are determinantal" differ by exactly this gap; conflating observation with constitution is the most common informal error in the field, and this makes it quantitative. A supplementary no-go (correct, and important for our own earlier work): a large fiber dimension does *not* imply a visible gap in any fixed local observable — matching decks through growing order forces all fixed cylinder statistics to converge, so fiber mass can hide entirely in observables whose support grows with N.

5.4 Marked homometry: complete diffraction misses the generator

The verified witness: u = (0,0,0,0,0,2,0,0,0,0,2,2) and v = (0,0,0,0,1,1,0,0,0,0,1,3) on $\mathbb{Z}/12$ have identical cyclic autocorrelation (12,4,0,0,0,4,8,4,0,0,0,4) — hence identical complete squared Fourier data, with no extinctions — yet u has *no* simple site and v has three; cubic moments (24 vs. 30) separate them at third order. Lifted through a Fibonacci cut-and-project scheme, this yields period-free weighted model sets with equal complete pure-point diffraction and simple-particle fractions 0 and ½. The sharp threshold is also right: with all marks $\leqslant 2$, mass plus complete autocorrelation *does* determine the simple-site count; mark 3 is the first escape, and v uses it. **Assessment:** the cleanest possible finite demonstration that a diffraction-only (power-spectrum-only) version of Dyson's quasicrystal program cannot recover local multiplicity structure — phase or a higher-order/bispectral observable is mandatory. The lift to genuine aperiodic model sets is what elevates it above a toy.

5.5 Prime labels and the Wilson fourth moment

On the free d-dimensional box with labelled unitary edge phases: one-edge windows are phase-blind; the fourth trace satisfies (verified on three box sizes to 10^{-8})

$$\operatorname{tr} A^4_{\text{flat}} - \operatorname{tr} A^4 = 16 \sum_{\text{faces}} w_f \sin^2(\varphi_f/2),$$

so within the fixed pair-window fiber the fourth trace is maximized exactly on the gauge-flat orbit; orientation needs an ordered word; and the generated C*-algebra is the full matrix algebra — Morita-trivial, K-theory $\mathbb{Z}$ — so *everything* of arithmetic interest (labels, holonomy, filtration) lives in the distinguished subalgebra structure, not in the abstract algebra. Together with the parity-interaction construction (every proper prime-marginal exactly product, global law not product, with d_{TV} computed), this packages the precise senses in which "Euler product" is labelled gluing data invisible to scalar spectra. As mathematics this is elementary-but-sharp; as a diagnostic for noncommutative-geometry approaches to RH it is well aimed: it says which structures a Connes-style finite-section model must *retain* to be nonvacuous.

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6. The one central open problem, and its exact perimeter

Everything above funnels into a single finite question, which the series has surrounded with verified no-go theorems without solving:

> **Open Problem (all-size Nyquist equality law).** For every N, does there exist a probability law on the hard-core face $E_N = \{m \in \{0,1,2\}^{\mathbb{Z}/2N} : \sum m = N,\ m_x m_{x+1} = 0\}$ with $E|p_k|^2 = k$ for all $1 \leqslant k < N$?

A positive answer makes the $\tfrac{1}{2} + 2/\pi^2$ certificate sharp at every size and — via the transfer principle for finite mimickers — would produce a multiplicity-carrying point process matching the full open finite CUE ramp, the strongest finite counterpart yet of Alternative-Hypothesis mimicry with collisions. A negative answer requires an explicit separating facet of the correlation polytope, which would itself be a new kind of obstruction.

The verified perimeter: (i) $|E_N| = C(2N,N)$ via the run-pairing bijection — configurations are plentiful; the problem is barycentric, not combinatorial (verified $N \leqslant 4$). (ii) The edge-charge recoding $q_x = m_x + m_{x+1} - 1$ turns E_N into the balanced three-tile cyclic language $\{(-1), (0,0), (+1,+1)\}$ (verified), with target spectrum $E|\hat{q}(k)|^2 = 4k \cdot \cos^2(\pi k/2N)$ — realizable by a free-fermion filter *outside* the language. (iii) Exact no-gos (all verified where finite): among half-grid projection-DPP atoms only the two alternating words lift; at $N = 3$ the liftable charge spectra are exactly $\{(0,0), (3,9), (9,3)\}$ while the positive-probability source atom 111000 filters to (12,0), outside their hull, killing atomwise repair; and every repair field h must satisfy the destructive cross-spectrum identity $2\operatorname{Re} E[\hat{\eta}(k)\hat{\bar{h}}(k)] = -E|\hat{h}(k)|^2$ mode by mode — independent decorations are impossible. (iv) The repair theorem (§ 4.2) lowers the bar to $o(N^{-2})$ feature error. The surviving constructive proposal — a Pfaffian/hard-rod source with a correlated doublon field tuned to the destructive cross-spectrum — is a program, not a theorem, and the series says so.

We regard this as a well-posed, genuinely open, and attackable problem in finite point-process moment theory, of independent interest even with every zeta motivation stripped away.

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7. What all this means for RH, the AH, and the ACUE — with corrections

7.1 The honest ledger

None of the verified theorems proves anything about zeta zeros, and the series never claims otherwise — indeed its most valuable habit is aggressive typing of claims. The genuine relations are these:

- **To the two-thirds theorem:** the Galerkin ladder (§ 2.2) gives the constant an exact finite meaning and a formal-verification target; the negative-square budget (§ 2.3) is its sharpest structural corollary; and the Gabor/Pick "common-signature" analysis locates the two missing bridges for any operator-theoretic strengthening — *observability* (a compressed observation can hide a negative square, quantitatively: $|\lambda_-| = O(a^2)$ for an off-line pair at distance a) and *cross-cutoff coherence* (self-moments at two cutoffs do not control mixed traces; the determinant-tail counterfeit, verified, refutes trace-to-determinant inference exactly).
- **To the Alternative Hypothesis / ACUE / Lagarias – Rodgers:** the Nyquist theorem is the exact finite mechanism by which multiplicity hides at the band edge; the quasi-free gap quantifies how far pair data are from a law; and the local-blindness theorem corrects the naive inference from fiber dimension to observable gaps. The Lagarias – Rodgers band-limited mimicry remains the decisive obstruction at the process level; the finite theorems are its exact, checkable shadows and sharpen where a counter-construction must operate (order $\geqslant 4$ data, growing support, or the Nyquist channel).
- **To Dyson's quasicrystal idea:** diffraction-only is now provably insufficient in the strongest finite sense (marked homometry with no extinctions, aperiodic lift); any viable version must carry phase (bispectrum) or Euler-labelled sign data, and the rational-rank obstruction forces infinite-dimensionality on any Lee – Yang-type realization of the prime spectrum.
- **To Connes-type programs:** the UV/IR separation ([A23] § 5) — the two-thirds proof is a *moving high-frequency* probe of the same Weil form of which CCM's construction is a *fixed-cutoff spectral-edge* problem — is an interpretation, not a theorem, but the finite Pontryagin extension lemma (verified) and the pollution/counterfeit battery give it teeth: they specify exactly which convergence statements are missing and what would falsify proposed bridges.

7.2 A correction to our own earlier synthesis

An earlier synthesis by the present authors asserted that 0.6725007··· "is the exact optimal value of a bandwidth-one LP" and that the Alternative Hypothesis adversary "is strictly suboptimal by 0.0301," citing the proximity of $½ + 2/\pi^2 - \delta\_MT$. The present series shows this framing conflates three feasible sets that must be kept apart ([A23] § 3.7): the Montgomery – Taylor profile space (where δ_MT is sharp *for that certificate class*), the marked open-Nyquist model (where $½ + 2/\pi^2$ lives, as a lower certificate for a different statistic), and the literal ACUE balanced-moment fiber (on which every state is binary and the simple fraction is identically 1, so the "LP value" is trivially degenerate there). Whether δ_MT is optimal over *all* bandwidth-one configurationwise certificates is open: the known ceiling for that broader class is ≈ 0.68183 (itself carrying an unresolved verification boundary — § 7.3), leaving a genuine gap. We consider the retraction itself informative: it is a live demonstration that the series' typing discipline catches real errors made by competent readers.

7.3 A verification boundary that should be closed

The PairCeiling component of the public Lean artifact reportedly kernel-checks its stability statement downstream of a hypothesis whose exact-rational witness is named by hash but absent from the public repository. Until that witness is published and replayed, the correct citation form is "a conditional stability ceiling of $\approx$ 0.6818287," not "the exact optimum of the bandwidth-one problem." This is a reproducibility observation, not a mathematical doubt; but in a field newly proud of formal verification, the distinction between kernel-checked-given-inputs and independently-replayable deserves exactly the care the series gives it.

7.4 The information ladder

The durable conceptual product, supported at every rung by a verified finite witness, is a strict hierarchy:

pair ramp < wider Fourier support or new arithmetic positivity < full correlation hierarchy < point-process law < spectral determinant,

with independent side-axes (phase, orientation, labels, constitution, realizability) that no amount of climbing substitutes for. The two-thirds theorem is remarkable precisely because it extracts a strong zero-count from the *lowest* rung by exploiting indefinite-form geometry and integrality; the verified counterexamples explain why silent promotion up the ladder is impossible; and the open problems (§ 6, plus growing arithmetic support beyond the unit band) say exactly what a further advance must purchase.

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8. Ranked assessment of novelty

Our referee's ranking of the verified contributions, by mathematical value:

1. **The Nyquist conservation-law package** (§ 3): exact, sharp, multi-faced, with a well-posed open endpoint problem. Publishable in a good probability or analysis journal on its own.
2. **The magnetic-holonomy TV formula and connected fixed-order mimicry** (§ 5.2): the sharpest known non-identifiability statement for DPPs; immediately relevant to the learning literature.
3. **The fourth-moment covariance / inradius / repair mechanism** (§ 4): a new exact computation with a genuinely useful reduction of the open problem.
4. **The quasi-free axiom gap with the necklace fiber formula and local-blindness theorem** (§ 5.3): the cleanest quantification of "observation $\neq$ constitution."
5. **The Galerkin closed form** (§ 2.2) and the **marked homometric aperiodic pair** (§ 5.4): smaller, but exact and well aimed.
6. The signature dictionary, budget corollary, Pontryagin extension, and counterfeit battery (§ 2.3, § 7.1): mostly interpretation and regression tests — but of a kind this subject conspicuously lacked.

Everything in this ranking passed independent verification; the claims we could not verify are quarantined in § 1.2 and § 7.3. All-size attainment, the thermodynamic sine – Bragg limit, and every zeta-side bridge remain open, as the source papers themselves state.

Appendix: reproducibility

The verification battery (`verify_codex.py` , plus the follow-up script with the three corrected tests) runs in under two minutes on commodity hardware: exact enumeration of determinantal laws through $2N = 12$ sites, all-atom total-variation computations for magnetic cycles through $g = 5$, quadratic programs to $n = 4000$ cells, and per-sample identity checks over random marked configurations. Deviations quoted in § 1.2 are worst-case over all tested instances. The three transcription corrections of § 1.3 are the only places where our reading of the source PDFs required repair; in each case the papers' mathematics was correct as intended.

Authors' note: the source manuscripts were produced by GPT5.6SOL-class agents under Codex orchestration; the verification and this synthesis were carried out independently in a separate session. Neither the sources nor this paper proves a new statement about the zeros of ζ(s); both aim to make the finite mathematics around the two-thirds theorem exact, checkable, and honestly typed.

Historical source 15: round3_synthesis

Source: `historical/riemann-rmt/round3_synthesis.md`. SHA-256: `ba768c0c81077b5ab5c355890130564c59a5579d6010e693171837eb82ac4836`.

Round-3 Synthesis: Ten Directions Coupling the Two-Thirds Method to the Finite Fiber Program

Bill (Qingyun) Sun · **GPT5.6SOL** · **Fable** — *August 11, 2026*

All ten agent reports complete; every quantitative claim below carries its origin (Direction n) and epistemic tag from the underlying report. Scripts in the session scratchpad; Lean citations to `anthropics/zeta-23-lean` @ 3635e74.

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0. The headline in three sentences

Reading Anthropic's method at the Lean source (not through paraphrase) and coupling it to the finite ACUE fiber machinery produced: **one near-miss improvement of the headline constant reduced to a single missing operator lemma with quantified payoff (+0.007 to +0.012); five new provable theorems about the method itself** (a rigidity upgrade of Lemma R, a strictly stronger distinct-zeros decoding, an exactly flat RH-independence frontier, an edge no-go, and a weight-freezing no-go); and **the first explicit closed-form resolution of the mimicker fiber** (the center-of-mass modulation family), together with an exact price list — in prime-correlation currency — for every escape route past 0.6725.

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1. The near-miss: 0.6725 is one operator lemma away from improvable (D1 + follow-ups)

The most consequential finding, established over three rounds of adversarial checking:

1. **The face identity is a flat-window accident.** The reason tr Q^3 appears worthless is that the *flat-window* sine – Gram data sits exactly on the {0,1,2} factorial face: $m_3 - 3m_2 + 2m_1 = 0$ (an identity holding at *every* finite N for ACUE — new rigidity fact). But at the **Montgomery – Taylor window itself**, $\Phi_3 = m_3 - 3m_2 + 2m_1 = \mathbf{-0.0117753128} \neq 0$ (closed trig form obtained; verified symbolically, numerically at 30 digits, and independently on the ACUE lattice to $3 \cdot 10^{-7}$). The third moment genuinely carries new information exactly where the proof already lives.
2. **The blocker is not arithmetic.** tr $\hat{A}^3$ at bandwidth one is licensed by the $\lfloor k/2 \rfloor \cdot L$ support staircase (D1; consistent with papers XI/XII). The blocker is that Lemma R's hypothesis class carries **no bound on the size of the compression's negative eigenvalues** — the in-window off-line blocks are constrained in count ($n_+ \leqslant p$) but not in magnitude, and the Φ_3 -deficit escapes through that channel.
3. **Window engineering cannot fix it** (adversarially settled): the RvM clump term (log T zeros per unit ordinate at zero kernel decay) forces the row-sum cap $M = A_0 \cdot \log T$ for *every* window; edge-vanishing windows only improve the negligible far field, cost $\geqslant 0.006$ in $2-q(w)$, and flip Φ_3 to the unmonetizable positive side.
4. **The surviving target, precisely:** a zero-side lemma $\|(C^{-1}\hat{A})_-\| \leq M_-$ uniform in T. Payoff at the MT window with the capped cubic decoder: δ **= 0.6796896 if M_- = 2, up to 0.6844924 if $M_- \leqslant 1$** — no window change, tr^3 the only new prime-side input. Required: depth-uniform control of off-line evaluation vectors (a cosh-weighted Poisson analogue +

overlap-angle bound). This is now the sharpest known formulation of "what would improve 0.6725 without new correlation ranges."

(Cross-check from D2: the true bandwidth-one ceiling is strictly above δ_MT — measured integrality gap ≈ 0.0093 — so headroom exists; the M₋ route targets exactly that headroom.)

2. New theorems about the method itself

2a. Stability/rigidity of Lemma R (D4) [PROVED]. The exact 7-term deficiency decomposition turns Lemma R into an identity (verified 7e-14; identically zero on the TightMult class), and the rigidity theorem with constant 1: an ε-near-equality configuration lies within Frobenius distance $\sqrt{\varepsilon}$ of an exact equality configuration (Löwdin frame). Zeta-side ledger: total deficiency over MT windows $\leqslant (q^*-1) \cdot N(T) = 0.3275 \cdot N(T)$, yielding a new unconditional pair-repulsion-type bound (on-line pairs with taper overlap $\geqslant \tau$ number $\leqslant 0.1637 \cdot N(T)/\tau^2$) and forcing hypothetical off-line zeros to look exactly like eigenvalue-2 pair blocks. Bonus reading: $q(\text{window}) = m_2$ (window-Gram) — **Montgomery – Taylor optimization *is* minimization of the RMT-predicted deficiency**, and under GUE the budget is exhausted by the Schur/von-Neumann term alone.

2b. The distinct-zeros upgrade $N\_d \geqslant 0.8362503 \cdot N + p$ (D7) [PROVED; ~5-line Lean edit]. Combining the c=2 inequality with the zero-count identity once (not twice) gives slope +1 in the off-line pair density — strictly stronger than the repo's own c=3 decoding (slope ½; pairs mispriced at $c^2=9$ vs forced value $k_3(2)=8$). Companion facts: the **simple-zero frontier is exactly flat** — $\delta(\pi) \equiv 0.6725007$ for all $\pi \in [0, 0.16375]$, hence **assuming RH buys nothing within this method** (the binding adversary is on-line doubles; exact 4p-budget/mass cancellation); at maximal off-line density all zeros are forced simple; the data forces pair-block spectra to (≈2, ≈0).

2c. The edge no-go (D5) [PROVED]. Every admissible bandwidth-one window has $\hat{r}(\pm 1) = 0$ (two-line Fejér – Riesz), so the weakest pointwise edge hypothesis $|F(1)-1| \leqslant \varepsilon$ certifies exactly 0.6725 — even at $\varepsilon = 0$. The finite Nyquist conservation law is powered by lattice aliasing that $\mathbb{R}$ lacks; the correct zeta-side edge object is the Cesàro mean $\bar{F}(\infty,T)$, with the exact collision identity $C(T) = N^* \cdot \bar{F}(\infty,T) - N(T)$. New reading: TightMult + PairCeiling/Signed = "the 2026 certificate is edge-blind by construction, and optimally so." Genuine new target: an unconditional *upper* bound on the edge-cell average of F via the Nyquist-blind null window ($r(0) = 0$).

2d. Weight-freezing (D9) [PROVED]. Derivative power sums are weight-homogeneous in the original power sums, so all bandwidth-one holomorphic statistics of the derivative process (ξ'-analog) are frozen on the fiber — "differentiate then pair-correlate at bandwidth one" cannot work, and the Lean ξ' gain (→ 0.86864) is purely arithmetic (the D_1 coefficient density). But the transport is real in *modulus/counting* observables: at N=6 the 1-point radial law of critical points resolves 7/10 fiber dimensions ($\varkappa = 2.14\times$ the best direct 4-point statistic), and **AH is not closed under differentiation** (exact N=3 counterexample). Finite Hardy identity: on-circle stationary points = roots of $zZ' - (N/2)Z$, all on the circle.

2e. The no-escape lemma for certificates (D2) [PROVED] — validity against the optimal period-M adversary forces $c_0 + \int r \cdot x \leqslant V\_M$ by a Riemann sum with no regularity assumption: the stability penalty buys rate, never limit. Quadratic-class certificates converge to exactly δ_MT (excess $0.31/M^2$): **MT is optimal in its class at every period, and the ≈0.0093 headroom above it is a pure integrality gap**, accessible only to integrality-aware certificates.

3. The fiber, finally explicit (D6 + D3 + D10)

3a. The center-of-mass modulation family (D6). $q\_g = q\_ACUE \cdot g(X \bmod N)$, $X = \Sigma x \cdot m\_x$, is an exact balanced-moment mimicker iff $E\,g = 1$ and $\hat{g}(\pm 1) = 0$ — an explicit **(N−3)-dimensional** family at every N (dims 0,0,2,3,4,5 at N=3..8, matching/entering the known fiber). The N=5 fiber is *completely solved*: a pentagon over $\mathbb{Q}(\sqrt{5})$; the known mimicker is

$|v|^2$ for a 5-sparse coherent superposition of translated Fermi seas; the destructive cross-spectrum constraint is the single syndrome $a^*Sa = 0$ (vanishing nearest-translate coherence). Mechanism: X mod N is exactly uniform under ACUE, and moment couplings to com-frequencies 2..N−2 vanish by a transport-cost bound (hop budget 2N vs cost $\geqslant$ 2N+2).

3b. The minimal detector theorem (D3). The cheapest observable with O(1) fiber leverage is $E|p\_a\ p\_b|^2$, a+b = N+1 — bandwidth exactly **1 + 1/N**; nothing cheaper exists (earthmover lower bound: fiber seas have displacement $\geqslant$ 2N+2). Dichotomy: bounded-likelihood mimickers (twists) cost bandwidth 2 − 4/N → 2 (twin-prime class); the cheap detectors work only on likelihood-degenerate directions. Prime-side price of the minimal detector: **h-averaged bipartite $E_2 \times E_2$ shifted correlations at length T^{1+1/N}** (dispersion-method-shaped, Linnik/Motohashi/BFI class). Open danger sign: leverage dropped 4.94 → 0.92 from N=5 to 6 — uniformity in N undetermined.

3c. The arithmetic shadow and the two-scale cost law (D10). The fiber's out-of-band profile: the first sliver past the edge kills 54 – 100% of the fiber with O(1) leverage, but the last direction survives to kill-degree d* = 3(N−3) ($\alpha^* \to 3$, conjectured law). Free structural kill: the functional equation's reflection symmetry eliminates the entire chiral half of the fiber (39/80 dims at N=7) at zero cost — and provably cannot do more. All squared-modulus data saturates below full rank. Marginal-value curve: pinning F on (1, 1+ε] buys $d\delta/d\varepsilon \approx 0.68$ at the edge; the AH deviation lives in the *constant term* of the ψ-variance at short shifts — the Siegel-zero price tag (Conrey – Iwaniec), consistent with everything above.

3d. Deficiency blindness dichotomy (D4). Lemma R's total deficiency is *exactly* frozen on the fiber (2-point functional); the fiber-detecting observables among spectral functionals are the interior clips $\Sigma((\mu-c')_+)^2$, c' strictly inside (1,2) — the boundaries are blinded by two new lattice facts (spectral cap $\mu \leqslant 2$ exactly; particle-hole symmetry $\mu \leftrightarrow 2-\mu$), while the continuum sine-Gram violates the cap (support [0,3] — the Christoffel hard-edge).

4. The updated quantitative map

object	value	status
headline constant δ_MT	0.672500703679	proved (repo)
conditional cubic upgrade (M_- = 2 / $\leqslant$ 1)	0.6796896 / 0.6844924	one lemma missing (D1)
true bandwidth-one ceiling	0.6818 ± 0.0015; candidate **15/22** exactly	D2 [CONJECTURE]; rigorous bracket [0.6725007, 0.6818287+]
quadratic-class ceiling	δ_MT exactly (excess $0.31/M^2$)	D2 [COMPUTED/PROVED]
lattice-union adversary cap	$1 - 3/\pi^2 = 0.69603$	D2 [PROVED] (totient forcing g = φ)
distinct zeros	N_d/N $\geqslant$ 0.8362503 + p/N	D7 [PROVED, unformalized]
off-line pair budget	p/N $\leqslant$ 0.1637496; at saturation all zeros simple	D7 [PROVED]
deficiency ledger	ΣD $\leqslant$ 0.3275 · N(T); rigidity $\sqrt{\varepsilon}$	D4 [PROVED]
doubles density of any Nyquist equality law	$\to 1/4 - 1/\pi^2 = 0.148679$	D6 [PROVED identities]
fiber's cheapest detector	bandwidth 1 + 1/N; robust directions 2 − 4/N	D3 [PROVED at N=5,6]
fiber kill-degree	d* = 3(N−3), α* → 3	D10 [CONJECTURE, verified N=5,6,7]
proved zeta rate	1/log T (two removable first-order losses; true tower $1/L^2$ conjectured)	D8 [read from repo + CONJECTURE]

object	value	status
the three N^{-2} towers	one object: the Nyquist cotangent tower (Bernoulli tower of $(\cot, \csc^2)$); exists only at the edge	D8 [PROVED forms + COMPUTED]

5. Corrections and errata from this round

- The naive premises of three directions were *refuted* by their own agents (the healthiest sign of the round): pointwise edge hypotheses buy nothing (D5); derivative pair correlation at bandwidth one is blind (D9); window engineering cannot cap the negative spectrum (D1 follow-up). Each refutation came with the corrected statement.
- D8 flagged a discrepancy in an earlier verification-script closed form for the Galerkin ladder (two variants circulating; both match the QP after the θ_n fix under their respective normalizations — final reconciliation recorded in `dir8_exact_form.py`, whose two-generator form $q_n = \tfrac{1}{2} + t_n \cdot \cot \varkappa_n$ is verified to 60 digits and supersedes both).
- The finite N=8 fiber dimension re-derived as 399 vs the earlier 403 at tighter rank tolerance (D4) — worth an exact-arithmetic recount.
- Support-field version of the destructive cross-spectrum is *infeasible* on the hard-core face (D6) — the constraint's true home is the com-syndrome on the original fiber.

6. Ranked program (what to actually do next)

1. **The M_- lemma** (D1): prove $\|(c^{-1}\hat{A})_-\| = O(1)$ via depth-uniform off-line vector control — worth +0.007..0.012 on the headline constant with tr^3 at bandwidth one. The single highest-value open item.
2. **Formalize `mult_two_pair`** (D7): $N_d \geqslant 0.83625 \cdot N + p$ — a five-line Lean edit yielding a strictly stronger published constant.
3. **The transport-cost lemma** (D6): finish the proof of the com-modulation family — an explicit $(N-3)$-dimensional mimicker polytope at every N, the paper-ready core of the fiber theory.
4. **Uniformity of the minimal detector** (D3): $N = 7, 8$ leverage of $E|p_a\, p_b|^2$, $a+b = N+1$; decides whether the dispersion-method target (bipartite E_2 correlations at $T^{1+1/N}$) is the real gateway.
5. **Edge-cell upper bound** (D5): the Nyquist-blind null-window route to the first unconditional upper edge datum for F.
6. **Max-ent Gibbs family at N ~ 30** (D6) + the D8 tightness evidence: the all-N Nyquist equality law now looks provable; its doubles density and $E|p_N|^2$ are already pinned by the frozen identities.
7. **$d^* = 3(N-3)$ kill-degree law** (D10) and the 15/22 ceiling identity (D2): two exact-arithmetic conjectures, both machine-attackable.

Every claim above traces to a tagged agent report with scripts in the scratchpad; the four refuted premises are retained in § 5 deliberately — the round's value lies as much in the closed doors as the opened ones.

Historical source 16: impostors_paper

Source: `historical/riemann-rmt/impostors_paper.md`. SHA-256: `760f8b362ab12bbf9d182c77f4b555983275a704de07927bc3a14d127fe0b8b4`.

Telling the Riemann Zeros from Their Impostors

A key step toward the alternative hypothesis, and a new bridge between zeta zeros and random matrix theory

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What this is about

The zeros of the Riemann zeta function are conjectured to be spaced like the eigenvalues of a large random unitary matrix. This is Montgomery's pair correlation conjecture, and half a century of work has not settled it. The obstruction has a name: the **alternative hypothesis**, the scenario in which the normalised gaps between zeros all lie asymptotically on a half-integer lattice rather than fluctuating like a random spectrum. It is consistent with everything currently provable, it has genuine arithmetic consequences, and — this is the hard part — it *imitates* the random-matrix answer for every statistic anyone has managed to test.

Tao made the obstruction concrete by moving it into random matrix theory. The **ACUE** is a lattice-supported measure on unitary spectra that reproduces the random-matrix two-point correlation exactly. The programme of "find the first statistic ACUE cannot fake" has been running ever since and keeps hitting the same wall: the set of measures matching any finite list of moments is a large convex body, so ruling out ACUE never rules out its neighbours.

This paper does four things.

We measure the wall exactly. The family of impostors — measures matching every balanced moment of degree $\leqslant N$ — has dimension 0, 0, 2, 10, 80, 403, 1804 for $N = 3, \cdots, 9$. We give closed-form families for every N, and show they remain invisible along the *entire* natural heat flow, because that flow is diagonal in the coefficients. Adding one more moment, or evolving the moments you have, is provably futile.

We find an instrument that sees past it. Not an average, but a *stopping time*: the finite de Bruijn – Newman depth Λ, the first moment at which two zeros collide when the heat flow is run backwards. It is not a polynomial statistic of any degree, so the impostors' freedom does not protect it. It separates the random-matrix and lattice scenarios not by a number but by a **universality class:** $-\Lambda \asymp$ N^{−8/3} against N^{−2}. The alternative hypothesis fails not by being fragile but by being **too stable** — it satisfies its own Riemann-hypothesis-analogue far too robustly.

We find the structure underneath, and it is the point of the paper. The static invisibility depth of the impostors and the dynamic relaxation spectrum of the flow turn out to be *the same operator*. Concretely, the quantity $\delta(N-\delta)$ that governs how deeply an impostor can hide is simultaneously the dimension of a Grassmannian, an affine Bruhat length, and — the new equality — the eigenvalue of the Jacobian of the zero dynamics at the clock configuration. This explains the whole phenomenon in one sentence: *impostors hide in exactly the modes the flow acts on most strongly*, which is why a hitting time sees what moments cannot.

We obtain an exact law for the refined instrument. Ordinary Λ is blind to eigenvectors, so isospectral impostors need a *marked* depth: perturb by a rank-one mark and differentiate. Its derivative turns out not to be the global quantity one expects from the determinant lemma, but a **rank-two contrast supported on the pair of eigenvalues that is about to collide** — plus one background term which, once identified, removes the last fitted constant from the theory. The resulting formula is parameter-free and matches measurement to eleven digits (§ 3).

We also relate Λ quantitatively to the Lagarias – Rodgers extremum μ, the existing formulation of the same question, and extract from that relation a single explicit number below which the alternative hypothesis is false.

What we do not claim. We have not proved the Riemann hypothesis, refuted the alternative hypothesis, or proved the pair correlation conjecture. § § 1 – 3 are theorems and computations about random matrices, plus a precise, falsifiable, currently unverified prediction about zeta. Where a statement is numerical rather than proved, it is labelled. A separate part of the paper reports three new unconditional records for prime clustering, which are proved and machine-certified.

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1. The alternative hypothesis, and an instrument that sees past it

1.1 Why the alternative hypothesis is hard to kill

Montgomery's pair correlation conjecture says the normalised gaps between zeta zeros follow the GUE law. The **alternative hypothesis** is the scenario in which those gaps all lie asymptotically in $(1/2)\mathbb{Z}$. It is consistent with everything provable about zeros, it has real arithmetic consequences, and it is precisely what current technique cannot exclude.

Tao's article *The alternative hypothesis for unitary matrices* moves the obstruction into random matrix theory, where it can be computed with. The **ACUE** is the measure on N-point configurations C inside the 2N-th roots of unity given by $\mu\_ACUE(C) = |\Delta(\zeta^C)|^2/(2N)^N$. It reproduces the CUE two-point correlation exactly. The natural programme — find the first moment at which they differ — keeps colliding with the same difficulty: the set of measures matching a moment list is a large convex body, so ruling out ACUE never rules out its neighbours. One must rule out a *fibre*, not a point.

How large the fibre is. We computed it exactly. The affine dimension of the set of measures on the ACUE support matching every balanced moment of degree $\leqslant$ N is

N	3	4	5	6	7	8	9
fibre dimension	0	0	2	10	80	403	1804

with an explicit closed-form subfamily for every N: $q\_g(C) = \mu\_ACUE(C) \cdot g(X(C))$, $X = \Sigma c \bmod N$, matching every balanced moment of degree $\leqslant$ N **iff** $E[g] = 1$ and $\hat{g}(\pm 1) = 0$ — an (N−3)-parameter family, verified at N = 5, 6, 7, 8 with worst moment error 10^{-12} and genuinely positive. A second family comes from the determinant character: μ_ACUE is the Born distribution $|\langle C|(\wedge^N F)A_0\rangle|^2$ of a single Slater determinant, and the tilts by $\det(U)^r$ are the interference patterns of two shifted Fermi seas — points of the first secant variety σ_2 (Gr(N,2N)) rather than the Grassmannian itself. At N = 8, of the 403 fibre directions, 401 are invisible to every pattern count of window width $\leqslant$ 2 and 383 to width $\leqslant$ 6.

And the natural flow does not help. Writing $P(z) = \det(I - zU) = \sum a_j z^j$, the finite analogue of the de Bruijn – Newman heat deformation is $P_r(z) = \sum a_j r^{j(N-j)} z^j$, $t = \log r$ — *diagonal in the coefficients*. Every balanced moment of degree $\leqslant N$ therefore stays frozen along the **entire** trajectory (verified to $4.5 \cdot 10^{-16}$). Evolving the moments you have is as futile as adding one more.

1.2 The depth

Define the **finite de Bruijn – Newman depth**: $-\Lambda(U)$ is the first time, running the heat flow backwards, at which two zeros collide. It is well defined because the flowed polynomial stays self-inversive, so a simple zero cannot leave the unit circle without first colliding. Equivalently the zeros obey the attracting circular Coulomb dynamics $\dot\theta_j = -\sum_{k\neq j} \cot((\theta_j - \theta_k)/2)$.

The right way to see it is geometric. Let R_N be the polynomials with all roots on the circle and $D_N = \{\mathrm{Disc} = 0\}$ its boundary. Then

> **$-\Lambda(P)$ is the distance from P to the discriminant hypersurface along the canonical heat ray.**

Static moments are coordinates *along* the mimicker fibre; the depth is a coordinate *transverse* to it. It is not a polynomial statistic of any degree — it is a first-passage time — which is exactly why the fibre's freedom does not protect it. This is the finite shadow of the picture in which $\Lambda_{dBN} \leqslant 0$ is the Riemann hypothesis and Rodgers – Tao's $\Lambda_{dBN} \geqslant 0$ makes RH, if true, *barely* true.

1.3 The separation

ACUE, exactly. Complete enumeration of all rotation orbits for $N = 3, \cdots, 10$ — 13,132 orbits, 184,756 configurations at $N = 10$ — with exact Vandermonde masses, validated to 40 digits. $P(\text{clock}) = 2^{1-N}$ **exactly** (Cauchy – Binet: $\sum_{|C|=N} |\Delta|^2 = (2N)^N$, each clock contributing N^N); clock polynomials $1 - cz^N$ are flow-invariant, so $\Lambda = -\infty$ there. Every non-clock configuration has minimal gap *exactly* π/N (pigeonhole). The law is $-\Lambda^{ACUE} \asymp N^{-2}$, fitted exponent **−2.0009**, with $N^2(-\Lambda)$ supported in $\approx[1.31, 1.99]$. In all 13,130 non-clock orbits the first collision occurs at a pair already adjacent at $t = 0$ — zero exceptions.

CUE, Monte Carlo to N = 256. $-\Lambda^{CUE} \asymp N^{-8/3}$, fitted exponent -2.678 ± 0.016, and the law is parameter-free: $8N^{8/3}(-\Lambda) \Longrightarrow G^2$ where $P(G > x) = \exp(-x^3/72\pi)$ is the sine-kernel smallest-gap law. The constant 72π was derived from the kernel, not fitted (measured 229 – 236 against 226.2, KS 0.035 – 0.041, median 29.6 – 30.5 against $(72\pi \ln 2)^{2/3} = 29.08$).

1.4 What the exponents mean

For an isolated close pair at gap δ the two-body reduction gives collision time $\delta^2/8 + o(\delta^2)$ — confirmed to 10^{-7}, and exact at $N = 2$ where $-\Lambda = -\log\cos(\delta/2)$. So the depth is governed by the smallest gap, and the smallest gap by the level repulsion exponent:

$p(s) \sim c\, s^\beta \Longrightarrow$ smallest of $\approx N$ gaps $\asymp N^{-1-1/(\beta+1)} \Longrightarrow$ $\mathbf{-\Lambda \asymp N^{-2-2/(\beta+1)}}$.

ensemble	β	predicted	measured
COE	1	-3	-3.064 (local slopes $-3.03 \cdots -3.10$)
CUE	2	$-8/3 = -2.667$	-2.678 ± 0.016 ($N \leqslant 256$)
CSE	4	$-12/5 = -2.4$	-2.510 ($N \leqslant 64$, still drifting)
ACUE lattice	∞	-2	-2.0009 (exact enumeration)

All measured slopes are slightly steeper than predicted, in the direction and of the size of the finite-N drift independently calibrated at $\beta = 2$, where the larger range shows the local slope converging to the prediction. The circular β-ensembles were sampled by the Killip – Nenciu CMV construction, validated against Haar CUE (KS 0.005) and direct COE = VV^T (KS 0.004); the localisation assumption holds in 95 – 99% of samples and in 100% of ACUE configurations.

So the depth is a scalar fingerprint of the microscopic repulsion exponent. And the interpretation inverts the naive expectation. One would guess a fake universe is caught by being *fragile* somewhere. The opposite: **ACUE's defect is that it is too stable.** CUE admits rare pairs at distance $N^{-4/3}$, a full factor $N^{-1/3}$ below the mean spacing, so it is always within $N^{-8/3}$ of losing real-rootedness; the lattice quantises every gap at π/N and forbids the accidents. A true random-matrix world is real-rooted but *microscopically fragile*; the alternative hypothesis satisfies its own Riemann-hypothesis-analogue far too robustly. That is Newman's dictum — "RH, if true, is only just true" — as a statement about a scaling exponent.

1.5 The depth is the Lagarias—Rodgers hard core in another coordinate

This is the sharpest consequence, and it says the depth is not a competitor to the existing formulation of the alternative hypothesis but that formulation in computable coordinates.

Lagarias and Rodgers study processes mimicking the sine process at coordinatewise bandwidth one and define $\mu = \sup\{c :$ some mimicker has minimum spacing $\geqslant c\}$, mean spacing 1. They prove $\mu \geqslant 1/2$ by the randomly shifted half-lattice — which is exactly ACUE — record the pair-only bound $\mu \leqslant 0.606894\cdots$, and conjecture $\mu = 1/2$. The alternative hypothesis is precisely the assertion that zeta's zeros realise a hard core of 1/2.

On the circle a hard core of c mean-spacings is $\delta_{min} = 2\pi c/N$, so with $-\Lambda = \rho \cdot \delta_{min}^2/8$,

$N^2(-\Lambda) = \rho \cdot \pi^2 c^2/2$**, and at the alternative-hypothesis value** $c = 1/2$**,** $N^2(-\Lambda) = \rho \cdot \pi^2/8$**.**

Every number in § 3.3 is this identity read one way or the other. The ACUE quantiles give

N	min $N^2(-\Lambda)$	ρ_min	median	ρ_med	max	ρ_max
6	1.353146	1.0968	1.419374	1.15050	1.952629	1.5827
8	1.330383	1.0784	1.418216	1.14956	1.976122	1.6018
10	1.314614	1.0656	1.412774	1.14515	1.985458	1.6094

An exactly computed configuration constant. Take the alternating clock (the zeros of $z^N + 1$), delete $e^{-i\pi/N}$ and insert 1, so the gap pattern becomes $1, 2, \cdots, 2, 3$ in half-lattice units and

$P_N(z) = (z - 1)(z^N + 1)/(z - e^{-i\pi/N})$.

Two independent routes agree. The lattice solver on this configuration gives $N^2(-\Lambda)$ rising monotonically to **1.41963827** at $N = 20$; the continuum limit $t = -s/N^2$, $z = e^{iu/N}$ yields

$G_s(u) = 2\cos(u/2) - 2\pi \int_0^{1/2} e^{s(1/4-y^2)} \cos((\pi+u)y)\, dy$,

whose first double zero ($G = \partial u\, G = 0$) *sits at* $s^* = 1.419640342\cdots$, $u^* = 1.812942145\cdots$ *with* $G''(u^*) = -0.3767 \neq 0$. *The two agree to* $2 \cdot 10^{-6}$ *at* $N = 20$*, consistent with an* $O(N^{-2})$ *approach. And* $s^* = \rho_\infty \cdot \pi^2/8$ with $\rho_\infty = 1.150717118\cdots$ — the constant is the hard core, dressed by the background. *(One correction to the record: s* is not the limit of the ensemble median. For $N \leqslant 7$ the two coincide, but the complete enumeration shows the median turns around at $N = 7$ and falls — 1.41822, 1.41520, 1.41277 at $N = 8, 9, 10$ — as the support spreads from [1.4147, 1.8246] to [1.3146, 1.9855]. s* belongs to the support of the limit law, via the single-dislocation stratum.)*

Equivalence of the two extremal problems. With $\mu_\Lambda = \sup\{\liminf N^2(-\Lambda)\}$ over mimickers,

$\mu_\Lambda \geqslant \pi^2\mu^2/2$ and $\mu \leqslant \sqrt{2\mu_\Lambda}/\pi$.

These are tight, not loose: LR's published $\mu \leqslant 0.606894\cdots$ is *exactly* the depth bound $\mu_\Lambda \leqslant 1.8177\cdots$. So **any improvement of the depth bound improves the Lagarias – Rodgers bound**, at an explicit exchange rate; a depth bound of 1.29 — barely below what ACUE configurations themselves realise — would already give $\mu \leqslant 0.511$.

A falsifiable number for the alternative hypothesis. Running the inequality the cheap way:

> **If the zeros of ζ have, in a window at height T containing $N = (\log T)/2\pi$ zeros, local Newman depth with liminf $N^2(-\Lambda) < \pi^2/8 = 1.2337\cdots$, the alternative hypothesis is false.**

This uses only the hard-core consequence of AH. Compare: AH predicts $\geqslant 1.2337$ with actual typical value 1.41964; CUE predicts $N^2(-\Lambda) \asymp N^{-2/3} \to 0$. The margin to the threshold is a factor 1.15; the margin from CUE is unbounded. And crucially, refuting AH does **not** require the full $N^{-8/3}$ law — only that the depth eventually drops below one explicit constant.

Why this reformulation may be more tractable. The LR extremum constrains a *minimum*, a hard combinatorial quantity, whereas Palm-type certificates — the row-sum square $\mathrm{Var}^0_{\mathrm{sine}}(S_f) \leqslant (M_h - A_f)(A_f - m_h)$ for band-limited f, and its multiwindow quadratic generalisation on the packing body K_h — apply naturally to smooth functionals. The depth is smooth in the configuration (we compute its gradient below), has a variational characterisation as a distance to a hypersurface, and has an explicit derivative law. Transporting those certificates from the row sum to the depth is the natural next attempt, and unlike the row-sum searches — which failed for an exact local reason, the pattern $\{-h, 0, h\}$ already forcing $S_0 = 2f(h) > 9/7$ for the first nonnegative Fejér profile — the depth has no such immediate obstruction.

Two lemmas, not one. It is worth separating them, because they have different status and do different jobs.

> **Hard-core lemma ($\rho \geqslant 1$).** Proved in § 1.7 directly from the adjacent-gap ODE. It gives the inequality in the direction used here, hence the falsification threshold: **$N^2(-\Lambda) < \pi^2/8 \Longrightarrow$ the alternative hypothesis is false.**
>
> **Localisation lemma ($\rho \to 1$ for finite β).** The remaining probabilistic estimate. It is what upgrades the exponents to distributional limits.

Only the second is open, and § 1.7 predicts its rate.

The leverage on μ, made explicit. Since $\mu_\Lambda \geqslant \pi^2\mu^2/2$, any upper bound $D_{dyn} \leqslant M$ for the depth of an admissible class gives

$\mu \leqslant \sqrt{(2M)}/\pi$.

$M = 1.29$ — barely below what ACUE configurations themselves realise — already gives $\mu \leqslant 0.511281\cdots$, a large improvement on the published $0.606894\cdots$. Crossing the alternative-hypothesis barrier itself requires $M < \pi^2/8 = 1.2337005501\cdots$. So the depth is not merely an equivalent coordinate for the Lagarias – Rodgers problem; it is a coordinate in which a *quantitative* target is visible, and the target is a single number.

1.6 A classification of impostors, with two computable criteria

Criterion I (transversality). For a configuration X and bandwidth r, with $M_r = (p_1, \cdots, p_r)$ and $\tau = -\Lambda$, ordinary Λ catches the class exactly when $\ker DM_r \not\subset \ker D\tau$. The relative residual of grad τ off the row space of DM_r:

configuration	r = 1	r = 2	r = 3	r ⩾ N/2
single dislocation, N = 8	0.991	0.941	0.737	rank saturates
random ACUE lattice, N = 7	0.996	0.963	0.569	rank saturates
generic non-lattice, N = 7	0.998	0.926	0.195	rank saturates

While rank $DM_r < N$ the kernel is nontrivial and the answer is decisively yes: 80 – 99% of the depth gradient lies in directions the first one or two moments cannot see. At $r \approx N/2$ the rank saturates at N, $\ker DM_r = 0$, and the residual drops to 10^{-16} — not because Λ fails but because the question becomes vacuous.

Class I, caught by Λ directly. Anything whose defect changes the geometry of the closest pair: collision-stratum families ($\delta = 0$ gives $\Lambda = 0$ outright, the mechanism by which function-field Newman constants are pinned by double roots), half-lattice adversaries with a hard lower spacing, and the centre-of-mass and secant families, which move the depth law by total variation 0.12 – 0.24 with every balanced moment frozen. Linear-programming tomography over the exact fibre at N = 6 gives $E[N^2(-\Lambda) \mid \text{non-clock}]$ ranging over **[1.3610, 1.4770]** against ACUE's 1.4336, while the mass at $\Lambda = -\infty$ can be pushed from 0 to **0.0975**, tripling ACUE's 0.03125, with every constrained moment held fixed.

Class II, invisible to Λ, caught by a marked depth. Λ is a function of the characteristic polynomial alone, so isospectral matrices have *identically* the same depth — verified to machine zero ($\tau(G_1) = \tau(G_2) = 0.068725421516$, difference $8 \cdot 10^{-17}$). The repair deforms $G \mapsto G + \eta\, uu^*$, transports to the circle by Cayley, and differentiates: $\chi(G;u) = \partial_\eta(-\Lambda)|_0$. It separates the isospectral pair immediately, median $|\chi(G_1;u) - \chi(G_2;u)| = 0.081$ against $|\chi|$ of order 0.01 – 0.2.

What χ actually measures is **not** the directional resolvent the determinant lemma suggests, but a rank-two contrast on the colliding pair, and the exact law — parameter-free, verified to eleven digits — is the subject of § 3.

Class III, immune to the linear flow. If the observable algebra misses a direction v, the heat operator — being diagonal — keeps $H_t v$ in the same hidden sector, so static invisibility persists for all t. **A linear flow does not destroy linear invisibility.** Λ escapes that argument only because it is not a linear observable transported by the flow but the first hitting time of the orbit against a variety: two orbits can share an invariant hidden sector and stand at different distances from D.

A test case at the boundary. The strongest known static escape is the pair of parity sectors $q_N^{\pm}(C) = \mu_N(C)(1 \pm (-1)^{\Sigma x})$, which are mutually singular, agree on every complete marginal on at most $N - 1$ sites, and match balanced trace moments of degree far beyond N. The depth *does* separate them at every finite N, but entirely through the atom: for even N both clocks have even slot-sum, so $P(\Lambda = -\infty) = 2^{2-N}$ in q^+ and **exactly 0** in q^-. The non-clock bulk is nearly identical — at N = 8 the quantiles are 1.3739/1.3959/1.4182/1.4554/1.4839 against 1.3752/1.3959/1.4196/1.4338/1.4788 — and the conditional means differ by $8.9 \cdot 10^{-3}$, $3.4 \cdot 10^{-3}$, $1.1 \cdot 10^{-3}$ at N = 6, 8, 10, shrinking. Since the separating mass 2^{2-N} vanishes, the parity sectors are a Class II/III object: caught at finite N, invisible in the limit, and a natural first target for the marked depth.

1.7 What is now proved: a two-body comparison theorem

The three scaling laws above were, until this point, numerical. They are not independent: all three follow from a local collision law plus a smallest-gap law, and the smallest-gap laws are known. What was missing was control of the *background* — the effect of the other N − 2 zeros on the collision time of the closest pair. That is now a theorem, and it is elementary.

Parameterise the flow by the depth $u = -t$, so the zeros obey $\theta_j' = -\sum_{k\neq j} \cot((\theta_j - \theta_k)/2)$ and collisions occur as u increases. Two facts:

Exact two-body solution. For N = 2, separating variables in $\delta' = -2\cot(\delta/2)$ gives $\cos(\delta(u)/2) = e^u \cos(\delta(0)/2)$, hence collision exactly at $u = -\log\cos(\delta(0)/2) = \delta(0)^2/8 \cdot (1 + \delta(0)^2/24 + \cdots)$.

> **Theorem A (two-body comparison).** Let (a,b) be an *adjacent* pair, i.e. no zero lies strictly inside the short arc from θ_b to θ_a, and let g be its gap. Then under the depth flow $g' \geqslant -2\cot(g/2)$. Consequently no collision occurs before $u = -\log\cos(\delta_{min}/2)$, so $-\Lambda \geqslant -\log\cos(\delta_{min}/2) \geqslant \delta_{min}^2/8$, **i.e.** $\rho \geqslant 1$ **always.**

Proof. Write $x_j^k = (\theta_j - \theta_k) \bmod 2\pi \in (0, 2\pi)$. Then

$$g' = \theta_a' - \theta_b' = -2\cot(g/2) - \sum_{k\neq a,b} [\cot(x_a^k/2) - \cot(x_b^k/2)].$$

Adjacency says that for every other zero k one has $x_a^k = x_b^k + g$ with both in $(0, 2\pi)$: going counterclockwise from k one meets b before a, and k is not inside the arc. Since $\cot(x/2)$ is strictly decreasing on $(0, 2\pi)$, each bracket is negative, so each term enters g' with a positive sign — **every other zero slows the collapse.** Hence $g' \geqslant -2\cot(g/2)$. The ordering of the zeros is preserved until the first collision, so each gap dominates the two-body solution started from its own initial value, and the earliest possible vanishing is at $u = -\log\cos(\delta_{min}/2)$. ∎

The sign claim was checked on 11,060 background terms across random and lattice configurations with zero exceptions, and the resulting inequality holds on every dataset in this paper: the minimum of $(-\Lambda)/(-\log\cos(\delta_{min}/2))$ is 1.0842, 1.0714, 1.0612 over the complete ACUE enumerations at N = 6, 8, 10, and 1.00019, 1.00021 over the CUE samples at N = 16, 64. (At N = 256 the measured minimum is $1 - 1.3 \cdot 10^{-5}$, an absolute discrepancy of $2 \cdot 10^{-11}$, at the solver's tolerance.)

Corollaries, all now rigorous.

- $-\Lambda^{ACUE} \geqslant \pi^2/(8N^2)$ **exactly**, since $\delta_{min} = \pi/N$ for every non-clock configuration (§ 1.3).
- **Any point process with hard core c has** $N^2(-\Lambda) \geqslant \pi^2c^2/2$. The exchange inequality $\mu_\Lambda \geqslant \pi^2\mu^2/2$ of § 1.5 — hence the Lagarias – Rodgers direction we actually use, and the falsification threshold $\pi^2/8 = 1.2337$ — is a theorem, not numerics. What remains conjectural there is only the *reverse* comparison.

Theorem B (matching upper bound where the background is subcritical). The background term is bounded by the mean value theorem: $|\sum_{k\neq a,b}[\cdots]| \leqslant g \cdot \sum_{k} ½\csc^2(x_b^k/2)$, and for a configuration all of whose gaps exceed c/N this is $\leqslant C \cdot N^2 g$, using $\sum_{k=1}^{N-1} ½\csc^2(\pi k/N) = (N^2-1)/6$. Integrating $\delta' \leqslant -2\cot(\delta/2) + CN^2\delta$ gives

$-\Lambda \leqslant (\delta_{min}^2/8)\cdot(1 + O(N^2\delta_{min}^2))$, hence $-\Lambda = (\delta_{min}^2/8)(1 + o(1))$ whenever $N^2\delta_{min}^2 \to 0$.

What this reduces the three laws to.

Law 1 (CUE). $\delta_{min} \asymp N^{-4/3}$ gives $N^2\delta_{min}^2 \asymp N^{-2/3} \to 0$, so Theorems A and B pin $-\Lambda = \delta_{min}^2/8 \cdot (1 + O(N^{-2/3}))$. The predicted relative correction exponent $-2/3 = -0.667$ matches the independently measured $0.598 \cdot N^{-0.729}$. Composing with the sine-kernel smallest-gap law $P(N^{4/3}\delta_{min} > x) \to \exp(-x^3/72\pi)$ — Ben Arous – Bourgade, and Feng – Wei for CβE — yields $\mathbf{8N^{8/3}(-\Lambda) \Rightarrow G^2}$. So Law 1 is reduced to a cited theorem plus Theorems A and B; what is still needed for a complete proof is that the error in Theorem B is uniform in probability, i.e. that the minimal gap is with high probability isolated and unaccompanied by a second anomalously small gap — which is precisely what the extreme-gap literature supplies.

Law 3 (β-universality). Identical, for every finite β: $\delta_{min} \asymp N^{-1-1/(\beta+1)}$ gives $N^2\delta_{min}^2 \asymp N^{-2/(\beta+1)} \to 0$, so $\mathbf{-\Lambda \asymp N^{-2-2/(\beta+1)}}$ reduces to the CβE extreme-gap law. The exponent is not an empirical fit; it is the image of the repulsion exponent under two elementary steps.

Law 2 (ACUE). Here $\delta_{min} = \pi/N$ exactly, so $N^2\delta_{min}^2 = \pi^2$ — **the background correction does not vanish**, which is precisely why $\rho^{ACUE} \in [1.05, 1.61]$ rather than $\to 1$. The lower bound $-\Lambda \geqslant \pi^2/(8N^2)$ is exact and proved; the matching upper bound needs $\rho \leqslant C$, which our complete enumeration establishes for $N \leqslant 10$ and to which the slowly decreasing minimising family of § 1.5 gives strong evidence, but which is not proved in general. This is the one remaining gap, and it is a lattice-specific constant problem rather than a universality problem.

The finite-β / β = ∞ dichotomy, and why it is singular. The background correction is governed by $\delta_{min}^2 \cdot S$ with $S \asymp N^2$ the background stiffness (the clock value is exactly $(N^2-1)/6$). Feng – Wei give $\delta_{min} \asymp N^{-1-1/(\beta+1)}$ for the circular β-ensembles, so

$\delta_{min}^2 \cdot S \asymp N^{-2-2/(\beta+1)} \cdot N^2 = \mathbf{N^{-2/(\beta+1)} \to 0}$ **for every finite β**,

whence $\rho_\beta \to 1$ and the two-body constant $1/8$ is asymptotically exact. But at $\beta = \infty$ the hard core pins $\delta_{min} \asymp 1/N$, so $\delta_{min}^2 \cdot S \asymp 1$: the background contributes at leading order and $\mathbf{\rho_\infty \neq 1}$. The lattice endpoint is therefore *not* the smooth limit of the finite-β formula — it is a singular hard-core endpoint, and that is exactly why the single-dislocation configuration realises

$\mathbf{s^* = 1.419640342\cdots = 1.150717118\cdots \times \pi^2/8}$,

a two-body collision time dressed by roughly 15% of many-body shielding, rather than $\pi^2/8$ itself.

This also predicts the *rate* at which ρ_β approaches 1, which is a sharper and independently testable statement than the leading exponent:

$\rho_\beta - 1 = O_{\mathbb{P}}(N^{-2/(\beta+1)+o(1)})$.

β	predicted rate	fitted over N ⩽ 128–256	local slope at the largest N
1 (COE)	−1	**−1.012**	−0.851
2 (CUE)	−2/3 = −0.667	−0.710	**−0.624**
4 (CSE)	−2/5 = −0.400	−0.501	**−0.407**

$\beta = 1$ lands on the prediction; for $\beta = 2$ and 4 the global fits are steeper but the local slopes decrease monotonically toward the predicted value and essentially reach it at the largest sizes, with the finite-N drift growing with β as expected (a slower predicted decay is more heavily contaminated).

The pipeline. The three laws are therefore not three empirical facts but one mechanism:

extreme-gap law (cited) + first-collision identity (§ 1.7) + background localisation (open) $\Longrightarrow -\Lambda = \rho \cdot \delta_{min}^2/8$ with $\rho \geqslant 1$ proved, $\rho \to 1$ for finite β, $\rho = O(1)$ but $\neq 1$ at $\beta = \infty \Longrightarrow$ $\boldsymbol{-\Lambda \asymp N^{-2-2/(\beta+1)}}$ across the whole family, lattice included.

So the honest status of the three laws changes from "numerical" to: **Law 1 and Law 3 reduced to published extreme-gap theorems plus the localisation lemma; Law 2 proved in the direction that matters for the alternative hypothesis, with its non-unit constant explained rather than fitted.**

1.8 The separation, proved on one side and reduced on the other

The two laws can now be separated by a *deterministic threshold* rather than by comparing fitted exponents.

> **Theorem C (CUE/ACUE separation). (i)** For every N and every non-clock ACUE configuration, $\boldsymbol{N^2(-\Lambda) \geqslant \pi^2/8} = 1.2337\cdots$ **(ii)** $P(N^2(-\Lambda^{CUE}) < \pi^2/8) \to 1$. Hence the two depth laws are eventually separated by the fixed constant $\pi^2/8$, almost surely on the ACUE side.

Part (i) is unconditional and needs nothing new: $\delta_{min} = \pi/N$ exactly for every non-clock ACUE configuration (§ 1.3, pigeonhole), and Theorem A gives $-\Lambda \geqslant \delta_{min}^2/8$. There is no probability in it — the bound holds configuration by configuration.

Part (ii) rests on two events, and it is worth separating what is cited from what we verified.

The gap event. The sine-kernel law $P(N^{4/3}\delta_{min} > x) \to \exp(-x^3/72\pi)$ gives $P(\delta_{min} > \pi/N) = P(N^{4/3}\delta_{min} > \pi N^{1/3}) \to \exp(-\pi^2 N/72)$: the CUE minimum gap exceeds the lattice gap with **superpolynomially small** probability. This is the cited input.

The regularity event. Theorem B needs the background slowdown coefficient $B := \sum_{k\neq a,b}[\cot(x_b^k/2) - \cot(x_a^k/2)]/g$ to satisfy $B \leqslant AN^2$. The clock value is exactly $(N^2-1)/6$, so $A \approx 1/6 = 0.1667$ is the natural scale. Measured on fresh Haar samples:

N	8	16	32	64	128
median B/N^2	0.1092	0.1202	0.1171	0.1202	0.1200
99th percentile	0.2192	0.2551	0.3572	0.2829	0.3273
maximum observed	0.2974	0.3106	0.6075	0.7948	0.3595

so $B \leqslant N^2$ comfortably, with a stable median of 0.120 and no observed excursion past 0.80. On that event Theorem B gives $N^2(-\Lambda) \leqslant (N^2\delta_{min}^2/8)(1 + O(N^2\delta_{min}^2))$, and the same samples give median $N^2\delta_{min}^2$ = 7.41, 4.66, 2.92, 1.85, 1.18 at N = 8, ⋯, 128 — a ratio of 6.28 across a factor 16 in N, against the predicted $N^{-2/3}$ scaling factor $16^{2/3} = 6.35$, agreement to 1%.

The conclusion, measured directly. The fraction of CUE samples with $N^2(-\Lambda)$ below the ACUE floor $\pi^2/8$, and the largest value observed:

N	8	16	32	64	128	256
fraction below $\pi^2/8$	0.565	0.823	0.967	0.9984	1.0000	1.0000
max $N^2(-\Lambda)$ observed	7.93	4.92	2.48	1.34	0.911	0.437
median $N^2(-\Lambda)$	1.083	0.627	0.385	0.231	0.146	0.0946

From N = 128 onward **every** sampled CUE configuration lies below the deterministic ACUE floor, and by N = 256 the median sits a factor 13 below it. The separation is not asymptotic bookkeeping: it is visible as a strict inequality between two computable quantities at accessible sizes.

What is proved and what is not. (i) is a theorem. In (ii), the gap event is a cited theorem and the regularity event is verified numerically rather than proved; converting the table above into a proof requires a high-probability bound on B, which is a standard rigidity statement for CUE (control of $\Sigma_k d_k^{-2}$ for the neighbourhood of the extremal pair) but one we have not carried out. That is the precise remaining gap, and it is narrower than the one we started with: the separation no longer depends on the exponent $-8/3$ at all, only on the CUE minimum gap being eventually smaller than π/N together with a bounded background.

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2. One operator: why a hitting time sees what moments cannot

This section is the structural core. Everything above says *that* the depth escapes the moment algebra; this says *why*, and the answer identifies two objects that had no known relation.

2.1 The static side: how deep an impostor can hide

Restrict to the gauge-invariant even shifts and let $|\Omega_s\rangle$ be the Slater state obtained from the Fermi sea by the shift 2s, $s \in \mathbb{Z}/N\mathbb{Z}$. Then $\langle C|\Omega_s\rangle = z(C)^s\langle C|\Omega_0\rangle$ where $z(C) = D_2\ (C)$ satisfies $z^N = 1$, so the entire secant family collapses to a single polynomial on $\mathbb{Z}_N$: for normalised coefficients a, the state $\Psi_a = \Sigma\, a_s|\Omega_s\rangle$ induces

$$q_a(C) = \mu_{ACUE}(C) \cdot |\Sigma_s\, a_s\, z(C)^s|^2 = \mu_{ACUE}(C) \cdot (1 + \Sigma_{\delta\neq 0}\, A_\delta\, z(C)^\delta),$$

with $A_\delta = \Sigma_s\, a_{s+\delta}\, \bar{a}_s$ the periodic autocorrelation. **The code object is not the support of a but its autocorrelation spectrum**, and the invisibility depth is

$$d_{vis}(a) = \min\{\ \delta(N-\delta) : \delta \neq 0,\ A_\delta \neq 0\ \}.$$

This is exact, not an analogy: the sector pairing $E_\mu[D_{2\delta}\ f\ \bar{g}]$ vanishes for degrees below $\delta(N-\delta)$ and first opens a rank-one channel at that degree, so q_a agrees with ACUE on all balanced observables of degree $\leqslant d$ **iff** $A_\delta = 0$ for every δ with $\delta(N-\delta) \leqslant d$. Since $d_N(\delta) = \delta(N-\delta)$ depends only on the cyclic Lee distance and is increasing, the pure max – min code problem is settled by pigeonhole: for L states the best possible minimum distance is $m(N-m)$ with $m = \lfloor N/L \rfloor$, maximised at L = 2 with value $\lfloor N^2/4 \rfloor$. Two Fermi seas are globally optimal; there is nothing to gain from five, and no compute should be spent looking.

The quantity $d_N(\delta) = \delta(N-\delta)$ is simultaneously the complex dimension of the Grassmannian $Gr(\delta,N)$, the pairing $\langle 2\rho, \omega_\delta\rangle$ — hence the affine Bruhat length $\ell(t_{\omega_\delta})$ of the dominant translation — and, up to the standard normalisation, the quadratic Casimir of the fundamental coweight. Those three identifications are representation theory. The fourth is new.

2.2 The dynamic side: the clock's stability operator

Its finite Fourier transform is elementary and exact: for $k \neq 0$,

$$\sum_{\delta=0}^{N-1} \delta(N-\delta)\, e^{-2\pi i k\delta/N} = \mathbf{-N / (2 \sin^2(\pi k/N))},$$

so d_N is the symbol of the long-range operator

$$(\mathcal{L}_N f)(x) = \sum_{k=1}^{N-1} [\, f(x) - f(x+k) \,] / (2 \sin^2(\pi k/N)), \quad \mathcal{L}_N e_\delta = \delta(N-\delta) \cdot e_\delta.$$

Now recall the dynamics. Backwards heat flow moves the zeros by the attracting Coulomb law $\dot\theta_j = -\sum_{k\neq j} \cot((\theta_j - \theta_k)/2)$, whose fixed point is the clock. Linearise about it, $\theta_j = 2\pi j/N + \varepsilon_j$, using $d/du[-\cot(u/2)] = \tfrac{1}{2}\csc^2(u/2)$:

$$\dot\varepsilon_j = \sum_{k\neq j} (\varepsilon_j - \varepsilon_k) / (2 \sin^2(\pi(j-k)/N)) = (\mathcal{L}_N \varepsilon)_j.$$

> **The operator controlling how deeply a static impostor can hide is the Jacobian of the zero dynamics at the clock.**

Verified to machine precision at $N = 4, 6, 8, 10, 12, 16, 20, 24$: the Fourier identity to 10^{-12}, the eigenvalues of $\mathcal{L}_N$ against $\delta(N-\delta)$ to 10^{-13}, and $\| \mathcal{L}_N - \text{Jacobian} \| \leqslant 2.4 \cdot 10^{-13}$. The spectrum is 0, and $\delta(N-\delta)$ with multiplicity two for $0 < \delta < N/2$, and the Nyquist mode $\delta = N/2$ alone at the top with eigenvalue $\mathbf{N^2/4 = \lfloor N^2/4 \rfloor}$ **— exactly the maximal invisibility depth obtained above by pigeonhole**, two entirely different derivations of the same number.

2.3 The mechanism, in one sentence

The eigenvalue $\delta(N-\delta)$ has two readings that now coincide: it is the degree at which a δ-mode becomes visible to balanced observables, and it is the rate at which that mode relaxes under the flow. Therefore

> **an impostor invisible to degree $\leqslant d$ observables is one whose deviation is supported on the modes that relax fastest — and those are exactly the modes the flow acts on most strongly.**

Hiding statically and being acted on dynamically are the same condition, read from opposite ends. That is why adding moments is futile while a hitting time is not: the moment algebra is blind precisely where the flow is loudest, and Λ is a functional of the flow. It also explains why the Nyquist channel keeps appearing on both sides of this project — as the top of the invisibility hierarchy and as the fastest-relaxing mode — and why ACUE, whose deviation from the clock lives at the *bottom* of this spectrum, sits so far from the discriminant.

2.4 What this opens

Three directions follow immediately, and each is computational rather than philosophical.

A quantum code. For $S \subset \mathbb{Z}_N$ the span $\mathcal{C}_S = \mathrm{span}\{\Omega_s : s \in S\}$ satisfies the Knill – Laflamme condition against the algebra of balanced observables of degree $< \min_{s\neq t} d_N(s-t)$: diagonal matrix elements are equal because D_{2s} is a phase, and off-diagonal ones are the sector pairings above. The ACUE cyclic shifts form an exact error-detecting code whose distance is the affine Bruhat length.

An interference problem that is genuinely open. Since $\hat{A}(k) = |\hat{a}(k)|^2 \geqslant 0$, the design space is $\{A : A_0 = 1$, A positive-definite on $\mathbb{Z}_N$, $A_\delta = 0$ on a low-Bruhat-energy region, $A_{\delta^*} \neq 0$ as high as possible$\}$ — a Delsarte-type linear program, and the natural name is a *zero-Bruhat-correlation-zone sequence*. Note the degenerate case: $A_\delta = 0$ for all $\delta \neq$

0 gives $q_a = \mu_{ACUE}$, no impostor at all.

A different semigroup. As $N \to \infty$ at fixed δ, $d_N(\delta)/N \to |\delta|$, so $N^{-1}\mathcal{L}_N \to (-\Delta)^{1/2}$, the half-Laplacian on the circle. The natural semigroup attached to the invisibility hierarchy is therefore the **Poisson** flow $e^{-t|k|}$, not the ordinary heat flow e^{-tk^2}. The $1/\sin^2$ kernel is also the defining interaction of the Haldane – Shastry chain, so the question of whether this hierarchy is a sector of an integrable spectrum is a concrete one, worth asking because an affirmative answer would replace brute-force moment computation by Bethe-type identities. We flag this as a lead, not a claim: sharing a kernel is not sharing a Hamiltonian.

A larger deformation class is available and largely unexplored. Any phase twist $u \in U(1)^{2N}$ acting on the Fermi sea leaves the Born distribution $|\langle C|\Omega_u\rangle|$ unchanged, and the Cauchy – Binet identity gives $E_\mu[(\prod_{c \in C} u_c)\, s_\lambda(x_C)\, \overline{s_\nu(x_C)}] = \det((T_u)_{A_\nu,A_\lambda})$ with $T_u = F^*D_u F$ circulant. So every phase-twisted moment is a circulant Toeplitz minor, hence automatically satisfies the Plücker relations and sits inside the KP/Toda τ-function formalism. The sharp question — whether vanishing of all such minors below degree $\lfloor N^2/4\rfloor$ forces u to be a linear phase up to gauge — would say that the determinant-character impostor is the canonical extremiser of its whole deformation class, which is a strictly stronger statement than exhibiting one counterexample.

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3. The marked depth: an exact rank-two law

Λ is a function of the characteristic polynomial alone, so isospectral matrices share it exactly — we verify $\tau(G_1) = \tau(G_2) = 0.068725421516$ to $8 \cdot 10^{-17}$. Impostors built to match rank, trace and Hilbert – Schmidt norm while differing in directional data therefore need a refinement. Deform by a rank-one **mark**, transport to the circle by Cayley, and differentiate:

$G_\eta = G + \eta uu^*$, $U_\eta = (G_\eta - iI)(G_\eta + iI)^{-1}$, $\chi(G;u) = \partial_\eta(-\Lambda(U_\eta))/\{\eta=0\}$.

It separates the isospectral pair at once — median $|\chi(G_1;u) - \chi(G_2;u)| = 0.081$ against $|\chi|$ of order 0.01 – 0.2. The interesting question is what it measures.

3.1 Not the resolvent

The determinant lemma $\det(zI - G - \eta uu^*) = \det(zI - G)(1 - \eta \cdot u^*(zI - G)^{-1}u)$ suggests that χ should track the directional resolvent and blow up wherever the inverse-Gram data is worst. **It does the opposite.** Rotating the mark onto the smallest-|eigenvalue| direction drives $u^*(z_0 I - G)^{-1}u$ from 1.615 to $6.667 = 1/|\lambda_{min}|$ while χ *falls* from 0.0267 to 0.0028 — an order of magnitude the wrong way. Condition-number susceptibility and collision susceptibility are different questions: the resolvent asks how much mass the mark puts on a spectral singularity, the depth asks how much *differential angular velocity* the mark induces on the pair that collides first. A near-null eigenvector orthogonal to that pair scores high on the first and nothing on the second.

3.2 The law

Let (a,b) be the first colliding pair with gap δ, and write

$q_j(u) = |\langle u, v_j\rangle|^2$, $c_j = -2/(1 + \lambda_j^2)$, $\mathbf{K_{ab} = c_a v_a v_a^* - c_b v_b v_b^*}$.

Hellmann – Feynman gives $d\lambda_j/d\eta = q_j$, the Cayley map contributes $d\theta_j/d\lambda_j = c_j$, so the first-order response of the critical gap is exactly the rank-$\leqslant 2$ quadratic form

$\delta'(u) = c_a q_a(u) - c_b q_b(u) = u^* K_{ab} u$.

The remaining ingredient is that the local collision law $-\Lambda = \varkappa \cdot \delta^2/8$ has $\varkappa = 8\tau/\delta^2$ a *functional of the configuration*, not a constant. Differentiating $\tau = \varkappa\delta^2/8$ therefore gives two terms, and this is the whole content of the earlier fitted factor:

> **Marked depth derivative law.** $D\tau\_G[uu^*] = (\varkappa(G) \cdot \delta(G)/4) \cdot u^*K_{ab}u + (\delta(G)^2/8) \cdot \varkappa'(u)$.

Measured at $N = 6$ with $\varkappa(G) = 1.280050$ taken directly from the configuration (not fitted): the correlation with measurement is **1.00000000**, the slope **1.00000**, and the residual **$5.7 \cdot 10^{-11}$** — finite-difference accuracy. The local term alone gives correlation 0.9958, slope 1.347, residual $8.1 \cdot 10^{-2}$. So the previously reported empirical constant $\rho \approx 1.72$ was never a constant: fitting $\chi = \rho(\delta/4)\delta'$ forces $\rho = \varkappa + (\delta/2)(\varkappa'/\delta')$, which varies from mark to mark and averages to something larger than $\varkappa$. **The law is now parameter-free**, and it decomposes cleanly into a local gap response and a background renormalisation response.

3.3 It is a polarisation detector, not an overlap detector

Write $T(u) = q_a + q_b$ for the mark's total mass in the critical two-plane and $I(u) = (q_a - q_b)/(q_a + q_b)$ for its imbalance. When the two critical eigenvalues are close, $c_a \approx c_b \approx c$ and

$\delta'(u) \approx c \cdot T(u) \cdot I(u)$.

In the critical two-plane $K_{ab} = \mathrm{diag}(c_a, -c_b) \approx c \cdot \sigma_z$. **The marked depth measures $P_a - P_b$, not $P_a + P_b$.** This corrects the natural guess: overlapping the critical pair is *not* sufficient. The mark $u = (v_a + v_b)/\sqrt{2}$ lies entirely inside the critical subspace, $T = 1$, yet has $I = 0$ and so gives no first-order response at all; the maximal-response marks are v_a and v_b themselves. The exact null cone is

$\mathcal{N}_{ab} = \{ u : c_a q_a = c_b q_b \} = \{ u : q_a/(1 + \lambda_a^2) = q_b/(1 + \lambda_b^2) \}$,

which reduces to $q_a = q_b$ only when $\lambda_a \approx \lambda_b$.

3.4 Three confirmations

Blind tomography. Sampling 60 random marks and solving $\chi(u_k) \approx u_k^*K u_k$ for a symmetric K, without telling the solver which pair collides: the recovered K has top-two energy fraction 0.986, and its leading two-dimensional eigenspace agrees with the true $\mathrm{span}\{v_a, v_b\}$ to principal angles of **$2.3 \cdot 10^{-6}$ and $3.0 \cdot 10^{-6}$ degrees**. The residual 1.4% of energy is the $\varkappa'$ term, which is not rank-two. So the critical pair can be recovered from depth measurements alone.

The null cone is reachable, and reveals a floor. Constructing marks with $c_a q_a = c_b q_b$ to 10^{-16} while keeping $q_a = 0.244$, $q_b = 0.694$ leaves $\chi = -3.3 \cdot 10^{-3}$ to $-3.9 \cdot 10^{-3}$ — the same order as the orthogonal control, $2.8 \cdot 10^{-3}$. This is a genuine background channel, not a residue of the local one, and it is worth noting *why*: for $u \perp v_a, v_b$ the perturbation leaves λ_a and λ_b **exactly** fixed, not merely to first order, since $(G + \eta uu^*)v_a = \lambda_a v_a$. So the floor cannot come from critical-pair eigenvalue motion at all; it is background renormalisation and trajectory shift, which is precisely the $\varkappa'$ term. The experiments therefore separate $\chi = \chi_{local} + \chi_{background}$ cleanly.

The resolvent returns at second order. It was not meaningless, only mis-ordered. Second-order rank-one perturbation theory gives $\lambda_j'' = 2q_j \sum_{k\neq j} q_k/(\lambda_j - \lambda_k)$, so the denominators $1/(\lambda_j - \lambda_k)$ — the resolvent structure — govern the *next* term. The correct hierarchy is: first order, critical-pair differential overlap; second order, spectral mixing through the resolvent; then global heat-flow background renormalisation. On the null cone the first channel is switched off by construction, which makes it the natural place to test the second.

At finite η the two-plane picture continues: the mark contributes off-diagonal coupling $\alpha_a \bar{\alpha}_b$ inside $\mathrm{span}\{v_a, v_b\}$, giving a 2×2 effective Hamiltonian whose gap has the usual square-root form. So the regimes are linear response, then an avoided-crossing (Landau – Zener) crossover as η approaches the original gap, then exact degeneracy where the off-diagonal term leads — a finite-dimensional local model, not an $N\times N$ resolvent computation.

3.5 The pattern this belongs to

Two very different compressions in this paper have the same shape. In the ACUE fibre, an enormous moment space collapses at the critical degree to a **rank-one** transport channel between conjugate rectangles. In the marked depth, an enormous spectral geometry collapses at first order to a **rank-two** contrast on the colliding pair. We record the pattern as a working principle rather than a theorem:

> **First-defect compression.** When a model is indistinguishable from its target under all low-complexity observables, the first observable that does distinguish them tends to act through a channel of very low rank.

If that is right, the search for detectors should be reorganised. The natural objects are not further class functions $\mathrm{Tr}\,\rho(U)$ or Schur polynomials $s_\lambda(U)$, but **marked matrix coefficients**: the quantity χ measures a dipole of the mark's spectral measure $\mu_u = \sum_j q_j(u)\, \delta_{\theta_j}$ across the critical pair, $c_a\, \mu_u(\theta_a) - c_b\, \mu_u(\theta_b)$. In the Langlands setting the analogous objects are periods and relative characters $\langle v, \pi(g)v\rangle$ rather than traces $\mathrm{Tr}\,\pi(g)$ — which is an argument, from this side, that a relative trace formula is the right home for an arithmetic version of this detector.

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4. What we can and cannot say about zero density

The other headline target adjacent to this circle of ideas is the zero-density exponent. Guth and Maynard's large-value estimate gives $N(\sigma,T) \leqslant T^{(30/13)(1-\sigma)+o(1)}$ and primes in intervals of length $x^{17/30+o(1)}$, the first improvement on the supremal exponent in four decades. Their argument bounds the top singular value of a Dirichlet-polynomial Gram matrix using its trace and a *centered third* spectral moment, and they note that estimates for fourth and higher powers are unavailable. It is tempting to conclude that supplying higher traces would improve the exponent.

We tested the pure moment-deflation channel exactly, and the answer is discouraging. Given only (p_1, p_r) and m nonnegative eigenvalues, the sharp bound on the largest is $\lambda_1^r + (p_1 - \lambda_1)^r/(m-1)^{r-1} \leqslant p_r$. For a spectrum consisting of one spike plus a flat bulk — the idealised large-value picture — **the r = 3 bound is already exactly tight, and r = 4, 6, 8, 10 buy precisely nothing** (gain 0.00% at m = 100, 1000, 10000, at spike fractions 0.5, 0.3, 0.15, 0.05). A gain appears only when the bulk itself carries secondary structure, and even then it is small: for a secondary cluster of 10 eigenvalues at 3% of the trace, going from r = 3 to r = 10 removes 1.1% of the overestimate; for 30 at 1.5%, 0.4%; for 100 at 0.5%, 0.1%.

The honest reading: whatever room exists in the Guth – Maynard framework is not in the moment hierarchy alone. It must come from the additive-energy input and the geometry of the large-value set, where higher traces could interact with the resonator structure rather than merely sharpening an already-tight scalar inequality. We have seen a proposed formula quantifying such a combined gain — a saving parameter $e = \eta + \varkappa$, a root y_e of $240y^2 + (72 - 10e)y - 13e = 0$, and an improved exponent $A(e) = 30/(13 + 10y_e) < 30/13$ — but we have not re-derived the Guth – Maynard argument and therefore neither endorse nor dispute it. It is recorded here as an unverified claim from a collaborating system, not as a result of this paper.

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5. Records for clustered primes

5.1 The mechanism

Fix a set of offsets $H = \{h_1, \cdots, h_k\}$ that is *admissible* — for each prime p the offsets miss a residue class mod p, so nothing forbids all of $n + h_1, \cdots, n + h_k$ being prime at once. Maynard's method weights each n by a square $w(n) = (\sum_d \lambda_d)^2$ built from divisor sums, and compares the total weight $S_1 = \sum w(n)$ against the prime-counting weight $S_2 = \sum w(n) \cdot \#\{i : n + h_i \text{ prime}\}$. If the primes are equidistributed to level θ and a variational constant

$$M_k = \sup_F k \cdot J(F)/I(F),\ I(F) = \int_{R_k} F^2,\ J(F) = \int_{R_{k-1}} \left(\int_0^{1-\Sigma} F\, dt_k\right)^2$$

over symmetric F on the simplex R_k exceeds $2m/\theta$, then every admissible k-tuple contains m + 1 primes infinitely often, hence $H_m := \liminf(p_{n+m} - p_n) \leqslant H(k)$, the diameter of the narrowest admissible k-tuple. Bombieri – Vinogradov gives $\theta = 1/2$ unconditionally, so the criterion is $M_k > 4m$.

5.2 What everyone had missed

For m = 1 the relevant k is 50 and M_k has been computed exhaustively. For $m \geqslant 2$ the relevant k runs from ten thousand to fifty million, and there M_k had never been computed at all: every record since 2014 used the same crude closed-form bound, $M_k \geqslant \log k - C$, obtained by truncating a product test function hard at the simplex boundary. The improvements of 2023 – 2025 raised the arithmetic input θ and left that bound untouched.

Its deficit is 2.3 to 2.9 units of log k, and about 1.1 units are recoverable elementarily. Since $M_k \approx \log k$ and $H(k) \approx k(\log k + 0.77)$, recovering 1.1 units is a factor $e^{1.1} \approx 3$ in k, hence nearly a factor 3 in H_m.

5.3 The engine

Take $F(t) = \prod_i g(t_i) \cdot 1[\Sigma t_i \leqslant k]$. Let X_i be i.i.d. with density g^2/c_2, $c_2 = \int g^2$, write S_j for partial sums and $G(u) = \int_0^u g$. Then, exactly,

$$I(F) = k^{-k} c_2^k \cdot P(S_k \leqslant k),\ J(F) = k^{-(k+1)} c_2^{k-1} \cdot E[G((k - S_{k-1})_+)^2],$$

and the truncation, discarded classically, becomes a genuine probability. The **layer-cake identity**

$$E[G((k - S)_+)^2] = \int 2G(u)\, g(u) \cdot P(S_{k-1} < k - u)\, du$$

converts it into an integral of true lower-tail probabilities, each bounded below rigorously (chord-majorised Chernoff, one-big-jump, Berry – Esseen with the safe non-i.i.d. constant 0.56); the bounds use only monotonicity of the true tail, so discretisation cannot invalidate them. Replacing the hard truncation by **shaped subexponential tails** $g(t) = e^{-(t/T_1)^\varkappa}/(1 + At)$ on a long support recovers the remainder: +0.12 units from exact truncation accounting, +0.49 from tail shaping, the rest from shape optimisation.

5.4 The results

	new bound	k	previous record
$H_2 = \liminf(p_{n+2} - p_n)$	**173,438**	15,856	396,504 (Stadlmann 2023/25)
H_3	**13,859,802**	923,601	24,797,814 (Polymath8b 2014)
H_4	**1,120,662,828**	56,000,000	1,431,556,072 (Polymath8b 2014)

Certificates $M_{15,856} \geqslant 8.013326752751$, $M_{923,601} \geqslant 12.006666706750$, $M_{56\cdot 10^6} \geqslant 16.065482942$, verified in ball arithmetic with outward rounding at every step, under three independent certification regimes (two Berry–Esseen constants × two tail routes). Tuples explicit: diameter 173,438 at k = 15,856 (admissibility re-verified by a second implementation); a repaired Hensley–Richards tuple of diameter 13,859,802 at k = 923,601, with a fully classical fallback of diameter 14,505,780; the primes-past-k tuple at $k = 5.6 \cdot 10^7$ with endpoints checked against published values of $\pi(x)$.

These are computer-assisted results produced by AI systems and have not been refereed; all certificates and scripts are published for replay. A further conditional improvement ($H_2 \leqslant 145{,}226$ via Deligne-strength equidistribution) is deliberately **not** claimed.

5.5 Why 246 did not move: five walls

1. **The ceiling is the tuple diameter.** No post-processing of Maynard–Tao output beats H(k_min); pair-correlation constants cannot lower H_1 at all.
2. **Scalar decoding is exactly optimal.** A two-point counterfeit ordered by convex order defeats every matrix, inertia or higher-moment decode; the threshold $f(m) = 2m/\theta$ is final.
3. **The weight cone is closed.** Enlarging squares to PSD forms gains nothing — rank-r sums of squares decouple by subadditivity — and the copositive relaxation is flat, certified by an explicit nine-pattern dual with residual $1.3 \cdot 10^{-14}$.
4. **Parity, made combinatorial.** A Liouville twist kills a set of pair-conclusions iff its kill-graph is bipartite (odd-cycle facets of the cut polytope), giving the floors $H_1 \geqslant 6$ and $k \geqslant 2m + 1$. *(The graph fact is Tao's, from his 2014 parity-obstruction note; the cut-polytope packaging is ours.)*
5. **The usable arithmetic frontier is not the published frontier.** The levels $\theta = 4/7$ (BFI), 3/5 (Maynard II) and 5/8 (Pascadi) are well-factorable or fixed-residue, while Maynard–Tao needs uniformity over a CRT-structured residue system varying with the modulus.

The k = 49 door, priced exactly. The pure constant obeys the classical bound $M_{49} \leqslant (49/48) \cdot \log 49 = \mathbf{3.97290 < 4}$, so *no enrichment of the test-function basis can ever reach the threshold in the pure class* — higher power-sum rows, semidefinite relaxations and generalised eigenvalue tests are all dead on arrival there, and running them is wasted compute. Only the ε-variant survives, where the bound relaxes to $(1 + \varepsilon) \cdot 3.97290$ and closes only for $\varepsilon \leqslant 0.00682$. Our best certified value is $M_{49,1/35} \geqslant 3.930490592$ with float optimum 3.959325169, against the threshold 4: gap 0.0407. That, and k = 47 (payoff $H_1 \leqslant 226$), are the only doors left.

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6. The signed sieve: an identity that closes a decade-old temptation

Maynard–Tao weights are squares, hence nonnegative, and positivity is what makes the decode valid. Chen's theorem, Iwaniec's linear-sieve weights $\lambda^\pm$, and Zhang's Landau–Siegel programme are all built from *signed* objects, so it is natural to ask what the signed enlargement buys. For signed w the decode fails — positive excess can be manufactured by $w(n) < 0$ at prime-poor n — and is repaired by paying the **debt** $D(w) = \sum_{\{w(n)<0\}} |w(n)| \cdot (m - \nu(n))_+$, where $\nu(n) = \#\{i : n + h_i \text{ prime}\}$.

Pointwise identity. For every integer $\nu \geqslant 0$ and every $m \geqslant 1$, $(\nu - m) + (m - \nu)_+ = (\nu - m)_+$.

Theorem (Signed No-Gain). With $w = w_+ - w_-$ of disjoint support,

$$S_2 - mS_1 - D(w) = \sum w_+(n)(\nu(n) - m) - \sum w_-(n)(\nu(n) - m)_+ \leqslant \sum w_+(n)(\nu(n) - m).$$

Proof. Substitute the identity in the w_- terms; the last step is $w_- \geqslant 0$ and $(\nu - m)_+ \geqslant 0$. ∎

Corollary. Any DHL(k, m+1) conclusion obtainable from a signed weight with the debt charged at face value is already obtainable from its positive part. The negative part is pure loss, of exactly the overshoot mass $\Sigma\, w_-(\nu - m)_+$.

The statement is pointwise: no model, no asymptotics, all k, m, tuples and weight classes at once. Verified additionally in exact rational arithmetic on a finite arithmetic microcosm — cellwise identity with zero violations, and the inequality on 1,000 random signed weights across five models.

What the numerics had been showing. Three phenomena looked like a phase structure; all three are corollaries. (i) Sweeping the debt price β in $\max\{\Phi - \beta D : S_1 = 1\}$, the value equals the classical optimum above a sharp β*, computed here in exact rational arithmetic with a verified dual: classical value 1087376209/3212440751, critical price β* = 23051796480/10991046857 = 2.0973249209⋯, with 85% of the optimiser's mass negative on 16 of 96 cells, and unboundedness below β_unb = 2.03265. Since β* > 1 always, the "phase" lives entirely at debt prices below face value. (ii) Imposing $\| w \|_1 \leqslant A$ gives λ(1) = λ_positive exactly and then a positive slope 0.32 – 0.89 — not a gain, since the decode is scale-invariant and what grows is the mass of w_+. (iii) At β = 1 every model is unbounded, which exposes the second, usually unremarked job positivity performs: for $w \geqslant 0$ the normalisation $S_1 = 1$ *is* the ℓ^1 norm, so **positivity is what makes the variational problem bounded.**

Hence exactly two escapes, neither variational. (a) Charge the debt below face value — requiring arithmetic information that bounds $(m - \nu)_+$ on a designed negative support below the truth, which is precisely the exceptional-character mechanism. By the theorem, **Zhang's programme is not one route among several; it is the only route that changes the variational picture at all.** (b) Keep w evaluable while w_+ is not — the positive part of a divisor-sum quadratic is not a divisor-sum quadratic, so a signed well-factorable λ can be evaluable at θ = 4/7, 3/5, 5/8 while its positive part is evaluable at no level beyond 1/2.

An independent argument from the switching side agrees. Switching never reduces the number r of exact-primality conditions in a debt term; Chen's debt has r = 1 and is payable by an upper sieve, while every DHL(k, m+1) with $m \geqslant 1$ forces residual debt with $r \geqslant 2$, and a two-vertex kill-graph is an edge, hence bipartite, hence parity-blocked wherever Liouville twists are admissible. Two disjoint analyses, one door.

The price list, if the debt were paid at level θ. All eight m = 1 crossings carry exact-rational certificates; H(k) is Engelsma-exact.

θ	threshold	k pure (certified M_k)	k with ε	$H_1 \leqslant$ pure / ε
1/2 (Bombieri – Vinogradov)	4	54	50	270 / 246 (unconditional)
4/7 (BFI)	3.5	31 (3.502015⋯)	29 (3.519881⋯)	140 / 130
7/12 (Maynard II)	24/7	29 (3.443305⋯)	26 (3.433616⋯)	130 / 114
3/5 (Maynard II)	10/3	26 (3.350647⋯)	23 (3.334616⋯)	114 / 94
5/8 (Pascadi)	3.2	22 (3.207656⋯)	20 (3.222666⋯)	90 / 80
1 (Elliott – Halberstam)	2	5 (M_5 = 2.007080)	—	12

None of BFI, Maynard II or Pascadi covers the sums the decode generates: all three fix one residue per modulus, while the decode's residues are CRT-composed from the shift set and vary with q. The sufficient intermediate statement is

> **(E_θ).** For coefficient systems c_q(a) jointly well-factorable with the residue selection — for every factorisation $q = q_1 q_2$ (resp. $q_1 q_2 q_3$) with ∏Q_j = x^{θ−ε} one can write c_q(a) = ∏j *γ_j(q_j, a mod q_j), |γ_j| ⩽ 1, with a mod p ∈ {h_i − h_j mod p : j ≠ i} for p | q — Σ{q* ⩽ x^{θ−ε}} Σ_{a ∈ A_i(q)} c_q(a) · E(x; q, a) ≪_{H,A,ε} x(log x)^{−A}.

(E_{4/7}) is "BFI Theorem 10, uniform over the CRT residue system of a fixed tuple polynomial". Polymath8a's MPZ[ϖ,δ] is its absolute-value cousin, proved by the same dispersion-plus-Deligne technology but only to level ≈ 0.5286, so (E_θ) interpolates two proved endpoints rather than crossing a parity barrier.

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7. Formalisation and machine verification

Every theorem in § 1.7, § 1.8 and § 2 is elementary enough to formalise, and we report precisely what has been machine-checked, by which tool, and what has not.

7.1 What could not be done here, and why

We could not compile Lean. The session's egress proxy returns a **403 policy denial** for `elan.lean-lang.org` and for `github.com` release downloads, and organisation policy denials are not to be retried. Public repositories are readable over git, but a Lean toolchain ships as a binary release, and building Lean plus Mathlib from source is not feasible here. The Lean file below is therefore **written but not compiled**; we present it as a formal specification rather than a verified artifact, and mark with `sorry` the two steps that need Mathlib's ODE comparison API, which we decline to guess at without a compiler.

7.2 What was machine-checked, and how

In place of Lean we discharged every algebraic and analytic step by exact symbolic computation (`sympy`, rational arithmetic throughout — no floating point) and, where the statement is a decision problem over the reals, by SMT (`z3`). Script `fab_verify_proof.py`: 18 checks, all passing.

step	statement	tool	status
1a	∂_s of the flow equals (N · D_z − D_z²) on each monomial z^j	sympy, exact	✔
1b	2z_j/(z_j − z_k) = 1 − i · cot((θ_j − θ_k)/2)	sympy, exact	✔
1c	the (N−1) terms cancel, leaving θ̇_j = −Σ cot	sympy, exact	✔
2a	cos(g/2) = e^s cos(g_0 /2) solves g′ = −2cot(g/2)	sympy, exact	✔
2b	the collision time is −log cos(g_0 /2), and the gap vanishes there	sympy, exact	✔
3	d/dx cot(x/2) = −½csc²(x/2), strictly negative on (0, 2π)	sympy, exact	✔

step	statement	tool	status
4a	$f(0) = 0$ and $f'(x) = \frac{1}{4}(2\tan(x/2) - x)$ for $f = -\log\cos(x/2) - x^2/8$	sympy, exact	✔
4b	$\tan t - t$ has nonnegative Taylor coefficients ($\frac{1}{3}$, 2/15, 17/315, 62/2835, 1382/155925, $\cdots$)	sympy, exact	✔
5	the sign lemma as a real-arithmetic decision problem: **unsat** for the negation	z3	✔
6	the ACUE pigeonhole giving $\delta_{min} = \pi/N$	exact integer reasoning	✔
7	$\Sigma_{k=1}^{N-1} \frac{1}{2}\csc^2(\pi k/N) = (N^2-1)/6$, at N = 4, 6, 8, 12, 20	sympy, exact	✔

Step 5 deserves comment, because it is the one place where a decision procedure is genuinely the right tool. Encoding the cosines and sines of the two half-angles as reals ca, sa, cb, sb subject to $ca^2 + sa^2 = 1$, $cb^2 + sb^2 = 1$, sa, sb > 0, together with the ordering constraint $sa \cdot cb - ca \cdot sb > 0$ (that is, $\sin(a - b) > 0$ for $0 < a - b < \pi$), z3 reports **unsat** for the negation $ca \cdot sb - cb \cdot sa \geqslant 0$ of the conclusion. The sign of every background term is therefore certified by a complete decision procedure over the reals rather than by inspection of a graph.

7.3 The Lean specification

```
/-- The cotangent, spelled directly so as not to depend on a particular Mathlib name. -/
noncomputable def cot (x : ℝ) : ℝ := Real.cos x / Real.sin x

lemma hasDerivAt_cot {x : ℝ} (hx : Real.sin x ≠ 0) :
    HasDerivAt cot (-(1 / Real.sin x ^ 2)) x

/-- `cot` is strictly decreasing on `(0, π)`: the sign input to Theorem A. -/
lemma cot_strictAntiOn : StrictAntiOn cot (Set.Ioo (0 : ℝ) π)

/-- **Background sign lemma.**  For an adjacent pair the other zeros satisfy
`0 < x_b < x_a < 2π`, and the resulting bracket is negative — so it enters the gap
derivative with a positive sign and slows the collapse. -/
lemma background_sign {xb xa : ℝ} (h0 : 0 < xb) (hlt : xb < xa) (h2 : xa < 2 * π) :
    cot (xa / 2) - cot (xb / 2) < 0

/-- **The depth inequality.** `-log (cos (x/2)) ≥ x²/8` on `[0, π)`. -/
lemma neg_log_cos_ge {x : ℝ} (h0 : 0 ≤ x) (h : x < π) :
    x ^ 2 / 8 ≤ -Real.log (Real.cos (x / 2))

/-- Exact solution of the two-body gap equation. -/
theorem two_body_solution
    (g : ℝ → ℝ) (hg : ∀ s, HasDerivAt g (-2 * cot (g s / 2)) s)
    (hrange : ∀ s, g s ∈ Set.Ioo (0 : ℝ) (2 * π)) (s : ℝ) :
    Real.cos (g s / 2) = Real.exp s * Real.cos (g 0 / 2)

/-- **Theorem A.** Every gap dominates the two-body solution started from the same
initial value, so no gap reaches `0` before `-log (cos (δ/2)) ≥ δ²/8`. -/
theorem depth_ge {ι : Type*} [Fintype ι] (g : ι → ℝ → ℝ) (δ : ℝ)
    (hδ0 : 0 < δ) (hδπ : δ < π) (hmin : ∀ i, δ ≤ g i 0)
```

```
    (hineq : ∀ i s, −2 * cot (g i s / 2) ≤ deriv (g i) s) :
    δ ^ 2 / 8 ≤ −Real.log (Real.cos (δ / 2))
```

The full file — proofs for the first four items, `sorry` for the last two — is `lean/DepthComparison.lean`. `cot_strictAntiOn` is proved from the derivative via `strictAntiOn_of_hasDerivWithinAt_neg`; `background_sign` follows at once; `neg_log_cos_ge` reduces to `t ≤ tan t` on `[0, π/2)`. What remains for a complete formalisation is `two_body_solution` (uniqueness for a scalar autonomous ODE) and the comparison step inside `depth_ge` (a Grönwall-type argument); both are standard and both are within Mathlib's reach.

7.4 The honest summary

The mathematics of Theorem A is verified: every algebraic identity by exact symbolic computation, the one inequality with real content by a complete decision procedure over the reals, and the numerical consequences on 11,060 background terms with zero exceptions. What is *not* verified is a Lean proof object, because no Lean toolchain could be obtained in this environment. The distinction is worth stating plainly, since "formalised" and "machine-checked" are often conflated and only the second is true here.

---

8. How this was done

The method is part of the result, so we describe it plainly.

Fleets, not oracles. Work proceeded in rounds. Each round posed one question and three to ten language-model agents attacked it in parallel from deliberately different angles, each writing and running its own code, each reporting line by line as *proved*, *computed*, *heuristic* or *conjecture*. Agents shared a written context file stating the current state of knowledge including known errors, but not conclusions: convergence of independent agents was treated as evidence, divergence as a bug report.

Adversarial defaults. Every context file instructed agents that the default assumption is that the idea has already been tried and failed, and that their first job is to find the reason. The prompt for the prime-gap round said outright that any improvement to 246 was 99% likely to be a misread constraint. Of the phenomena that looked like discoveries during this project, most were arithmetic or normalisation errors; the survivors survived because they were attacked first by their own authors.

Exact arithmetic at every threshold. Nothing near a decision boundary was accepted in floating point: ball arithmetic with outward rounding for the Maynard certificates, exact-rational Rayleigh quotients for the variational crossings, exact Vandermonde masses with 40-digit spot checks for the ACUE enumeration, exact rationals for the signed-sieve identity and the critical price β^*. The most common failure mode in machine-generated mathematics is a plausible float, and the remedy is cheap.

Two implementations or it did not happen. Every headline number here was produced twice by independent code, usually by agents that could not see each other's work: record tuples re-verified by a second admissibility checker; CUE depth by an ODE integrator and by coefficient bisection agreeing to 10^{-6}; the single-dislocation constant by a lattice enumeration and by a transcendental double-root equation, agreeing to six digits. Where two implementations disagreed — an engine crossing at k = 15,856 against another at 29,500 — the discrepancy was tracked to its source before anything was claimed.

The human supplies the questions. Every genuinely new direction came from a human mathematician's judgement: to stop optimising 246 and look at H_2; to ask whether removing the square opens a phase; to propose the de Bruijn – Newman depth as a dynamic observable at all; to insist that the constant 72π be factorised rather than fitted; to recognise the single-dislocation configuration as the right object to compute exactly. The models supplied speed, breadth, exact arithmetic, and the willingness to write and discard fifty scripts a day.

Negative results are the main product. Five walls, one no-go theorem, one empty phase, one refuted extrapolation, one refuted mechanism for the marked depth, one refuted hope for the moment-deflation channel. A method that only reports successes cannot be trusted about them. The single most valuable output of the signed-sieve round is that the route is closed and we can say exactly what would open it.

9. Open problems

1. **Complete the CUE depth law**: Theorems A and B (§ 1.7) reduce $8N^{8/3}(-\Lambda) \Longrightarrow G^2$ to the cited smallest-gap law; what remains is a high-probability bound on the background coefficient $B \leqslant AN^2$ (§ 1.8) — a CUE rigidity statement controlling $\Sigma_k d_k^{-2}$ near the extremal pair. Measured: median $B/N^2 = 0.120$, max observed 0.79. This same bound completes Theorem C(ii).
2. **Bound ρ for the lattice:** prove $\rho^{ACUE} \leqslant C$, completing Law 2's upper bound. The lower bound $-\Lambda \geqslant \pi^2/(8N^2)$ is now exact (§ 1.7). This is a constant problem for one specific configuration family, not a universality problem.
3. **Make $s^* = 1.419640342\cdots$ a theorem** for the single-dislocation family, and compute the separated-defect constant $\rho\infty = 1.19120\cdots$ exactly; the natural setting is an infinite clock with one localised defect under the Coulomb dynamics.
4. **The marked depth law**: prove $D\tau\_G[uu^*] = (\varkappa\delta/4) \cdot u^*K_{ab}u + (\delta^2/8) \cdot \varkappa'(u)$ as an asymptotic identity with controlled error, and characterise the background term $\varkappa'$. Numerically the law is exact (correlation 1.00000000, residual $5.7 \cdot 10^{-11}$); what is missing is the same localisation estimate as problem 1, differentiated. A natural sub-target: on the null cone $c_a q_a = c_b q_b$, show the leading behaviour is governed by $\lambda_j'' = 2q_j \Sigma_{k\neq j} q_k/(\lambda_j - \lambda_k)$.
5. **A transversality theorem:** with static observables M and fibre F_m, prove $\tau|$ *{F_m} generically non-constant, adding marks until ∩ ker DΛ*{u_j} ∩ ker DM = {0}.
6. **Transport the Palm certificates to the depth**, aiming at a bound $\mu\_\Lambda < 1.8177$ — which by § 3.5 would improve the Lagarias – Rodgers upper bound on μ.
7. **(E_θ)**, the tuple-residue well-factorable estimate, for any $\theta > 1/2$: the single statement between the certified price list and $H_1 \leqslant 130$.
8. **The k = 49 door**, in the ε-class only: certified 3.930490592, float 3.959325169, threshold 4, upper bounds closing only $\varepsilon \leqslant 0.00682$.

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Historical source 17: depth_scaling_theorem

Source: `historical/riemann-rmt/depth_scaling_theorem.md`. SHA-256: `742b21590598a8b4aa4d73bad590be85946f670433e6b682552a6b0d18c8930a`.

The depth scaling law $-\Lambda \asymp N^{-2-2/(\beta+1)}$

Bill (Qingyun) Sun · **GPT5.6SOL** · **Fable** — proof document, August 2026

We prove the scaling law for the finite de Bruijn – Newman depth across the circular β-ensembles and the lattice (ACUE) endpoint, isolating exactly which step is deterministic, which is cited, and which remains open. Statuses: **[P]** proved here, **[C]** cited, **[O]** open.

0. Setup and conventions

For a monic $P(z) = \prod_{j=1}^N (z - e^{i\theta_j}) = \sum_j a_j z^j$ define the **flow**

$P_s(z) = \sum_{j=0}^N a_j e^{s \cdot j(N-j)} z^j,\ s \geq 0,$

and the **depth**

$D(P) = \inf\{ s > 0 : \mathrm{disc}(P_s) = 0 \} \in (0, \infty].$

This is the finite de Bruijn – Newman depth with $D = -\Lambda$; the exponent $j(N-j)$ is the centred heat weight, and $s > 0$ amplifies the middle coefficients (the backward direction). All zeros start on the unit circle, and P_s remains self-inversive, so no simple zero can leave the circle without a prior collision; hence D is the first collision time.

Write $g_i(s)$ for the i-th cyclic gap, $\delta_{min} = \min_i g_i(0)$. Call a pair (a,b) **adjacent** if no zero lies strictly inside the short arc from θ_b to θ_a.

1. The root dynamics [P]

Lemma 1. Until the first collision, the zeros of P_s satisfy

$$\mathbf{d\theta_j/ds = -\sum_{k\neq j} \cot((\theta_j - \theta_k)/2).}$$

Proof. With $D_z = z\,\partial_z$ one has $\partial_s P_s = (N D_z - D_z^2) P_s$, since the symbol of D_z on z^j is j. Let $z_j(s)$ be a simple zero. Differentiating $P_s(z_j(s)) = 0$ gives $\dot z_j = -(\partial_s P)(z_j)/P'(z_j)$. At a zero, $P(z_j) = 0$, so

$(N D_z - D_z^2)P\,|_{z_j} = N z_j P'(z_j) - [z_j P'(z_j) + z_j^2 P''(z_j)] = (N-1) z_j P'(z_j) - z_j^2 P''(z_j),$

hence $\dot z_j = -(N-1) z_j + z_j^2 \cdot P''(z_j)/P'(z_j)$. For a simple zero of a product, $P''(z_j)/P'(z_j) = 2 \sum_{k\neq j} (z_j - z_k)^{-1}$, so

$\dot z_j = -(N-1) z_j + 2 z_j^2 \sum_{k\neq j} (z_j - z_k)^{-1}.$

Now put $z_j = e^{i\theta_j}$ and $\varphi = \theta_j - \theta_k$. Since $e^{i\theta_j} - e^{i\theta_k} = e^{i(\theta_j+\theta_k)/2} \cdot 2i \sin(\varphi/2)$ and $e^{i\theta_j} = e^{i(\theta_j+\theta_k)/2} e^{i\varphi/2}$,

$2 z_j /(z_j - z_k) = e^{i\varphi/2} / (i \sin(\varphi/2)) = 1 - i \cdot \cot(\varphi/2).$

Summing, $\sum_{k\neq j} 2z_j/(z_j-z_k) = (N-1) - i \sum_{k\neq j} \cot(\varphi_{jk}/2)$. Substituting and using $\dot\theta_j = \dot z_j/(i z_j)$:

$\dot\theta_j = [-(N-1) + (N-1) - i \sum \cot(\varphi_{jk}/2)] / i = - \sum_{k\neq j} \cot((\theta_j - \theta_k)/2)$. ∎

The clock configuration is the unique fixed point up to rotation (all cotangent sums vanish by antisymmetry), and linearising about it returns the operator $\mathcal{L}_N$ with eigenvalues $\delta(N-\delta)$.

2. The two-body problem, solved exactly [P]

Lemma 2. The scalar equation $g' = -2\cot(g/2)$, $g(0) = g_0 \in (0, 2\pi)$, has the exact solution

$$\cos(g(s)/2) = e^{s} \cos(g_0/2),$$

so g reaches 0 precisely at $s = -\log \cos(g_0/2)$. Moreover, for $x \in [0, \pi)$,

$$-\log \cos(x/2) \geqslant x^2/8,$$ with equality only at $x = 0$.

Proof. Separating variables, $\tan(g/2)\, dg = -2\, ds$ integrates to $-2 \log \cos(g/2) = -2s + C$; matching at $s = 0$ gives $\cos(g/2) = e^s \cos(g_0/2)$. Setting $g = 0$ gives $s = -\log \cos(g_0/2)$. For the inequality put $f(x) = -\log \cos(x/2) - x^2/8$; then $f(0) = 0$ and $f'(x) = \tfrac{1}{2}\tan(x/2) - x/4 = \tfrac{1}{4}(2\tan(x/2) - x) \geqslant 0$ on $[0,\pi)$ because $\tan t \geqslant t$ for $t \in [0, \pi/2)$. ∎

(For $N = 2$ this is the whole story: $D = -\log \cos(\delta/2)$ exactly.)

3. The comparison theorem [P]

Theorem A (two-body comparison). For every configuration and every adjacent pair with gap g,

$$g' \geqslant -2 \cot(g/2)$$ for all s before the first collision.

Consequently

$$D \geqslant -\log \cos(\delta_{min}/2) \geqslant \delta_{min}^2/8.$$

Equivalently, writing $D = \rho \cdot \delta_{min}^2/8$, one has $\rho \geqslant 1$ always.

Proof. Let (a,b) be adjacent with $g = \theta_a - \theta_b > 0$ measured across the short arc. By Lemma 1,

$$g' = \theta_a' - \theta_b' = -2\cot(g/2) - \sum_{k\neq a,b} [\cot(x_a^k/2) - \cot(x_b^k/2)],\quad x_j^k := (\theta_j - \theta_k) \bmod 2\pi \in (0, 2\pi).$$

Fix $k \notin \{a,b\}$. Adjacency means k does not lie in the open arc from θ_b to θ_a, so travelling counterclockwise from θ_k one meets θ_b before θ_a, i.e. $x_a^k = x_b^k + g$ with $0 < x_b^k < x_a^k < 2\pi$. On $(0, 2\pi)$ the map $x \mapsto \cot(x/2)$ is strictly decreasing (its derivative is $-\tfrac{1}{2}\csc^2(x/2) < 0$), so $\cot(x_a^k/2) - \cot(x_b^k/2) < 0$ and the whole bracket enters g' with a **positive** sign. Hence $g' \geqslant -2\cot(g/2)$.

Zeros cannot cross without colliding, so the cyclic order — and therefore the adjacency of each consecutive pair — is preserved up to the first collision. Applying the differential inequality to each gap and comparing with Lemma 2's solution started from the same initial value, every gap satisfies $g_i(s) \geqslant G(s; g_i(0))$ where $G(\cdot\,; g_0)$ vanishes first at $-\log \cos(g_0/2)$. Hence no gap vanishes before $\min_i(-\log \cos(g_i(0)/2)) = -\log \cos(\delta_{min}/2)$, which is the claim. ∎

Numerical check. The sign claim was tested on 11,060 background terms across random and lattice configurations: zero exceptions. The conclusion was tested on every dataset in this project; the minimum of $D/(-\log \cos(\delta_{min}/2))$ is 1.0842, 1.0714, 1.0612 over the complete ACUE enumerations at N = 6, 8, 10 and 1.00019, 1.00021 over CUE samples at N = 16, 64.

4. The matching upper bound [P, under a regularity hypothesis]

Define the **background stiffness of the critical pair**

$$S := \sum_{k\neq a,b} \tfrac{1}{2} \csc^2(x_b^k/2), \text{ so that } 0 \leqslant -\sum_k[\cot(x_a^k/2) - \cot(x_b^k/2)] \leqslant g \cdot S$$

by the mean value theorem, since $-(d/dx)\cot(x/2) = \tfrac{1}{2}\csc^2(x/2)$ and $x_a^k = x_b^k + g$. At the clock, $S = \sum_{k=1}^{N-1} \tfrac{1}{2}\csc^2(\pi k/N) = (N^2-1)/6$ exactly, so $S \asymp N^2$ is the natural scale.

Theorem B. Suppose $S(s) \leqslant A N^2$ throughout the collision window. Then

$$-2\cot(g/2) \leqslant g' \leqslant -2\cot(g/2) + A N^2 g,$$

and integrating the right-hand inequality from $g = \delta_{min}$ down to 0,

$$\mathbf{D \leqslant (\delta_{min}^2/8) \cdot (1 + O(A N^2 \delta_{min}^2))},$$

uniformly while $A N^2 \delta_{min}^2 < 2$. In particular $\mathbf{D = (\delta_{min}^2/8)(1 + o(1))}$ **whenever** $\mathbf{N^2\delta_{min}^2 \to 0}$.

Proof. For small g, $2\cot(g/2) = 4/g + O(g)$, so $g' \leqslant -4/g + AN^2g + O(g)$ and $ds \leqslant g\, dg/(4 - AN^2g^2 + O(g^2))$. Integrating from 0 to δ_{min} gives $D \leqslant -(2AN^2)^{-1} \log(1 - AN^2\delta_{min}^2/4) = (\delta_{min}^2/8)(1 + AN^2\delta_{min}^2/8 + \cdots)$. ∎

Status of the hypothesis. For CUE it is verified numerically with a uniform constant: the median of S/N^2 is 0.109, 0.120, 0.117, 0.120, 0.120 at N = 8, 16, 32, 64, 128 (against the clock value 1/6 = 0.1667), the 99th percentile stays below 0.36 and the largest observed value is 0.79. Turning this into a high-probability bound is the one **[O]** step; it is a standard rigidity statement (control of $\sum_k d_k^{-2}$ in the neighbourhood of the extremal pair).

5. The extreme-gap input [C]

Feng – Wei (Ann. Probab. 49 (2021)) for the circular β-ensemble, and **Ben Arous – Bourgade** (Ann. Probab. 41 (2013)) for β = 2: the smallest gap satisfies

$\boldsymbol{\delta_{min} \asymp N^{-1-1/(\beta+1)}}$ in probability,

with the k-th smallest gap density $\propto x^{k(\beta+1)-1}e^{-cx^{\beta+1}}$. For $\beta = 2$ the limit law is parameter-free: $P(N^{4/3} \delta_{min} > x) \to \exp(-x^3/72\pi)$.

For the lattice endpoint no citation is needed:

Lemma 3 [P]. Every non-clock ACUE configuration has $\delta_{min} = \pi/N$ *exactly.*

Proof. The support is the 2N-th roots of unity, so every gap is a positive multiple of π/N, and the N gaps sum to 2π, i.e. to N multiples of $2\pi/N$. If every gap were $\geqslant 2\pi/N$ they would all equal $2\pi/N$, which is the clock. Hence some gap equals π/N; and no gap can be smaller, since π/N is the minimal positive multiple. ∎

6. The theorem

Theorem. Let D_N be the depth of an N-point configuration.

(i) Finite β. For the circular β-ensemble with $\beta \in (0,\infty)$, assuming the regularity hypothesis of Theorem B,

$$\mathbf{D_N = (\delta_{min}^2/8)(1 + O_{\mathbb{P}}(N^{-2/(\beta+1)+o(1)})) \asymp N^{-2-2/(\beta+1)}}.$$

(ii) β = 2 distributionally. Composing with Ben Arous – Bourgade and Slutsky,

$$\mathbf{8 N^{8/3} D_N \Longrightarrow G^2,\ P(G > x) = \exp(-x^3/72\pi)}.$$

(iii) Lattice endpoint (ACUE, $\beta = \infty$). Unconditionally, for every non-clock configuration,

$\mathbf{D_N \geqslant \pi^2/(8N^2)}$, i.e. $N^2D_N \geqslant \pi^2/8 = 1.2337005501\cdots$, and with $\rho = O(1)$ — verified for $N \leqslant 10$ by complete enumeration, where $\rho \in [1.049, 1.610]$ — **$D_N \asymp N^{-2}$**, which is the exponent $-2-2/(\beta+1)$ at $\beta = \infty$.

Proof. Theorem A gives $D \geqslant \delta_{min}^2/8$ in every case. For (i), Feng – Wei give $\delta_{min} \asymp N^{-1-1/(\beta+1)}$, hence $N^2\delta_{min}^2 \asymp N^{-2/(\beta+1)} \to 0$, so Theorem B's error term vanishes at that rate and the two bounds match. (ii) is (i) at $\beta = 2$ together with the parameter-free gap law and the continuous mapping theorem. For (iii), Lemma 3 makes the lower bound exact and pins $\delta_{min} = \pi/N$, so $N^2\delta_{min}^2 = \pi^2$: the error term in Theorem B does **not** vanish, and ρ is genuinely bounded away from 1. ∎

7. Why β = ∞ is a singular, not a limiting, endpoint

The correction in Theorem B is governed by $\delta_{min}^2 \cdot S$ with $S \asymp N^2$. Since $\delta_{min} \asymp N^{-1-1/(\beta+1)}$,

$\delta_{min}^2 \cdot S \asymp N^{-2/(\beta+1)}$,

which tends to 0 for **every finite β** — so $\rho_\beta \to 1$ and the two-body constant 1/8 is asymptotically exact. At $\beta = \infty$ the hard core pins $\delta_{min} \asymp 1/N$ and $\delta_{min}^2 \cdot S \asymp 1$: the background contributes at leading order and $\rho_\infty \neq 1$. The lattice endpoint is therefore not the smooth limit of the finite-β formula, and this is precisely why the single-dislocation configuration realises

$s^* = 1.419640342\cdots = \mathbf{1.150717118\cdots} \times \pi^2/8$,

a two-body collision time dressed by ≈15% of many-body shielding.

The same computation predicts the *rate*, **$\rho_\beta - 1 = O_{\mathbb{P}}(N^{-2/(\beta+1)+o(1)})$**, which is a sharper statement than the leading exponent and is confirmed at three points:

β	predicted	fitted	local slope at largest N
1 (COE)	−1	−1.012	−0.851
2 (CUE)	−2/3	−0.710	−0.624
4 (CSE)	−2/5	−0.501	−0.407

8. Claim ledger

statement	status
Lemma 1, root dynamics $\dot\theta_j = -\sum \cot((\theta_j-\theta_k)/2)$	**[P]**
Lemma 2, exact two-body solution and $-\log\cos(x/2) \geqslant x^2/8$	**[P]**
Theorem A, $g' \geqslant -2\cot(g/2)$ and $\rho \geqslant 1$	**[P]**
Lemma 3, $\delta_{min} = \pi/N$ for non-clock ACUE	**[P]**
Theorem B, upper bound given $S \leqslant AN^2$	**[P]** given the hypothesis
the regularity hypothesis $S \leqslant AN^2$ w.h.p. for CβE	**[O]** — verified numerically, median $S/N^2 = 0.120$
$\delta_{min} \asymp N^{-1-1/(\beta+1)}$ for CβE	**[C]** Feng – Wei; Ben Arous – Bourgade at $\beta = 2$
$\rho_\infty = O(1)$ for ACUE at all N	**[O]** — proved for $N \leqslant 10$ by enumeration

statement	status
the resulting law $-\Lambda \asymp N^{-2-2/(\beta+1)}$, lattice included	follows from the above

The single open analytic ingredient is the regularity hypothesis; it closes (i), (ii) and the CUE side of the CUE/ACUE separation simultaneously. The lattice upper bound needs the separate, purely deterministic bound $\rho_\infty = O(1)$.

Historical source 18: newman_depth_note

Source: `historical/riemann-rmt/newman_depth_note.md`. SHA-256: `959ce6ac7697de8bf679fd44f78065ffe0d5112ad3fa77c1e5dc9aecb3addd5f`.

The finite de Bruijn–Newman depth: an exact dynamic observable separating CUE from ACUE

Bill (Qingyun) Sun · GPT5.6SOL · Fable

August 18, 2026 — research note; complete numerical evidence, key identities proved

Summary

For a unitary $N \times N$ matrix U with $P(z) = \det(I - zU) = \Sigma\, a_j z^j$, define the **backward heat flow** $P_r(z) = \Sigma\, a_j r^{j(N-j)} z^j$ ($t = \log r \leqslant 0$) and the **finite Newman depth** $\Lambda(U) = -t^*$, where t^* is the first time the discriminant vanishes (first root collision). This is the exact finite-dimensional avatar of the de Bruijn – Newman constant: RH $\Longleftrightarrow \Lambda_{dBN} \leqslant 0$ becomes "roots on the circle until collision", and $-\Lambda$ measures how far below criticality a configuration sits. Two independent computations (exact weighted enumeration of the ACUE ensemble for $N \leqslant 10$, 13,132 rotation orbits; Monte Carlo for Haar CUE up to $N = 256$, ~44,500 samples) establish:

1. **CUE is asymptotically barely-subcritical:** $-\Lambda^{CUE} \asymp N^{-8/3}$, with the parameter-free law $8N^{8/3}(-\Lambda) \Longrightarrow G^2$, where $P(G > x) = \exp(-x^3/72\pi)$ is the sine-kernel smallest-gap law (constant 72π confirmed to ~1.5%, KS ≈ 0.04, median 29.6 – 30.5 vs predicted $(72\pi \ln 2)^{2/3} = 29.08$). Fitted exponent -2.678 ± 0.016.
2. **ACUE is over-stable:** $-\Lambda^{ACUE} \asymp N^{-2}$ (fitted slope -2.0009), with $N^2(-\Lambda)$ supported in $\approx [1.31, 1.99]$, median $\to \approx 1.41$. The alternative $N^{-8/3}$ is decisively ruled out on the lattice, and the alternative N^{-2} is decisively ruled out for CUE: the depth ratio falls as $\approx 3.6 \cdot N^{-0.70}$ (theory $N^{-2/3}$), reaching $10\times$ separation at $N^* \approx 165$.
3. **Clock configurations:** $P_{ACUE}(\text{clock}) = 2^{1-N}$ **exactly** (proved by Cauchy – Binet: $\Sigma_{|C|=N} |\Delta(C)|^2 = (2N)^N$, each clock contributing N^N), and clocks have $\Lambda = -\infty$ ($P = 1 - cz^N$ is flow-invariant). These are the unique flow-stationary configurations.
4. **Local collision law:** $-\Lambda = \delta^2_{min}/8 \cdot (1 + o(1))$ holds in the continuum regime $\delta_{min} \ll$ other gaps (Richardson-verified to 10^{-7}; $N=2$ exact: $-\Lambda = -\log\cos(\delta/2)$). For CUE the correction is positive and vanishes as $\approx 0.60 \cdot N^{-0.73}$ (the background *delays* collision). On the ACUE lattice $\delta_{min} = \pi/N$ always (proved: all gaps $\geqslant 2\pi/N$ forces the clock), neighboring gaps are the same scale, and the constant renormalizes: $\rho = 8(-\Lambda)/\delta^2_{min} \in [1.066, 1.609]$, with the isolated-defect-pair value $\rho_\infty \approx$ **1.1912** (N-independent to 4 digits by $N = 8$).
5. **The depth escapes the weight-freezing no-go.** The flow is diagonal on coefficients, so every balanced moment of degree $\leqslant N$ remains frozen along the flow (dynamic extension of the Round-3 freezing theorem; verified to 4.5e−16). Yet Λ — an "infinite-degree" observable — separates the ACUE fiber at $O(1)$: a mimicker in the $N=5$ $\mathbb{Q}(\sqrt{5})$ family shifts $E[N^2(-\Lambda); \text{non-clock}]$ by -0.093 and the clock atom from 0.0625 to 0.1398; at $N = 8$ mimicker families move the Λ-law by total variation 0.12 – 0.24, with the dependence concentrated on the center-of-mass class $X = 0$. **Λ is the first verified dynamic anti-ACUE observable.**

Why this matters

The ACUE (Tao's alternative-hypothesis ensemble) matches CUE on all pair statistics available to bandwidth-limited detectors — the wall behind every Montgomery-type approach. All static balanced moments of degree $\leqslant N$ are frozen across the mimicker fiber, so no polynomial statistic of low degree can tell the ensembles apart. The Newman depth is dynamic and non-polynomial: it sees the full collision geometry of the heat flow. The scaling gap $N^{-8/3}$ vs N^{-2} says the two hypotheses predict *different universality classes of subcriticality* — GUE-repulsion ($P(\delta) \sim \delta^3 \Longrightarrow$ extreme gap $N^{-4/3} \Longrightarrow$ depth $N^{-8/3}$) versus lattice rigidity ($\delta_{min} = \pi/N$ deterministic $\Longrightarrow$ depth N^{-2}). Any future zeta-side quantity that mimics finite depth (e.g. flowed pair-counts under the actual de Bruijn – Newman backward flow on Ξ) inherits this separation — a concrete new place to look for a >2/3 simplicity proportion or an anti-AH statistic, orthogonal to the bandwidth war.

Key identities (proved)

- Root dynamics under the flow: $\dot{\theta}_j = -\Sigma_{k\neq j} \cot((\theta_j - \theta_k)/2)$ (attracting Coulomb gas on the circle); a simple root cannot leave $|z| = 1$ before a collision (self-inversive invariance).
- Generator on power sums: $\mathbf{L\, p_m = -m(N-m)\, p_m - m\, \Sigma_{a=1}^{m-1}\, p_a\, p_{m-a}}$ (verified to 4.1e−28 across N = 5..10; closes the polynomial-moment hierarchy and proves the freezing corollary).
- $N = 2$: $\Lambda = \log(|\mathrm{tr}\, U|/2)$, i.e. $-\Lambda = -\log \cos(\delta/2)$.
- $P_{ACUE}(\text{clock}) = 2^{1-N}$; clocks are the exact $\Lambda = -\infty$ atoms; ACUE(8) median($-\Lambda$) cross-validated by both implementations (2.214e−2 vs 2.216e−2).

Verification ledger

item	status
Exact enumeration N = 3..10 (13,132 orbits, Vandermonde masses; $\Sigma\mu$ = 1 to 3e−15)	complete, mpmath 40-digit spot checks worst 1.3e−9
CUE Monte Carlo N = 2..256, two independent Λ-solvers (ODE vs coefficient bisection)	agree to 1e−6; N=2 analytic law to 8.7e−13
First collision always at an initially-adjacent pair (ACUE)	13,130/13,130 orbits, zero surprises
Clock probability 2^{1-N}, $\Lambda = -\infty$	proved (Cauchy – Binet; flow invariance)
$\delta_{min} = \pi/N$ for non-clock ACUE	proved (pigeonhole) + verified all orbits
72π smallest-gap constant, G^2-law for CUE depth	fitted c = 229 – 236 vs 226.2; KS 0.035 – 0.041
Freezing along flow (balanced degree $\leqslant$ N)	proved (diagonal flow) + 4.5e−16 numeric

Open target (sharp): the closed form of the lattice defect constant $\rho_\infty \approx 1.1912$ (two-defect reduction of the Coulomb ODE against the rigid period-2 background); a block-decomposition of the same computation would upgrade the N^{-2} law to proved-with-constants.

Scripts and data: session archive `dyn1_*.py`, `dyn1_results_N{3..10}.npz`, `dyn2_*.py`, `dyn2_data_N{2..256}.npz`. *References: Ben Arous – Bourgade (extreme gaps), Feng – Wei (CβE small gaps), Vinson; Rodgers – Tao ($\Lambda_{dBN} \geqslant 0$); Tao (ACUE).*

Historical source 19: signed_sieve_nogo

Source: `historical/riemann-rmt/signed_sieve_nogo.md`. SHA-256: `8802c586ba28180b3a3eed9ef61bd232355de5b30585f0c0fd21c426a16d038d`.

The signed Maynard–Tao sieve: a no-gain theorem, and where the door actually is

Bill (Qingyun) Sun · GPT5.6SOL · Fable

August 18, 2026 — research note. Answers the question "does removing positivity from the Maynard – Tao variational problem open a new phase?" The answer is no, for a reason that is elementary, exact, and — this is the useful part — tells you precisely what would have to replace positivity.

0. The question

Maynard – Tao weights are squares, $w(n) = (\sum_d \lambda_d)^2$, hence pointwise nonnegative. Positivity is what makes the decode valid: if $w \geqslant 0$ and $S_2 - mS_1 > 0$, where $S_1 = \sum_n w(n)$ and $S_2 = \sum_n w(n)\nu(n)$ with $\nu(n) = \#\{i : n + h_i \text{ prime}\}$, then some n has $\nu(n) \geqslant m + 1$. Zhang's Landau – Siegel program ("摆脱平方" — remove the square) and Iwaniec's well-factorable $\lambda^\pm$ both suggest that signed weights are the natural next class, and that positivity is a self-imposed restriction. The concrete question we were asked to settle: **after allowing signed/indefinite sieve weights, does the variational optimum exhibit a phase the classical positive family cannot see?**

For a signed w the decode fails as stated — the positive excess can be manufactured by $w(n) < 0$ at prime-poor n. It is repaired by paying the **debt**

$D(w) = \sum_{\{n : w(n) < 0\}} |w(n)| \cdot (m - \nu(n))_+$,

so that $S_2 - mS_1 - D > 0$ does imply the conclusion. The experiment is then to optimize the repaired functional $\Phi(w) - \beta D(w)$ over the signed class, with $\beta = 1$ the true price, and look for a phase transition in β or in any enlargement parameter.

1. The theorem

Pointwise identity. For every integer $\nu \geqslant 0$ and every $m \geqslant 1$,

$(\nu - m) + (m - \nu)_+ = (\nu - m)_+$.

(Both sides equal $\nu - m$ when $\nu \geqslant m$, and 0 when $\nu < m$.)

Theorem (Signed No-Gain). Write $w = w_+ - w_-$ with $w_+, w_- \geqslant 0$ of disjoint support. Then

$$S_2 - mS_1 - D(w) = \sum_n w_+(n)(\nu(n) - m) - \sum_n w_-(n)(\nu(n) - m)_+ \leqslant \sum_n w_+(n)(\nu(n) - m).$$

Proof. $S_2 - mS_1 = \sum w_+(\nu-m) - \sum w_-(\nu-m)$ and $D = \sum w_-(m-\nu)_+$; subtract and apply the identity to the w_- terms. The final inequality holds because $w_- \geqslant 0$ and $(\nu-m)_+ \geqslant 0$. ∎

Corollary. Any DHL(k, m+1) conclusion obtainable from a signed weight w with the debt charged at face value is already obtainable from the nonnegative weight w_+: if the repaired functional at w is positive, then the plain functional at w_+ is positive. **The negative part is pure loss**, and the loss is exactly $\sum w_-(n)(\nu(n) - m)_+$ — the overshoot mass sitting on the negative support.

The theorem is pointwise, so it needs no model, no cell structure, no asymptotics, and it holds for every k, m, tuple, and weight class simultaneously. It subsumes the weaker fact we proved first (that the *naked* ratio sup Tr(BQ)/Tr(AQ) over indefinite Q equals the PSD value by rank-one extremality): the naked-ratio statement is about dropping positivity from the objective; this one says that even after correctly repairing the decode, the enlargement is empty.

Verification. Exact rational arithmetic on the finite arithmetic microcosm (n uniform over ℤ_W, W = ∏ feature primes × ∏ big primes; ν = coprimality count on the tuple; weights in the span of level-L features): the cellwise form of the identity holds with zero violations on all 96/150/180/24-cell models, and the inequality holds on 1,000 random signed weights across five models (k = 3, 4, 5; m = 1, 2; L = 1, 2). Script `fab_theorem.py`.

2. What the numerics were really showing

Before the identity was noticed, the LP experiments produced three facts that looked like a phase structure. All three are corollaries of the theorem, and each is instructive.

(a) The critical-price kink. Sweeping β in max{Φ(w) − βD(w) : S_1 (w) = 1}, the value is *exactly* the classical positive optimum for β above a sharp β*, then acquires a signed optimizer below it. Certified example (k = 3, m = 1, features {2,3,5,7}), exact-rational simplex with verified dual: val_pos = **1087376209/3212440751** = 0.3384891094603102, signed vertex Φ = 1.2082816957⋯, D = 0.4147152297⋯, giving the exact critical price

β* = (Φ − val_pos)/D = 23051796480/10991046857 = 2.0973249209031⋯,

the optimizer carrying 85% of its mass negative on 16 of 96 cells. Certified on both sides: at $β^* - 10^{-3}$ the optimum is 0.3389038246899884 > val_pos with a genuinely signed optimizer; at $β^* + 10^{-3}$ it equals val_pos as exact rationals. Below β_unb = 2.03265 the LP is unbounded (in particular at β = 2, certified). Across eight model variants β* ranged 1.44 – 6.72, and in five of the eight the window (β_unb, β*) was numerically **empty** (width < 10^{-6}): the value falls off the classical plateau directly into unboundedness. Table in `fab_phase_results.json`.

The theorem explains this: β* > 1 always, because at β = 1 the signed class is dominated. The "phase" lives entirely in the unphysical region β < 1, where the debt is charged at less than face value.

(b) The apparent linear gain in the coefficient budget. Adding the constraint $\lVert w \rVert_1 \leqslant A$ gives λ_signed(A) with λ_signed(1) = λ_positive exactly (since $\lVert w \rVert_1 = S_1 = 1$ forces w ⩾ 0), and then a strictly positive slope: dλ/dA at $A = 1^+$ equals 0.320, 0.504, 0.894, 0.394 in the four models (60 – 170% of the classical value per unit of budget), with the constraint active at every A and asymptotically linear growth. This looks like a large gain and is not one: the decode criterion is scale-invariant (only the *sign* of $S_2 - mS_1 - D$ matters), and by the theorem the repaired value is $Φ(w_+)$ minus a loss, so what grows with A is simply the total mass of w_+, not the truth of the conclusion. `fab_norm_results.json`.

(c) Unboundedness at the true price. At β = 1 every model is unbounded. Same cause: the affine slice {S_1 = 1} does not bound $\lVert w_+ \rVert$ once negative mass is allowed, and $Φ(w_+)$ scales with it. For w ⩾ 0 the normalization $S_1 = 1$ *is* the ℓ^1 norm **— that is the second, usually unremarked job positivity does in Maynard – Tao: it makes the variational problem bounded.** Remove it and boundedness must be re-imposed by hand; no choice of norm bound creates a gain, by the theorem.

3. Where the door actually is

The theorem has exactly two hypotheses, so there are exactly two escapes, and they are not variational.

(i) Charge the debt at less than face value ($\beta < 1$). This requires arithmetic information that bounds $(m - \nu)_+$ on the *designed* negative support by strictly less than the truth — i.e. an input that says "on this set, primes are more common than the trivial bound allows". This is precisely the exceptional-character mechanism: near a Landau – Siegel zero the distribution of primes in the relevant progressions is anomalously rigid, and the shortfall on a set designed around the exceptional modulus can be bounded below face value. **Zhang's program is not one route among several; by the theorem it is the only route that changes the variational picture at all.**

(ii) Keep w evaluable while w_+ is not. The theorem compares w with w_+, which presumes both are usable. In the arithmetic setting they are not interchangeable: the positive part of a divisor-sum quadratic is not itself a divisor-sum quadratic, so a signed, well-factorable λ can be evaluable at level $\theta = 4/7, 3/5, 5/8$ while its positive part is not evaluable at any level beyond 1/2. **This is the entire content of the signed route: the gain is never variational, it is exclusively about which weights the arithmetic can evaluate.** It is therefore correctly priced by the conditional ledger (§ 4), not by any optimization over weight cones.

Independent confirmation from the switching side. The Chen-switch audit reaches the same conclusion by a disjoint argument. Switching is a re-indexing bijection: it reduces the anatomical depth of one linear form but never the number r of exact-primality conditions a debt term carries. Chen's debt has $r = 1$ and is payable by an upper sieve (parity-free). Every DHL(k, m+1) conclusion with $m \geqslant 1$ forces residual debt with $r \geqslant 2$; a two-vertex kill-graph is an edge, hence bipartite, hence parity-blocked for every input class admitting Liouville twists. The escape is not a cleverer switch but an input class in which the twist is inadmissible — the exceptional-character regime again, or full GEH. Two independent analyses, one door.

4. The conditional ledger (what the door is worth)

If the debt is paid at level θ, the price list is fully computed. All eight $m = 1$ crossings below carry exact-rational certificates produced this session (Galerkin engine, full-vector Rayleigh quotients, no truncation loss); H(k) values are Engelsma-exact and were independently re-proved here for $k \leqslant 62$.

θ	threshold 2/θ	k_min pure (certified M_k)	k_min ε (certified M_{k,ε})	$H_1 \leqslant$ pure / ε
1/2 (Bombieri – Vinogradov)	4	54	50	270 / 246 (unconditional, current record)
4/7 (BFI)	3.5	31 (3.502015495⋯)	29 (3.519881249⋯)	140 / 130
7/12 (Maynard II, linear sieve)	24/7 ≈ 3.4286	29 (3.443305315⋯)	26 (3.433616497⋯)	130 / 114
3/5 (Maynard II)	10/3 ≈ 3.3333	26 (3.350647068⋯)	23 (3.334615948⋯)	114 / 94
5/8 (Pascadi)	3.2	22 (3.207656229⋯)	20 (3.222665844⋯)	90 / 80
1 (Elliott – Halberstam)	2	5 (M_5 = 2.007080)	—	12

$m = 2$ rows (p9 product-profile engine, certified-lower-bound path, rigorous primes-past-k diameters): $\theta = 4/7 \to k = 5{,}647$, $H_2 \leqslant 58{,}058$; 7/12 → 4,835, 48,988; 3/5 → 3,931, 38,878; 5/8 → 3,022, 29,180; EH → 221, 1,498. $m = 3$: 4/7 → 202,528; 7/12 → 160,703; EH → 1,978 ($H_3 \leqslant 18{,}144$). Every row is parity-consistent (all $\geqslant$ the floor 6, and $\geqslant$ the EH value 12).

The missing estimate, stated precisely. None of BFI (Acta Math. 156, Thm 10), Maynard II (Mem. AMS 1543), or Pascadi (arXiv:2505.00653) covers the sums the decode generates: all three fix one residue a for every modulus, while the decode's residues $a \in A_i(q)$ are CRT-composed from the fixed shift set $\{h_i - h_j\}$ and therefore vary with q. What would suffice is the intermediate statement

> **(E_θ).** Fix an admissible tuple H, an index i, and $A, \varepsilon > 0$. For coefficient systems $c_q(a)$ jointly well-factorable with the residue selection — for every factorization $q = q_1 q_2$ (resp. $q_1 q_2 q_3$) with $\prod Q_j = x^{\theta-\varepsilon}$ one can write $c_q(a) = \prod_j \gamma_j(q_j, a \bmod q_j)$, $|\gamma_j| \leqslant 1$, *with* $a \bmod p \in \{h_i - h_j \bmod p : j \neq i\}$ *for all* $p \mid q$ — $\sum_{q \leqslant x^{\theta-\varepsilon}} \sum_{a \in A_i(q)} c_q(a) \cdot E(x; q, a) \ll_{H,A,\varepsilon} x (\log x)^{-A}$.

$(E_{4/7})$ is "BFI Theorem 10, uniform over the CRT residue system of a fixed tuple polynomial". Evidence that the gap is bridgeable rather than parity-hard: Polymath8a's MPZ[ϖ,δ] is exactly this statement's absolute-value cousin, proved by the same dispersion + Deligne technology, but only to level $1/2 + 2\varpi \leqslant 0.5286$ and for densely-divisible moduli. So (E_θ) interpolates two proved endpoints — bilinear strength with one residue, and residue flexibility at weak level.

5. Verdict

No new phase. The signed enlargement of the Maynard – Tao variational problem is empty: with the decode debt charged at face value, every signed weight is dominated by its own positive part, by an exact pointwise identity. This kills the route cheaply and completely, as intended — and it converts an open-ended search into two sharply stated arithmetic requirements, (E_θ) for evaluability and the exceptional-character input for sub-face-value debt. The variational side of the signed programme is closed; everything of value in it is arithmetic.

Artifacts: `fab_theorem.py` *(identity, exact verification),* `fab_phase.py` / `fab_phase_results.json` *(β-structure, β^*, β_{unb} across eight models),* `fab_norm.py` / `fab_norm_results.json` *(ℓ^1-budget ramp),* `sgn1_*` *(finite model + LP engines),* `sgn2_*` *(ledger certificates,* `sgn2_certificates.json`, `sgn2_mk_large.json`*).*

Historical source 20: H2_H3_record_announcement

Source: `historical/riemann-rmt/H2_H3_record_announcement.md`. SHA-256: `0c2bebc885a4250015b5a81270f41afeda4dbc4dd8afb7454cd3229fb3774223`.

New bounds for gaps between consecutive primes with two and three primes between

$H_2 \leqslant 173{,}438$, $H_3 \leqslant 13{,}859{,}802$, $H_4 \leqslant 1{,}120{,}662{,}828$

Bill (Qingyun) Sun · GPT5.6SOL · Fable

August 15, 2026 (updated same day) — research announcement; independent expert review invited

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Statements

Theorem A. $\liminf_{n\to\infty} (p_{n+2} - p_n) \leqslant$ **173,438.** *(Previous record: 396,504, Stadlmann 2023/2025; before that 398,130, Polymath8b 2014.)*

Theorem B. $\liminf_{n\to\infty} (p_{n+3} - p_n) \leqslant$ **13,859,802.** *(Previous record: 24,797,814, Polymath8b 2014.)*

Theorem C. $\liminf_{n\to\infty} (p_{n+4} - p_n) \leqslant$ **1,120,662,828.** *(Previous record: 1,431,556,072, Polymath8b 2014.)* Certificate $M_{56,000,000} \geqslant 16.0655 > 16$; tuple = the first $56 \cdot 10^6$ primes exceeding $56 \cdot 10^6$ (first 56,000,003, last 1,176,662,831; π-anchors verified against published values).

Hypothesis chain (complete)

All three theorems follow from exactly three published inputs plus machine-verified certificates:

1. **Maynard's theorem** [Maynard, *Small gaps between primes*, Ann. of Math. 181 (2015), Prop. 4.1 + Thm 3.1 pipeline]: if the primes have level of distribution θ and $M_k > 2m/\theta$, then DHL(k, m+1): every admissible k-tuple has infinitely many translates containing $\geqslant$ m+1 primes. Our certificates use test functions supported in the **closed standard simplex** — no ε-enlargement, no vanishing-marginal variant, no equidistribution beyond (2) — so the criterion applies verbatim.
2. **Bombieri – Vinogradov** (1965): every $\theta < 1/2$ is a level of distribution; hence $M_k > 4m$ suffices (our margins 0.013 and 0.0067 leave room to fix $\theta < 1/2$).
3. **Berry – Esseen inequality** with the safe non-iid constant C = 0.56, used inside the certified lower bound for the simplex-truncation probability; applied in a regime using only elementary normal-tail bounds $\bar{\Phi}(z) \leqslant \min(\varphi(z)/z, e^{-z^2/2}/2)$ (no special-function dependence).

Certificates. $M_{15,856} \geqslant$ **8.013326752751** and $M_{923,601} \geqslant$ **12.006666706750**, produced by a one-dimensional product-profile engine (below) and verified in ball/interval arithmetic with outward enclosures (all integrands polynomial → exact rational integrals; exponentials majorized by chords/Taylor enclosures; every inequality rounded against the result;

certificate files `p9_exact_cert_k15856.json`, `p9_exact_cert_k923601.json`, `p9_exact_cert_k56000000.json` serialize the full chains). Three independent certification regimes (two Berry – Esseen constants × two tail-bound routes) pass at both k; Monte Carlo upper-consistency checks and small-k sanity bounds (the engine stays below the known M_2, M_5, M_{54}, M_{105}) all pass.

Tuples. k = 15,856: an explicit admissible tuple of diameter **173,438** (file `p9_tuple_k15856.npy`), admissibility re-verified independently by a second implementation (every prime $p \leqslant 15{,}856$ misses a class). k = 923,601: a repaired Hensley – Richards tuple of diameter **13,859,802** (symmetric window $\{\pm 1\} \cup \{\pm q : q \text{ prime} \in [45{,}007, 6{,}929{,}899]$, 5,692 mid-size primes deleted by a deletion-fixpoint repair$\} \cup \{+6{,}929{,}903\}$; file `p9_tuple_k923601.npy`, sha256 d5fe6890···6f02), admissibility over all 73,001 primes $p \leqslant k$ verified by two independent implementations; the simpler primes-past-k tuple (diameter 14,505,780, verified by direct sieve) provides a fully classical fallback. Combining certificate + tuple through Maynard's theorem yields Theorems A – C.

The method (what is actually new)

The variational lower-bound engine, not the framework. For $F = \prod_i g(x_i) \cdot 1[\sum x_i \leqslant k]$ the Maynard functionals reduce exactly to one-dimensional objects:

- $I(F) = k^{-k} c_2^{\,k} \cdot P(S_k \leqslant k)$, $J(F) = k^{-(k+1)} c_2^{\,k-1} \cdot E[G((k - S_{k-1})_+)^2]$, where X_i iid with density g^2/c_2, $c_2 = \int g^2$, $G(u) = \int_0^u g$;
- the **layer-cake identity** $E[G((k-S)_+)^2] = \int 2G(u)g(u) \cdot P(S_{k-1} < k-u)\,du$ turns the simplex truncation into an integral of *true lower-tail probabilities*, each bounded below by $1 - \beta(u)$ with rigorous β (chord-majorized Chernoff / one-big-jump / Berry – Esseen) — per-piece bounds use only the monotonicity of the true tail, so grid artifacts cannot invalidate them;
- **shaped subexponential tails** $g = e^{-(t/T_1)^{\varkappa}}/(1 + At)$ on a long support replace the hard truncation of the classical treatment.

Against the crude closed-form truncation bound used for all $m \geqslant 2$ records since 2014 (deficit from log k: $\approx 2.3 – 2.9$), the exact layer-cake accounting (+0.12) and shaped tails (+0.49) recover ≈ 1.1 units of deficit, i.e. a factor ≈ 3 in k, i.e. the factors 2.29 (m=2) and 1.71 (m=3) above. Head-to-head reconciliation against an independently written engine reproducing the 2014 structure confirms both the numbers and the provenance of the gap; the 2023/2025 record improvements had upgraded only the *arithmetic* input while retaining the crude variational bound, which is why this gain was still on the table.

Verification ledger

item	status
Engine identities (I, J, layer cake) and correction directions	proved, re-derived independently in the audit
$M_{15,856} \geqslant 8.0133$, $M_{923,601} \geqslant 12.0067$, $M_{56M} \geqslant 16.0655$	ball-arithmetic certificates, 3 regimes each
k=15,856 tuple: admissible, diameter 173,438	verified by two independent implementations
k=923,601 H-R tuple: admissible, diameter 13,859,802	verified by two independent implementations; classical fallback 14,505,780 also verified
k=56,000,000 tuple: primes-past-k, diameter 1,120,662,828	segmented sieve, π-anchors match published values

item	status
Threshold chain (Maynard + BV, closed simplex, zero extra charges)	audited against the theorem statement
Engine-vs-engine discrepancy (15,856 vs 29,500 crossing)	fully explained (layer-cake + tail shape)

Declared limitations. (i) These are computer-assisted results by AI systems; the certificate files and scripts are published for expert replay, and formal peer review has not yet occurred. (ii) A further conditional improvement ($H_2 \leqslant 145{,}226$ via $k = 13{,}476$ with Polymath8a/Deligne-strength equidistribution) is *not* claimed: it depends on a cap-normalization in the truncated-variational criterion that we have reconstructed but not verified verbatim against Polymath8b's theorem. (iii) $H_1 = 246$ is untouched: our parallel investigations proved the relevant walls (ceiling = tuple diameter; decode optimality; weight-cone closure) and located the $k = 49/47$ variational doors, which remain open but uncrossed (best certified $M_{49,\varepsilon} \geqslant 3.9305$, float 3.9593 vs threshold 4).

Artifacts. All engines, certificates, tuples, and audit scripts in `research/riemann-rmt/` and the session archive: `p9_mk_engine.py`, `p9_certify_hp.py`, `p9_exact_cert_k*.json`, `p9_tuple_k15856.npy`, verification scripts, and the companion surveys (*The Walls and the Doors*; Codex handoff note) documenting the framework analysis.

References

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Historical source 21: prime_gap_survey

Source: `historical/riemann-rmt/prime_gap_survey.md`. SHA-256: `706f0de654e61de32016d45637bf887994393b6911647095378782325a6a7a66`.

The Walls and the Doors: Exact Theorems on the Bounded–Prime–Gaps Framework

A survey of five companion investigations, with a complete computational compendium

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Abstract

Twelve years after Polymath8b established that infinitely many pairs of primes differ by at most $H_1 = 246$, that constant is unmoved. This survey synthesizes five companion investigations that, rather than optimizing once more inside the Maynard – Tao framework, determine its exact boundaries — and in doing so locate the only places a smaller constant can come from. The results are of two kinds. **Walls (proved):** the admissible-tuple side of 246 has exactly zero slack (exhaustive re-proof that H(50) = 246 and H(49) = 240 are minimal); the information-theoretic ceiling of the framework's data class equals the tuple diameter exactly, so no post-processing of sieve output can beat H(k_min); the scalar Maynard – Tao decode is precisely optimal against all available data, including every one-sided correlation bound — the matrix/inertia layer buys counting constants but never a smaller gap; the sieve-weight cone cannot be usefully enlarged — rank-r sum-of-squares weights decouple by a subadditivity theorem, and the strictly larger copositive cone is *flat* against the sieve's evaluation pencil (machine-verified dual certificate); and the 2024 Guth – Maynard large-value revolution is provably orthogonal to the level-of-distribution inputs the framework consumes. Along the way we prove a structural theorem of independent interest: **a family of pairwise prime-correlation conclusions can be simultaneously annihilated by a Selberg-type parity twist if and only if its kill-graph is bipartite** — the odd-cycle facets of the cut polytope are exactly what makes bounded gaps provable while twin primes are not. **Doors (quantified):** the variational door at k = 49 (payoff 6) and especially k = 47 (payoff 20 — the payoff is convex, a fact previously unremarked); the conversion of Maynard's uniform-residue equidistribution trilogy into shell-support gain, needing only 8 – 13% of the full-shell variational surplus and undecided in print; and the m = 2 record — corrected here to the current $H_2 \leqslant 396{,}504$ (Stadlmann) — whose admissible tuple is "almost surely not minimal" and whose variational optimization is a decade staler than H_1's. A final section compiles, in compact machine-usable form, every formula, constant, threshold, and construction needed to compute in this framework.

No result in this survey improves $H_1 = 246$. The claim is sharper: we prove where improvement is impossible, and price where it is possible.

Status convention. [P] proved (including computer-assisted exhaustive proofs with independent verification); [C] computational (reproducible, residuals stated); [H] heuristic; [Q] conjecture/open.

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1. Introduction: one constant, one pipeline

Write DHL(k, m+1) for the assertion that every admissible k-tuple $\mathcal{H}$ has infinitely many translates n + $\mathcal{H}$ containing at least m+1 primes. The modern pipeline, due to Goldston – Pintz – Yıldırım, Zhang, Maynard [May15], Tao, and Polymath8b [Pol14a], factors every bounded-gap theorem into three independent components:

(arithmetic) $\theta \longrightarrow$ (variational) $M\_k > 2m/\theta \Longrightarrow$ DHL(k, m+1) $\longrightarrow$ (combinatorial) $H\_m \leqslant H(k)$,

where θ is a level of distribution of the primes (Bombieri – Vinogradov gives $\theta = 1/2$ unconditionally), M_k is the Maynard – Tao variational constant (§ 9.1), and H(k) is the minimal diameter of an admissible k-tuple. The records: $H_1 \leqslant 246$ (k = 50, via the ε-refined variational threshold at $\theta = 1/2$) [Pol14a]; $H_1 \leqslant 12$ under Elliott – Halberstam; $\leqslant 6$ under generalized EH — where the parity obstruction provably stops all sieve-type methods. For m = 2 the record is $H_2 \leqslant 396{,}504$ (Stadlmann [Sta23], superseding Polymath8b's 398,130 — a correction to several summary accounts, including our own earlier one); $H_3 \leqslant 24{,}797{,}814$; $H\_m \ll e^{3.8075m}$ [Sta23].

The five companion investigations surveyed here each interrogate one component with the toolset developed in our finite random-matrix program (exact LP/SOS duality, convex-order counterfeits, positive-inertia decoding, exhaustive combinatorial search): Paper I the combinatorial floor, Paper II the information-theoretic ceiling, Paper III the decode layer, Paper IV the weight cone, Paper V the arithmetic frontier. Their common finding is that the framework is far more rigid than the folklore suggests — rigid enough that each "wall" is now a theorem with an explicit certificate, and the surviving "doors" carry exact price tags.

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2. Paper I — The combinatorial component is closed (and the payoff is convex)

(Exhaustive admissible-tuple computation; scripts `p4_rho_exhaust.c` and companions.)

Theorem 1.1 [P]. H(50) = 246 and H(49) = 240 are minimal: no admissible 50-tuple of diameter $\leqslant$ 245 and no admissible 49-tuple of diameter $\leqslant$ 239 exists.

The proof is a complete enumeration (not a failed search): after two structural reductions — parity monochromaticity (any admissible tuple is monochromatic mod 2, so one works in half-coordinates) and complete branching over the avoided residue classes mod 3, 5, 7 — a DFS over residue-coverage bitmasks with ρ^*-table pruning exhausts the decision instances ($\approx$1.6 – 1.85 million nodes each) in seconds. An independent implementation with a different search tree, and exact-arithmetic re-verification of every witness, confirm the values; they agree with Engelsma's classical tables (minimality known for all k $\leqslant$ 342) and OEIS A008407. The significance is not novelty of the values but the *closure statement*: **the tuple side of H_1 = 246 has exactly zero slack — any improvement must come from the analytic side, i.e. from lowering k below 50.**

The convexity observation. The verified payoff ladder near the record is H(49) = 240, H(48) = 236, **H(47) = 226**, H(46) = 216: gains of 6, 4, **10**, **10** per unit of k. (Several circulating summaries state H(47) = 232; this is wrong.) The strategic consequence, previously unremarked: the variational effort should not be spent exclusively on the marginal door k = 49 — the k = 47 door pays more than three times as much, and the threshold distance grows only linearly (§ 6). Conversely a regression to k = 51 costs only 6.

The m = 2 opportunity [C/H]. For k = 35,265 (Stadlmann's DHL(35265, 3)), the record tuple of diameter 396,504 is flagged by its own author as "almost surely not minimal." Rigorous lower bounds (Montgomery – Vaughan Brun – Titchmarsh: 216,632; Selberg large sieve: 202,249 — computed exactly) leave a factor-1.8 void, and the same bounds run $\approx$50% low at k = 50 where the truth is known. A reproduction of the Sutherland pipeline (shifted Schinzel sieve → greedy → local optimization) reaches within 3.0% of the record in ~30 CPU-minutes; the record's last percent lives in iterated merging and residue-class annealing. Estimated realistic slack at fixed k: 0.03 – 0.3%, i.e. 100 – 1,200 units of H_2 **— each single unit an immediate new record on liminf(p_{n+2} − p_n), requiring zero new analytic input.** A full-scale attack (with an LP-based repair step unavailable in 2014) is in progress.

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3. Paper II — The exact ceiling, and the bipartite structure of parity

(The counterfeit theory of the bounded-gaps data class; scripts `p8_moment_lp.py`, `p8_parity_fiber.py`.)

This is the round's most conceptually novel contribution: a two-layer adversary theory that computes, exactly, what any argument consuming the framework's data can and cannot conclude.

3.1 Layer 1: the ceiling equals the tuple diameter

Model the framework's knowledge about the prime-indicator vector $(X_1,\cdots,X_k)$ at a translate of the tuple: **(R1)** the weighted marginals $E[X_i]$ (delivered two-sidedly by the sieve at level θ), and **(R2)** *upper* bounds on all joint moments $E[\prod_{i\in S}X_i]$, $|S| \geqslant 2$ (Selberg/large sieve; the best published pair constant is 3.3996 · Hardy – Littlewood [Wu04], improved to ≈ 3.30 by Lichtman [Lic25]). Crucially **(R3):** no lower bound on any joint moment is available — that absence *is* the parity wall.

Theorem 2.1 (ceiling = diameter) [P in the model]. Over all joint laws consistent with (R1) – (R2): (i) $\min P(X_1+\cdots+X_k \geqslant m+1) = \max(0, (s-m)/(k-m))$ where $s = \sum E[X_i]$ — the pigeonhole bound is the *entire* provable content, and no strengthening of the upper-bound rows moves the threshold $s > m$; (ii) for any $H < \mathrm{diam}(\mathcal{H})$, the minimum probability that some pair at distance $\leqslant H$ is simultaneously prime is exactly 0, jumping to $(s-1)/(k-1)$ at $H = \mathrm{diam}(\mathcal{H})$. Hence the class ceiling is $H_{ceiling} = H(k_{min})$ exactly: **no post-processing of Maynard – Tao output — matrix, moment, spectral, or otherwise — can ever certify a gap smaller than the tuple diameter**, and improved pair-correlation upper bounds (Wu → Lichtman → oracle) can never lower H_1, though they do raise the guaranteed *density* of prime-pair translates.

3.2 Layer 2: the parity fiber is a slice of the cut polytope

Encode a Selberg-type parity twist as a reweighting by $f(\lambda(n+h_1),\cdots,\lambda(n+h_k)) \geqslant 0$ (λ = Liouville) constrained to preserve all one-marginal data; the reachable pair-correlation multipliers y_{ij} then form a polytope B_k.

Theorem 2.2 (bipartite kill-graph) [P]. A set K of pair-conclusions can be simultaneously annihilated ($y_{ij} = 0$ for all $ij \in K$) by a parity twist **iff the graph K is bipartite.** (Bipartite: weight the two sides ±; odd cycle: an independent set meets C in < |C|/2 vertices, contradicting the forced marginals.) Verified exhaustively against all 63 kill-patterns at k = 4 and all 1023 at k = 5: LP-feasibility coincides with bipartiteness without exception.

The odd-cycle facets of the correlation/cut polytope are thus precisely the boundary between what sieves can and cannot see. With no tuning, the model reproduces every known frontier fact: Selberg's classical k = 2 counterexample (the unique kill-all solution, $f = 2 \cdot 1\{\lambda_1 \lambda_2 = -1\}$); the factor-2 parity floor on twin-prime constants in both directions (the coordinate-range [0,2] of B_k; the gap from Lichtman's 3.30 to the floor 2 is the framework's remaining "integrality gap"); the parity-blindness floor H_1 **⩾ 6 for every parity-immune argument** (certifying "some gap ⩽ H" needs a non-bipartite close-pair graph, i.e. a triangle, and the narrowest admissible triangle is {0,2,6}) — exactly matching "GEH gives 6, and 4, 2 are blocked"; and the m-floor: DHL(k, m+1) is parity-blocked iff $k \leqslant 2m$. A conservation law at k = 3: killing two pair-conclusions forces the third to exactly twice Hardy – Littlewood — what parity takes at one gap it repays doubled at another.

3.3 The Siegel dichotomy, audited

Under strong Siegel zeros the Tao – Teräväinen order-$\leqslant 2$ Hardy – Littlewood – Chowla rows [TT22] pin every pair coordinate $y_{ij} = 1$ — the parity fiber's pair projection collapses to a point, which is the mechanism behind Heath-Brown's "Siegel zeros $\Longrightarrow$ twin primes" and Wright's $H_m \ll e^{1.9828m}$ [Wri21]. But the audit of the tempting two-branch strategy is negative [P in the model]: on the no-Siegel branch the data class is asymptotically identical to the unconditional one (Siegel-freeness enters Maynard – Tao only through effectivity, never through θ or M_k), so **min(branch A, branch B) equals the unconditional record — the dichotomy currently yields nothing new for H_1 or H_2**. Its real content is a target: any lemma of the form "no Siegel zeros $\Longrightarrow$ BV-averaged level $1/2 + \delta'$" would immediately combine with Wright's branch into an unconditional improvement.

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4. Paper III — The decode layer is exactly optimal (a convex–order theorem)

(Positive-inertia decoding of the Maynard – Tao form; scripts `p5_functionals*.py` *,* `p5_decode_lp.py` *.)*

Motivated by the rank – trace ("Lemma R") decoding that powers the 2026 two-thirds theorem for zeta zeros, one asks: does the Maynard – Tao quadratic form contain more than the one scalar inequality traditionally extracted from it?

The availability matrix [P, bookkeeping]. For the sieve weight $w = (\sum\lambda_d)^2$ at level $R = N^r$, the computable entries are: everything linear in at most one prime indicator (two-sided, at BV level, requiring $2r < \theta$); everything built from majorants/divisor functions (two-sided, elementary); and all *quadratic-in-primes* entries only **one-sidedly** (upper bounds by majorant swap, with overshoot $\varkappa = 16/(1-2r)^2 \geqslant 64$ at the threshold weight; lower bounds only trivial $\geqslant 0$). The entire cross block is unknown in exactly the direction that matters.

Theorem 3.1 (decode optimality) [P at the decode layer]. Let $A = E_\mu[\sum 1_P(n+h_i)] = (\theta/2)M_k$ be the sieve's certified mean. Against the data class consisting of all exact linear-in-primes data together with valid upper bounds on $E\,\varphi(X)$ for *every convex φ* — a class containing all falling moments and all r-fold correlation upper bounds however sharp — the largest certifiable r in "$\exists n$: $X \geqslant r$" is exactly $\lfloor A\rfloor+1$. *Proof:* the counterfeit law with X two-point supported on $\{\lfloor A\rfloor, \lceil A\rceil\}$ at mean A is the minimum in convex order among laws on $\mathbb{Z}_{\geqslant 0}$ with that mean; hence it satisfies every constraint in the class and admits no n with $X \geqslant \lfloor A\rfloor+2$. ∎

Consequently **$f(m) = 2m/\theta$ is decode-optimal: no matrix, inertia, or moment refinement of the Maynard – Tao decode can lower any H_m within the available data.** The two natural loopholes are closed quantitatively: designing sign-cancelling profiles to suppress the pair functional fails (the pair-to-mean² ratio γ is pinned within 1.2% of 1 along the entire Pareto frontier at k up to $1.6 \cdot 10^6$ [C]); and multi-translate simultaneity reduces to the scalar bound on the union tuple by first-moment budget conservation [P].

What survives [C]. Above threshold ($A = m + \delta$) the sharp two-point-adversary bound

$$\Pr_\mu[X \geqslant m+1] \geqslant \delta \,/\, (\min(k, 2B/\delta - m + 1) - m),\ B = \varkappa\gamma A^2/2 - m(m-1)/2,$$

improves the certified *measure* of prime-rich translates by factors up to 23× (m = 2) and 505× (m = 3, large δ) once pair upper bounds are used — an exponential-in-m gain in the counting constant, honestly assessed as a sharp LP form of a technique already present in Banks – Freiberg – Maynard, and worth exactly zero for H_m itself. The parity audit confirms the decode never touches a specified-shift pair; the counterfeit takes pair mass *below* Hardy – Littlewood, which no available row forbids.

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5. Paper IV — The weight cone cannot be enlarged (decoupling and flatness)

(SOS-certified enlargements; scripts `p7_*.py`, certificate `p7_dual_certificate.npz`.)

The Maynard – Tao weights are single squares. Two enlargements suggest themselves: sums of several squares (rank $r > 1$), and signed corrections certified pointwise-nonnegative on the divisor lattice (a copositive, rather than PSD, Gram matrix). Both are closed.

Theorem 4.1 (decoupling) [P]. Every DHL criterion in the pipeline has the form $\Phi(w) > 0$ with Φ subadditive on the weight cone (main terms and error budgets are linear in w; Cauchy – Schwarz penalties $-\sqrt{(B(w)C(w))}$ are superadditive). Hence a rank-r witness implies a rank-1 witness in the same per-square class: **rank never helps, for any legal variant** — the ε-trick and vanishing-marginal refinements enlarge the per-square class, not the useful rank. (Rank could only pay under an aggregate-cap constraint nonlinear in w; no sieve quantity has that form, since everything ever bounded is $\Sigma_n\, a(n)w(n)$.) A companion collapse [P]: PSD certificates plus the level constraint force long-support rows to vanish (Schur), so SOS + level = the classical class exactly.

Theorem 4.2 (flatness of the copositive cone) [C, with a machine-verified closure certificate]. The copositive cone is strictly larger as a cone — the LP finds wildly non-PSD legal Gram matrices — but its extra directions are orthogonal to the sieve's evaluation pencil $(\tilde{J}, \tilde{I})$. Measured optimal-value gaps at $k = 3, 4, 5$ over spans up to $r = 8$: $\leqslant 2.6 \cdot 10^{-7}$ relative, in an exact fixed-R model cross-validated against 10^6 real integers (where the exact finite-cone optimum is attained by a PSD matrix). At $k = 3$ the closure is certified: nine explicit realizable divisor patterns P with weights $y_P > 0$ satisfy $\tilde{J} = t \cdot \tilde{I} - \Sigma\, y_P v_P v_P^T$ to $1.3 \cdot 10^{-14}$ with t exceeding the PSD optimum by $1.4 \cdot 10^{-6}$ — since $\langle v_P, C v_P\rangle \geqslant 0$ for every copositive C, this bounds the *entire* enlarged cone. The cross-only long-support experiment (the sharpest corner: components beyond the level admitted only in cross terms) recovers ≈ 0 of the 10.2% headroom that illegal full support would give. Mechanism [H]: truncated divisor patterns realize all sign-alternating finite differences of the profiles — a family dense enough to pin the pencil's negative eigenspace; nothing in it weakens as k grows.

Consequence. The weight-cone axis is closed. Combined with Paper III, all conceivable "optimize harder inside the same data" strategies are now excluded by theorem or certificate; movement requires evaluation-side inputs.

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6. Paper V — The arithmetic frontier, 2014 → 2026

(Level-of-distribution audit and the exact payoff function; script `p6_payoff.py`.)

The MT-usable frontier has moved 0.5233 → 0.5263 in twelve years. The sieve consumes equidistribution with two properties: all moduli (its error moduli are generic squarefree products) and residue uniformity (its residues are the $k^{\omega(q)}$ roots of $\prod(n+h_i)$, varying with q). Everything beyond Bombieri – Vinogradov's $\theta = 1/2$ exists only in restricted forms, sorted by compatibility: Polymath8a's 1/2 + 7/300 (densely divisible moduli, poly-root residues — usable on a truncated support shell); Maynard's trilogy culminating in 11/21 with **completely uniform residues** [MayIII] — the first such object since Zhang; Stadlmann's 1/2 + 1/40 (smooth moduli) [Sta23]; Pascadi's minorant-level 10/19 [Pas25] — while the numerically larger levels (BFI 4/7, Maynard II 3/5, Pascadi 5/8) are fixed-residue or well-factorable-weight results that the MT error term, carrying absolute values over varying residues, provably cannot consume. The mechanism by which restricted inputs failed to move H_1 in 2014 is reconstructed exactly: the input enters *only through the shape of the accessible support polytope Ω*, the shell beyond the unit simplex must be truncated to coordinates the modulus class can guarantee, and the ε-trick already harvests shell value at zero arithmetic cost — a structured input must beat the ε-trick on its own shell. Net 2014 gain at $m = 1$: ≈ 0. **Coverage density in modulus mass is the wrong parameter [P]: non-geometrically-aligned partial coverage is worth exactly nothing; the right parameter is the variational surplus ρ_geom of the accessible sub-shell.**

The payoff function, calibrated [C]. With the master criterion $\mathrm{DHL}(k, m+1) \Longleftarrow M\_k(\Omega) > 2m/\theta$ and $M\_k(\lambda\Omega) = \lambda M\_k(\Omega)$: tipping $k = 49$ (worth 6) needs full-BV $\delta \approx 0.0014 - 0.0023$, or ρ_geom $\approx 8 - 13\%$ of the full-shell surplus at Maynard-III strength; $k = 48$ (worth 10): $\delta \approx 0.003 - 0.005$; $k = 47$ (worth 20): $\delta \approx 0.005 - 0.008$. Sensitivity at the record: $dH_1/d\delta \approx -2400$. The pure-simplex route is dead by the rigorous upper bound $M\_k \leqslant (k/(k-1))\log k$: $M_{50} \leqslant 3.99186 < 4$ [P] — every road to < 246 passes through enlarged variants.

Guth – Maynard is orthogonal to H_1 [P at proof-structure level]. The classical BV proof consumes no zero-density input; full GRH itself gives only $\theta = 1/2$ pointwise — everything beyond comes from dispersion + Weil/Deligne + spectral Kloosterman machinery in which zero-density estimates appear nowhere (dependency audit of Maynard I – III, Stadlmann, Pascadi). The 2024 large-value theorem [GM24] therefore feeds short-interval results, not bounded gaps. The genuine open computation this paper isolates: **translate Maynard III's divisor-window conditions into shell-polytope constraints and compute $M_{49}(R \cup \Omega^*)$ with the ε-trick layered on — an undecided, decidable question that no one has published,** and exactly the kind of 10^{-2}-precision constrained Rayleigh-quotient problem our exact-rational SDP/Krylov machinery certifies.

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7. Synthesis: the no-go map and the two open doors

The five papers compose into a single statement:

> **Within the Maynard – Tao data class, $H_1 = H(k\_min)$ exactly, and every route to a smaller H_1 must lower k_min — by enlarging the variational value through new *evaluation-side* arithmetic (uniform-residue level beyond 1/2 on an accessible shell), and in no other way.** Post-processing (Paper II), decode refinement (Paper III), weight-cone enlargement (Paper IV), tuple search (Paper I), and zero-density technology (Paper V) are all excluded by theorem or certificate.

The doors, priced: **(D1)** the $k = 49/48/47$ variational doors, payoffs 6/10/20, with the Maynard-III shell computation as the concrete undecided question; **(D2)** the $m \geqslant 2$ records — softer in *both* factors: the variational optimization at $k \approx 35{,}000$ never received 2014's $k = 50$-level effort (threshold $M\_k > 8$; every unit of variational gain converts at $dH_2/dk \approx 11$ via the tuple tables), and the record tuple itself has measurable slack (each unit a new world record); **(D3)** parity-sensitive rows short of Siegel zeros — order-2 Liouville correlation rows in natural density would pin pair coordinates exactly as Tao – Teräväinen's do conditionally; the bipartite theorem says precisely which conclusions each new row unlocks.

Work in progress (five further investigations, reports pending): the exact-rational variational baseline at $k = 45 - 55$; the ε-variant attack at $k = 49/47$; certified upper bounds closing or opening those doors; the large-k ($m = 2, 3$) variational engine; and the additive – Mellin Type-II program.

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8. Computational compendium

Everything needed to compute in this framework, compact but complete. All notation as above; [P]/[C] status inherited from the sections cited.

8.1 The variational problem

- Simplex $R_k = \{t \in [0,\infty)^k : \Sigma t_i \leqslant 1\}$; symmetric F: $I(F) = \int \{R_k\} F^2, J^{\{(m)\}}(F) = \int\{R_{\{k-1\}}\}(\int F\, dt_m)^2$; $M_k = \sup_F k \cdot J(F)/I(F)$ (all $J^{\{(m)\}}$ equal by symmetry).
- Bilinear polarization (legal at no cost): $I(F_1, F_2) = \int F_1 F_2$, $J^{\{(m)\}}(F_1, F_2) = \int (\int F_1\, dt_m)(\int F_2\, dt_m)$.
- Polynomial bases in power sums $P_j = \Sigma t_i^j$ reduce I, J to 1-D Beta-type integrals; exact rational arithmetic feasible through $k \sim 10^2$ at useful degree. Product-profile 1-D reduction for $k \sim 10^4$ (optimal $F \approx \prod g(t_i)$, $g \approx 1/(1+At)$ truncated).
- Threshold: **DHL(k, m+1) ⟸ $M_k(\Omega\_accessible) > 2m/\theta$**; $\theta = 1/2$ unconditional ⇒ $M_k > 4m$. Scaling law $M_k(\lambda\Omega) = \lambda \cdot M_k(\Omega)$.
- ε-trick: support $(1+\varepsilon)R_k$ with marginals vanishing outside $(1-\varepsilon)R_{\{k-1\}}$; zero arithmetic cost; this is how k = 54 → 50.
- Upper bound [P]: $M_k \leqslant (k/(k-1)) \cdot \log k$. Values: $M_{48} \leqslant 3.95357$, $M_{49} \leqslant 3.97290$, $M_{50} \leqslant 3.99186$ (pure problem dead ⩽ 50); first non-excluded k = 51 (4.01046).
- Anchors: $M_5 = 2.007080 > 2$ (EH ⇒ $H_1 \leqslant 12$); $M_4 = 1.845401 < 2$ (provably, so pure class cannot give DHL(4,2) under EH); $M_3 = 1.646440$; $M_2 = 1.385933$; $M_{105} > 4$ (Maynard); $M_{54} > 4.00238$ (Polymath8b); ε-variant at k = 50 > 4 ⇒ 246.
- Asymptotics: $M_k = \log k - c_0 + o(1)$; calibrated slope $dM/dk \approx 1/k$ near k = 50.

8.2 Records and thresholds (August 2026)

quantity	value	source/condition
H_1	**246** (k = 50)	unconditional [Pol14a]
H_1 under EH / GEH	12 / 6	6 is the parity floor [P, § 3.2]
H_2	**396,504** (k = 35,265)	Stadlmann [Sta23] — not 398,130
H_3	24,797,814 (k = 1,649,821)	[Pol14a]
H_m	$\ll e^{\{3.8075m\}}$	[Sta23] — not 3.815
m-threshold	$M_k > 4m$ ($\theta = 1/2$); m = 2 ⇒ > 8	
pair-correlation upper constants	3.3996 [Wu04] → ≈ 3.30 [Lic25]; parity floor 2	cannot affect H_1 [P, § 3.1]

8.3 Admissible tuples (all proven minimal for k ⩽ 342; witnesses on file for k ⩽ 62)

H(k), k = 3..62: 6, 8, 12, 16, 20, 26, 30, 32, 36, 42, 48, 50, 56, 60, 66, 70, 76, 80, 84, 90, 94, 100, 110, 114, 120, 126, 130, 136, 140, 146, 152, 156, 158, 162, 168, 176, 182, 186, 188, 196, 200, 210, 212, 216, **226, 236, 240, 246**, 252, 254, 264, 270, 272, 278, 282, 288, 300, 304, 310, 320. Key ladder: k = 46/47/48/49/50 → 216/226/236/240/246 (payoffs 30/20/10/6 relative to 246; **H(47) = 226**, not 232). Admissibility test: for every prime p ⩽ k, some residue class mod p is missed (p > k automatic). Structural facts for search: monochromatic mod 2; complete mod-3,5,7 class branching is exhaustive for k ⩾ 7; ρ*-pruning by smaller widths is sound (subsets/translates of admissible are admissible). Proven-optimal witnesses (diam 246, k=50): 0 4 6 10 16 18 24 28 34 36 60 64 66 70 76 78 84 88 90 94 100 106 108 114 120 126 130 136 144 148 154 156 160 168 174 178 186 190 196 198 204 214 216 220 226 234 238 240 244 246. Rigorous lower bounds at large k (both run ~50% low): Brun – Titchmarsh H(35265) ⩾ 216,632; large sieve ⩾ 202,249. m=2 exchange rate: $\Delta H_2/\Delta k \approx 11.2$ near k ≈ 35,300.

8.4 Levels of distribution (what the MT sieve can and cannot eat)

input	θ	moduli	residues	MT–usable
Bombieri – Vinogradov	1/2	all	uniform (max_a)	**yes — the workhorse**
Polymath8a	1/2 + 7/300	densely divisible	poly roots	shell-truncated
Maynard III [MayIII]	11/21	factorable windows	**uniform**	shell-truncated; window widths = the open question
Stadlmann [Sta23]	1/2 + 1/40	smooth	MPZ-type	shell-truncated
Pascadi minorant [Pas25]	10/19	smooth	minorant of 1_P	best usable restricted level
BFI / Maynard II / Pascadi	4/7, 3/5, 5/8	well-factorable λ	**fixed a**	**no** (two structural reasons: residue-variation; absolute values kill factorable weights)

Tipping table (full-BV equivalent): $k = 49 \Leftarrow \delta \approx 0.0014 - 0.0023$; $k = 48 \Leftarrow 0.003 - 0.005$; $k = 47 \Leftarrow 0.005 - 0.008$; $dH_1/d\delta \approx -2400$. Shell-coverage form at $\theta' = 11/21$: $k = 49$ needs ρ_geom $\approx 8 - 13\%$. GRH gives only $\theta = 1/2$; zero-density estimates (incl. Guth – Maynard) appear nowhere in the beyond-1/2 dependency chains.

8.5 The adversary/counterfeit toolkit

- **Ceiling LP (Layer 1):** data $\{E[X_i] = \mu_i$ (equality), $E[\prod_{S} X] \leqslant \beta_S$ ($|S| \geqslant 2$)$\}$; min $P(\Sigma X \geqslant m+1) = \max(0, (s-m)/(k-m))$, dual $1\{S \geqslant m+1\} \geqslant (S-m)/(k-m)$; specified-gap conclusions have LP value 0 below the diameter. Quantitative pair-density: with all-pairs cap β, min $\#\{$translates with $\geqslant 2$ primes$\} = (s-1)/(r^*-1)$, $r^* = \beta k(k-1)/(s-1)$.
- **Convex-order counterfeit (decode killer):** X two-point on $\{\lfloor A \rfloor, \lceil A \rceil\}$, mean A — minimal in convex order on $\mathbb{Z}_{\geqslant 0}$, hence satisfies every convex-functional upper bound; kills all decodes beyond $\lfloor A \rfloor+1$. Feasibility margins of real certificates above the (unusable) firing boundary: 97 – 129× (level-constrained), 2.0 – 4.0× even at the parity floor.
- **Parity twist fiber (Layer 2):** distributions q on minus-sets $T \subseteq [k]$ with all vertex-marginals 1/2; pair multipliers $y_{ij} = 4P(i,j \in T) \in [0,2]$. **Kill-set feasible $\Longleftrightarrow$ bipartite.** $B_3 = \mathrm{conv}\{(2,0,0),(0,2,0),(0,0,2),(2,2,2)\}$; $\Sigma y_{ij} \geqslant 2(k-2)$ tight; DHL(k,m+1) parity-blocked $\Longleftrightarrow k \leqslant 2m$; parity floor for any "some gap $\leqslant$ H" certificate: a triangle in the close-pair graph $\Rightarrow H \geqslant 6$ via $\{0,2,6\}$. Siegel rows (order-$\leqslant$2 HLC [TT22]) pin all $y_{ij} = 1$; then 3-primes conclusions revive iff pair-sum > s/2.
- **Sieve availability matrix:** two-sided = linear-in-primes (needs $2r < \theta$) and majorant/divisor blocks ($2r + \Sigma 2s_j < 1$); one-sided-upper = all quadratic-in-primes blocks, overshoot $\varkappa = 16/(1-2r)^2$ (= 64 at r = 1/4); pair-suppression impossible: γ = J-pair/($A^2/2$) pinned in [0.988, 1.015] along the Pareto frontier ($\gamma = (k-1)/k$ exactly for products).
- **Weight-cone closure:** subadditivity ⇒ rank-1 suffices (§ 5); copositive-cone flatness certificate: $\tilde{J} = t \cdot \tilde{I} - \Sigma_9\, y_P v_P v_P^T$ (residual $1.3 \cdot 10^{-14}$). To test any proposed enlargement: check whether it changes a linear functional of w or only the positivity certificate — only the former can matter.
- **Counting bound (the one thing the matrix layer buys):** $\Pr[X \geqslant m+1] \geqslant \delta/(\min(k, 2B/\delta - m + 1) - m)$, $B = \varkappa\gamma A^2/2 - \mathrm{binom}(m,2)$; beats pigeonhole iff $\delta > \varkappa\gamma m^2/k$; gains 23× (m=2), 505× (m=3); worth 0 for H_m.

8.6 Strategic invariants

1. All H_1 improvement factors through k_min; $dH_1/dk \in \{6,4,10,10,\ldots\}$ (convex) below 50.
2. All k_min improvement factors through $M(\Omega)$ on enlarged supports; all Ω-enlargement factors through uniform-residue arithmetic beyond $\theta = 1/2$ (or unconditional shell access à la Maynard III).
3. $m \geqslant 2$ records are soft in both factors (variational at $k \sim 10^4$ under-optimized; tuples non-minimal) with conversion 11.2 : 1 (analytic : combinatorial).
4. Parity floors: $H_1 \geqslant 6$ for the entire method class; $k \geqslant 2m+1$ for DHL; pair constants $\geqslant 2 \cdot$ HL both directions.
5. The only presently-undecided decidable question worth a large compute budget: **M_{49} ($R \cup \Omega$_MaynardIII) vs 4 —** and its $k = 47$ sibling.

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9. Open problems

1. **The shell computation** (§ 6): extract Maynard III's divisor-window parameters, build Ω^*, certify $M_{49}(R \cup \Omega^*)$ against 4 with exact-rational inner/outer bounds. Undecided in print; decidable.
2. **The H_2 tuple record**: any admissible 35,265-tuple of diameter $\leqslant$ 396,503. (In progress.)
3. **The $m = 2$ variational re-optimization** with Stadlmann-level inputs and 2014-vintage care at k slightly below 35,265.
4. **The continuum flatness theorem** (§ 5): $k \cdot \tilde{J} - \rho^*\tilde{I} \in -\mathrm{cone}\{v\_Pv\_P^T\}$ for arbitrary smooth spans — closing the copositive cone at all k by proof rather than certificate.
5. **The missing dichotomy lemma**: "no Siegel zeros up to scale $\Longrightarrow \theta = 1/2 + \delta'$ for MT-averaged moduli" — the one statement that would make the Siegel two-branch strategy yield unconditional records via Wright's branch.
6. **New parity-sensitive rows in natural density**: any unconditional lower bound on a single joint moment E[X_iX_j] (however weak) breaks Layer 2 at that edge; the bipartite theorem then computes exactly which package of conclusions unlocks.

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Companion scripts and machine-verified certificates for every [C] claim are archived in the project scratchpad (`p4_*`, `p5_*`, `p6_*`, `p7_*`, `p8_*`); exhaustive-search witnesses re-verified in exact arithmetic by independent implementations.

Historical source 22: stopping_times_paper

Source: `historical/riemann-rmt/stopping_times_paper.md`. SHA-256: `8f7715642343c32bb8d39d30cae1d6b56bb0a0ec98ab2b5f7b26c1f6960b418a`.

Stopping Times See What Moments Cannot

Prime gaps, signed sieves, and a dynamic separation of CUE from the alternative hypothesis

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Abstract

We report on a research programme carried out by a human mathematician working with several large language models in a tight adversarial loop. Three groups of results are described.

(1) **Prime gaps.** New unconditional bounds for gaps containing several primes: liminf(p_{n+2} − p_n) ⩽ 173,438, liminf(p_{n+3} − p_n) ⩽ 13,859,802, and liminf(p_{n+4} − p_n) ⩽ 1,120,662,828, improving the previous records 396,504 / 24,797,814 / 1,431,556,072 by factors 2.3 / 1.8 / 1.3. The gain is purely variational: the records since 2014 had upgraded the arithmetic input while retaining a crude closed-form bound on the Maynard – Tao functional. We also give a no-go map for H_1 = 246 with five proved walls, and price the two doors that remain open.

(2) **Signed sieves.** A one-line pointwise identity shows that the signed (indefinite) enlargement of the Maynard – Tao variational problem is *empty*: with the decode debt charged at face value, every signed weight is dominated by its own positive part. This closes the variational side of a programme suggested by Zhang's Landau – Siegel work and by Iwaniec's well-factorable weights, and —more usefully—shows that everything of value in the signed route is arithmetic, not variational. We give the precise missing estimate (E_θ) for each level θ ∈ {4/7, 7/12, 3/5, 5/8} together with the conditional price list it would buy (H_1 ⩽ 130, 114, 94, 80).

(3) **A dynamic separation.** Tao's alternative hypothesis ensemble ACUE matches CUE on every low-order statistic that bandwidth-limited detectors can see. We first measure the obstruction: the exact fibre of measures matching all balanced moments of degree ⩽ N has dimension 0, 0, 2, 10, 80, 403, 1804 for N = 3, ⋯, 9, contains an explicit (N−3)-parameter family q_ACUE(C) · g(Σc mod N) with ĝ(±1) = 0, and — because the heat flow is diagonal in the coefficients — stays frozen to that algebra along the *entire* flow. We then show that a *stopping time* escapes it: the finite de Bruijn – Newman depth Λ, the first collision time of the zeros under backwards heat flow, separates CUE from ACUE at the level of universality class, −Λ^CUE ≍ N^{−8/3} against −Λ^ACUE ≍ N^{−2}. The exponent gap is the extreme-value statistics of the spectrum in scalar form: CUE admits rare pairs at distance N^{−4/3} while the lattice quantises every gap at π/N, so the alternative hypothesis satisfies its own RH-analogue *too robustly*. More generally, for the circular β-ensembles we find and confirm −Λ ≍ N^{−2−2/(β+1)} at β = 1, 2, 4, with the rigid lattice as the β = ∞ endpoint. The methodological content is a slogan:

> **moment matching does not imply stopping-time matching.**

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1. What this paper is about

Three problems, three walls, and one way through.

The first wall is the bounded gaps problem. Since Zhang, Maynard and Polymath8, the record $\liminf(p_{n+1} - p_n) \leqslant 246$ has stood for twelve years. We did not move it, and we explain in § 3 why we now believe it is protected by a specific and provable structure rather than by a lack of effort. But the same analysis showed that the *higher* records — two, three, four primes in a bounded window — were protected by nothing at all, and those we did move.

The second wall is positivity. Maynard – Tao weights are squares. Everybody who has worked with them has wondered what happens if you drop the square; Zhang's Landau – Siegel programme is built around exactly this, and Iwaniec's linear-sieve weights $\lambda^{\pm}$ are signed and consume levels of distribution that squares cannot reach. We settle the variational half of that question completely, in the negative, by an identity that fits on one line.

The third wall is the alternative hypothesis. Tao's ACUE ensemble is a lattice-supported measure on unitary spectra that reproduces CUE's pair correlation and, as we and others have verified, an entire algebra of low-degree statistics. The programme of "find a moment ACUE cannot fake" has been running for years and keeps hitting the same obstruction: the fibre of measures matching a given set of moments is large — we compute it exactly, and it reaches dimension 403 by $N = 8$ — so adding one more moment enlarges the search rather than ending it. Our contribution is to stop adding moments. A *stopping time* is not a polynomial statistic, and the fibre's freedom does not protect it: what it protects is the algebra of observables, and the first-passage time of a flow is not in that algebra.

The paper is written to be read by mathematicians, not by machines; but the last section describes how the work was actually done, because the method is part of the result.

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2. Prime gaps: what moved and why

2.1 The framework in one paragraph

Fix an admissible k-tuple $H = \{h_1, \cdots, h_k\}$ (admissible: for each prime p, the h_i do not cover all residues mod p). Maynard's method attaches to each n a weight $w(n) = (\sum_d \lambda_d)^2$ built from divisor sums, and compares $S_1 = \sum w(n)$ with $S_2 = \sum w(n) \cdot \#\{i : n + h_i \text{ prime}\}$. If the primes have level of distribution θ and the variational constant

$M_k = \sup_F k \cdot J(F)/I(F), I(F) = \int_{R_k} F^2, J(F) = \int_{R_{k-1}} (\int_0^{1-\Sigma} F\, dt_k)^2,$

taken over symmetric F on the simplex $R_k = \{t \in [0,\infty)^k : \Sigma t_i \leqslant 1\}$, exceeds $2m/\theta$, then every admissible k-tuple contains m+1 primes infinitely often, and hence $H_m := \liminf(p_{n+m} - p_n)$ is at most the diameter H(k) of the narrowest admissible k-tuple. Bombieri – Vinogradov gives $\theta = 1/2$ unconditionally, so the criterion is $M_k > 4m$.

Two quantities therefore control everything: the analytic constant M_k, and the combinatorial constant H(k). The record $H_1 = 246$ comes from $k = 50$ and a variant of M_k tuned by an ε-trick.

2.2 The observation

For $m = 1$, k is small (50) and M_k has been computed to death. For $m \geqslant 2$, k is large — tens of thousands to millions — and there M_k has never been computed at all. Every record since 2014 has used the same crude closed-form lower bound, essentially $M_k \geqslant \log k - C$ with an explicit but generous C, obtained by truncating a product test function hard at the simplex boundary. The improvements of 2023 – 2025 raised θ, not M_k.

The deficit in that bound is between 2.3 and 2.9 units of $\log k$, and about 1.1 of those units are recoverable by elementary means. Since $M_k \approx \log k$, recovering 1.1 units is a factor $e^{1.1} \approx 3$ in k, and $H(k) \approx k(\log k + 0.77)$ is nearly linear in k, so a factor 3 in k is close to a factor 3 in H_m.

2.3 The engine

Take $F(t) = \prod_i g(t_i) \cdot 1[\sum t_i \leqslant k]$ (working on the dilated simplex). Let X_i be i.i.d. with density g^2/c_2 where $c_2 = \int g^2$, write S_j for partial sums, and $G(u) = \int_0^u g$. Then exactly

$I(F) = k^{-k} c_2^k \cdot P(S_k \leqslant k)$, $J(F) = k^{-(k+1)} c_2^{k-1} \cdot E[G((k - S_{k-1})_+)^2]$.

The simplex truncation, which the classical treatment handles by throwing it away, is now a genuine probability, and the second factor is unwrapped by the **layer-cake identity**

$E[G((k - S)_+)^2] = \int 2G(u)\, g(u) \cdot P(S_{k-1} < k - u)\, du$,

turning it into an integral of true lower-tail probabilities. Each of those is bounded below rigorously (chord-majorised Chernoff, one-big-jump, Berry – Esseen with the safe non-i.i.d. constant 0.56); because the bounds use only monotonicity of the true tail, discretisation cannot invalidate them. Finally, replacing the hard truncation by **shaped subexponential tails** $g(t) = e^{-(t/T_1)^\varkappa}/(1 + At)$ on a long support recovers the rest.

Exact accounting of the truncation contributes +0.12 units, tail shaping +0.49, and the remainder comes from optimising the shape parameters — about 1.1 units in total, as predicted.

2.4 The results

	bound	k	previous
$H_2 = \liminf(p_{n+2} - p_n)$	**173,438**	15,856	396,504 (Stadlmann 2023/25)
H_3	**13,859,802**	923,601	24,797,814 (Polymath8b 2014)
H_4	**1,120,662,828**	56,000,000	1,431,556,072 (Polymath8b 2014)

The certificates $M_{15,856} \geqslant 8.013326752751$, $M_{923,601} \geqslant 12.006666706750$ and $M_{56\cdot 10^6} \geqslant 16.065482942$ are verified in ball arithmetic with outward rounding at every step; three independent certification regimes (two Berry – Esseen constants $\times$ two tail routes) pass at each k. The tuples are explicit: for $k = 15{,}856$ an admissible tuple of diameter 173,438 verified by two independent implementations; for $k = 923{,}601$ a repaired Hensley – Richards tuple of diameter 13,859,802 (with a fully classical fallback of diameter 14,505,780 from the primes past k); for $k = 5.6 \cdot 10^7$ the primes-past-k tuple, its endpoints checked against published values of $\pi(x)$.

These are computer-assisted results produced by AI systems and have not been refereed. All certificates and scripts are published for replay. A further conditional improvement ($H_2 \leqslant 145{,}226$ via Polymath8a/Deligne-strength equidistribution) is deliberately **not** claimed: it depends on a cap normalisation we reconstructed but could not verify verbatim.

2.5 Why 246 did not move: five walls

We spent a comparable effort on H_1 and failed, but the failure is informative. Five statements, each proved during this work, close five different escape routes.

1. **The ceiling is the tuple diameter.** No post-processing of Maynard – Tao output can produce a bound below H(k_min). In particular, pair-correlation constants of the Wu/Lichtman type — which look like they should help — cannot lower H_1 at all.
2. **Scalar decoding is exactly optimal.** One might hope to replace "$S_2 - mS_1 > 0$" by a matrix, inertia, or higher-moment decode. A two-point counterfeit construction, ordered by convex order, defeats every such decode; the scalar threshold $f(m) = 2m/\theta$ is final.
3. **The weight cone is closed.** Enlarging squares to arbitrary PSD quadratic forms Q gains nothing: rank-r sums of squares decouple by subadditivity, so the supremum is attained at rank one. The copositive relaxation is flat as well, certified by an explicit nine-pattern dual with residual $1.3 \cdot 10^{-14}$.
4. **Parity, made combinatorial.** A parity (Liouville) twist kills a set of pair-conclusions if and only if the associated *kill-graph* is bipartite; odd-cycle facets of the cut polytope are exactly the obstructions. This yields the floors $H_1 \geqslant 6$ and $k \geqslant 2m + 1$ for the whole method class.
5. **The usable arithmetic frontier is not the published frontier.** The post-2014 levels $\theta = 4/7$ (Bombieri – Friedlander – Iwaniec), 3/5 (Maynard II) and 5/8 (Pascadi) are *well-factorable* or *fixed-residue*. Maynard – Tao needs uniformity over a CRT-structured system of residues varying with the modulus, which none of them provides. The usable frontier is Maynard III's 11/21 with a shell truncation, Stadlmann's 1/2 + 1/40, and Pascadi's minorant 10/19 — all shell-restricted.

What remains open is narrow and precisely priced. The $k = 49$ door needs M_{49}-variant > 4; the pure constant is $M_{49} \in [3.891258, 3.97290]$ (lower bound exact-rational), and the best certified ε-variant is $M_{49,1/35} \geqslant 3.930490592$ with float optimum 3.959325169. Known upper-bound technology closes the door only for $\varepsilon \leqslant 0.00682$, so the door is open, exactly as Polymath left it. Crossing it would give $H_1 \leqslant 240$; $k = 47$ would give 226.

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3. The signed sieve: a one-line no-go

3.1 The question

Since Maynard – Tao weights are squares, and since Chen's theorem and Iwaniec's $\lambda^\pm$ are both built from *signed* objects, it is natural to ask what a signed sieve would buy. Zhang's Landau – Siegel programme is organised around removing the square. The hope is a variational phase invisible to the positive family.

For signed w the decode fails: the positive excess can be manufactured by $w(n) < 0$ at prime-poor n. It is repaired by paying the **debt**

$D(w) = \Sigma_{\{n : w(n) < 0\}} |w(n)| \cdot (m - \nu(n))_+, \nu(n) = \#\{i : n + h_i \text{ prime}\}$,

after which $S_2 - mS_1 - D > 0$ does imply that some n has $\nu(n) \geqslant m + 1$. So the honest experiment is to optimise $\Phi(w) - \beta D(w)$ with $\beta = 1$ the true price, and look for a phase transition.

3.2 The identity

For every integer $\nu \geqslant 0$ and every $m \geqslant 1$,

$(\nu - m) + (m - \nu)_+ = (\nu - m)_+$,

both sides being $\nu - m$ when $\nu \geqslant m$ and 0 otherwise. Writing $w = w_+ - w_-$ with $w_+, w_- \geqslant 0$ of disjoint support and substituting:

> **Theorem (Signed No-Gain).** $S_2 - mS_1 - D(w) = \Sigma\_n\ w_+(n)(\nu(n) - m) - \Sigma\_n\ w_-(n)(\nu(n) - m)_+ \leqslant \Sigma\_n\ w_+(n)(\nu(n) - m)$.

Since $w_- \geqslant 0$ and $(\nu - m)_+ \geqslant 0$, the negative part is pure loss, and the loss is exactly the overshoot mass $\Sigma\ w_-(\nu - m)_+$ sitting on the negative support.

Corollary. Any DHL(k, m+1) conclusion obtainable from a signed weight w with the debt charged at face value is already obtainable from the nonnegative weight w_+. The signed class is redundant.

The statement is pointwise, so it holds for every k, m, tuple and weight class at once, needs no asymptotics, and subsumes the weaker fact that the naked ratio sup Tr(BQ)/Tr(AQ) over indefinite Q equals the PSD value by rank-one extremality.

3.3 What the numerics had been showing

We found the identity only after a substantial LP campaign on a finite arithmetic microcosm (n uniform on $\mathbb{Z}\_W$, ν the coprimality count on the tuple, weights in the span of level-L features, everything in exact rational arithmetic). Three phenomena had looked like a phase structure; all three are corollaries.

- **A critical price.** The value of $\max\{\Phi - \beta D : S_1 = 1\}$ is *exactly* the classical positive optimum for β above a sharp β^*, and acquires a signed optimiser below it. In the base model (k = 3, features {2,3,5,7}) an exact-rational simplex with certified dual gives the classical value 1087376209/3212440751 = 0.3384891094603102 and the critical price

  $\beta^* = 23051796480/10991046857 = 2.0973249209031\cdots$,

  the signed vertex having $\Phi = 1.2082816957\cdots$, $D = 0.4147152297\cdots$, with 85% of its mass negative on 16 of 96 cells. The transition is certified on both sides: at $\beta^* - 10^{-3}$ the optimum is strictly above the plateau (0.3389038246899884 > 0.3384891094603102) and genuinely signed, while at $\beta^* + 10^{-3}$ it equals the plateau as exact rationals. At $\beta = 2$ the programme is unbounded. The theorem explains why $\beta^* > 1$ always: at the true price the signed class is dominated.

- **An apparent linear gain.** Imposing $\| w \|_1 \leqslant A$ gives $\lambda(1) = \lambda\_\text{positive}$ exactly (for $w \geqslant 0$ the normalisation $S_1 = 1$ *is* the ℓ^1 norm) and then a strictly positive slope 0.32 – 0.89 per unit of budget. This is not a gain: the decode is scale-invariant, and what grows with A is the mass of w_+, not the truth of the conclusion.

- **Unboundedness at $\beta = 1$.** In all eight model variants. Same cause. This exposes a second, usually unremarked, job that positivity performs in Maynard – Tao: *it makes the variational problem bounded*, by tying the ℓ^1 norm to the normalisation. Remove it and boundedness must be re-imposed by hand — and by the theorem, no choice of norm bound creates a gain.

3.4 Where the door actually is

The theorem has exactly two hypotheses, hence exactly two escapes, and neither is variational.

(i) Charge the debt below face value. This needs arithmetic information bounding $(m - \nu)_+$ on a *designed* negative support by strictly less than the truth — that is, an input asserting that primes are anomalously common on a prescribed set. This is precisely the exceptional-character mechanism. By the theorem, Zhang's programme is not one route among several; it is the only route that changes the variational picture at all.

(ii) Keep w evaluable while w_+ is not. The theorem compares w with w_+, presuming both usable. Arithmetically they are not interchangeable: the positive part of a divisor-sum quadratic is not a divisor-sum quadratic, so a signed well-factorable λ can be evaluable at $\theta = 4/7$, $3/5$, $5/8$ while its positive part is evaluable at no level beyond $1/2$. **The signed route's entire value is which weights the arithmetic can evaluate, never the shape of the optimum.**

An independent argument from the switching side reaches the same conclusion. Switching is a re-indexing bijection: it lowers the anatomical depth of one linear form but never the number r of exact-primality conditions in a debt term. Chen's debt has $r = 1$ and is payable by an upper sieve (parity-free). Every DHL(k, m+1) conclusion with $m \geqslant 1$ forces residual debt with $r \geqslant 2$; a two-vertex kill-graph is an edge, hence bipartite, hence parity-blocked for every input class admitting Liouville twists. The escape is not a cleverer switch but an input class in which the twist is inadmissible — the exceptional-character regime again. Two disjoint analyses, one door.

3.5 The price list

If the debt is paid at level θ, the consequences are fully computed; all eight $m = 1$ crossings carry exact-rational certificates, and H(k) is Engelsma-exact.

θ	2/θ	k pure (certified M_k)	k with ε	$H_1 \leqslant$ pure / ε
1/2 (Bombieri – Vinogradov)	4	54	50	270 / 246 (unconditional)
4/7 (BFI)	3.5	31 (3.502015⋯)	29 (3.519881⋯)	140 / 130
7/12 (Maynard II, linear sieve)	24/7	29 (3.443305⋯)	26 (3.433616⋯)	130 / 114
3/5 (Maynard II)	10/3	26 (3.350647⋯)	23 (3.334616⋯)	114 / 94
5/8 (Pascadi)	3.2	22 (3.207656⋯)	20 (3.222666⋯)	90 / 80
1 (Elliott – Halberstam)	2	5 ($M_5 = 2.007080$)	—	12

The missing statement, in the form we would want to see proved:

> **(E_θ).** Fix an admissible tuple H, an index i, and $A, \varepsilon > 0$. For coefficient systems $c_q(a)$ jointly well-factorable with the residue selection — for every factorisation $q = q_1 q_2$ (resp. $q_1 q_2 q_3$) with $\prod Q_j = x^{\theta-\varepsilon}$ one can write $c_q(a) = \prod_j \gamma_j(q_j, a \bmod q_j)$ *with* $|\gamma_j| \leqslant 1$ *and* $a \bmod p \in \{h_i - h_j \bmod p : j \neq i\}$ *for all* $p \mid q$ — one has $\sum_{q \leqslant x^{\theta-\varepsilon}} \sum_{a \in A_i(q)} c_q(a) \cdot E(x; q, a) \ll_{H,A,\varepsilon} x (\log x)^{-A}$.

($E_{4/7}$) is "BFI Theorem 10, uniform over the CRT residue system of a fixed tuple polynomial". Polymath8a's MPZ[ϖ,δ] is its absolute-value cousin, proved by the same dispersion-plus-Deligne technology but only to level ≈ 0.5286 and for densely divisible moduli, so (E_θ) interpolates two proved endpoints rather than crossing a parity barrier.

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4. Stopping times: the finite de Bruijn–Newman depth

4.1 Why the alternative hypothesis is worth attacking

Montgomery's pair correlation conjecture says the normalised gaps between zeros of ζ follow the GUE law. Half a century later it remains open, and the reason is sharper than "it is hard". The **alternative hypothesis** (AH) is the scenario in which the normalised gaps all lie asymptotically in $(1/2)\mathbb{Z}$ — the zeros sitting on a half-integer lattice rather than fluctuating like a random matrix spectrum. AH is not idle: it is consistent with everything currently provable about zeros, and it has real arithmetic consequences (it would force strong bounds on class numbers, and would be incompatible with certain conjectures on Landau – Siegel zeros). It is precisely the scenario that the known techniques cannot exclude.

Tao's blog article *The alternative hypothesis for unitary matrices* makes the obstruction concrete by moving it into random-matrix theory, where it can be computed with. Define the **ACUE** — the alternative circular unitary ensemble — as the measure on N-point configurations $C \subset \{2N\text{-th roots of unity}\}$, $|C| = N$, with

$\mu_{ACUE}(C) = |\Delta(\zeta^C)|^2 / (2N)^N$, Δ = Vandermonde, $\zeta = e^{2\pi i/2N}$.

This is a lattice-supported measure whose eigenvalue gaps are all multiples of half the mean spacing — a perfect finite model of AH. Tao's point is that ACUE reproduces the CUE two-point correlation exactly, so any argument that only sees pair statistics cannot distinguish them, and he asks what does. The natural programme — find the first moment or correlation at which they differ — has been running ever since, and it keeps colliding with the same difficulty: the set of measures matching a given list of moments is a large convex body, so ruling out ACUE itself never rules out its neighbours. One must rule out a *fibre*, not a point.

There is a second thread. The Riemann ξ function has a canonical heat deformation ξ_t, and de Bruijn and Newman showed there is a constant Λ_{dBN} such that ξ_t has only real zeros exactly when $t \geqslant \Lambda_{dBN}$. The Riemann hypothesis is $\Lambda_{dBN} \leqslant 0$; Rodgers and Tao proved $\Lambda_{dBN} \geqslant 0$. So RH, if true, is *barely* true: ξ sits exactly on the boundary of heat stability, which is Newman's dictum that the Riemann hypothesis, if true, is only just true. Crucially, Rodgers and Tao's proof is a statement about *dynamics*: assuming $\Lambda_{dBN} < 0$ forces the zeros, run backwards, into an increasingly rigid local equilibrium — a clock — which contradicts what is known about their fluctuations.

This paper joins the two threads. The second suggests the observable that resolves the first.

4.2 How large the blind spot really is: explicit mimicker families

Before looking for a distinguishing statistic it is worth measuring the enemy. We constructed, for each N, the exact convex body of probability measures on the ACUE support matching every balanced moment $E[p_\lambda \bar{p}_\nu]$ of degree $\leqslant N$ — the *mimicker fibre*. Everything here is exact (rational Vandermonde masses, rank computed with a spectral gap of 10^6 or better).

The fibre. Its affine dimension at the level of rotation orbits is

N	3	4	5	6	7	8	9
dim of mimicker fibre	0	0	2	10	80	403	1804

So for $N \leqslant 4$ the moment data pins the measure — ACUE is rigid and any statistic distinguishes it. From $N = 5$ the fibre opens, and it grows explosively. At $N = 8$ there are 403 independent directions in which one may deform ACUE without changing a single balanced moment of degree $\leqslant N$.

An explicit family for every N. Inside the fibre there is a clean closed-form subfamily. Let $X(C) = \Sigma_{c \in C}\, c \bmod N$ be the centre-of-mass class and set

$$q_g(C) = \mu_{ACUE}(C) \cdot g(X(C)),\ g : \mathbb{Z}/N \to \mathbb{R}_{\geqslant 0}.$$

Then q_g matches all balanced moments of degree $\leqslant N$ **iff** $E[g] = 1$ and the Fourier coefficients $\hat{g}(\pm 1)$ vanish; all other frequencies are free. Hence a family of affine dimension $\mathbf{N - 3}$, nonempty for every $N \geqslant 4$ and verified directly at $N = 5, 6, 7, 8$ (null dimensions 2, 3, 4, 5; worst balanced-moment error 10^{-12}; positivity radius in random null directions up to 1.0 – 2.2, so genuinely positive measures, not signed deformations).

N	family dim	forced-zero frequencies	free frequencies
5	2	$j = \pm 1$	$j = 2$
6	3	$j = \pm 1$	$j = 2, 3$
7	4	$j = \pm 1$	$j = 2, 3$
8	5	$j = \pm 1$	$j = 2, 3, 4$

A second, representation-theoretic family. Writing $D_r(C) = \prod_{c \in C} \zeta^{rc} = \det(U_C)^r$ for the determinant character, the tilts

$$q(C) = \mu_{ACUE}(C) \cdot [1 + c \cdot \mathrm{Re}(\eta\, D_r(C))],\ |\eta| = 1,\ 0 \leqslant c \leqslant 1,$$

also lie in the fibre. These have an exact quantum-mechanical description: μ_{ACUE} is the Born distribution $|\langle C \mid (\wedge^N F)A_0\rangle|^2$ of a single Slater determinant (F the DFT matrix, A_0 the Fermi sea $e_0 \wedge \cdots \wedge e_{N-1}$) — a re-proof that ACUE is a projection determinantal process — and the tilted family is the interference pattern of *two* shifted Fermi seas, a coherent superposition $(|\Omega_0\rangle + a|\Omega_r\rangle)/\sqrt{(1+a^2)}$ with $c = 2a/(1+a^2)$. In Plücker coordinates ACUE is a point of the Grassmannian $Gr(N, 2N)$ and the mimickers are points of its first secant variety $\sigma_2(Gr(N,2N)) \setminus Gr(N,2N)$: the minimal non-Gaussian deformations of a fermionic Gaussian state.

How invisible are these directions? At $N = 8$, of the 403 fibre directions, 401 are invisible to *every* pattern count of window width $\leqslant 2$, 397 to width $\leqslant 3$, 396 to width $\leqslant 4$, 392 to width $\leqslant 5$, and 383 to width $\leqslant 6$. The blind spot is not a thin exceptional set; it is almost the whole fibre.

And the flow does not help. Writing $P(z) = \det(I - zU) = \Sigma_j\, a_j z^j$, the finite analogue of the de Bruijn – Newman deformation is

$$P_r(z) = \Sigma_j\, a_j\, r^{j(N-j)} z^j,\ t = \log r,$$

which is *diagonal in the coefficients*. Consequently every balanced moment of degree $\leqslant N$ is a t-dependent linear combination of frozen quantities, so it stays frozen along the **entire** trajectory — we verified this to $4.5 \cdot 10^{-16}$ on explicit mimickers. Evolving the moments you already have is provably futile, and so is adding one more.

This is the situation the rest of the section is about: a large, explicitly parameterised family of impostors, invisible to a whole algebra of observables, and invisible to that algebra for all time.

4.3 The observable

Define the **finite depth**

$-\Lambda(U) = t^*$, the first time (going backwards, $t < 0$) at which the discriminant of P_t vanishes,

i.e. the first collision of two zeros. Because the flowed polynomial stays self-inversive, a simple zero cannot leave the unit circle without first colliding, so this is well defined. Equivalently the zeros move by the attracting circular Coulomb dynamics

$\dot{\theta}_j = -\sum_{k \neq j} \cot((\theta_j - \theta_k)/2)$,

derived from $\partial_t P = (ND - D^2)P$ with $D = z\, d/dz$. The right way to think of $-\Lambda$ is geometric:

> **$-\Lambda(P)$ is the distance from P to the discriminant hypersurface along the canonical heat ray.**

Static moments are coordinates *along* the mimicker fibre; Λ is a coordinate *transverse* to it, pointing at the boundary. It is not a polynomial statistic of any degree — it is a first-passage time — and that is exactly why the fibre's freedom does not protect it.

4.4 The separation

Two independent computations, one exact and one Monte Carlo, settle the scaling.

ACUE (exact). Complete enumeration of all rotation orbits for $N = 3, \cdots, 10$ — 13,132 orbits, 184,756 configurations at $N = 10$ — with exact Vandermonde masses, validated to 40 digits.

- $P(\text{clock}) = 2^{1-N}$ **exactly**. By Cauchy – Binet, $\sum_{|C|=N} |\Delta(C)|^2 = \det(AA^*) = (2N)^N$ for the DFT matrix A, while each of the two clock configurations has $|\Delta|^2 = N^N$. Clock polynomials are $1 - cz^N$, hence flow-invariant, hence have $\Lambda = -\infty$: they are the exact stationary atoms.
- For every non-clock configuration the minimal gap is exactly π/N (pigeonhole: all gaps $\geqslant 2\pi/N$ forces the clock).
- $-\Lambda^{ACUE} \asymp N^{-2}$, with fitted exponent **−2.0009**; $N^2(-\Lambda)$ is supported in $\approx[1.31, 1.99]$. The alternative $N^{-8/3}$ is decisively excluded.
- In all 13,130 non-clock orbits the first collision occurs at a pair that was already adjacent at $t = 0$ — zero exceptions.

CUE (Monte Carlo, N up to 256). $-\Lambda^{CUE} \asymp N^{-8/3}$, fitted exponent -2.678 ± 0.016. More than the exponent, the *law* is parameter-free:

$8N^{8/3}(-\Lambda) \Longrightarrow G^2$, where $P(G > x) = \exp(-x^3/72\pi)$

is the sine-kernel smallest-gap law. The constant 72π was derived from the kernel, not fitted; measured 229 – 236 against 226.2, with KS distances 0.035 – 0.041 and median 29.6 – 30.5 against the prediction $(72\pi \ln 2)^{2/3} = 29.08$.

The ratio of depths therefore grows like $N^{2/3}$ — the two hypotheses do not merely differ, they sit in **different universality classes of subcriticality**. The $10\times$ separation point is $N^* \approx 165$, matching an a-priori estimate of ≈ 160.

The interpretation is the striking part. ACUE's defect is not that it is fragile in some direction; it is that it is **too robust**. CUE is barely stable under reverse heat flow, in exactly Newman's sense; ACUE is over-equilibrated, sitting far from the discriminant because its lattice rigidity forbids the rare very close pairs ($\delta_{min} \asymp N^{-4/3}$) that CUE produces. A fake RH universe survives too long under a natural deformation.

4.5 What the exponents mean

The numbers 8/3 and 2 are not two readings on a dial; they encode a structural difference, and it is worth unpacking exactly what.

Λ measures a distance to a hypersurface. Let R_N be the set of degree-N polynomials with all roots on the circle, and $D_N = \{\mathrm{Disc} = 0\}$ the discriminant hypersurface, which is exactly the boundary of R_N: a configuration leaves the "all roots on the circle" locus precisely by two roots first colliding. The heat flow gives a canonical vector field on coefficient space. Then

$-\Lambda(P)$ = how far one must travel from P, against the heat field, to reach D_N.

So Λ is a *margin*: the distance from a configuration to the failure of its own Riemann-hypothesis analogue. This is the exact finite-dimensional shadow of the de Bruijn – Newman picture, in which $\Lambda_{dBN} \leqslant 0$ is RH and $\Lambda_{dBN} = 0$ says ξ sits on the boundary.

Why the exponent is a fingerprint of level repulsion. For an isolated close pair at angular gap δ, the two-body reduction of the Coulomb dynamics gives collision time $\delta^2/8 + o(\delta^2)$ — a purely local law we confirmed to 10^{-7} in the continuum regime, and exactly at N = 2 where $-\Lambda = -\log\cos(\delta/2)$. Hence the depth is governed by the *smallest gap in the configuration*, and the smallest gap is governed by the level repulsion exponent:

$p(s) \sim c\, s^\beta \Longrightarrow$ min of $\approx N$ gaps $\asymp N^{-1/(\beta+1)}$ (normalised) $\asymp N^{-1-1/(\beta+1)}$ (angular) $\Longrightarrow -\Lambda \asymp N^{-2-2/(\beta+1)}$.

Reading the chain for the two cases in question:

- **CUE** ($\beta = 2$). Quadratic repulsion still permits rare, anomalously close pairs: the smallest of N gaps is of order $N^{-4/3}$, a full factor $N^{-1/3}$ *below* the mean spacing N^{-1}. One such accident is enough, and it produces depth $N^{-8/3}$.
- **ACUE** ($\beta = \infty$). The lattice forbids accidents. Every gap is a multiple of π/N, so the smallest gap is *exactly* π/N — deterministically, with no fluctuation at all (we prove this: any configuration all of whose gaps exceed $2\pi/N$ must be a clock). Depth N^{-2}.

So the ratio $-\Lambda^{ACUE} / -\Lambda^{CUE} \asymp N^{2/3}$ is precisely the ratio between "the closest pair among N random ones" and "the closest pair when closeness is quantised". The exponent gap *is* the extreme-value statistics of the spectrum, converted into a single scalar.

The interpretive punchline. One might have expected a fake RH universe to be caught out by being fragile somewhere. The opposite happens. ACUE's defect is that it is **too stable**: it sits $N^{2/3}$ times farther from the discriminant than a genuine random-matrix spectrum does. A true GUE/CUE world is real-rooted but *microscopically fragile* — it is always within $N^{-8/3}$ of losing the property, because it always contains one near-collision. ACUE is over-equilibrated: it satisfies its RH-analogue far too robustly, because its rigidity rules out the near-collisions.

This is the finite-N precise form of Newman's dictum. "RH, if true, is only just true" is not a piece of rhetoric; it is a statement about a scaling exponent, and the alternative hypothesis fails it by getting the exponent wrong in the direction of excessive safety.

It also reframes what one must prove about ζ. The static programme asks for a correlation statistic at frequencies beyond the reach of current technique. The dynamic programme asks instead: *is the zeta zero configuration, locally, as fragile as a random matrix?* Those are different questions, and § 4.10 argues the second may be the easier one — because refuting ACUE needs only that the depth be **$o(N^{-2})$**, with any rate whatsoever, not the full $N^{-8/3}$ law.

4.6 A universality law

The mechanism suggests, and the data confirm, a general law. If the normalised nearest-neighbour gap density behaves like $p(s) \sim c\, s^\beta$ as $s \to 0$, then the smallest of $\approx N$ gaps is of order $N^{-1/(\beta+1)}$, the physical gap is $N^{-1-1/(\beta+1)}$, and since an isolated pair collides in time $\delta^2/8 + o(\delta^2)$,

> **Dynamic Newman universality.** $-\Lambda_N \asymp N^{-2-2/(\beta+1)}$.

ensemble	β	predicted	measured
COE	1	−3	−3.064 (local slopes −3.03 ⋯ −3.10)
CUE	2	−8/3 = −2.667	−2.678 ± 0.016 ($N \leqslant 256$)
CSE	4	−12/5 = −2.4	−2.510 ($N \leqslant 64$, still drifting)
ACUE lattice	∞	−2	−2.0009 (exact)

All measured slopes are slightly steeper than predicted, in the direction and of the size of the finite-N drift independently calibrated at $\beta = 2$, where the larger N range shows the local slope converging to the prediction from below. The circular β-ensembles were sampled by the Killip – Nenciu CMV construction, validated against Haar CUE (KS 0.005) and against direct COE = VV^T (KS 0.004). The localisation assumption behind the law — that the first collision is governed by the minimal initial gap — holds in 95 – 99% of samples, and in 100% of ACUE configurations.

So the depth is a scalar fingerprint of the microscopic repulsion exponent. That is a clean random-matrix statement independent of anything about zeta.

4.7 A configuration constant, computed two ways

Inside the ACUE law there is a distinguished stratum. Take the alternating clock (the zeros of $z^N + 1$), delete the point $e^{-i\pi/N}$, and insert 1; the gap pattern becomes 1, 2, 2, ⋯, 2, 3 in half-lattice units and the polynomial is exactly

$$P_N(z) = (z - 1)(z^N + 1)/(z - e^{-i\pi/N}).$$

Two completely independent routes give the same asymptotic depth.

Lattice route. Our exact solver on this configuration, $N = 3 \cdots 20$, gives $N^2(-\Lambda)$ increasing monotonically 1.41473945, 1.41821551, ⋯, 1.41963827 at $N = 20$.

Continuum route. Setting $t = -s/N^2$, $z = e^{iu/N}$ and passing to the limit yields the local function

$$G_s(u) = 2\cos(u/2) - 2\pi \int_0^{1/2} e^{s(1/4 - y^2)} \cos((\pi + u)y)\, dy,$$

whose first double zero ($G = \partial_u G = 0$) is at

$s^* = 1.419640342\cdots$, $u^* = 1.812942145\cdots$, with $G''(u^*) = -0.3767 \neq 0$ (a generic tangency).

The two agree to $2 \cdot 10^{-6}$ at $N = 20$, consistent with an $O(N^{-2})$ approach. This is Proposition-grade: a lattice enumeration and a transcendental double-root equation meeting to six digits.

One correction to the record. It is tempting to read s^* as the limit of the ensemble median, because for $N \leqslant 7$ the two coincide: the median of $N^2(-\Lambda)$ rises 1.41474, 1.41908, 1.41950. The complete enumeration shows the median **turns around at $N = 7$** and falls — 1.41822 ($N = 8$), 1.41520 ($N = 9$), 1.41277 ($N = 10$) — as the support spreads from [1.4147, 1.8246] to [1.3146, 1.9855]. At small N the single-dislocation orbit dominates the ACUE measure; from $N = 8$ the ensemble outgrows it. So s^* is a *configuration* constant belonging to the support of the limit law, not the median's limit. The neighbouring constant for a well-separated defect pair is $N^2(-\Lambda) \to 1.46946$ (i.e. $\rho_\infty = 8N^2(-\Lambda)/\pi^2 \to 1.19120$), the other end of the same defect family.

4.8 The depth escapes the freezing theorem

The point of the whole exercise. Return to the mimicker families of § 4.2 — the 403-dimensional fibre at $N = 8$, the centre-of-mass family q_ACUE · g(X), the secant tilts by det(U)^r — every one of which is frozen to the moment algebra for all time. Λ separates them at $O(1)$:

- an $N = 5$ mimicker in the $\mathbb{Q}(\sqrt{5})$ family moves the clock atom from 0.0625 to 0.1398 and shifts $E[N^2(-\Lambda) \mid \text{non-clock}]$ by -0.093;
- at $N = 8$, mimicker families move the entire Λ-law by total variation 0.12 – 0.24, with the dependence concentrated on the centre-of-mass class $X = 0$;
- linear-programming tomography over the exact fibre at $N = 6$ (affine dimension 10 at orbit level) gives $E[N^2(-\Lambda) \mid \text{non-clock}]$ ranging over **[1.3610, 1.4770]** against ACUE's 1.4336, while the clock atom — the mass at $\Lambda = -\infty$ — can be pushed from 0 to **0.0975**, i.e. tripled from ACUE's 0.03125, with every constrained moment held fixed.

The conclusion deserves stating twice.

> **Moment matching does not imply stopping-time matching.** Two ensembles can be indistinguishable to a whole algebra of observables, and remain indistinguishable along an entire natural flow, yet have first-passage laws in different scaling universality classes.

4.9 A classification of counterexamples, and two computable criteria

Not every impostor is caught by the same instrument. The counterexample families that arose in this project fall into three classes, and — this is the useful part — membership is decidable by two computations rather than by taste.

Criterion I (transversality). Fix a configuration X and a bandwidth r; let $M_r(X) = (p_1, \cdots, p_r)$ be the moment map and $\tau = -\Lambda$ the depth. The class is caught by ordinary Λ precisely when $\ker DM_r \not\subset \ker D\tau$, i.e. when grad τ has a component off the row space of DM_r. We compute the relative residual directly.

configuration	r = 1	r = 2	r = 3	r ⩾ N/2
single dislocation, N = 6	0.976	0.800	—	rank saturates
single dislocation, N = 8	0.991	0.941	0.737	rank saturates

configuration	r = 1	r = 2	r = 3	r ⩾ N/2
random ACUE lattice, N = 7	0.996	0.963	0.569	rank saturates
generic non-lattice, N = 7	0.998	0.926	0.195	rank saturates

Read this correctly. While rank $DM_r < N$ the moment map has a kernel and the question is meaningful; in that entire regime the answer is decisively yes — 80% to 99% of the depth gradient lies in directions the first one or two moments cannot see. Once $r \approx N/2$ the rank saturates at N, ker $DM_r = 0$, and there are no moment-blind directions left to test: the residual drops to 10^{-16} not because Λ fails but because the question becomes vacuous. So: **moment-null does not imply depth-null**, quantitatively, for every bandwidth at which the distinction exists.

Class I — caught by Λ directly. Anything whose defect changes the geometry of the closest pair. This includes the collision-stratum families (a configuration sitting on $\delta = 0$ has $\Lambda = 0$ outright, the same mechanism by which function-field Newman constants are pinned by double roots), the half-lattice and PairCeiling adversaries (§ 4.5: a hard lower spacing $\delta_{min} \gtrsim c/N$ forces $-\Lambda \gtrsim N^{-2}$, incompatible with CUE's $N^{-8/3}$), and the centre-of-mass and secant families of § 4.2, which move the depth law by total variation 0.12 – 0.24 while every balanced moment stays frozen.

Class II — invisible to Λ, caught by the marked depth. Λ is a function of the characteristic polynomial alone, so it cannot see eigenvectors. Two isospectral matrices have *identically* the same depth; we verify this to machine zero ($\tau(G_1) = \tau(G_2) = 0.068725421516$, difference $8 \cdot 10^{-17}$). Counterexamples built to match rank, trace and Hilbert – Schmidt norm while differing in a directional Schur complement live here. The repair is a **marked depth**: deform $G \mapsto G + \eta\, uu^*$, transport to the circle by Cayley, and differentiate,

$$\chi(G; u) = \partial\eta \left(-\Lambda(U\{\eta,u\}) \right) |_{\{\eta=0\}}.$$

It separates the isospectral pair immediately — median $|\chi(G_1;u) - \chi(G_2;u)| = 0.081$ over random marks, against $|\chi|$ itself of order 0.01 – 0.2.

What drives χ is not what one would guess. The determinant lemma $\det(zI - G - \eta uu^*) = \det(zI - G)(1 - \eta \cdot u^*(zI - G)^{-1}u)$ suggests the marked depth should track the directional resolvent, and hence blow up wherever the inverse-Gram pathology is worst. It does not. Rotating the mark onto the smallest-|eigenvalue| direction drives the resolvent $u^*(z_0 I - G)^{-1}u$ from 1.615 to 6.667 = $1/|\lambda_{min}|$ while χ *falls* from 0.0267 to 0.0028 — an order of magnitude the wrong way. The correct mechanism is local to the collision. Rank-one perturbation moves λ_j by $\eta|\langle u, v_j\rangle|^2$, the Cayley map contributes $d\theta/d\lambda = -2/(1 + \lambda^2)$, and the depth responds through the critical gap, giving

$$\chi(G; u) \approx \rho \cdot (\delta/4) \cdot \left(c_a |\langle u, v_a\rangle|^2 - c_b |\langle u, v_b\rangle|^2 \right), \quad c_j = -2/(1 + \lambda_j^2),$$

where (a, b) is the pair that collides first, δ their gap, and ρ a background renormalisation of the same kind as the ACUE constant of § 4.7. Measured against random marks the correlation with this formula is **0.9958** (best-fit $\rho = 1.72$ in the instance tested, against $\tau/(\delta^2/8) = 1.28$ for the value itself — derivative and value renormalise differently). The control confirms the mechanism: a mark orthogonal to *both* critical eigenvectors gives $\chi = 2.8 \cdot 10^{-3}$, two orders below typical.

So the marked depth is a **differential alignment detector**: it measures how asymmetrically the mark couples to the two eigenvectors about to collide, and vanishes on marks that couple to them equally. That is a sharper instrument than a resolvent probe, and it says exactly which marks to choose when testing a Class II family — those overlapping the critical pair, not those exploring the ill-conditioned directions.

Class III — genuinely immune to the linear flow. Suppose the observable algebra misses a direction v. Because the heat operator is diagonal in the coefficient basis, $H_t v$ generally stays in the same hidden sector, so if $F(X) = F(X + \varepsilon v)$ for all observables F, then $F(H_t X) = F(H_t(X + \varepsilon v))$ for all t as well. **A linear flow does not destroy linear invisibility** — this is a genuine negative result and it is why "evolve the moments you have" fails, as § 4.2 records.

The point is that Λ is not in that argument's scope. It is not a linear observable transported by the flow; it is the first hitting time of the orbit against the discriminant variety. Two orbits can live in the same invariant hidden sector and still stand at different distances from $D = \{\mathrm{Disc} = 0\}$. Class III is therefore not "counterexamples the depth cannot see" but "counterexamples for which the *linear* dynamic argument gives nothing" — and Criterion I is what decides whether the nonlinear stopping time recovers them.

The programme in one line. With static observables M and fibre $F_m = \{X : M(X) = m\}$, the statement to prove is that τ restricted to F_m is generically non-constant, with marks added, $\Lambda_{u_1}, \cdots, \Lambda_{u_k}$, until $\cap_j \ker D\Lambda_{u_j} \cap \ker DM = \{0\}$. Static moments supply coarse coordinates; marked Newman constants supply the directional ones. The tables above are the first evidence that this is achievable rather than merely well-posed.

4.10 The depth is the Lagarias–Rodgers hard core in another coordinate

The sharpest consequence of the depth law is that it is not a new statistic competing with the existing formulation of the alternative hypothesis — it is that formulation, in coordinates where our machinery computes exactly.

Lagarias and Rodgers study the class $\mathcal{T}_1$ of point processes that mimic the sine process at coordinatewise bandwidth one — for every r and every Schwartz η with supp $\hat{\eta} \subset [-1,1]^r$, the factorial correlation statistics agree — and define

$\mu = \sup\{ c : \text{some } X \in \mathcal{T}_1 \text{ has minimum spacing} \geqslant c \}$, mean spacing normalised to 1.

They prove $\mu \geqslant 1/2$ by the randomly shifted half-lattice (which is exactly ACUE), record the pair-only upper bound $\mu \leqslant 0.606894\cdots$, and conjecture $\mu = 1/2$. The alternative hypothesis is precisely the assertion that zeta's zeros realise a hard core of 1/2.

The bridge. On the circle with N points the mean angular spacing is $2\pi/N$, so a hard core of c mean-spacings is $\delta_{min} = 2\pi c/N$. Combined with the local collision law $-\Lambda = \rho \cdot \delta_{min}^2/8$ this gives

$N^2(-\Lambda) = \rho \cdot \pi^2 c^2 / 2$, and at the alternative-hypothesis value $c = 1/2$, $N^2(-\Lambda) = \rho \cdot \pi^2/8$.

Here $\rho \geqslant 1$ is the background delay factor of § 4.5 and § 4.7. Every number we computed is this identity read in one direction or the other. The ACUE quantiles at $N = 3, \cdots, 10$ give

N	min $N^2(-\Lambda)$	ρ_{min}	median	ρ_{med}	max	ρ_{max}
6	1.353146	1.0968	1.419374	1.15050	1.952629	1.5827
8	1.330383	1.0784	1.418216	1.14956	1.976122	1.6018
10	1.314614	1.0656	1.412774	1.14515	1.985458	1.6094

and the single-dislocation constant of § 4.7 is exactly the median branch: $s^* = 1.41964034\cdots = \rho_\infty \cdot \pi^2/8$ with $\rho_\infty =$ **1.1507015⋯**. The minimising family $\{0,\cdots,N-3\} \cup \{N+3, N+4\}$ — a maximally packed block beside a void — has ρ falling slowly (1.0656, 1.0611, 1.0579, ⋯, 1.0495 at N = 18) toward a limit near 1.03 – 1.05; the block is nearly stationary under the flow, so only its ends move, which is why it resists collapse and keeps $\rho > 1$. The maximising symmetric three-block family gives $\rho \to 1.6094$, i.e. $N^2(-\Lambda) \to 1.9861$.

The equivalence of extremal problems. Define the depth extremum $\mu_\Lambda = \sup\{ \liminf N^2(-\Lambda) : X \in \mathcal{F}_1 \}$. The identity gives, in both directions,

$\mu_\Lambda \geqslant \pi^2\mu^2/2$ and $\mu \leqslant \sqrt{(2\mu_\Lambda)}/\pi$.

These are not loose: LR's published bound $\mu \leqslant 0.606894\cdots$ is *exactly* the depth bound $\mu_\Lambda \leqslant \pi^2(0.606894)^2/2 = 1.8177\cdots$, and the conjecture $\mu = 1/2$ is exactly $\mu_\Lambda = \pi^2/8 \cdot \rho$ for the half-lattice's own ρ. So **any improvement of the depth bound is an improvement of the Lagarias – Rodgers bound**, at the explicit exchange rate above; a depth bound of 1.29 — barely below what the ACUE configurations themselves realise — would already give $\mu \leqslant 0.511$.

A falsifiable threshold for the alternative hypothesis. Running the inequality in the direction that costs nothing:

> **If the zeros of ζ have, in a window at height T with $N = (\log T)/2\pi$ zeros, local Newman depth satisfying $\liminf N^2(-\Lambda) < \pi^2/8 = 1.2337\cdots$, then the alternative hypothesis is false.**

This uses only the hard-core consequence of AH — not the lattice structure, not any correlation beyond it — and it replaces the vague target "beat the bandwidth wall" with one number. Compare the two sides: AH predicts $N^2(-\Lambda) \geqslant 1.2337$ with the half-lattice's actual typical value 1.41964, while CUE predicts $N^2(-\Lambda) \asymp N^{-2/3} \to 0$. The gap between prediction and threshold is a factor 1.15, and between CUE and threshold it is unbounded.

Why this reformulation may be easier to attack. The LR extremum is a constraint on a *minimum*, a hard combinatorial quantity; Palm-type certificates — the row-sum square $\mathrm{Var}^0\_\mathrm{sine}(S\_f) \leqslant (M\_h - A\_f)(A\_f - m\_h)$ for band-limited f, and its multiwindow quadratic generalisation on the packing body K_h — apply naturally to smooth functionals and awkwardly to a minimum. The depth is smooth in the configuration (§ 4.9 computes its gradient), it has a variational characterisation as a distance to a hypersurface, and its derivative has the explicit critical-pair law of § 4.9. Transporting the Palm/packing certificates from the row sum to the depth is therefore the natural next attempt, and unlike the row-sum searches — which failed for an exact local reason, the pattern $\{-h, 0, h\}$ already forcing $S_0 = 2f(h) > 9/7$ for the first nonnegative Fejér profile — the depth has no such immediate obstruction.

One missing lemma. The bridge is rigorous except for $\rho \geqslant 1$, i.e. that the background never *accelerates* the first collision below the isolated two-body time $\delta\_\mathrm{min}^2/8$. Every computation in this project supports it: on the CUE side the correction is positive with median $0.598 \cdot N^{-0.729}$, on the ACUE side $\rho \in [1.049, 1.610]$ over all configurations and sizes tested, and the mechanism is clear — the neighbours of a close pair pull its members outward. We have not proved it. It is the single lemma standing between the numerology above and a theorem, and it is exactly the same localisation estimate that Open Problems 1 and 3 require.

4.11 What this could mean for zeta, stated carefully

We claim no theorem about ζ. But the structure suggests a target that is strictly weaker than what the static programme requires. ACUE predicts $N^2(-\Lambda) \asymp 1$; CUE predicts $N^2(-\Lambda) \asymp N^{-2/3} \to 0$. Therefore, to exclude the ACUE universality class for zeta one does not need the full 8/3 law: it suffices to prove that the local depth of ξ under the true de Bruijn – Newman flow, suitably normalised at height T, satisfies

$(\log T)^2 \cdot D\_T \to 0$,

with any rate. That is a substantially weaker statement than a full extreme-gap theorem, and Rodgers – Tao's method — averaged information plus a heat-flow energy argument, rather than the full GUE law — is evidence that such intermediate statements are reachable. Making the truncation rigorous (one may not simply cut a window of zeros and call the result a polynomial; the Polymath15 machinery for computing H_t is the honest route) is the first obstacle.

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5. How this was done

The methodology is not incidental, so we describe it plainly.

Fleets, not oracles. Work proceeded in rounds. Each round posed one question, and between three and ten language-model agents attacked it in parallel from deliberately different angles, each writing and running its own code, each producing a report tagged line by line as *proved*, *computed*, *heuristic* or *conjecture*. Agents shared a written context file stating the current state of knowledge, including known errors. They did not share conclusions; convergence of independent agents was treated as evidence, and divergence as a bug report.

Adversarial defaults. Every context file instructed agents that the default assumption is that the idea has already been tried and failed, and that their first job is to find the reason. The prompt for the prime-gap round said outright that any improvement to 246 was 99% likely to be a misread constraint. This is not modesty; it is calibration. Of the phenomena that looked like discoveries during this project, most were arithmetic or normalisation errors, and the ones that survived did so because they were attacked first by their own authors.

Exact arithmetic at every threshold. Nothing near a decision boundary was accepted in floating point. The Maynard – Tao certificates are ball-arithmetic with outward rounding; the variational crossings are exact-rational Rayleigh quotients of rounded eigenvectors; the ACUE enumeration uses exact Vandermonde masses with 40-digit spot checks; the signed-sieve identity was verified in exact rationals on a thousand random weights. The single most common failure mode in machine-generated mathematics is a plausible float, and the remedy is cheap.

Two implementations or it did not happen. Every headline number here was produced twice by independent code, usually by agents that could not see each other's work: the record tuples were re-verified by a second admissibility checker; the CUE depth was computed by an ODE integrator and by coefficient bisection agreeing to 10^{-6}; the single-dislocation constant was obtained from a lattice enumeration and from a transcendental equation, agreeing to six digits. Where two implementations disagreed — an engine crossing at k = 15,856 against another at 29,500 — the discrepancy was tracked to its source before anything was claimed.

The human supplies the questions. Every genuinely new direction in this paper came from a human mathematician's judgement: to stop optimising 246 and look at H_2; to ask whether removing the square opens a phase; to propose the de Bruijn – Newman depth as a dynamic observable in the first place; to insist that 72π be factorised rather than fitted; to recognise that the single-dislocation configuration was the right object to compute exactly. The models supplied speed, breadth, exact arithmetic, and — importantly — the willingness to write and discard fifty scripts a day. The division of labour was not "human checks machine" but "human chooses the question, machine exhausts it, human reads the residue".

Negative results are the main product. Five walls, one no-go theorem, one empty phase, one refuted extrapolation. A method that only reports successes cannot be trusted about them. The most valuable single output of the signed-sieve round is that it is closed, and that we can say exactly what would open it.

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6. Open problems

1. **Rigorise the CUE depth law.** Prove 8N^{8/3}(−Λ) ⟹ G². The hard half — the smallest-gap limit law — exists in the literature (Ben Arous – Bourgade, Feng – Wei). What is needed is the localisation lemma: that the remaining N − 2

zeros perturb the two-body collision time by $1 + o(1)$. Our data show the correction is positive and of size $\approx 0.60 \cdot N^{-0.73}$.

2. **Make $s^* = 1.419640342\cdots$ a theorem** for the single-dislocation family, and compute the separated-defect constant $\rho_\infty = 1.19120\cdots$ exactly. The natural framework is an infinite clock with one localised defect under the Coulomb dynamics, i.e. a Calogero – Moser problem; the target is a closed form in terms of theta or Bessel data.
3. **The marked depth law.** Prove $\chi(G;u) = \rho \cdot (\delta/4) \cdot (c_a|\langle u,v_a\rangle|^2 - c_b|\langle u,v_b\rangle|^2)$ with the background constant ρ, and determine ρ for the lattice families. Empirically the correlation is 0.9958 (§ 4.9); what is missing is the same localisation estimate as in problem 1, differentiated.
4. **A transversality conjecture.** With static observables M and fibre $F_m = \{X : M(X) = m\}$, prove that $\tau = -\Lambda$ restricted to F_m is generically non-constant; equivalently $\ker DM \not\subset \ker D\Lambda$, with marks added until the intersection is trivial.
5. **Function-field depth.** Frobenius conjugacy classes give characteristic polynomials, hence depths. Does equidistribution of Frobenius push through to the depth law — a genuinely arithmetic Newman-depth universality theorem, rather than a random-matrix analogy? The $g = 1$ case is the exact $N = 2$ law in arithmetic clothing, $\Lambda = \log(|a_p|/2\sqrt{p})$.
6. **(E_θ), the tuple-residue well-factorable estimate**, for any $\theta > 1/2$. This is the single statement standing between the certified price list of § 3.5 and $H_1 \leqslant 130$.
7. **The k = 49 door.** Decide whether some legal Maynard – Tao variant has $M_{49} > 4$. Certified 3.930490592, float optimum 3.959325169, upper bounds closing only $\varepsilon \leqslant 0.00682$.

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All engines, certificates, tuples and audit scripts are in the accompanying repository.

Historical source 23: joint_context_v2

Source: `historical/riemann-rmt/joint_context_v2.md`. SHA-256: `ed977e3f56ffedd8e777663a44df40a4fc03734a8819967953f96f8bebb2576b`.

Shared context v2 — Anthropic method (from Lean source) × finite RMT fiber program

Read this fully before starting your task. You are one of 10 parallel agents; your final text is your report.

A. Anthropic's actual method (read from the real Lean repo, NOT paraphrase)

The full Lean 4 formalization is cloned at **/workspace/anthropics/zeta-23-lean** (read it!). Key facts extracted:

1. **Lemma R (rank – trace), k-form** (`Zeta23/ZeroSide/RankTraceMult.lean`): For P = Σ_j m_j v_j v_j* (on-line zero atoms, multiplicities m_j, near-unit vectors), Q Hermitian with $n_+(Q) \leqslant b$: `2c·tr(P+Q) − ‖P+Q‖_F² ≤ Σ_j k_c(m_j) + c²·b`, where **k_c(p)** = $c^2 - ((c-p)_+)^2$ (= 2cp − p² for p ⩽ c, saturates at c² for p ⩾ c) — a concave clipped secant. Decoding values: $k_2(1)=3$, $k_2(\geqslant 2)=4$ (⇒ simple-zero counts, c=2); $k_3(1)=5$, $k_3(2)=8$, $k_3(\geqslant 3)=9$ (⇒ distinct counts, c=3, gives 5/6). Proof: spectral split $Q = Q_+ - Q_-$, drop $\mathrm{tr}(PQ_+) \geqslant 0$, von Neumann trace inequality, elementary secants $(x-c/2)^2 \geqslant 0$, $(x-c)^2 \geqslant 0$.
2. **Tightness** (`Zeta23/ZeroSide/TightMult.lean`, `lemmaR_tight`): for on-line atoms with integer m_j ⩽ c on orthonormal vectors plus b pair-blocks of eigenvalue c, EQUALITY holds: 2c · tr(P+Q) − ‖P+Q‖_F² = Σ_j k_c(m_j) + c² · b. So the certificate cannot be improved using only (tr, ‖ · ‖_F, rank, inertia).
3. **Prime side**: tr G and tr G² = ‖G‖_F² of the Gabor/taper compression are computed unconditionally from the explicit formula (`Zeta23/PrimeSideA,B`, `Poisson.lean`, `Taper/`), with Montgomery – Vaughan Hilbert inequality (`Zeta23/MV/`) giving near-orthogonality of zero vectors (Gram ≈ I). Bandwidth-one support; Weyl perturbation for tails.
4. **The constant**: Theorem D optimal window (ThmD/Functional.lean) = Montgomery – Taylor: $v^*(s) \propto \cos(\sqrt{2}\, s)$, $q^* = \tfrac{1}{2}+(1/\sqrt{2})\cot(1/\sqrt{2})$, δ_MT = $2-q^*$ = 0.672500703679.
5. **PairCeiling** (`Zeta23/PairCeiling/Stability.lean`): stability inequality (two integrations by parts): any valid bandwidth-one certificate (c_0, r), $r \in C^1[0,1]$, against a configuration with grid form-factor masses s_j and simple fraction p: `c₀ + ∫₀¹ r(x)·x dx ≤ p + |r(1)|·|D(1)| + |r′(1)|·|E(1)| + sup|E|·∫₀¹|r″|`. Instance at explicit 256-periodic marked law (`LawN256.lean`): every bandwidth-one certificate certifies ⩽ **0.6818287 + $2.55 \cdot 10^{-6} \cdot (|r'(1)| + \int |r''|)$**. The only unverified input is `EnclOK` (256 integer enclosures from an exact-rational certificate, hash-named, not in repo). Note 2.55e-6 ≈ $1/(6 \cdot 256^2)$ (grid/Bernoulli fingerprint).
6. **ξ′ theorems** (`Zeta23/XiPrime/`): same rank – trace device on the Farmer – Gonek/Montgomery argument for ξ′ gives unconditionally 0.85838 simple-on-line (flat window) and **0.86864 with a QUARTIC window** — so quartic windows already pay when the arithmetic input exists (ξ′ explicit formula supplies it).

B. Verified finite results (our program; verify scripts in this scratchpad)

- ACUE = rank-N Fourier projection DPP on ℤ/2N (Tao 2019); mimicker fiber (all laws matching balanced moments $E[p_\lambda \bar{p}_\nu]=\delta z_\lambda$, $|\lambda|=|\nu|\leqslant N$): dims 0,0,2,10,80,403,1804 for N=3..9. N=5 exact over $\mathbb{Q}(\sqrt{5})$.

- Rigidity: all 2-point (proved, aliasing) and 3-point (verified 1e-14) correlations frozen on the fiber; first freedom at 4-point. Fiber gap $\delta^*(N)$ for word 01010101: 0.0102, 0.0123, 0.0138, 0.0151 (N=6..9), transfer theorem: uniform gap ⇒ Lagarias – Rodgers non-uniqueness.
- **Nyquist conservation law** (verified): with marks $m_x \geq 0$, $\Sigma m = N$ on $\mathbb{Z}/2N$ and open rows $E|p_k|^2=k$ ($k<N$): $E\Sigma m(m-1) = (E|p_N|^2-N)/2N$; closing row k=N forces binary. Slack identity: $Es_1 - (N/2 + \csc^2(\pi/2N)/2N) = E\Sigma m_x m_{x+1} + E\Sigma h(m_x)$, h=0 on {0,1,2}; the combination is Nyquist-blind ($1+\cos \pi = 0$). Bound $Es_1/N \geq ½ + 2/\pi^2 = 0.70264$. Equality face E_N = hard-core {0,1,2}, $|E_N| = C(2N,N)$. Open: all-N attainment ("Nyquist equality law").
- **Fourth moments of the sine – Gram features** (verified N⩽6): for $V_k=|p_k|^2$ under ACUE, $\mathrm{Cov}(V_k,V_\ell) = k^2 1\{k=\ell\} - 2(k+\ell-N)_+$, $\lambda_{min} = 1$ exactly. Inradius $\rho_N \geq 1/(N^2-N/2)$; repair theorem: feature error $o(N^{-2})$ on the equality face ⇒ exact law with o(N) slack.
- Holonomy: DPP laws on g-cycles agreeing to order <g with $dTV = (2r)^g|\cos g\varphi-\cos g\psi|$ exactly. Homometry: marked pairs equal complete $|\mathrm{Fourier}|^2$, different simple fractions. Local blindness: exponential fibers can have zero projection on all fixed local words.
- **Phase twists** (new paper): multiplying by center-of-mass characters translates the Fermi sea; rank-4 twists are stochastically homometric to ACUE (equal Patterson autocorrelation at all frequencies, marginals through order 3, separated at 4-point); the full moment-null space has an exact Fourier – Slater description, exponential dimension; matched-block compressions are forced = identity (all residual in the cross block).
- **Christoffel trace hierarchy through degree 12** (new papers XI/XII): exact continuum moments of the sine – Gram matrix m9=11166011/151200, m10=83443081/554400, m11=852071287/2721600, m12=1033020076559/1556755200; sharp forced-rank (Christoffel at 0) R10 = 0.9385985⋯, R12 = 0.9482017⋯ (exact rationals; Gauss – Radau adversaries attain). Marked benchmarks: 80.39% (deg 6), 84.94% (8), 87.72% (10), 89.64% (12). Odd – even Schur law: odd trace = coupling coordinate (no standalone gain), next even trace = innovation energy, gain = squared ratio. Bandwidth: order-k collision channel has max Fourier weight $\lfloor k/2 \rfloor \cdot L$ (quotient-MaxCut). All-order N^{-2} parity (Gorenstein). Hard-edge conjecture: support [0,3], Christoffel $\Lambda_n(0) \sim (3n)^{-1}$.
- Third-moment escape (series synthesis): $E(M)_3 = r$ without cap gives NO gain (mass escapes to high marks); fourth factorial moment restores $p_1 - (2-q) \geq r^2/(s+2r)$.
- **Scalar-compression moment code** (Prop 8.1, series synthesis): if subspace V satisfies $P_V D_j P_V = b_j P_V$ for diagonal moment observables D_j, then every unit $v \in V$ gives an honest law $q_v(o) = |v(o)|^2$ matching all moments. (Quantum-marginal-style realization tool.)
- Inner-SOS ⊆ exact LP ⊆ outer-SDP sandwich; signed PSD pseudo-laws are not laws. Inertia alone gives no local gap: exponentially large word-invisible positive tangent exists (rank ceiling $2^{\ell-1}-1$ for length-ℓ words).
- Two constants NEVER to conflate: $\delta_{MT} = 0.6725007$ (zeta certificate) vs $½+2/\pi^2 = 0.70264$ (marked open-Nyquist model). Literal ACUE fiber is binary ($s_1 \equiv N$).

C. Environment

- python3 + numpy/scipy in this scratchpad; write scripts here, name by your topic. Exact rational/cyclotomic arithmetic via sympy (pip install sympy if needed).
- Lean repo at /workspace/anthropics/zeta-23-lean — READ the relevant files for your task (grep docstrings; they cite paper §§).
- arxiv.org / anthropic.com are egress-blocked; WebSearch summaries work.

- Report format: English, precise; tag every claim [PROVED] / [COMPUTED] (with script path + numbers) / [HEURISTIC] / [CONJECTURE]. State explicitly what is NEW relative to sections A/B above. End with the single most promising concrete next step.

Historical source 24: prime_gap_context

Source: `historical/riemann-rmt/prime_gap_context.md`. SHA-256: `f46b6cffca37508e2d90a966befbdc80dcc07d39aaf68654adb04cf1a7edc77f`.

Shared context — Round 4: attacking the bounded prime gap record $H_1 = 246$

You are one of 10 parallel agents. Read fully. Your final text is your report. English, tagged [PROVED]/[COMPUTED]/[HEURISTIC]/[CONJECTURE], scripts in this scratchpad, end with the single most promising next step. BE ADVERSARIAL WITH YOURSELF: the default assumption is that Polymath already tried your idea — your job is to find what they could not do in 2014 (modern SDP/SOS solvers, our new machinery, post-2014 arithmetic inputs) or to prove a clean no-go.

A. The state of the art (verified via search, August 2026)

- H_1 = **liminf(p_{n+1}−p_n) ⩽ 246** — Polymath8b (2014), unconditional, still the record. Under Elliott – Halberstam (EH): 12. Under generalized EH: 6 (parity blocks further).
- H_2 = **liminf(p_{n+2}−p_n) ⩽** 398,130; H_3 ⩽ 24,797,814; H_m ≪ e^{3.815m} — much less optimized than H_1 !
- Pipeline: DHL(k,2) ⟸ M-type variational threshold > 4/θ · ... — with Bombieri – Vinogradov level θ = 1/2, the requirement is **M_k** > **4** (pure) or **M_k^{[ε]}-variant** > 4 (ε-trick). Then H_1 ⩽ H(k) = diameter of the narrowest admissible k-tuple.
- Numbers: Maynard: M_{105} > 4 → H=600. Polymath8b: M_{54} > 4.00238 → H(54)=270; the **ε-trick** got the k=50 threshold → **H(50)=246**. They stopped at 50, writing that 50-or-49 was the limit; the k=49 door was left numerically closed but (to our knowledge) NOT rigorously closed for the most general variant. **H(49) = 240, H(48) = 236, H(47) = 232** (verify these diameters yourself from the admissible-tuple databases/searches — do not trust this line blindly).
- The pure variational problem: M_k = sup over symmetric F ∈ L²(simplex R_k = {t ∈ [0,∞)^k : Σt_i ⩽ 1}) of k · J(F)/I(F), I(F) = ∫ F², J(F) = ∫ (∫ F dt_k)² dt₁ ..dt_{k−1}. Known: M_k ⩾ log k − C; M_k ⩽ (k/(k−1)) · log k (so no barrier at 4 from above for k=49: log 49 = 3.892 < 4 — the PURE M_{49} is provably < 4.06-ish and numerically ≈ 3.85? — establish the true value). The ε-trick enlarges support to (1+ε) · simplex with a vanishing-marginal condition, trading against a (1−kε-type) penalty; Polymath also studied M'_k variants (unrestricted support with vanishing marginals) and proved upper bounds. KEY QUESTION for this round: what is the sharp upper bound for the BEST legal variant at k = 49, and is it above or below 4?
- Recent arithmetic inputs (post-2014): Polymath8a equidistribution θ = 1/2 + 7/300 for smooth/densely-divisible moduli (restricted residues — does NOT plug directly into MT sieve, which needs all residue classes; Polymath8b Section on "using equidistribution" got only ε-improvements and did not lower 246); Stadlmann (2023+) improved smooth-moduli exponents; Guth – Maynard (2024) large-value estimates for Dirichlet polynomials (improved zero-density; feeds short-interval prime results, NOT directly the BV level); Lichtman's sieve refinements. Verify current versions via WebSearch as needed.

B. Our machinery inventory (from Rounds 1—3; all in this scratchpad + /workspace/anthropics/zeta-23-lean)

1. **Decoder/LP theory**: generalized concave-score Lemma R; capped decoder LPs with escape-channel analysis; exact adversary construction (moment LPs with spread/pair/negative channels). The MT variational problem is a Rayleigh quotient = exactly our territory.
2. **SOS/Gram certificates**: multivariate trigonometric-SOS strictly beating scalar Fejér – Riesz cones (colleague's result: N=3 exact constant $(67-2\sqrt{22})/81$; N=4 exact-rational); inner-SOS/exact-LP/outer-SDP sandwich.
3. **Positive-inertia decoding**: rank – trace inequalities turning quadratic-form data into COUNTS (many positive directions → many prime-rich configurations) — the "matrix score $n_+(B-A)$" idea; the $\sqrt{\varepsilon}$ rigidity theorem (near-equality forces structure).
4. **Krylov/fermionic transfer**: half-power trick ⟨v,B^{2m}v⟩ = ‖ B^m v ‖ ²; second-quantized transfer for symmetric-function operators — high-order moment computations at otherwise impossible sizes.
5. **Non-identifiability/counterfeit methodology**: exact finite fibers, identity-compression, parity-type obstructions made quantitative in LP language.
6. **Euler – Maclaurin/cotangent tower**: exact finite-size ladders for variational discretizations (know your discretization error EXACTLY instead of eyeballing convergence).

C. Honesty rules for this round

- 246 has stood for 12 years against Tao, Maynard, and a swarm of Polymath optimizers. Any "improvement" you find is 99% likely a bug or a misread constraint: re-derive the exact threshold inequality (including the $\theta = 1/2$ factor and the ε-penalty terms) from the Polymath8b paper structure before claiming anything.
- The deliverable hierarchy (in decreasing value): (i) a verified new bound $H_1 < 246$ (extraordinary claim, extraordinary checking); (ii) a verified improvement of $H_2 = 398{,}130$ or higher-m records (much softer targets!); (iii) a rigorous closure ("the k=49 door is closed for variant class X, here is the dual certificate") — a real theorem; (iv) a quantified reduction ("improving BV level to $\theta = 1/2 + \delta$ for all-residue averaged moduli would give $H_1 = f(\delta)$") with the exact function f; (v) honest no-go maps.
- Compute budget: sympy/numpy/scipy/mpmath available; pip install cvxpy/OSQP if needed for SDPs. Krylov bases for the variational problem are 1-D integrals of polynomial bases — exact rational arithmetic is feasible; use it for anything near a threshold.
- python3 in this scratchpad; WebSearch works (arxiv.org itself is egress-blocked for fetch; search snippets work; the Polymath wiki michaelnielsen.org and Tao's blog terrytao.wordpress.com are ALSO blocked for direct fetch — rely on search result snippets and your own derivations).

Historical source 25: signed_context

Source: `historical/riemann-rmt/signed_context.md`. SHA-256: `f9d18ed12b12644f97574bf2efc9613815334722879a693dce0995adf7530fae`.

Shared context — Round 5: the SIGNED sieve experiment (Zhang × Maynard—Tao × ACUE, one language)

You are one of 3 parallel agents executing the first concrete experiment of the unified program: **does removing the square/positivity constraint from the Maynard – Tao variational problem produce a genuinely new phase (λ_signed ≫ λ_positive), and at what arithmetic price?** English, tags [PROVED]/[COMPUTED]/[HEURISTIC]/[CONJECTURE]; scripts in this scratchpad with your prefix; final text = your report; end with the single most promising next step. Be adversarial with yourself: the trivial version of this experiment has a KNOWN null answer (see § 3) — your job starts where the trivial version ends.

1. The three formalisms, written as one

Level 0 (decode). Weights w: n ∈ [N,2N) → ℝ; ν(n) = #{i : n+h_i prime} for an admissible k-tuple H. $S_1 = \Sigma$ w(n), $S_2 = \Sigma$ w(n)ν(n). Classical DHL decode: if **w ⩾ 0 pointwise** and $S_2 - mS_1 > 0$ then some n has ν(n) ⩾ m+1. If w is signed the implication FAILS: the positive excess can be generated by w(n) < 0 at prime-poor n. Decode repair requires paying the **debt** D := Σ_{w(n)<0} |w(n)| · (m − ν(n))$_+$ — either bound D directly, or (Zhang-style) support w_- on a set where ν is controlled by independent means (exceptional-character / bilinear estimates).

Level 1 (divisor-sum data). MT weights: v(n) = vector of divisor sums, w(n) = Q(v(n)) for a quadratic form Q on λ-space. Classical MT: Q = rank-1 PSD, i.e. w = (Σ_d λ_d)². MT asymptotics make S_1, S_2 LINEAR in Q: $S_1 \sim$ Tr(A Q) · X, $S_2 \sim \Sigma_i$ Tr(B_i Q) · X (A, B Gram forms; in the F-language these are the I- and J-integrals). The variational constant is the generalized eigenvalue problem M_k = sup Tr(BQ)/Tr(AQ).

Level 2 (finite/ACUE certificates — our machinery). All of the above have exact finite-N avatars: LPs/SDPs over the ACUE fiber with ball-arithmetic certificates; the marked-ACUE operator whose λ gave 0.702642 (free) vs 0.672500703679 (Anthropic's constrained value); kill-graph parity obstructions as odd-cycle facets of the cut polytope. Finite models are where "new phase?" can be answered EXACTLY.

2. Our proved walls — what they do and do NOT forbid (critical: read carefully)

- **P5 (decode wall)**: scalar decode $S_2 - mS_1$ exactly optimal — proved WITHIN the class of decodes consuming linear-sieve (type-I) data with pointwise-nonnegative weights. Convex-order two-point counterfeit construction.
- **P7 (weight-cone wall)**: sup over PSD Q equals sup over rank-1 Q (SOS decoupling by subadditivity); the copositive relaxation is flat (9-pattern dual certificate). So the entire PSD cone gains ZERO over Maynard's square.
- **Parity wall**: a parity twist kills a set of pair-conclusions iff its kill-graph is bipartite; floors $H_1 \geqslant 6$, k ⩾ 2m+1 for the positive-weight linear-sieve class.
- **Level wall (#6)**: the post-2014 levels θ = 4/7 (BFI), 3/5 (Maynard II), 5/8 (Pascadi) are **well-factorable / fixed-residue**: structurally unusable by CLASSICAL MT precisely because positive squares cannot be factored into the bilinear shapes those theorems require.

The signed enlargement exits the hypothesis class of ALL FOUR walls simultaneously. That is its entire point, and why it is not already dead. Indefinite Q is outside P7's cones; a signed decode with debt is outside P5's class; signed weights are the only known door past parity (Selberg's examples block positive linear-sieve data only — the χ-twisted counterexamples are themselves SIGNED objects); and well-factorable λ's are exactly what BFI/Maynard-II/Pascadi levels consume.

3. The trivial null result (do not rediscover it; start beyond it)

If you only drop PSD from the RATIO objective, nothing happens: sup over ALL symmetric Q of Tr(BQ)/Tr(AQ) (with Tr(AQ) > 0) is attained at rank-1 extreme points and EQUALS the PSD value = λ_max of the pencil (B, A) = M_k. [PROVED — eigendecomposition makes the ratio a weighted mean of Rayleigh quotients.] So **λ_signed = λ_positive for the naked ratio; no new phase there.** The genuinely new physics can only enter through the two things that CHANGE when w may be signed: (i) the decode inequality acquires the debt term D (a new, NONLINEAR-in-Q constraint: D depends on the negative part of the function $n \mapsto Q(v(n))$, not of the matrix Q), and (ii) the ARITHMETIC INPUT set enlarges: signed/well-factorable weights may consume $\theta \in \{4/7, 3/5, 5/8\}$ bilinear estimates that positive squares cannot touch. The experiment is therefore: optimize the REPAIRED functional $\Phi\_\beta(Q) = \Sigma_i \operatorname{Tr}(B_i Q) - m \cdot \operatorname{Tr}(AQ) - \beta \cdot D(Q)$ (or the constrained version $D(Q) \leqslant \delta \cdot \operatorname{Tr}(AQ)$) as the control parameter β (resp. δ) sweeps from the classical limit ($\beta = \infty \Longleftrightarrow w \geqslant 0$ forced) to full bilinear control (β small), and detect whether the value exhibits a PHASE TRANSITION (kink/jump, optimizer structure change, parity-floor crossing) rather than a smooth ramp.

4. Threshold ledger (why the payoff is enormous if the phase exists)

DHL(k, m+1) needs M-value > 2m/θ. Ladder for m = 1: $\theta = 1/2 \to 4$ (k = 50, H = 246); $\theta = 4/7 \to 3.5$ (M_k > 3.5 at k $\approx$ 32, H(32) = 140 [verify]); $\theta = 3/5 \to 10/3 \approx 3.333$ (k $\approx$ 28, H(28) = 120 [verify]); $\theta = 5/8 \to 3.2$ (k $\approx$ 25, H(25) = 104 [verify]); θ = 1 (EH) $\to$ 2 (k = 5, H = 12). Every entry between 246 and 12 is currently blocked ONLY by the positivity/well-factorable incompatibility. M_k values: $M_4 = 1.845401$, $M_5 = 2.007080$, $M_{25} \approx 3.10?$, M_{28}/M_{32} compute yourself (1-D engine or symmetric Galerkin, both described in codex_handoff.md § 3); $M_{50} \geqslant 3.9071$, $M_{105} > 4$. H(k) table: H(25..32) verify from Engelsma/OEIS A008407; H(50) = 246.

5. Known signed precedents (mine them, don't reinvent)

- Chen's theorem: the ONE historical success of a signed decode — w = weighted sieve minus penalty terms; the debt was paid by the switching principle (bilinear form of the remainder).
- Zhang's Landau – Siegel program: negative part supported where the exceptional character forces ν-control; "摆脱平方" = replace $(\Sigma\lambda)^2$ by $\Sigma\lambda \cdot (\Sigma\mu$ shifted) cross terms whose bilinear structure matches dispersion estimates.
- Well-factorable λ (Iwaniec): the linear sieve's $\lambda\pm$ ARE signed and DO consume θ = 4/7-type levels — but only for m = 0-type conclusions (upper/lower bounds on singletons), never DHL(k, m+1), m $\geqslant$ 1. The precise obstruction to lifting Iwaniec's trick to tuples = the debt term. Quantify it.
- Selberg's parity examples & Bombieri's asymptotic sieve: the no-go boundary. Any claimed λ_signed $\gg$ λ_positive at β small must be checked against a parity counterfeit: construct the Liouville-twisted measure explicitly in the finite model and evaluate your repaired functional on it. If your optimizer survives the counterfeit, THAT is the headline.

6. Compute discipline

python3 + numpy/scipy/sympy/mpmath in this scratchpad; exact rational arithmetic near any threshold; LP/SDP via scipy.optimize.linprog (HiGHS) or cvxpy if you pip-install it. Small-k pencils: the Dirichlet-integral Galerkin bases from codex_handoff.md § 3(a) (files p1_/p2_ conventions). Finite-N ACUE: enumeration code patterns in dir*/p8_ files.

WebSearch works (snippets only; arxiv/ wordpress/michaelnielsen direct fetch blocked). Do NOT re-run the record-chain engines (p9_*) — that program is finished and committed.

Historical source 26: codex_handoff

Source: `historical/riemann-rmt/codex_handoff.md`. SHA-256: `2f6f5c614cbd639bbcd23231698a1833243aff058277ac388df040db563e571a`.

CODEX HANDOFF — Bounded prime gaps: formulas, methods, current state, and open computations

Session handoff, August 15, 2026. Everything below is machine-actionable; file paths relative to the session scratchpad. Status tags: [P] proved, [X] exact-rational certified, [C] computed (float/hp, uncertified), [R] recalled-needs-verification.

1. The framework in five formulas

1. **Variational constant.** $R_k = \{t \in [0,\infty)^k : \Sigma t_i \leqslant 1\}$; symmetric F; $I(F) = \int_{R_k} F^2\, dt$, $J^{(m)}(F) = \int_{R_{k-1}} (\int_0^{1-\Sigma} F\, dt_m)^2$; $M_k = \sup k \cdot J/I$.
2. **Threshold.** $DHL(k, m+1) \Longleftarrow M_k\text{-variant} > 2m/\theta$. Unconditional $\theta = 1/2$ (Bombieri – Vinogradov): $> 4m$. With $MPZ[\varpi,\delta]$ (Polymath8a Deligne strength: $600\varpi + 180\delta < 7$): threshold $2m/(1/2 + 2\varpi)$ but support truncated by δ (see § 4). Then $H_m \leqslant H(k)$ = min diameter of admissible k-tuple.
3. **ε-trick.** Support $(1+\varepsilon)R_k$; J-integrals truncated to fibers with $\Sigma_{j\neq i} t_j \leqslant 1-\varepsilon$ (vanishing marginal). Zero arithmetic cost. This is how Polymath8b got $k = 54 \to 50$.
4. **Upper bounds [P].** $M_k \leqslant (k/(k-1)) \cdot \log k \Rightarrow M_{49} \leqslant 3.97290$, $M_{50} \leqslant 3.99186$ (pure door dead $\leqslant 50$). ε-variants: $M_{k,\varepsilon} \leqslant (1+\varepsilon)(k/(k-1))\log k \Rightarrow k=49$ door *rigorously* closed only for $\varepsilon \leqslant 0.00682$. No upper-bound technology beyond this exists for larger ε — the k=49 door is OPEN (Polymath: "undecided").
5. **Tuple side.** Admissible: for every prime $p \leqslant k$ some class mod p is missed. H(k) proven minimal $\forall k \leqslant 342$ (Engelsma + our independent exhaustive re-proof $\leqslant 62$). Ladder: H(46..50) = 216, 226, 236, 240, 246. Payoffs from k=50: −6/−10/−20/−30. Empirical $H(k) \approx k(\log k + 0.77)$ at $k \sim 10^4$ (fits (50,246), (35410,398130)).

2. Current numerical state (the live numbers)

object	value	status	file
M_{49} (pure, d=20 power-sum basis $p\leqslant 5$, dim 1125)	$\geqslant 3.891257590916$	[X] exact rational	`mt_hp_k49_p5.json`
M_{50} (pure, same basis)	$\geqslant 3.907113699811$	[X]	`hpE.log`
$M_{\{49, \varepsilon=1/35\}}$ (d=18 basis, n=1597)	float opt 3.959325169; **certified $\geqslant$ 3.930490592**	[C]/[X]	`p2_arb_d18_k49.log`, `p2_cert_d18_float.log`
$M_{\{49, \varepsilon=1/25\}}$ (d=14)	3.915989908	[C]	`p2_arb_d14_epsscan.log`
gap to the k=49 door	4 − 3.9593 = 0.0407 (float), 0.0695 (certified)	—	—
m=2 pure-BV crossing ($M_k > 8$)	**k = 15,856** (certified 8.00108 by the 1-D engine)	[C — ENGINE UNVERIFIED]	`p9_pure2.log`

object	value	status	file
m=2 Deligne crossing	**k = 13,467** (thresh 7.8273 at δ=0.0205, ϖ=0.00552)	[C — ENGINE+THRESHOLD UNVERIFIED]	`p9_del2.log`
m=3 pure-BV / Deligne	k = 923,601 / **660,985**	[C — same caveats]	`p9_pure3.log`, `p9_del3.log`
m=2 tuple search (k=35,265)	best diameter **397,352** vs record 396,504	[C], search interrupted	`p4_run_*_d3973*.txt`
record H_2	396,504 @ k=35,265 (Stadlmann θ=1/2+1/40)	[verified via search]	—
P3's independent product-ansatz m=2 crossing	k ≈ 29,500	[C, weaker engine]	`p3_m2_machine2.py`

If P9's engine and threshold chain survive verification, H_2 drops to ≈ H(13,467) ≈ 138,000 (Deligne) or ≈ H(15,856) ≈ 165,000 (pure BV) — a ~2.5 – 2.9× record improvement. This is the highest-value open verification on the board.

3. The variational engines (how to compute M_k)

(a) Exact symmetric-polynomial Galerkin (small k ⩽ ~60). Basis: monomials $(1-P_1)^a P_2^b P_3^c \cdots$ (power sums P_j = Σt_i^j), degree ⩽ d. All I- and J-integrals reduce via the Dirichlet formula *∫ {R_k} ∏t_i^{a_i}(1−Σt)^b dt = ∏Γ(a_i+1) · Γ(b+1)/Γ(k+Σa_i+b+1) — assemble exact rational Gram matrices A (for kJ) and B (for I); solve the generalized eigenproblem;* ***certify*** *by exact rational Rayleigh quotient of the rounded eigenvector (numerator/denominator sizes ~600 bits at d=18). Engines: P1's (*`hpE/hpD/hpM.log` *conventions, precision-boosted inverse iteration 544→1570 bits) and P2's (*`p2_*` *— includes the ε-trick truncation: J-fiber integrals with upper limit 1−ε; assembly is the bottleneck, ~25 min for n=1597).* ***(b) ε-scan protocol.*** *ε = 1/den rational for exact arithmetic; scan den ∈ {35, 31, 28, 25}; observed optimum near ε ≈ 0.0286 – 0.04 at d=14 – 18. Shift-invert accelerator: "[shift → 0.078...]" lines mark the spectral shift used.* ***(c) 1-D product-profile engine (large k ~ 10^4 – 10^6) — P9's, UNVERIFIED.*** *Ansatz F = ∏g(t_i) · 1*{R_k}; reduce I, J to 1-D integrals of g plus a truncation correction for the simplex constraint (Chernoff/exponential-tilt bound on P(Σt_i > 1) under the g-density). *The soundness of the certified lower bound hinges entirely on the direction of the truncation correction in both I (upper-bounded) and J (lower-bounded) — this is exactly where a sign error would fabricate records.* Read `p9_*.py` before trusting; re-derive the two-sided correction independently. P3's slower-but-cleaner variant (`p3_m2_machine2.py`, mpmath 30-digit) crossed at 29,500 — the 15,856 vs 29,500 gap between the two engines MUST be explained before any claim. **(d) MPZ/Deligne thresholds (P9's del runs).** Parameters scanned: (δ, ϖ) ∈ {(0.008, 0.00927), (0.0143, 0.00739), (0.0205, 0.00552)} — the constraint is 600ϖ + 180δ < 7 [R: verify exact form + whether the k-dependent dense-divisibility truncation cost was correctly charged into the certified value; this is the weakest link].

4. Walls — do not spend compute here (all proved this session)

1. Ceiling = tuple diameter: no post-processing of MT output beats H(k_min). Pair-correlation upper-bound constants can never lower H_1.
2. Scalar decode optimal: convex-order two-point counterfeit kills all matrix/inertia/moment decodes; f(m) = 2m/θ final.
3. Weight cone closed: rank-r SOS decouples (subadditivity); copositive cone flat (9-pattern dual certificate, residual 1.3e-14).
4. Parity: kill-graph bipartite ⟺ killable; H ⩾ 6 absolute floor for the method class; DHL(k,m+1) parity-blocked iff k ⩽ 2m.

5. Guth – Maynard orthogonal to H_1 (zero-density inputs appear nowhere in the beyond-1/2 dependency chains); GRH gives only $\theta = 1/2$.
6. Fixed-residue / well-factorable levels (BFI 4/7, Maynard II 3/5, Pascadi 5/8) structurally unusable by MT. Usable-restricted frontier: Maynard III 11/21 (uniform residues), Stadlmann 1/2+1/40, Pascadi minorant 10/19 — all shell-truncated.

5. Open computations, ranked (with method + acceptance criteria)

1. **VERIFY-P9 (record-critical).** (i) Re-derive the 1-D engine's truncation control two-sidedly; (ii) reconcile 15,856 (P9) vs 29,500 (P3); (iii) re-derive the exact DHL(k,3) criterion from Polymath8b (Thm 3.5-family) including all ε/truncation charges at MPZ[ϖ,δ]; (iv) re-certify the crossing k in exact arithmetic. Accept: an exact-rational certificate M-variant(k*) > threshold with every constant re-derived. Then: narrow-tuple search at k* (engine: § 6) ⇒ new H_2.
2. **TUPLE-35265.** Beat 396,504: restart from `p4_run_*` state files (best 397,352). Methods that got there: shifted-Schinzel + greedy + class-swap annealing (mod-p re-choices, $p \in [50, 2000]$, min-cost repair). Missing (2014's edge): iterated merging of two good tuples on overlapping windows; LP-based repair. Any diameter $\leqslant$ 396,503 with exact admissibility verification (every $p \leqslant$ 35,265) = world record.
3. **K49-DOOR.** Push (α)-engine to d = 20 – 22 at ε-scan granularity den $\in \{30\cdots40\}$ (warm-start from `p2_vec_hp_k49_d18_e35.npy`); if the float optimum plateaus < 3.98, run the vanishing-marginal variant (larger legal class, unexplored at k=49 — Polymath blog IX stopped at single-digit k). Non-separable dual certificates (weights on (t_i, u_i, P_2 -fiber)) are the only known route to CLOSE the door above $\varepsilon = 0.0068$.
4. **SHELL-M49 (undecided in print).** Extract Maynard III divisor-window parameters → shell polytope Ω*; compute M_{49} (R ∪ Ω*) with ε-trick layered; needs only 8 – 13% of full-shell surplus. Inner/outer sandwich decides vs 4.
5. **P10 wrap-up.** `p10_corner_eig.log` calibrated; sideband corners give conditional zeta-side gains (+0.022 at B=4) — finish the BFI-range map and the honest verdict.

6. Tuple–search machinery (P4's, all in scratchpad)

`p4_rho_exhaust.c` (exact minimal diameters $\leqslant$ 62, ρ*-pruning + mod-3,5,7 branching + parity halving), `p4_verify_witnesses.py` (exact admissibility checker — run on EVERYTHING before claiming), `p4_bigtuple*.py` (Schinzel/greedy/anneal pipeline), state files `p4_run_{A,B,C,D}2*.{log,txt}` (four annealing chains, best 397,352). Exchange rates: $\Delta H_2 / \Delta k \approx 11.2$ near k = 35,300; tuple-side slack estimate 0.03 – 0.3%.

7. Key constants (verified this session)

M_2 = 1.385933, M_3 = 1.646440, M_4 = 1.845401, M_5 = 2.007080 (EH ⇒ $H_1 \leqslant 12$); M_{54} > 4.00238 [Polymath]; H_1 = 246 (k=50, proven-minimal tuple); H_2 = 396,504 (k=35,265, Stadlmann); H_3 = 24,797,814 (k=1,649,821); pair upper constants 3.3996 (Wu) → ≈3.30 (Lichtman), parity floor 2; H(k) k=3..62 full list in `p4_payoff_table.txt`; $dH_1 / d\delta \approx -2400$; k=49/48/47 tipping $\delta \approx$ 0.002/0.004/0.006.

Historical source 27: rmt_zeta_survey

Source: `historical/riemann-rmt/rmt_zeta_survey.md`. SHA-256:
`a60e1a3fb0e3b6db037446a59867c14de0bc2b3b8df38ff4a77738a10278ca2b`.

The Bandwidth-One Ceiling: Finite Random-Matrix Fibers, Moment Rigidity, and the Optimality of the 67.25% Critical-Line Bound

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Abstract

The recent unconditional theorem that more than two thirds — precisely 67.2500⋯% — of the nontrivial zeros of the Riemann zeta function are simple zeros on the critical line [9] closes, to within a hair, the fifty-year-old gap between unconditional zero-density technology and the Montgomery – Taylor bound previously known only under the Riemann Hypothesis. This survey develops a structural explanation for the constant 0.6725007⋯: it is the exact optimal value of a *Chebyshev-type extremal problem over probability measures constrained only by two-point (pair-correlation) data of bandwidth one*, and the proof of [9] is, in disguise, the dual certificate of that optimization problem. We make this precise by building a finite model — the Alternative Circular Unitary Ensemble (ACUE) on $\mathbb{Z}/2N$ and its moment fiber of "CUE mimickers" [13] — and proving a sequence of rigidity and duality results in it. The main findings surveyed here are: (i) an exact **two-point rigidity theorem** (all mimickers share all two-point correlations, at every distance, via a Fourier aliasing argument), and a **three-point rigidity phenomenon** (exact at $N = 5$, machine-verified to 10^{-14} for $N \leqslant 8$) with first freedom appearing at four points; (ii) **finite uniqueness for $N \leqslant 4$** and super-exponential fiber growth thereafter (dim 2, 10, 80, 403, 1804 for $N = 5,\cdots,9$); (iii) a **duality dictionary** identifying the rank – trace/inertia argument of [9] with the dual of a linear program whose primal adversaries are mimicker measures and whose dual optimum is the Montgomery – Taylor window, with closed form $0.6725007\cdots = 3/2 - (1/\sqrt{2}) \cdot \cot(1/\sqrt{2})$; (iv) a proof that the **Alternative Hypothesis point process is a strictly suboptimal adversary**, losing to the free band-limited adversary by 0.0301; and (v) a **hierarchy plateau**: positivity (sum-of-squares) constraints of any order do not move the ceiling, while genuinely new correlation *data* does — quantifying exactly what arithmetic input (pair correlation beyond bandwidth one, i.e., prime-pair information) would be required to prove more than 67.25%. We close with implications for the Lagarias – Rodgers uniqueness question, where our results localize any possible non-uniqueness to a four-point phenomenon.

Conventions on epistemic status

Because this survey mixes formal proof, exact symbolic computation, high-precision numerics, and heuristics, every result is tagged:

- **[P]** proved (human-checkable proof, or exact arithmetic in a number field);
- **[C]** computational fact (machine-verified numerics, residuals at working precision, reproducible scripts);

- **[H]** heuristic (argument we believe but have not closed);
- **[Q]** conjecture / question.

1. Introduction

Let $N(T)$ count nontrivial zeros $\rho = \beta + i\gamma$ of $\zeta(s)$ with $0 < \gamma \leqslant T$, and let $N_0^*(T)$ count those that are simple and on the critical line $\beta = \frac{1}{2}$. The Riemann Hypothesis (RH) asserts $N_0(T) = N(T)$; the folklore strengthening asserts all zeros are simple. Unconditionally, the proportion of zeros known to lie on the line rose slowly for a century: Selberg's positive proportion, Levinson's 1/3, Conrey's 2/5 [3], and $5/12 \approx 41.7\%$ by Pratt – Robles – Zaharescu – Zeindler [4].

In August 2026 a research model of Anthropic announced, with a complete Lean 4 formalization (repository `anthropics/zeta-23-lean`, no `sorry`, no extra axioms), the unconditional bounds [9]:

> **Theorem (Claude, 2026).** $\liminf N_0^*(T,2T)/N(T,2T) \geqslant 2 - c_1^* = 0.6725007\cdots$, where $c_1^* = \frac{1}{2} + (1/\sqrt{2}) \cdot \cot(1/\sqrt{2}) = 1.3274992963\cdots$ is the Montgomery – Taylor constant. The same proportion holds for simple critical zeros, and $\geqslant 83.625\% = (1 + 0.6725)/2$ of all zeros are distinct.

The striking feature is that $2 - c_1^*$ is exactly the constant Montgomery and Taylor obtained *under RH* from the pair-correlation approach (see Cheer – Goldston [2]). The proof of [9] removes the hypothesis at zero cost in the constant, by replacing Montgomery's conditional pair-correlation asymptotic with an unconditional argument built from the Weil explicit-formula quadratic form (following Bombieri [8]), the Montgomery – Vaughan generalized Hilbert inequality [14], and a rank – inertia – trace inequality in place of diagonalization; the "narrow box" hypothesis of Baluyot – Goldston – Suriajaya – Turnage-Butterbaugh [5, 6] is dismantled rather than assumed.

This survey asks, and largely answers, a structural question:

> *Why this constant? Is $0.6725007\cdots$ an artifact of one proof, or the exact value of an intrinsic obstruction?*

Our answer, developed through a finite random-matrix model, is the latter:

> **Thesis.** $0.6725007\cdots$ is the optimal value of a linear program: *minimize the fraction of simple points over all point processes consistent with bandwidth-one two-point correlation data*. The proof of [9] constructs the dual optimal certificate (the Montgomery – Taylor window); the primal near-optimal adversaries are Alternative-Hypothesis-like measures, which are however **strictly** suboptimal — the true extremal adversary is free. Consequently no method whose arithmetic input is confined to bandwidth-one two-point data can prove a proportion larger than $0.6725007\cdots$, and no amount of higher-order *positivity* (sum-of-squares, Lasserre relaxation, higher inertia bookkeeping) can change this; only new *data* can.

The finite model that makes this precise is the ACUE of Tao [11] and the "finite CUE mimicker" framework of the companion paper [13], which proved that the finite moment problem for the ACUE is *non-unique* for $N \geqslant 5$, with explicit algebraic counterexamples. The body of this survey is organized around six groups of results obtained in that model and their translation back to ζ.

2. Background

2.1 Montgomery's pair correlation and the Montgomery—Taylor window

Montgomery [1] introduced, for the ordinates γ of zeta zeros,

$F(\alpha) = F(\alpha, T)$: the Fourier transform of the pair-correlation measure of normalized ordinates $\tilde{\gamma} = \gamma \cdot (\log T)/2\pi$,

and proved (under RH, later made unconditional in restricted ranges [5]) that $F(\alpha) = |\alpha| + o(1)$ for $|\alpha| \leqslant 1$, while F is *unknown* for $|\alpha| > 1$ — knowledge there is equivalent to strong information about prime pairs (Goldston – Montgomery). The classical application: for even test functions r with supp $\hat{r} \subset [-1, 1]$,

Σ over pairs of zeros of $r(\tilde{\gamma} - \tilde{\gamma}')$ is computable from F on $[-1,1]$ alone,

and choosing r optimally bounds the fraction of non-simple zeros. Montgomery's Fejér-kernel choice gives $\geqslant 2/3$ simple (under RH); the optimal choice — the **Montgomery – Taylor window** — improves this to $2 - c_1^*$, where

$c_1^* = ½ + (1/\sqrt{2}) \cdot \cot(1/\sqrt{2})$, so that $2 - c_1^* = 3/2 - (1/\sqrt{2}) \cdot \cot(1/\sqrt{2}) = 0.6725007297\cdots$ [P]

(The extremal function is the solution of a one-dimensional Sturm – Liouville-type variational problem on $[-1,1]$; we verified the closed form to 50 digits against the variational optimum, and it is the unique critical point [C].)

2.2 The unconditional proof of [9], in one paragraph

The proof works with the Weil explicit-formula quadratic form Q on a space of band-limited test functions: critical-line zeros contribute positive-semidefinite directions, off-line zeros contribute hyperbolic (signature (+,−)) planes (Bombieri [8]). Rather than diagonalizing Q, the argument controls the **inertia** of Q through the elementary inequality $n_+(Q) \geqslant (\mathrm{tr}\, Q)^2/\mathrm{tr}(Q^2)$ and its rank-restricted refinements, evaluating tr Q and tr Q^2 unconditionally via the Montgomery – Vaughan Hilbert inequality [14] (quasi-orthogonality of the vectors $x^{(i\gamma)}$), Chebyshev – Mertens bounds, Riemann – von Mangoldt, and Stirling estimates for Γ'/Γ. The optimization over the test-function space reproduces exactly the Montgomery – Taylor extremal problem — which is the first hint that the theorem computes the value of an optimization problem, not merely a bound.

2.3 The ACUE and finite mimickers

The Alternative Hypothesis (AH) is the logical foil to the pair-correlation conjecture: the (unlikely, but unrefuted) scenario in which normalized zero gaps concentrate on the half-integer lattice $½\mathbb{Z}$. Lagarias – Rodgers [10] showed AH is consistent with all known band-limited correlation data, and Tao [11] constructed the **ACUE**: a determinantal point process on the 2N-th roots of unity (rank-N Fourier projection) whose scaling limit realizes AH and matches CUE band-limited statistics.

The companion paper [13] studies the finite moment problem: call a rotation-invariant probability law q on N-subsets of $\mathbb{Z}/2N$ a **CUE mimicker** if $E_q[p_\lambda \bar{p}_\nu] = \delta_{\lambda\nu} z_\lambda$ for all partitions $|\lambda| = |\nu| \leqslant N$, where p_k are the power-sum character statistics and the right-hand side is the Diaconis – Shahshahani [12] CUE moment table. The ACUE law μ_N is one solution; [13] proved the solution set (the **moment fiber**) is non-unique for $N \geqslant 5$, with an explicit counterexample over $\mathbb{Q}(\sqrt{5})$ at N = 5, machine certificates at N = 9, 10, and a transfer theorem (their Thm 6.2): a uniform correlation gap along a mimicker family would produce a second point process matching the sine kernel in all band-limited correlations — answering the Lagarias – Rodgers uniqueness question in the negative. Their Thm 7.1 shows uniform gaps force exponentially large density ratios (via strong Rayleigh concentration, Pemantle – Peres [15]).

3. The finite moment fiber: dimension and finite uniqueness

Write Ω_N for the N-subsets of $\mathbb{Z}/2N$ modulo rotation, A_N for the balanced-moment constraint matrix on orbit space, and $X_N = \{q \geqslant 0 : A_N q = b_N\}$ for the fiber.

Theorem 3.1 (Finite uniqueness for small N). [P] For $N \leqslant 4$ the fiber is a single point: the ACUE is the *unique* CUE mimicker. (rank A_N equals the number of orbits: 4 at $N = 3$, 10 at $N = 4$; exact rank computation in cyclotomic arithmetic.)

Theorem 3.2 (Fiber growth). [C] The affine dimension of X_N for $N = 5, \cdots, 9$ is

N	3	4	5	6	7	8	9
orbits	4	10	26	80	246	810	2704
rank A_N	4	10	24	70	166	407	900
fiber dim	0	0	2	10	80	403	1804

At $N = 9$ the fiber occupies two thirds of the orbit space: the moment constraints, complete to full bandwidth, pin down a vanishing proportion of the degrees of freedom.

The $N = 5$ fiber is 2-dimensional and entirely algebraic over $\mathbb{Q}(\sqrt{5})$; we re-verified the explicit mimicker of [13] by exact arithmetic in $\mathbb{Q}(\zeta_{10})$ — all 88 balanced-moment pairs, positivity, and the correlation gap — confirming rank 24 and fiber dimension 2 [P].

The growth $2 \to 10 \to 80 \to 403 \to 1804$ is super-exponential and tracks the excess of the orbit count over the number of independent symmetric-function constraints; heuristically $\dim X_N \asymp |\Omega_N| \cdot (1 - o(1))$ **[H]**. Interpretation: *at fixed bandwidth, moment data is an exponentially small window into the space of point processes* — the natural finite-N sharpening of the Lagarias – Rodgers observation that band-limited correlations cannot see AH.

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4. Rigidity: what the fiber cannot vary

The fiber is huge, yet almost nothing observable varies over it. This is the technical heart of the survey.

Theorem 4.1 (Two-point rigidity). [P] Every mimicker $q \in X_N$ has *identical* two-point correlations at every lattice distance $b \in \mathbb{Z}/2N$ — not merely the band-limited ones: $E_q[\#\{\text{pairs at distance } b\}] = E_{\mu_N}[\cdots]$ for all b.

Proof. The pair-count at distance b is a linear statistic whose Fourier expansion involves only $E|p_k|^2$, $k \in \mathbb{Z}/2N$. Aliasing on the cyclic group folds each frequency k to $\min(k, 2N - k) \leqslant N$, which lies inside the balanced-moment window; so every coefficient is pinned by the constraints. ∎

This is a genuinely finite phenomenon with no continuum analogue: on $\mathbb{R}$, band-limiting *loses* information at high frequencies; on $\mathbb{Z}/2N$, aliasing *returns* it. The discrete model is therefore strictly more rigid than the Montgomery problem — a fact quantified in § 5.

Result 4.2 (Three-point rigidity). All three-point correlations are constant across the fiber: **[P]** at N = 5 (exact, in $\mathbb{Q}(\zeta_{10}))$; **[C]** for N = 6, 7, 8 (null-space projection residuals $\leqslant 6\times 10^{-14}$, versus O(1) four-point projections). No aliasing argument covers three points; empirically the Diaconis – Shahshahani identities and rotation invariance conspire — at N = 5, 88 constraints crush the 3-point statistics onto the row space. A conceptual proof is open **[Q]**.

Result 4.3 (First freedom at four points). [P for N = 5, C for N ⩽ 9] Four-point correlations genuinely vary: at N = 5 the first distinguishable pattern is the word `0000` (four consecutive empty sites), and the pattern `01010101` has fiber range [−0.0263, +0.0255] around its ACUE value 0.04434. The LP-extremal gap of the word `01010101` over the fiber grows monotonically:

$\delta^*(N)$ = 0.01019, 0.01232, 0.01375, 0.01514 for N = 6, 7, 8, 9;

the hand-built N = 9 certificate of [13] (0.014738) attains 97% of the LP optimum, and their density-capped search box is far smaller than the fiber's actual extent.

Consequence for Lagarias – Rodgers. Combining Theorem 4.1, Result 4.2, and Thm 6.2 of [13]: *any* second point process produced by the finite-to-infinite transfer agrees with the AH/ACUE process in **all** one-, two-, and three-point correlations — at every frequency, not only band-limited ones. If the Lagarias – Rodgers uniqueness question has a negative answer, non-uniqueness is a **four-point phenomenon**. This sharply narrows where to look, and explains why no two-point (spectral/pair-correlation) technique can ever detect it.

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5. The duality dictionary: rank—trace = LP dual, MT window = certificate

We now state the central structural claim: the proof of [9] and the finite fiber sit on the two sides of one linear program.

The finite LP. Allow multisets (multiplicities model non-simple zeros). Over probability measures q on N-multisets of $\mathbb{Z}/2N$ satisfying the balanced CUE moments up to degree $D = \alpha N$, maximize the expected non-simple mass:

ε_N(α) = max { E_q[fraction of repeated points] : q ∈ X_{N,D}^{multiset} }.

Result 5.1 (Bandwidth phase diagram). [C] At N = 5: ε = 0.60, 0.27, 0.12, 0 for α = 0.4, 0.6, 0.8, 1.0 (similarly N = 6). The continuum Montgomery prediction is ε_∞(α) = 1/α + α/3 − 1 (= 1/3 at α = 1). Full-bandwidth discreteness gives **complete rigidity** (ε = 0, forced simplicity!) by the aliasing of Theorem 4.1 — the discrete model is strictly harder to fool than the continuum.

Thesis 5.2 (Dictionary). [P in the finite model; H as a reading of [9]] The following correspondences hold:

	Proof of [9] (dual side)	Finite model (primal side)
data	$F(\alpha)$, $\lvert\alpha\rvert < 1$, unconditional	balanced moments, degree ⩽ N, exact
orthogonality	Montgomery – Vaughan Hilbert inequality	Diaconis – Shahshahani identities
certificate	Montgomery – Taylor window	LP dual optimal variables

	Proof of [9] (dual side)	Finite model (primal side)
adversary	AH-type point processes	mimicker fiber
rank tool	inertia vs $(\operatorname{tr} Q, \operatorname{tr} Q^2)$	Fock rank sandwich (rank 1043 at $N = 10$)
$\mathbb{Z}/2$ symmetry	$\rho \leftrightarrow 1 - \bar{\rho}$ (functional equation)	particle – hole $C \mapsto -C^c$

In the finite model this is literal: the rank – trace inequality $n_+ \geqslant (\operatorname{tr} Q)^2/\operatorname{tr} Q^2$ is the Lagrangian-dual bound of the LP, and the dual optimal solution, restricted to the two-point sector and interpolated as $N \to \infty$, converges to the Montgomery – Taylor window **[C]** (clean convergence requires the two-point-only LP; in the full-moment LP the two-point sector still carries $\approx$ 52% of the dual weight).

Result 5.3 (Scaling limit). [C, with closed forms P] The bandwidth-one discrete LP value converges to the continuum Montgomery – Taylor value with a computable finite-size correction; matching one-parameter families give exact finite-N formulas of the shape $\frac{1}{2} - 1/(2N^2 \sin^2(\pi/2N))$, consistent with $O(1/N^2)$ convergence to the continuum extremal value.

The dictionary recasts the theorem of [9] as: *the dual certificate exists unconditionally, therefore the primal value is an unconditional upper bound on non-simple mass.* Its optimality (the subject of § 6 – § 7) recasts the question "can we beat 67.25%?" as "is the primal optimum attained?" — and the answer is yes, by explicit near-feasible adversaries.

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6. The Alternative Hypothesis is a strictly suboptimal adversary

It is folklore to regard AH — gaps on $\frac{1}{2}\mathbb{Z}$ — as *the* enemy of pair-correlation methods. Our second structural finding is that this is quantitatively false.

Theorem 6.1 (AH suboptimality). [P] In the bandwidth-one extremal problem, adversaries supported on the half-integer lattice $\frac{1}{2}\mathbb{Z}$ force the simple fraction no lower than

$$\frac{1}{2} + 2/\pi^2 = 0.7026423\cdots,$$

(equivalently, the AH repeated-mass parameter satisfies $\beta \leqslant \frac{1}{2} - 2/\pi^2$, attained by an explicit two-parameter family). The free adversary attains 0.6725007. The gap is $0.0301423\cdots$.

Proof idea. Parametrize AH-admissible pair-correlation measures by their masses on $\frac{1}{2}\mathbb{Z}$ and impose the bandwidth-one constraints; the resulting LP over $\ell^1(\frac{1}{2}\mathbb{Z})$ has an explicit optimum via complementary slackness — the dual certificate would have to be equioscillatory on $\frac{1}{2}\mathbb{Z}$, and the Fourier-analytic obstruction evaluates to the Fejér-kernel value $2/\pi^2$. The free optimum's extremal measure is instead supported on a *non-lattice* configuration $\{1.057, 2.030, 3.020, \cdots\}$ of near-integers **[C]** — close to, but distinctly off, both $\mathbb{Z}$ and $\frac{1}{2}\mathbb{Z}$.

Three consequences. (a) The binding constraint at 0.6725 is **information (bandwidth), not lattice structure**: deleting AH from the adversary class would only improve the constant to 0.7026, and nothing in the bandwidth-one data justifies deleting it. (b) Conversely, any technique that could exclude near-integer-lattice adversaries — a much weaker statement than excluding AH — would already improve on [9]. (c) The extremal configuration's near-integer (rather than half-integer) spacing explains an old numerical puzzle in the optimization literature: the MT window's equioscillation nodes do not sit on $\frac{1}{2}\mathbb{Z}$.

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7. Positivity does not help; data does

Could one push past the ceiling with a stronger *proof technique* over the same data — higher moments of Q, deeper inertia bookkeeping, sum-of-squares certificates? In the finite model this question is decidable, and the answer is no.

> **Result 7.1 (Hierarchy plateau). [C]** Impose, on top of the bandwidth-limited two-point data, positive-semidefiniteness of higher-order moment (Lasserre/SOS) matrices of any degree. The LP/SDP optimal value does not move (changes $< 10^{-3}$ through the levels computable at $N \leqslant 8$, with degeneracy identifiable as solver conditioning). The reason is structural: the AH/ACUE-type near-optimizers *satisfy every higher positivity constraint* — they are genuine point processes, so all their moment matrices are automatically PSD. Positivity cannot separate what probability cannot.

> **Result 7.2 (Data cuts). [C]** By contrast, adjoining higher-correlation **data** (not positivity) moves the value sharply: in the truncated-bandwidth multiset LP, adding the exact three-point correlation table cuts the maximal non-simple mass from ≈ 0.294 to ≈ 0.12 in the tested configuration. Data and positivity play categorically different roles: positivity fixes the geometry of the feasible cone; data slices it.

For ζ, Result 7.1 is the finite shadow of a fact forced by Lagarias – Rodgers [10]: the AH process matches *all* band-limited n-level correlations of the sine process, so no unconditional method whose arithmetic input is band-limited correlation data — of **any** order, processed by **any** positivity argument — can prove a simple-zero proportion exceeding what the AH-inclusive LP allows. Our contribution is the quantitative complement: within two-point bandwidth one, that LP value is exactly 0.6725007, it is *attained* (§ 6), and the ceiling for two-point-based certificates of any window is 0.6818287 (the `PairCeiling` bound formalized in [9]), leaving at most 0.0094 accessible to windowcraft — and our computations indicate even that slack is not reachable without new data **[H]**.

What new data would suffice. The escape routes, in decreasing directness **[H]**: (i) any unconditional lower-bound information on $F(\alpha)$ for $\alpha > 1$ — this is prime-pair (twin-prime-scale) information via Goldston – Montgomery; even an ε-sliver beyond $\alpha = 1$ breaks the LP's tightness; (ii) arithmetic positivity beyond correlations, i.e., Weil-form amplifiers carrying prime structure that is not a function of F (Bombieri's program [8] continued); (iii) in the finite model, four-point data — the first order at which the fiber opens (Result 4.3).

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8. Spectral barriers and the uniform-gap problem

The transfer theorem of [13] requires a *uniform* correlation gap δ along $N \to \infty$. Whether such a family exists is their central open problem, and our computations sharpen both sides of it.

For existence **[C]**: $\delta^*(N)$ is monotonically increasing through $N = 9$ (§ 4), and the exponential barrier of [13, Thm 7.1] is quantitatively toothless in this range — the forced density ratio $\| dq/d\mu \|\_\infty \geqslant (\delta/2) \cdot \exp(\delta^2 N/8\ell^2)$ does not exceed 10 until $N \approx 1.8\times10^7$. The observed LP extremals already use density ratios ≈ 211 at $N = 8$ (LP vertices with 416 of 810 orbits vacated) — far outside the capped search boxes used in [13], so the certificate landscape is much larger than previously explored.

Against existence **[H]**: the constraint transfer matrices show top-eigenvalue growth consistent with e^{cN}, and a block-Toeplitz/Szegő-type analysis (incomplete) suggests the feasible gap directions rotate out of any fixed word's dual cone, which would force $\delta^*(N) \to 0$ along subsequences unless the extremal word is allowed to vary with N. A Fock-space rank computation at $N = 10$ reproduces the rank-1043 sandwich of [13] exactly **[C]**, confirming no hidden degeneracy that would make uniform gaps cheap.

We record the sharpest finite question our methods isolate:

> **Question 8.1. [Q]** Is $\liminf \delta^*(N) > 0$, where $\delta^*(N)$ is the fiber-LP gap of the word `01010101`? An affirmative answer, by [13, Thm 6.2], resolves Lagarias – Rodgers uniqueness in the negative; by § 4 the resulting second process would be indistinguishable from AH below four points.

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9. Assessment: what is new here

Relative to the literature, the surveyed program contributes, in our view, five items.

1. **A theorem-level explanation of a constant.** The identification $0.6725007\cdots$ = value of a bandwidth-one LP, with the proof of [9] as its dual certificate, converts "the best known constant" into "the exact value of a named obstruction." Constants with this status are rare (compare 1/3 and Levinson's method, where no matching adversary is known).
2. **Exact finite rigidity theorems.** Two-point rigidity by aliasing (Thm 4.1) is, to our knowledge, the first *exact* rigidity statement for ACUE moment fibers, and the three-point rigidity / four-point freedom dichotomy is new; it localizes Lagarias – Rodgers non-uniqueness at four points.
3. **AH demoted from obstruction to bystander.** Theorem 6.1's strict 0.0301 gap corrects a widely repeated heuristic: the pair-correlation ceiling is *not* "because AH might be true." The extremal adversary is a near-integer, non-lattice configuration.
4. **Positivity/data separation.** Result 7.1/7.2 gives the first quantitative demonstration in this problem that relaxation hierarchies plateau at the information ceiling while data strictly cuts — a clean instance, in analytic number theory, of a phenomenon familiar in combinatorial optimization.
5. **A calibrated target list.** The 0.0094 window between 0.6725007 and the PairCeiling 0.6818287, the $\alpha > 1$ threshold, and Question 8.1 constitute concrete, falsifiable next steps, each with an exact finite-model analogue on which methods can be rehearsed cheaply.

We emphasize scope: results tagged [C] rest on reproducible computation (exact where stated, else with residuals at 10^{-14}); the reading of [9] through the dictionary is proved in the finite model and remains an interpretation — though a predictive one — of the infinite-dimensional proof.

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10. Open problems

1. Prove three-point rigidity (Result 4.2) for all N, ideally by exhibiting the algebraic identity behind the numerical conspiracies [Q].
2. Question 8.1 (uniform gaps ⇒ Lagarias – Rodgers non-uniqueness).
3. Determine whether any window-plus-positivity certificate can exceed 0.6725007 within bandwidth one, i.e., close the 0.0094 slack from above (our evidence says no [H]).
4. Find an arithmetic source of $F(\alpha)$, $\alpha > 1$: even $F(\alpha) \geqslant c > 0$ on $(1, 1+\varepsilon)$ unconditionally would beat the ceiling; quantify the constant gained as a function of (c, ε) via the LP (the machinery of § 5 computes this).
5. Formalize Theorems 4.1 and 6.1 in Lean 4 against the `zeta-23-lean` infrastructure; both are finite-dimensional and within reach of current tooling.

References

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Reproducibility: all computations tagged [C] are reproducible from short Python scripts (NumPy/SciPy, exact cyclotomic arithmetic where stated) developed alongside this survey; the $N = 5$ verifications are exact in $\mathbb{Q}(\zeta_{10})$.

Historical source 28: rmt_zeta_popular

Source: `historical/riemann-rmt/rmt_zeta_popular.md`. SHA-256: `d22dbc5e35dfa46a2883a15d78d91434bdb2ee7343a4a9fc2106723d9ee5c0fb`.

Why 67.25% Is Not Just a Number

A plain-language companion to "The Bandwidth-One Ceiling"

Bill (Qingyun) Sun · **GPT5.6SOL** · **Fable** — *for readers with general graduate-level mathematics, no analytic number theory assumed*

The headline result it explains. In 2026 an Anthropic research model proved — unconditionally, with a machine-checked Lean proof — that at least **67.2500⋯%** of the zeros of the Riemann zeta function lie on the critical line and are simple. The Riemann Hypothesis (RH) says the answer is 100%; before this, the best unconditional result was 41.7%. The odd thing is the constant itself: 0.6725007⋯ is *exactly* the number Montgomery and Taylor had derived in the 1970s – 90s **assuming RH**. Getting the same constant without the hypothesis looks like a coincidence. Our paper shows it is not.

The one-sentence thesis. The number $0.6725007\cdots = 3/2 - (1/\sqrt{2}) \cdot \cot(1/\sqrt{2})$ is the exact optimal value of an *optimization problem about limited information* — and the new proof is, structurally, the optimal certificate of that optimization, which is why no proof using the same information can ever do better.

What "limited information" means. All progress on counting simple critical zeros flows through one measurement: the *pair correlation* of zeros — the statistics of gaps between pairs — and crucially, we only possess this measurement at "bandwidth $\leqslant 1$" (Fourier frequencies $|\alpha| \leqslant 1$). Beyond bandwidth 1, pair correlation encodes information about *pairs of primes* — twin-prime territory — which nobody can currently reach. So ask a clean question: among **all conceivable point processes** whose bandwidth-1 pair statistics match the zeta zeros, which one has the *fewest* simple points? That is a linear program. Its value is 0.6725007⋯. A proof that only consumes bandwidth-1 pair data can never certify more, because an "adversary" process achieving exactly that value exists and is indistinguishable from the truth using that data. The 2026 theorem reaches the ceiling; our result is that it *is* a ceiling, and we locate the exit doors.

Where random matrices enter. Montgomery's famous conversation with Dyson revealed that zeta zeros statistically resemble eigenvalues of large random unitary matrices (the CUE). But there is a rival scenario — the *Alternative Hypothesis* (AH) — in which zero gaps live on the half-integer lattice ½, 1, 3/2, 2, ⋯, wildly unlike random matrices, yet (a theorem of Lagarias – Rodgers) matching every band-limited correlation measurement of every order. To study this rivalry concretely, we work in a finite model: Tao's **ACUE**, a determinantal process choosing N points among the 2N-th roots of unity, and the space of all "mimickers" — probability laws matching its CUE-style moments. Three discoveries:

1. **The mimicker space is enormous** — dimension 0 for $N \leqslant 4$ (the moment problem has a unique solution!), then 2, 10, 80, 403, 1804 for $N = 5,\cdots,9$. Moment data is an exponentially thin window into what a point process can be.
2. **Yet it is rigid where it counts:** we *prove* every mimicker has identical two-point statistics at every distance (a Fourier aliasing argument special to finite groups), and verify the same for all three-point statistics. The first genuine freedom appears at **four points**. So if the zeta-zero process has a hidden impostor — the Lagarias – Rodgers uniqueness

question — the impostor differs from AH only in 4-point (and higher) patterns. That is a new, sharp localization of a well-known open problem.

3. **AH is not actually the enemy.** Folklore says the 2/3-ish barrier exists "because AH might be true." We prove AH-type adversaries are strictly *suboptimal*: they only force the bound $1/2 + 2/\pi^2 \approx 0.7026$, while the true extremal adversary — a non-lattice configuration with near-integer spacings — forces exactly 0.6725. The barrier is about information, not about the half-integer lattice.

What is mathematically new. (i) A *proved* rigidity theorem for ACUE moment fibers (2-point, via aliasing) and the 3-point/4-point dichotomy; (ii) the identification of the rank – inertia – trace argument in the 2026 proof with linear-programming duality, the Montgomery – Taylor window appearing as the dual optimal solution — turning a virtuoso inequality into the canonical certificate of a named optimization problem; (iii) a clean separation, new in this subject, between *positivity* and *data*: adding sum-of-squares/Lasserre-type positivity constraints of any order does not move the ceiling (the adversaries are genuine probability measures, so they pass every positivity test), while adding genuinely new correlation data slices it immediately.

Why it matters for RH. It converts "try to push the constant" into a map. To prove more than 67.25% one must add *data*, not cleverness: either (a) any unconditional foothold on pair correlation beyond bandwidth 1 — equivalent to progress on prime pairs, with the LP machinery converting any ε-sliver directly into an improved percentage; or (b) arithmetic positivity that is not a function of pair correlation at all (Bombieri's Weil-quadratic-form program); or (c) in the finite model, four-point information — the exact order where the impostor space opens up. Meanwhile the same framework gives the random-matrix side a concrete finite target: if the mimicker gap $\delta^*(N)$ (already computed to be increasing: $0.0102 \to 0.0151$ for $N = 6\cdots9$) stays bounded away from zero, Lagarias – Rodgers uniqueness *fails*, and the zeta zeros' statistics would admit a genuine impostor visible only at four points.

The takeaway. $0.6725007\cdots$ is not a scoreboard entry. It is the precise price of knowing only pair correlations at bandwidth one — a Chebyshev-type extremal value with a closed formula, a dual certificate, and an explicit adversary. Barriers of this kind are valuable exactly because they say where the next theorem *cannot* come from — and therefore where it must.

Historical source 29: README

Source: `historical/riemann-rmt/README.md`. SHA-256:
`25ce7119647fc7b9311ed71dd2047cf2a5ca47e458690f1e27a0217d3742c269`.

Riemann zeta / random-matrix / prime-gap research artifacts

Session outputs (Bill (Qingyun) Sun, GPT5.6SOL, Fable — August 2026):

- `rmt_zeta_survey.md` — The Bandwidth-One Ceiling (survey, round 1)
- `rmt_zeta_popular.md` — one-page companion
- `final_verified_paper.md` — Finite Spectral Certificates and Their Fibers (verified synthesis of the Codex series)
- `tao_ah_notes.pdf` — notes on Tao's Alternative Hypothesis problem
- `round3_synthesis.md` — ten directions coupling the two-thirds method to the finite fiber program
- `prime_gap_survey.md` — The Walls and the Doors (bounded prime gaps round)
- `verify_codex.py` — independent verification battery
- `joint_context_v2.md`, `prime_gap_context.md` — shared agent contexts

Full computational scripts and certificates remain in the session scratchpad archive.

Historical source 30: HANDOFF_GPT6_ASTRA

Source: `historical/riemann-rmt/handoff/HANDOFF_GPT6_ASTRA.md`. SHA-256: `34950171df1d33390c98ad052be08ad0f7bac881ac409d9afc257e5e048e1cbc`.

HANDOFF — The Riemann / Random-Matrix / Prime-Gap Programme

Complete state of the research, for GPT6 Astra to take over

Authors of the programme: Bill (Qingyun) Sun · GPT5.6SOL · Fable **Handoff written:** 5 September 2026, by Fable (Claude), at the end of a ~3-week multi-agent session **Repository:** `galpha-ai/Alpha-devbox`, branch `claude/riemann-zeta-random-matrix-udxp3f`, directory `research/riemann-rmt/` (PR #11, draft). Standalone companion package: `research/riemann-rmt/riemann-impostors/`.

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0. How to read this document

This is written for a successor system that will attempt a *major* theorem. It is therefore long, and deliberately so: it records not only what was proved and computed, but what was tried and failed, what the failures taught, and where every number came from. Nothing here is padded; if a section seems long it is because the corresponding wall took weeks to map.

Status tags, used throughout, with strict meanings.

tag	meaning
[P]	proved, with a proof written down in the cited file; elementary steps machine-checked where noted
[X]	exact: certified in exact rational / ball / cyclotomic arithmetic, no floating point in the chain
[C]	computed: verified numerically, usually by two independent implementations, not proved
[R]	reduced: follows from a *cited published theorem* plus an explicitly stated open lemma
[H]	heuristic / conjecture, with the evidence stated
[✗]	tried and refuted; the refutation is itself a result
[U]	unverified claim received from a collaborating system; recorded, not endorsed

Trust rules that we learned the hard way. (1) Every headline number in this programme was produced twice by independent code before being believed; where the two disagreed, the discrepancy was tracked to source before any claim. (2) Nothing near a decision boundary is accepted in floating point. (3) The default assumption for any "improvement" is that it is a misread constraint; five of the most exciting-looking phenomena in this programme were exactly that. (4) "Machine-checked" and "formalised" are different things; only the first is true of most of this work.

What this programme did NOT do, so that the successor does not inherit a false belief. We did not prove the Riemann hypothesis, the Alternative Hypothesis or its negation, the pair correlation conjecture, the density conjecture, the twin prime conjecture, Lagarias – Rodgers $\mu = 1/2$, or any improvement of $H_1 \leqslant 246$. We did not compile any Lean. We could not read three URLs the human asked us to incorporate at the very end (Anthropic's post on formalising Fermat's Last Theorem and two `t.co` links): the session's egress proxy blocked `www.anthropic.com` and `t.co`. **Astra should read those three sources first**; nothing in this document reflects their content.

A note on a number the human mentioned. The handoff request refers to a prime-gap bound of "186" attributed to Weijie Su and GPT6 Astra (and, once, "286"). We have no record of either. Two facts worth knowing before touching it: **H(40) = 186 exactly** in the Engelsma table of minimal admissible-tuple diameters (independently re-proved minimal by us for $k \leqslant 62$), so a bound of 186 would mean DHL(40, 2); and **286 is not the diameter of any minimal admissible tuple** (H(57) = 282, H(58) = 288), so "286" is presumably a typo for 186 or 246. What DHL(40,2) would require is analysed in § 9.5.

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1. Timeline: what each round asked and what it delivered

The programme ran in rounds. Each round posed one question to a fleet of 3 – 10 parallel agents with a shared written context (state of knowledge, known errors, honesty rules) but no shared conclusions; convergence was treated as evidence and divergence as a bug report.

round	question	main deliverables
1	Analyse Anthropic's 2026 unconditional theorem "$\geqslant$ 67.25% of zeta zeros are simple and on the line" (Lean repo `anthropics/zeta-23-lean`, constant δ_MT $= 3/2 - (1/\sqrt{2})\cot(1/\sqrt{2}) = 0.672500703679$) from random-matrix angles; connect to Tao's ACUE	RMT survey; popular note; identification of the bandwidth-one wall; first finite ACUE fibre computations
2	Verify five Codex (GPT5.6SOL-ULTRA) PDFs; extract real mathematics	`verify_codex.py` battery; `final_verified_paper.md`; several Codex claims corrected (e.g. transcription errors θ_n $= \arccos(1 - n^{-2}/a\_n)$; free-box requirement for the Wilson trace); our own overclaim retracted ("0.6725 is the exact bandwidth-one LP optimum")
3	Read Anthropic's method *at the Lean source*, couple it to the finite fibre machinery, ten directions	`round3_synthesis.md`: seven-term deficiency identity + $\sqrt{\varepsilon}$ rigidity; $N\_d \geqslant 0.8362503N + p$; flat RH-frontier; edge no-go; weight freezing; centre-of-mass mimicker family; $M\_$ lemma with quantified payoff; ceiling candidate 15/22; kill-degree $d^* = 3(N-3)$
4	Attack the bounded-gap record $H_1 = 246$ with $\geqslant$ 10 agents; improve *any* record concretely	Five proved walls; the k = 49 door priced; **new records H_2, H_3, H_4** via the layer-cake engine; `prime_gap_survey.md` ("The Walls and the Doors"); Codex handoff note

round	question	main deliverables
5	Zhang Yitang's directive: unify Maynard – Tao + Zhang's signed construction + ACUE certificates, run the signed-sieve phase experiment	**Signed no-gain theorem** (one-line identity); exact critical price β*; conditional price list with (E_θ); Chen-switch obstruction; Bourgain scale diagnosis of the 0.0301 gap
6 (P0)	The de Bruijn – Newman finite depth Λ as a dynamic anti-ACUE observable	Exact ACUE enumeration N ⩽ 10; CUE Monte Carlo N ⩽ 256; **−Λ^CUE ≍ N^{−8/3} vs −Λ^ACUE ≍ N^{−2}**; β-universality; s* = 1.419640342; the operator unification 𝓛_N = Jacobian at the clock; marked-depth law; **Theorem A** (ρ ⩾ 1) **proved**; Theorem C separation; LR bridge and the π²/8 threshold
7	Consolidation: papers, machine verification (18/18 sympy + z3), Lean specification, companion repository, this handoff	`impostors_paper.md`, `depth_scaling_theorem.md`, `riemann-impostors/`

Two things happened operationally that the successor should plan around. Agents were killed twice mid-run: once by a session usage limit, once by exhaustion of usage credits (six agents at once). In both cases the scratchpad survived and partial results were harvested from disk; several of the best results in this document (the exact β*, the FUSION-ACUE negative finding in § 3.6) were recovered that way. **Design every long computation so that its partial output is on disk in a form that a stranger can resume.**

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2. The zeta side: Anthropic's 2/3 theorem, read at the source, and what we added

2.1 The method as it actually is (from the Lean, not from paraphrase)

The theorem: at least $\delta\_MT = 3/2 - (1/\sqrt{2}) \cdot \cot(1/\sqrt{2}) = 0.672500703679\cdots$ of the nontrivial zeros of ζ are simple and lie on the critical line, unconditionally. The Lean repository `anthropics/zeta-23-lean` (key files `Zeta23/ZeroSide/RankTraceMult.lean`, `TightMult.lean`, `Mult.lean`, `PairCeiling/*.lean`, `XiPrime/`, `Taper/`, `Poisson.lean`, `MV/`, `Tail/`) organises the proof around:

- **Lemma R (rank – trace, k-form).** For Hermitian P, Q and c > 0: $2c \cdot \mathrm{tr}(P+Q) - \|P+Q\|^2_F \leqslant \sum_j k_c(m_j) + c^2 \cdot b$, with $k_c(p) = c^2 - ((c-p)_+)^2$. This converts a quadratic form built from zero data into a *count* of positive directions. It is tight (`lemmaR_tight`).
- **PairCeiling.** An upper constant $0.6818287 + 2.55 \cdot 10^{-6} \cdot R$ under the hypothesis `EnclOK`; this is the "bandwidth-one ceiling" of the method's own data.
- **ξ′ quartic windows and TightMult**: the Montgomery – Taylor-type window optimisation, done with Fejér – Riesz-style nonnegativity at bandwidth one.

The core structural fact, which every later result respects: the method consumes exactly the pair-correlation data of the zeros at Fourier bandwidth ⩽ 1 (Montgomery's proven range), and its constant is set by a Rayleigh quotient in that data.

2.2 What we proved about the method (Round 3, all in `round3_synthesis.md`)

[P] Seven-term deficiency identity and $\sqrt{\varepsilon}$ rigidity (D4). Lemma R is an identity up to an explicit seven-term deficiency (verified to $7 \cdot 10^{-14}$, identically zero on the TightMult class); an ε-near-equality configuration lies within Frobenius distance $\sqrt{\varepsilon}$ of an exact equality configuration (Löwdin frame). Zeta-side ledger: total deficiency over MT windows $\leqslant (q^*-1) \cdot N(T) = 0.3275 \cdot N(T)$, giving an unconditional pair-repulsion-type bound (on-line pairs with taper overlap $\geqslant \tau$ number $\leqslant 0.1637 \cdot N(T)/\tau^2$). Reading: Montgomery – Taylor optimisation *is* minimisation of the RMT-predicted deficiency, and under GUE the budget is exhausted by the Schur/von Neumann term alone.

[P] The distinct-zeros upgrade $N\_d \geqslant 0.8362503 \cdot N + p$ (D7). Combining the c = 2 inequality with the zero-count identity once (not twice) gives slope +1 in the off-line pair density — strictly stronger than the repository's own c = 3 decoding (slope ½; pairs mispriced at $c^2 = 9$ against the forced value $k_3(2) = 8$). This is an unformalised **five-line Lean edit** to `mult_two_pair` and would yield a strictly stronger published constant for distinct zeros. Nobody has made the edit.

[P] The RH-frontier is exactly flat. $\delta(\pi) \equiv 0.6725007$ for all off-line proportions $\pi \in [0, 0.16375]$: **assuming RH buys nothing within this method** (the binding adversary is on-line double zeros; exact 4p-budget/mass cancellation). At maximal off-line density all zeros are forced simple; pair-block spectra are forced to $(\approx 2, \approx 0)$.

[P] The edge no-go (D5). Every admissible bandwidth-one window has $\hat{r}(\pm 1) = 0$ (two-line Fejér – Riesz), so the weakest pointwise edge hypothesis $|F(1) - 1| \leqslant \varepsilon$ certifies exactly 0.6725 even at $\varepsilon = 0$. The finite Nyquist conservation law is powered by lattice aliasing that $\mathbb{R}$ lacks. The correct zeta-side edge object is the Cesàro mean $\bar{F}(\infty,T)$, with the exact collision identity $C(T) = N^* \cdot \bar{F}(\infty,T) - N(T)$.

[P] Weight freezing. All balanced moments of degree $\leqslant N$ are frozen across the mimicker fibre — later (Round 6) shown to be frozen along the *entire* heat flow, because the flow is diagonal on coefficients.

[C] Φ_3 at the MT window. The flat-window sine-Gram data sits on the {0,1,2} factorial face ($m_3 - 3m_2 + 2m_1 = 0$, an identity at every finite N for ACUE), but at the Montgomery – Taylor window itself $\Phi_3 = m_3 - 3m_2 + 2m_1 =$ **−0.0117753128** (closed trig form; verified symbolically, at 30 digits, and on the lattice to $3 \cdot 10^{-7}$). The third moment carries information exactly where the proof lives; the blocker is not arithmetic but that Lemma R's hypothesis class carries **no bound on the magnitude of the compression's negative eigenvalues**.

[✗] Window engineering cannot fix it. The Rudnick – Sarnak – style clump term (log T zeros per unit ordinate at zero kernel decay) forces the row-sum cap $M = A_0 \cdot \log T$ for *every* window; edge-vanishing windows improve only the negligible far field, cost $\geqslant 0.006$ in $2 - q(w)$, and flip Φ_3 to the unmonetisable positive side. Adversarially settled.

The surviving zeta-side target, precisely (D1). A lemma $\|(c^{-1}\hat{A})_-\| \leqslant M_-$ **uniform in T** — depth-uniform control of off-line evaluation vectors (a cosh-weighted Poisson analogue plus an overlap-angle bound). Payoff at the MT window with the capped cubic decoder: δ **= 0.6796896 if $M_- = 2$, up to 0.6844924 if $M_- \leqslant 1$**, with tr^3 the only new prime-side input. This remains the sharpest known formulation of "what would improve 0.6725 without new correlation ranges."

[H] The true bandwidth-one ceiling. Bracket [0.6725007, 0.6818287+], measured integrality gap ≈ 0.0093, candidate exact value **15/22 = 0.681818⋯**. Two exact-arithmetic conjectures were left machine-attackable: the 15/22 identity and the kill-degree law $d^* = 3(N-3)$ (verified N = 5, 6, 7).

[P forms + C] The Nyquist cotangent tower. The three N^{-2} convergence towers seen in different places are one object — the Bernoulli tower of $(\cot, \csc^2)$ — and exist only at the edge. Doubles density of any Nyquist equality law → $1/4 - 1/\pi^2 =$ 0.148679.

2.3 The 0.702642 vs 0.672500 anomaly, closed (Round 5, agent BRG-SCALE, files `brg_*.py`)

The free finite optimum of the marked-ACUE pencil is $\delta\_free(N) = ½ + csc^2(\pi/2N)/(2N^2) \to ½ + 2/\pi^2 = 0.70264237$, with exact saddle $\zeta^* = N^2 + N - csc^2(\pi/2N)$; the Anthropic-feasible restriction gives $\delta\_MT = 0.672500704$; gap 0.03014166.

[C] The gap is multiscale in Fourier but single-site in position. Band ledger (stable from N = 128 to 1024): DC atom +20.33%, low band ($0 < \alpha \leqslant ¼$) **−12.73%**, mid +20.20%, edge quarter +72.21%, Nyquist row **exactly 0.00%**; the density $\lambda(\alpha)$ vanishes linearly at $\alpha = 1$ — no boundary layer, so no single sharp frequency inequality closes it. The whole free advantage is certified by **one lattice inequality $S_1 \geqslant 0$** — nonnegativity of the pair count at displacement $x = \pm 1$, i.e. at half the mean zero spacing — whose bandwidth-one shadow $\hat{c}\_k = cos^2(\pi k/2N)$ is full-band coherent. Decoupling-type bounds misprice the Rayleigh value by 15 – 77× (constructive alignment; both extremisers are Perron vectors of a positive kernel), against a 4.5% target: **ℓ^2-decoupling is structurally the wrong tool here.**

[C] Calibration that changes what one should chase. Of the 0.030142, only ≈ **0.0093** is even in principle recoverable in the continuum (to the ceiling ≈ 0.68183); the remaining ≈ 0.0208 is pure lattice aliasing. "0.6725 → 0.7026" via any continuum inequality is hopeless; the real continuum target is "0.6725 → 0.6818" and it needs integrality-aware (point-evaluation / measure-valued) certificates — the $\mathbb{R}$-analogue of $S_1 \geqslant 0$ at half mean spacing.

2.4 The moment-deflation audit (Round 5, `fab_deflation.py`) — relevant to Guth–Maynard

Guth – Maynard bound the top singular value of the Dirichlet-polynomial Gram matrix using tr B and a centred third moment, noting that estimates for powers ⩾ 4 are unavailable. The exact extremal problem "given (p_1, p_r) and m nonnegative eigenvalues, bound λ_1" — sharp bound λ_1^r + (p_1 − λ_1)^r/(m−1)^{r−1} ⩽ p_r — shows that for a spike-plus-flat-bulk spectrum **the r = 3 bound is already exactly tight and r = 4, 6, 8, 10 gain 0.00%** (m = 10^2, 10^3, 10^4; spike fractions 0.5 ⋯ 0.05); with a secondary cluster the gain is 0.1 – 1.1%. Whatever room exists in Guth – Maynard is not in the moment hierarchy alone but in the additive-energy input and the resonator geometry. A formula received from Codex ($e = \eta + \varkappa$, $240y^2 + (72 - 10e)y - 13e = 0$, $A(e) = 30/(13 + 10y\_e)$) is recorded as **[U]**: we never re-derived their argument.

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3. Tao's Alternative Hypothesis and the ACUE impostor programme

3.1 Why AH matters and why it resists

Montgomery's pair-correlation conjecture: normalised zero gaps follow the GUE law. The Alternative Hypothesis (AH): the gaps lie asymptotically in $(1/2)\mathbb{Z}$. AH is consistent with every proven zero statistic, has arithmetic consequences (class numbers; Landau – Siegel), and is precisely what current technique cannot exclude. Tao's ACUE — the measure on N-subsets C of the 2N-th roots of unity with $\mu\_ACUE(C) = |\Delta(\zeta^C)|^2/(2N)^N$ — reproduces CUE pair correlation exactly and is a finite model of AH. The obstruction to "find the statistic ACUE cannot fake" is that one must rule out a *fibre*, not a point: the measures matching any finite moment list form a large convex body.

3.2 The fibre, measured exactly [X]

Affine dimension (rotation-orbit level) of {measures on the ACUE support matching every balanced moment $E[p\_\lambda \bar{p}\_\nu]$ of degree ⩽ N}:

N	3	4	5	6	7	8	9
dim	0	0	2	10	80	403	1804

(N = 6 and 8 re-confirmed by the tomography code: ranks 70 and 407 with spectral gaps > 10^6.) Structural facts: 2- and 3-point statistics are rigid, 4-point free (N = 5 onward); the fibre at N = 5 is completely solved — a pentagon over $\mathbb{Q}(\sqrt{5})$, the known mimicker being $|v|^2$ for a 5-sparse coherent superposition of translated Fermi seas with the single syndrome a*Sa = 0. The reflection symmetry of the functional equation kills the entire chiral half of the fibre (39/80 dims at N = 7) at zero cost and provably cannot do more. At N = 8, of 403 directions, **401 are invisible to every pattern count of window width ⩽** 2, 397 to ⩽ 3, 396 to ⩽ 4, 392 to ⩽ 5, 383 to ⩽ 6.

3.3 Explicit families [X]

Centre-of-mass family (`d6_allN_family.py`, `mimicker_fibre.py`): with X(C) = Σc mod N, q_g(C) = μ_ACUE(C) · g(X(C)) matches every balanced moment of degree ⩽ N **iff** E[g] = 1 and **ĝ(±1) = 0**; all other frequencies free. Hence an (N−3)-parameter family of honest probability laws for every N ⩾ 4: null dims 2, 3, 4, 5 at N = 5, 6, 7, 8 (forced-zero frequency j = ±1; free j = 2 / 2,3 / 2,3 / 2,3,4), worst moment error ⩽ $1.5 \cdot 10^{-11}$, positivity radius 1.0 – 2.2. Mechanism: X mod N is exactly uniform under ACUE and couplings to com-frequencies 2..N−2 vanish by a transport-cost bound (hop budget 2N vs cost ⩾ 2N+2).

Determinant-character (secant) family. D_r(C) = ∏ *{c ∈ C} ζ^{rc} = det(U_C)^r. Tilts q = μ · [1 + c · Re(η D_r(C))] lie in the fibre. Exact quantum-mechanical description (from the colleague's message, verified consistent with our data): μ_ACUE(C) = |⟨C|(∧ ^N F)A₀ ⟩|² is the Born distribution of a single Slater determinant (F = DFT, A₀ = e₀ ∧ ⋯* ∧ e{N−1}) — re-proving ACUE is a projection DPP; ⟨C|Ω_r⟩ = ε_r D_r(C)⟨C|Ω_0⟩; the tilt with c = 2a/(1+a²) is the interference pattern (|Ω₀⟩ + a|Ω_r⟩)/√(1+a²) — our c = ½ corresponds to a = 2 − √3. In Plücker coordinates ACUE is a point of Gr(N,2N); the mimicker lies on σ_2 (Gr(N,2N)) \ Gr: the minimal non-Gaussian deformation of a fermionic Gaussian state.

Parity sectors (from the uploaded "Finite All-Depth Escape" paper; tested by us): for even N ⩾ 6, q_N^± = μ_N(1 ± (−1)^{Σc}) are mutually singular, match balanced trace moments of degree d < N²/4, agree on every complete binary marginal on ⩽ N−1 sites, yet have the same local limit. That paper also proves pair rigidity for simple half-lattice processes (first spatial escape is three-body) and constructs, for every s ⩾ 3 and large N, honest laws matching bandwidth-one rows through order s while the three-site decoder P(0,1,2 ∈ C) stays nonzero; the missing piece for a second infinite LR mimicker is one uniform positivity ratio Γ*_{N,s}.

3.4 The autocorrelation—Bruhat structure [P (identities) + X]

Restricting to even shifts, |Ω_s⟩ := |Ω_{2s}⟩, z(C) := D₂(C) with z^N = 1, so the entire secant family collapses to a polynomial P_a(z) = Σ a_s z^s on ℤ_N:

q_a(C) = μ(C) · |P_a(z(C))|² = μ(C) · (1 + Σ_{δ≠0} A_δ z(C)^δ), A_δ = Σ_s a_{s+δ} ā_s.

The code object is the autocorrelation spectrum, not the support. The invisibility depth is d_vis(a) = min{δ(N−δ) : δ ≠ 0, A_δ ≠ 0}, because the sector pairing B_δ(f,g) = E_μ[D_{2δ} f ḡ] vanishes for deg < δ(N−δ) and opens a rank-one channel at exactly that degree between the conjugate rectangles (N−δ)^δ → δ^{N−δ}. So q_a agrees with ACUE on all balanced observables of degree ⩽ d iff A_δ = 0 for all δ with δ(N−δ) ⩽ d. The critical detector Φ_{N,s}(U) = det(U)^{2s} · s_{(N−s)^s}(U) · conj s_{s^{N−s}}(U) is the character of W_{N,s} = V_R ⊗ V_S^∨ ⊗ det^{2s}, has E_ACUE Φ = (−1)^{s(N−s)}, and E_CUE Φ = 0 because W^{U(N)} = 0. The pure max – min code problem is solved by pigeonhole: L states on the N-cycle give at best $m(N-m)$, $m = \lfloor N/L \rfloor$, maximised at $L = 2$ with $\lfloor N^2/4 \rfloor$ — **two Fermi seas are globally optimal; do not spend compute on five.** The open problem is the *interference* code: A positive-definite on $\mathbb{Z}_N$, $A_0 = 1$, $A_\delta = 0$ on a low-Bruhat-energy zone, $A_{\delta^*} \neq 0$ as high as possible (a Delsarte LP; "zero-Bruhat-correlation-zone sequences"; the CAZAC extreme $A \equiv 0$ gives $q = \mu$, no impostor).

$d_N(\delta) = \delta(N-\delta)$ is simultaneously $\dim_{\mathbb{C}} Gr(\delta,N)$, the pairing $\langle 2\rho, \omega_\delta\rangle$ (hence the affine Bruhat length $\ell(t_{\omega_\delta})$), $N \cdot \|\omega_\delta\|^2$, and (up to convention) the quadratic Casimir $(N+1)/N \cdot \delta(N-\delta)$. The fourth identification — with the *dynamics* — is § 4.7 and is the structural centre of the programme.

3.5 Monodromy separator: real Frobenius families are det-sector dark [C, exact averages]

Agent L3 (`monsep_*.py`; artifact "Monodromy Separator"). Over $F_q[T]$ Dirichlet families ($N = 4$: $q \in \{3,\cdots,23\}$; $N = 5$: $q \leqslant 13$; $N = 6$: $q \leqslant 11$; ~40M characters, exact FFT character sums): every winding $\neq 0$ observable satisfies $|E_\chi| \leqslant 1.7 \cdot q^{-1/2}$, with observed decay slopes −2 to −5 (near the family-noise floor $q^{-(N+1)/2}$). ACUE is bright on the same detectors: $E_\mu[p_N^2] = N$, $E_\mu[p_N^2|p_1|^2] = N-2$, $E_\mu[\det^4 |p_{N-1}p_{N-3}|^2] = 1$; the rank-4 twist mimicker has $E[\det^4] = ½$ (N=4), ¼ (N=5,6). The weight-(N+1) Gram is exactly Haar − 1 in every entry (Haar closed form $E[p_\lambda \bar{p}_\nu] = z_\lambda \delta - \mathrm{sgn} \cdot \mathrm{sgn}$ **[P]**). Families converge to Haar with a $+c/q$ correction — the *opposite* side from ACUE's −1. $\dim(V_\lambda \otimes V_\mu^\vee \otimes \det^r)^{U(N)} = 0$ for all detectors **[P]**; det is not pinned (per-class $E[\det] = 0$). Certificate: $|E_{family}| <$ bright/2 at every computed q except $\det^4 |p_4 p_2|^2$ at $(N,q) = (5,3)$. **Sign correction discovered:** $\Sigma_{\deg f = m} \Lambda(f)\chi(f) =$ **−**$q^{m/2} \mathrm{tr}(\Theta^m)$. What remains for the colleague's Conjecture G2 (minimal derived-L separator) is to convert $W^{G_{geom}} = 0$ into a uniform $O(q^{-1/2})$ trace estimate via Katz's big-monodromy theorem (IMRN 2013), and to extend the bright gap to the 7 non-twist fibre directions at N = 6.

3.6 FUSION-ACUE: a negative result recovered from a dead agent's logs [✗, exact]

Agent L1 (`fus_*.py`, killed by credit exhaustion; logs harvested). **Task 1 verified**: the ACUE matrix coefficients $M^{(r)}_{\lambda,\mu}$ equal det of a 0/1 congruence matrix with phase +1 and $Z_N = N!$ (N = 2,3,4 exhaustive; N = 5 spot-checked, 0 mismatches), all values in {−1, 0, +1}. **Verlinde S-matrix = Kac – Walton at sl(N) level N verified** (N = 2, 3, 4: 18/550/2100 triple coefficients, 0 mismatches). **But the identification "KW fusion = Σ_t ACUE triple pairings (charge decomposition)" FAILS at N = 3: 22 mismatches out of 550** (it holds at N = 2). Examples: KW (1,1)⊗(3,1)→() gives 0, ACUE gives 1; KW (2,1)⊗(2,1)→() gives 1, ACUE gives 2. So the naive form of the colleague's FUSION-ACUE Conjecture A is false; the correct statement, if any, needs a different charge bookkeeping. **This was never reported to the human before this document.**

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4. The finite de Bruijn–Newman depth: the P0 programme

4.1 Definition and conventions (get these right; two earlier write-ups had sign slips)

For monic $P(z) = \prod(z - e^{i\theta_j}) = \Sigma\, a_j z^j$ define $P_s(z) = \Sigma\, a_j e^{s \cdot j(N-j)} z^j$ for $s \geqslant 0$ and the **depth** $D(P) = \inf\{s > 0 : \mathrm{disc}(P_s) = 0\}$. $D = -\Lambda$. P_s stays self-inversive, so D is the first collision time. **Lemma 1 [P, machine-checked]:** the zeros obey $\dot{\theta}_j = -\Sigma_{k\neq j} \cot((\theta_j - \theta_k)/2)$ (attracting circular Coulomb gas; derived from $\partial_s P = (N D_z - D_z^2)P$ and $2z_j/(z_j - z_k) = 1 - i \cdot \cot((\theta_j-\theta_k)/2)$, with the (N−1) terms cancelling exactly). N = 2: $D = -\log\cos(\delta/2)$. Clocks ($P = 1 - cz^N$) are flow-invariant, $D = \infty$.

4.2 ACUE, exact enumeration N = 3..10 [X] (`dyn1_*`, data `dyn1_results_N*.npz`)

13,132 rotation orbits (184,756 configurations at N = 10), exact Vandermonde masses; 40-digit spot checks worst $1.3 \cdot 10^{-9}$. Results:

- **P(clock) = 2^{1−N} exactly [P]** (Cauchy – Binet: Σ_{|C|=N}|Δ|² = det(AA*) = (2N)^N; each clock has |Δ|² = N^N).
- **δ_min = π/N for every non-clock configuration [P]** (pigeonhole).
- **−Λ ≍ N^{−2}**, fitted slope −2.0009; N²(−Λ) ∈ [1.31, 1.99].
- First collision always at an initially-adjacent pair: 13,130/13,130.
- Generator identity L p_m = −m(N−m)p_m − m Σ_{a<m} p_a p_{m−a} (verified $4.1 \cdot 10^{-28}$).
- Quantiles of N²(−Λ) (min / median / max): N=6: 1.353146 / 1.419374 / 1.952629; N=8: 1.330383 / 1.418216 / 1.976122; N=10: 1.314614 / 1.412774 / 1.985458.
- **The median turns around at N = 7** (1.41474, 1.41822, 1.41908, 1.41937, 1.41950, then 1.41822, 1.41520, 1.41277) — so the single-dislocation constant is NOT the ensemble median limit (a colleague's extrapolation from N ⩽ 7 was corrected).
- Minimising family {0..N−3} ∪ {N+3, N+4}: ρ = 1.0656 (N=10) ⋯ 1.0495 (N=18), slowly decreasing toward ≈ 1.03 – 1.05. Maximising symmetric 3-block: ρ → 1.6094, N²(−Λ) → 1.9861.

4.3 CUE, Monte Carlo N ⩽ 256 [C] (`dyn2_*`, data `dyn2_data_N*.npz`)

Two independent Λ-solvers (ODE vs coefficient bisection) agree to 10^{-6}; N = 2 law to $8.7 \cdot 10^{-13}$. **−Λ ≍ N^{−8/3}**: fitted −2.691 ± 0.010 (N ⩾ 16), −2.678 ± 0.016 (N ⩾ 32). Distributional law: **8N^{8/3}(−Λ) ⟹ G², P(G > x) = exp(−x³/72π)** (72π derived from the sine kernel, not fitted: measured 229 – 236 vs 226.2; KS 0.035 – 0.041; median 29.6 – 30.5 vs (72π ln 2)^{2/3} = 29.08). Local law R = 8(−Λ)/δ_min² > 1 with med(R−1) = 0.598 · N^{−0.729}. Depth ratio to ACUE falls as 3.64 · N^{−0.704}; 10× separation at N* ≈ 165 (predicted ≈ 160).

4.4 β-universality [C, three-point + endpoint] (`beta_*`)

Killip – Nenciu CMV sampling validated (β=2 vs stored CUE KS 0.0236; vs Haar spacing KS 0.0048; β=1 vs direct COE = VVᵀ KS 0.0035). Prediction −Λ ≍ N^{−2−2/(β+1)}:

β	predicted	fitted	local slopes
1 (COE)	−3	−3.064	−3.080, −3.026, −3.100
2 (CUE)	−8/3	−2.678	→ −2.627 at 128→256
4 (CSE)	−12/5	−2.510	−2.546, −2.480, −2.515
∞ (ACUE)	−2	−2.0009	exact enumeration

Localisation (first collision = minimal initial gap) holds in 95 – 99% of CβE samples and 100% of ACUE. Convergence *rate* prediction ρ_β − 1 = O(N^{−2/(β+1)}): fitted −1.012 / −0.710 / −0.501 vs −1 / −2/3 / −2/5, local slopes at largest N −0.851 / −0.624 / −0.407.

4.5 The single-dislocation constant [C, two independent routes]

Alternating clock minus $e^{-i\pi/N}$ plus 1: $P_N(z) = (z-1)(z^N+1)/(z-e^{-i\pi/N})$, gap pattern $1,2,\cdots,2,3$. Lattice solver: $N^2(-\Lambda) = 1.41473945$ ($N=3$) $\cdots$ **1.41963827** ($N=20$), monotone. Continuum limit $t = -s/N^2$, $z = e^{iu/N}$: $G_s(u) = 2\cos(u/2) - 2\pi \int_0^{1/2} e^{s(1/4-y^2)}\cos((\pi+u)y)\,dy$, first double zero at *$s = 1.419640342$, u* = 1.812942145**, $G''(u^*) = -0.3767$. Agreement $2 \cdot 10^{-6}$ at $N = 20$ ($O(N^{-2})$ approach). *$s = \rho_\infty \cdot \pi^2/8$ with $\rho_\infty = 1.150717118022$** (corrected from an earlier misprint 1.1507015): a two-body collision time dressed by $\approx 15\%$ many-body shielding.

4.6 Theorems A, B, C and the Lagarias—Rodgers bridge [P / R] (`depth_scaling_theorem.md`)

Theorem A [P]. For an adjacent pair, $g' = -2\cot(g/2) - \sum_{k\neq a,b}[\cot(x_a^k/2) - \cot(x_b^k/2)]$ with $x_j^k = (\theta_j - \theta_k) \bmod 2\pi$; adjacency forces $x_a^k = x_b^k + g$ in $(0,2\pi)$ and $\cot(\cdot/2)$ is strictly decreasing there, so every bracket is negative and enters with a plus sign: **every other zero slows the collapse**. Hence $g' \geqslant -2\cot(g/2)$, and since order is preserved until collision, **$-\Lambda \geqslant -\log\cos(\delta_{min}/2) \geqslant \delta_{min}^2/8$ ($\rho \geqslant 1$)**. Checked: 11,060/11,060 background terms positive; ratio minima 1.0842 / 1.0714 / 1.0612 (ACUE $N = 6, 8, 10$), 1.00019 / 1.00021 (CUE $N = 16, 64$). Exact two-body solution: $\cos(g(s)/2) = e^s \cos(g_0/2)$; and $-\log\cos(x/2) \geqslant x^2/8$ on $[0,\pi)$.

Theorem B [P given hypothesis]. With background stiffness $S = \sum_k \tfrac{1}{2}\csc^2(x_b^k/2)$ (clock value $(N^2-1)/6$ exactly), if $S \leqslant AN^2$ then $-\Lambda \leqslant (\delta_{min}^2/8)(1 + O(AN^2\delta_{min}^2))$. Measured S/N^2 on CUE: median 0.109 / 0.120 / 0.117 / 0.120 / 0.120 ($N = 8\cdots128$), q99 $\leqslant 0.36$, max 0.79. **The high-probability bound $S \leqslant AN^2$ is the single open analytic ingredient**; it closes Law 1, Law 3 and Theorem C(ii) at once. Because $N^2\delta_{min}^2 \asymp N^{-2/(\beta+1)} \to 0$ for finite β but $= \pi^2$ at $\beta = \infty$, the lattice is a *singular* endpoint where $\rho \neq 1$.

Theorem C. (i) **[P]** $N^2(-\Lambda) \geqslant \pi^2/8 = 1.2337005501$ for every non-clock ACUE configuration. (ii) **[R]** $P(N^2(-\Lambda^{CUE}) < \pi^2/8) \to 1$, from $P(\delta_{min} > \pi/N) \approx \exp(-\pi^2 N/72)$ plus the S-bound. Measured fraction below the floor: 0.565, 0.823, 0.967, 0.9984, **1.0000, 1.0000** ($N = 8\cdots256$); sample maxima 0.911 ($N=128$), 0.437 ($N=256$).

LR bridge [P modulo the reverse comparison]. A hard core of c mean spacings gives $\delta_{min} = 2\pi c/N$, so $N^2(-\Lambda) = \rho \cdot \pi^2 c^2/2$. Hence **$\mu_\Lambda \geqslant \pi^2\mu^2/2$** and **$\mu \leqslant \sqrt{(2\mu_\Lambda)}/\pi$**: Lagarias – Rodgers' $\mu \leqslant 0.606894$ is exactly $\mu_\Lambda \leqslant 1.8177$; a depth bound 1.29 would give $\mu \leqslant 0.511281$; the AH barrier is $M < \pi^2/8$. **Falsifiable threshold: $\liminf N^2(-\Lambda_{\zeta,local}) < \pi^2/8 \Longrightarrow$ AH false.**

4.7 The operator unification [P + X] (`fab_halflaplacian.py`)

$\sum_{\delta=0}^{N-1} \delta(N-\delta)\, e^{-2\pi ik\delta/N} = -N/(2\sin^2(\pi k/N))$ for $k \neq 0$, so $(\mathcal{L}_N f)(x) = \sum_{k=1}^{N-1}[f(x) - f(x+k)]/(2\sin^2(\pi k/N))$ has $\mathcal{L}_N e_\delta = \delta(N-\delta)e_\delta$. **Linearising the Coulomb flow at the clock, $\dot{\varepsilon}_j = \sum_{k\neq j}(\varepsilon_j - \varepsilon_k)/(2\sin^2(\pi(j-k)/N)) = (\mathcal{L}_N \varepsilon)_j$** — $\| \mathcal{L}_N - \text{Jacobian} \| \leqslant 2.4 \cdot 10^{-13}$ for $N = 4\cdots24$, top eigenvalue $N^2/4 = \lfloor N^2/4 \rfloor =$ the pigeonhole maximal invisibility depth. **Mechanism: impostors hide in exactly the fastest-relaxing modes of the clock's stability operator**, which is why adding moments fails and a hitting time does not. As $N \to \infty$, $N^{-1}\mathcal{L}_N \to (-\Delta)^{1/2}$: the invisibility hierarchy's natural semigroup is Poisson, not heat. The $1/\sin^2$ kernel is Haldane – Shastry's; whether the hierarchy is a sector of an integrable spectrum is a lead, not a claim.

4.8 The marked depth: exact rank-two law [C, parameter-free]

Λ is blind to eigenvectors (isospectral G_1, G_2: τ equal to $8 \cdot 10^{-17}$). Marked depth $\chi(G;u) = \partial_\eta(-\Lambda(\mathrm{Cayley}(G + \eta uu^*)))|_0$ separates them (median difference 0.081). Falsified guess: χ does *not* track the resolvent $u^*(zI-G)^{-1}u$ (resolvent $1.615 \to 6.667$ while χ $0.0267 \to 0.0028$). Law:

Dτ[uu] = (ϰδ/4) · uK_ab u + (δ²/8) · ϰ′ (u), K_ab = c_a v_a v_a − c_b v_b v_b, $c_j = -2/(1+\lambda_j^2)$,**

with (a,b) the first colliding pair and $\varkappa = 8\tau/\delta^2$ a *functional*. Correlation 1.00000000, slope 1.00000, residual $5.7 \cdot 10^{-11}$ (local term alone: 0.9958 / 1.347 / $8.1 \cdot 10^{-2}$) — the earlier "$\rho \approx 1.72$" was $\varkappa + (\delta/2)(\varkappa'/\delta')$ varying per mark. It is a **polarisation detector** ($P_a - P_b$, $\approx c \cdot \sigma_z$): $u = (v_a + v_b)/\sqrt{2}$ gives zero response; null cone $c_a q_a = c_b q_b$; orthogonal and null-cone floors $\approx 3 \cdot 10^{-3}$ are background (orthogonal marks fix λ_a, λ_b exactly); the resolvent returns at second order via $\lambda_j'' = 2q_j \sum_{k\neq j} q_k/(\lambda_j - \lambda_k)$. Blind rank-2 tomography from 60 random marks recovers $\mathrm{span}\{v_a, v_b\}$ to principal angles $2.3 \cdot 10^{-6}$, $3.0 \cdot 10^{-6}$ degrees (top-2 energy 0.986).

4.9 Classification of impostors [C]

Criterion I (transversality): relative residual of grad τ off row(DM_r): single dislocation N=8: 0.991 / 0.941 / 0.737 (r = 1,2,3); random ACUE N=7: 0.996 / 0.963 / 0.569; generic N=7: 0.998 / 0.926 / 0.195; collapses to 10^{-16} once rank DM_r = N (question becomes vacuous). Class I (caught by Λ): collision strata, half-lattice adversaries, centre-of-mass and secant families (TV 0.12 – 0.24 at N=8); fibre tomography at N=6: $E[N^2(-\Lambda)\,|\,\text{non-clock}] \in$ **[1.3610, 1.4770]** vs ACUE 1.4336, clock atom $\in$ [0, **0.0975**] vs 0.03125. Class II (needs marked Λ): isospectral families. Class III: linear invisibility survives the linear flow (heat operator is diagonal), and Λ escapes only because it is a nonlinear stopping time. **Parity sectors**: separated by Λ at every finite N *only through the atom* (2^{2-N} vs exactly 0; N=8 bulk quantiles 1.3739/1.3959/1.4182/ 1.4554/1.4839 vs 1.3752/1.3959/1.4196/1.4338/1.4788; conditional-mean differences $8.9 \cdot 10^{-3} \to 1.1 \cdot 10^{-3}$) — Class II/III in the limit.

4.10 Two ideas from the colleague not yet executed

- **Dynamic inertia profile.** Hermite – Sylvester: real-rootedness ⟺ positivity of a Hermite form H(p); along the flow, $I_p(t)$ = signature $H(p_t)$; Λ = first loss of full signature. Inertia counts *how many* negative directions, depth *how far to the first* — same geometry, two coordinates. Ties the Anthropic inertia (Lemma R) programme to the Newman programme; untested.
- **Arithmetic heat flow (Dobner).** $\zeta_t(s) = \sum \exp((t/4)\log^2 n)\, n^{-s}$: the dBN flow is diagonal on Dirichlet coefficients, $a_n \mapsto a_n e^{(t/4)\log^2 n}$. Log-Gaussian weights inserted into sieve/mollifier/character sums — untested "Newmanized parity" idea (§ 9.7).

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5. Bounded gaps between primes

5.1 Framework and the two controlling constants

Admissible k-tuple H; Maynard weights $w(n) = (\sum_d \lambda_d)^2$; $S_1 = \sum w$, $S_2 = \sum w \cdot \nu(n)$. If primes have level of distribution θ and $M_k = \sup_F k \cdot J(F)/I(F)$ (symmetric F on the simplex R_k) exceeds 2m/θ, then DHL(k, m+1) and $H_m \leqslant H(k)$. Bombieri – Vinogradov: θ = 1/2, criterion $M_k > 4m$. Upper bound [P]: $M_k \leqslant (k/(k-1)) \cdot \log k$; ε-variants $M_{k,\varepsilon} \leqslant (1+\varepsilon)(k/(k-1))\log k$. Ladder of minimal diameters (Engelsma, proven minimal ⩽ 342; our independent exhaustive re-proof ⩽ 62, `p4_rho_exhaust.c`, `p4_payoff_table.txt` with witnesses): H(40..50) = 186, 188, 196, 200, 210, 212, 216, 226, 236, 240, 246; H(54) = 270; H(105) = 600. Empirical $H(k) \approx k(\log k + 0.77)$ for $k \sim 10^4$ (fits (35410, 398130)).

5.2 The records [X + P] — `H2_H3_record_announcement.md`, `p9_*`

	bound	k	previous
H_2	173,438	15,856	396,504 (Stadlmann; earlier 398,130 Polymath8b)
H_3	13,859,802	923,601	24,797,814 (Polymath8b)
H_4	1,120,662,828	56,000,000	1,431,556,072 (Polymath8b)

Chain: Maynard's theorem (closed simplex, no ε, no extra equidistribution) + Bombieri – Vinogradov (margins 0.013 / 0.0067 / 0.065 leave room for $\theta < 1/2$) + Berry – Esseen C = 0.56 + ball arithmetic. Certificates $M_{15,856} \geqslant 8.013326752751$, $M_{923,601} \geqslant 12.006666706750$, $M_{56\cdot 10^6} \geqslant 16.065482942209$ (`p9_exact_cert_k*.json`, `p9_certify_hp.py`; three regimes each: two BE constants × two tail routes; small-k sanity: engine stays below known M_2, M_5, M_{54}, M_{105}). Tuples: k = 15,856 diameter 173,438 (`p9_tuple_k15856.npy`, second-implementation admissibility check); k = 923,601 repaired Hensley – Richards tuple diameter 13,859,802 (symmetric window $\{\pm 1\} \cup \{\pm q :$ q prime $\in [45{,}007, 6{,}929{,}899]$, 5,692 mid-size primes deleted by a deletion-fixpoint repair that converged in 4 iterations$\} \cup \{+6{,}929{,}903\}$; sha256 d5fe6890⋯6f02; classical fallback primes-past-k diameter 14,505,780); $k = 5.6 \cdot 10^7$ primes-past-k, endpoints 56,000,003 → 1,176,662,831, π-anchors matched.

The engine (`p9_mk_engine.py`). $F = \prod g(t_i) \cdot 1[\Sigma t_i \leqslant k]$; X_i iid density g^2/c_2; exactly $I = k^{-k} c_2^{\,k} \cdot P(S_k \leqslant k)$, $J = k^{-(k+1)} c_2^{\,k-1} \cdot E[G((k-S_{k-1})_+)^2]$; **layer-cake identity** $E[G((k-S)_+)^2] = \int 2G(u)g(u) \cdot P(S_{k-1} < k-u)du$ with rigorous lower tails (chord-majorised Chernoff / one-big-jump / Berry – Esseen; monotonicity-only, so grids cannot invalidate); **shaped tails** $g = e^{-(t/T_1)^{\varkappa}}/(1+At)$. Recovers ≈ 1.1 units of log k (exact accounting +0.12, tail shape +0.49, rest by shape optimisation) = factor ≈ 3 in k. The 2014 crude closed-form bound had a deficit of 2.3 – 2.9 units; the 2023 – 25 improvements raised θ only. **Engine-vs-engine reconciliation:** an independent product-ansatz engine (P3) crossed m = 2 at k ≈ 29,500 vs P9's 15,856; the difference was fully explained (layer-cake + tail shape) before anything was claimed. **Pitfall fixed:** P9's first HP run at k = 923,601 gave 9.61 because the λ grid was not scaled with T — fixed by $\lambda \propto 1/T$.

Not claimed: $H_2 \leqslant 145{,}226$ via k = 13,467 (Deligne-strength MPZ[ϖ,δ] at $(\delta,\varpi) = (0.0205, 0.00552)$, threshold 7.8273): depends on a cap-normalisation in the truncated variational criterion that we reconstructed but never verified verbatim against Polymath8b. Also computed but not claimed: Deligne crossings k = 13,467 (m=2), 660,985 (m=3).

5.3 Five walls for H_1 = 246 [P] — `prime_gap_survey.md`

1. **Ceiling = tuple diameter.** No post-processing of Maynard – Tao output beats H(k_min); pair correlation constants (Wu 3.3996 → Lichtman ≈ 3.30, parity floor 2) cannot lower H_1.
2. **Scalar decode exactly optimal.** Convex-order two-point counterfeit kills all matrix / inertia / moment decodes; $f(m) = 2m/\theta$ is final.
3. **Weight cone closed.** rank-r SOS decouples by subadditivity; copositive cone flat (9-pattern dual, residual $1.3 \cdot 10^{-14}$). Matrix-Maynard reformulation (uploaded doc) agrees: the SDP has a rank-one optimiser; even at level 1, $M_2/2 = 0.692965 < 1$ with $M_2 = 1/(1 - W(1/e)) = 1.38593$ — no positive eigenvalue at the twin threshold.
4. **Parity, combinatorial.** Kill-graph bipartite ⟺ killable (cut-polytope odd-cycle facets); floors $H_1 \geqslant 6$, $k \geqslant 2m+1$. (The graph fact is Tao's 2014 note; the packaging is ours.)
5. **Level wall.** BFI 4/7, Maynard II 3/5, Pascadi 5/8 are well-factorable / fixed-residue and structurally unusable by classical MT; usable-restricted frontier: Maynard III 11/21 (uniform residues, shell-truncated), Stadlmann 1/2 + 1/40,

Pascadi minorant 10/19. Guth – Maynard zero density is orthogonal to H_1; GRH gives only $\theta = 1/2$.

The doors, priced. k = 49 → H = 240 (pays 6); k = 47 → 226 (pays 20). Pure $M_{49} \in$ [3.891257590916 exact (d = 20 power-sum basis p ⩽ 5, dim 1125, `mt_hp_k49_p5.json`), 3.97290]; $M_{50} \geqslant 3.907113699811$. ε-variant M_{49,1/35} (d = 18, n = 1597): float 3.959325169, **certified ⩾ 3.930490592**; M_{49,1/25} (d=14) 3.915989908. Upper bounds close the door only for ε ⩽ 0.00682; Polymath left it "undecided". Tipping: k = 49/48/47 need δ ≈ 0.002/0.004/0.006 in θ, or 8 – 13% of the Maynard-III shell surplus (open computation SHELL-M49). Tuple search at k = 35,265: best 397,352 vs record 396,504 (P4 annealing chains; missing: iterated merging, LP repair). **Pure-support kill test for k = 49 is unnecessary:** (49/48)log 49 = 3.97290 < 4 already kills every basis enrichment (higher power sums, SDP); only the ε-class is alive.

5.4 The signed sieve [P + X] — `signed_sieve_nogo.md`, `sgn1_*`, `sgn2_*`, `fab_theorem.py`

Identity $(\nu-m) + (m-\nu)_+ = (\nu-m)_+ \Longrightarrow$ **$S_2 - mS_1 - D(w) = \Sigma w_+(\nu-m) - \Sigma w_-(\nu-m)_+ \leqslant \Sigma w_+(\nu-m)$:** the signed class is redundant at face-value debt. Diagnostics on a finite microcosm (n uniform on ℤ_W, ν = coprimality count, weights in level-L feature span, exact rationals): value = classical plateau for β > β* with *β = 23051796480/10991046857 = 2.0973249209**, classical value 1087376209/3212440751 = 0.3384891094603102 (verified dual; signed vertex Φ = 1.2082816957, D = 0.4147152297, 85% negative mass on 16/96 cells; certified on both sides; unbounded below β_unb = 2.03265 and at β = 2). Across eight variants β* ∈ [1.44, 6.72], and in five of eight the window (β_unb, β*) is empty (< 10^{-6}). ℓ^1-budget ramp: λ(1) = λ_positive, slope 0.32 – 0.89 — gauge, not gain. **Positivity's second job: it makes the variational problem bounded** ($\|w\|_1 = S_1 = 1$). Escapes: (i) debt below face value (exceptional character — Zhang's programme is the *only* route changing the variational picture); (ii) w evaluable while w_+ is not (well-factorable λ). Chen-switch obstruction [P-level]: switching never reduces the number r of exact-primality conditions; Chen's debt has r = 1; DHL(k, m+1), m ⩾ 1 forces r ⩾ 2 = edge = bipartite = parity-blocked.

Conditional price list [X certificates] (`sgn2_certificates.json`, `sgn2_mk_large.json`):

θ	2/θ	k pure (M_k)	k ε (M_{k,ε})	H_1 pure/ε	m=2: k, H_2 ⩽	m=3: k, H_3 ⩽
1/2	4	54	50	270 / 246	15,856 → 173,438	923,601 → 13,859,802
4/7	3.5	31 (3.502015496)	29 (3.519881250)	140 / 130	5,647 → 58,058	202,528 → 2,856,288
7/12	24/7	29 (3.443305315)	26 (3.433616498)	130 / 114	4,835 → 48,988	160,703 → 2,226,804
3/5	10/3	26 (3.350647068)	23 (3.334615948)	114 / 94	3,931 → 38,878	120,497 → 1,632,566
5/8	3.2	22 (3.207656229)	20 (3.222665844)	90 / 80	3,022 → 29,180	80,165 → 1,051,602
1 (EH)	2	5 (2.007080)	—	12	221 → 1,498	1,978 → 18,144

(m ⩾ 2 rows use the p9 engine's certified-lower-bound path and primes-past-k diameters.) All rows are parity-consistent (⩾ 12 for m = 1). **Missing estimate (E_θ)** — tuple-residue well-factorable estimate: for c_q(a) jointly well-factorable with the CRT residue selection a mod p ∈ {h_i − h_j}, Σ_{q⩽x^{θ−ε}} Σ_{a ∈ A_i(q)} c_q(a)E(x;q,a) ≪ x(log x)^{−A}. None of BFI Thm 10 / Maynard II / Pascadi covers it (all fix one residue per modulus); MPZ[ϖ,δ] is its absolute-value cousin at level ⩽ 0.5286. (E_{7/12}) is structurally cheapest (λ can literally be Iwaniec's λ±).

5.5 EH → 8 via one signed cross block [P conditional] (uploaded `EH8_OBJECTIVE_ALIGNED_CROSSBLOCK.md`)

For $H = (0,2,6,8,12)$, with the frozen exact trial $Q_5 = 1562651575013110693/778568621714732244 = 2.00708265325605$ ($M_5 > 2$), EH plus one Maynard-weighted scalar $\Sigma\, \nu\_x(n)C_8(n)(\Theta\_H(n) - \log 3x) = o(\log x \cdot \Sigma\nu\_x)$, $C_8 = \lambda(n)\lambda(n+2) - \lambda(n)\lambda(n+12) - \lambda(n+2)\lambda(n+12)$, gives $H_1 \leqslant 8$ (tolerance $-1/500$, or $-7/2000$ under full EH). $W_8 = (1-ac)(1-bc)$ is the *unique* parity-neutral blocker (Theorem 4.1); connected bipartite bad graph on r vertices imports even Walsh orders through $2\lfloor r/2 \rfloor$ (Theorem 4.2). Ladder: gap 10 needs one 2-point Liouville correlation; 8 needs the degree-2 block; 6 needs one quartic; 4 is odd-cycle impossible (parity wall). RMT corollary: GRH + Dirichlet-WPC_β with $\beta > 2 - Q_5/2 = 0.996458673\cdots$ + $OMC_8 \Longrightarrow H_1 \leqslant 8$. Novelty is narrow (objective-aligned one-scalar compression); Ford – Maynard's primal – dual theory and Murty – Vatwani are the benchmarks; OMC_8 is not known to be easier than GEH.

5.6 The Lagarias–Rodgers μ programme (uploaded LR docs, verified in part)

μ = sup hard core of bandwidth-one sine mimickers: $1/2 \leqslant \mu \leqslant 0.606894$ (LR); conjecture $\mu = 1/2$. Our contributions there: exact signed Fermi-path formula for DPP targets; hard-core necklace enumeration; **Theorem 4.1** (unbounded exact feasible branch $\Longrightarrow$ LR process, one-way); **Theorem 11.1** exact all-cardinality 20-cycle separator over $\mathbb{Q}(2\cos(\pi/10))$ (positive on all 109 orbits, Fermi target $-1/500$) and its exact non-transfer to the continuous circle ($L(X\_\triangle) = -954123/256000000$; seam obstructions $\{0,19\}$, $\{0,18\}$); the smooth nine-coordinate ansatz collapses to zero margin jointly at $N = 4,5,6,7$ (a lattice-trained vector already fails at $N = 8$: min $-1.08 \cdot 10^{-3}$ at $\{0,3,\cdots,36\} \subset \mathbb{Z}/40$). **Palm row-sum square (Theorem 13.1 / Prop 5.1):** for band-limited f (supp $\hat{f} \subset [-½,½]$), $\mathrm{Var}^0\_\mathrm{sine}(S\_f) \leqslant (M\_h - A\_f)(A\_f - m\_h)$ is necessary for an h-hard-core mimicker; the multiwindow quadratic version on the packing body K_h. The first nonnegative Fejér profile fails for an exact local reason ($\{-h,0,h\}$ forces $S_0 = 2f(h) > 9/7$); signed sinc-power searches found nothing. **The depth bridge (§ 4.6) is the new lever on μ.**

5.7 Agendas received and their status

- **Zhang Yitang (2026).** Landau – Siegel; "remove the square". Verdict: the variational half is closed (§ 5.4); the arithmetic half — sub-face-value debt via the exceptional character — is the only door. His "first experiment" (λ_signed vs λ_positive) is answered: no new phase.
- **Bourgain toolkit.** Exp 3 (ACUE × decoupling) done (§ 2.3, negative for decoupling); Exp 2 merged with Zhang's; Exp 1 (Type III × additive energy) and Exp 4 (function-field affine sieve) never launched.
- **Cross-field tools list** (log-Chowla bridge, exponent-calculus compiler, hypermetric facets, magic functions): approved in principle, superseded, never launched.

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6. Failed trials, refutations, and what each one taught

These are first-class results. Each closed a direction that a fresh system would otherwise re-open.

trial	outcome	insight
"0.6725 is the exact bandwidth-one LP optimum; AH is suboptimal by 0.0301" (our own Round-1 claim)	[✗] retracted (feasible-set conflation)	the free lattice optimum and the Anthropic-feasible optimum live in different cones; 0.0208 of the 0.0301 is aliasing artifact

trial	outcome	insight
Improve 0.6725 by window engineering	[✗]	the clump term forces $M = A_0 \log T$ for every window; only the M_- lemma can help
Pointwise edge hypothesis $\lvert F(1)-1 \rvert \leqslant \varepsilon$	[✗] buys exactly 0 even at $\varepsilon = 0$	$\hat{f}(\pm 1) = 0$ for every admissible window; the edge object is the Cesàro mean
Derivative pair correlation at bandwidth one	[✗] blind	—
tr Q^3 as a free upgrade	[✗] (worthless at flat window; $\Phi_3 = -0.01178$ at MT window but unmonetisable without M_-)	the blocker is the unbounded negative spectrum of the compression, not arithmetic
ℓ^2-decoupling on the 0.0301 gap	[✗] misprices by 15 – 77×	gap is single-site in position ($S_1 \geqslant 0$ at half spacing), full-band in Fourier
Higher traces (tr $B^4 \cdots B^{10}$) improving Guth – Maynard by pure deflation	[✗] 0% in the idealised regime, 0.1 – 1.1% with secondary clusters	room must come from additive energy / resonator geometry
Matrix / inertia / moment decode for Maynard – Tao	[✗] rank-one optimiser; scalar decode optimal	convex-order two-point counterfeit
PSD / copositive enlargement of the weight cone	[✗] flat	rank-r SOS decouples
Signed weights open a new variational phase (Zhang's experiment)	[✗] one-line identity	positivity also *bounds* the problem; the value of signed weights is purely arithmetic
$k = 49$ pure-support basis enrichment / SDP kill test	unnecessary	classical upper bound $3.97290 < 4$ already kills it; only ε-class alive
Higher-order (5+ Fermi sea) secant adversaries for deeper invisibility	[✗] $L = 2$ is globally optimal by pigeonhole ($\lfloor N^2/4 \rfloor$)	the open problem is the *interference* code (autocorrelation zeros), not the support
FUSION-ACUE "KW fusion = ACUE triple pairings"	[✗] at $N = 3$ (22 mismatches) though Verlinde = KW holds	charge bookkeeping in Conjecture A is wrong as stated
Marked depth tracks the directional resolvent	[✗] an order of magnitude the wrong way	first order is a rank-2 critical-pair contrast; resolvent returns at second order
$s^* = 1.419640342$ as the ACUE median limit	[✗] median turns at $N = 7$	s^* is a configuration constant of the single-dislocation stratum
$\rho \approx 1.72$ as a universal constant in the marked-depth law	[✗]	it is $\varkappa + (\delta/2)\varkappa' / \delta'$ per mark; law is parameter-free with $\varkappa$ measured
Linear heat flow "mixing" hidden fibre directions into visibility	[✗] the flow is diagonal; invisibility persists for all t	only the nonlinear stopping time escapes
Smooth nine-coordinate LR separator uniform in N	[✗] zero margin at $N = 4..7$ jointly; fails at $N = 8$	continuum stability is a two-parameter (mesh × volume) limit; fixed-cardinality certificates can be spurious
Nonnegative Fejér profile in the Palm row-sum bound	[✗] exact local failure ($\{-h,0,h\}$)	signed f with certified packing bounds is the right search
Tuple search at $k = 35{,}265$ beating 396,504	not achieved (397,352)	needs iterated merging / LP repair

trial	outcome	insight
Deligne-route $H_2 \leqslant 145{,}226$	**not claimed**	cap-normalisation unverified verbatim
Shared-context data errors ($H(47) = 232$, $H_2 = 398{,}130$, exponent 3.815, $H(26) = 120$)	corrected by agents to 226 / 396,504 / 3.8075 / 114	keep a living errata list in the context file

Meta-lessons. (a) Each refutation came with the corrected statement; that is the standard to keep. (b) Five different "phase transitions" turned out to be normalisation artifacts (β-kink, ℓ^1 ramp, unboundedness at $\beta = 1$, $\rho = 1.72$, the median $\to s^*$). Always ask whether the decode is scale-invariant before reading a growth as a gain. (c) When an idea uses only static polynomial information about a family that is moment-frozen, it will fail; move to stopping-time / hitting-time / marked observables.

7. Verification status, Lean, and where everything lives

7.1 Machine verification (`fab_verify_proof.py` / `verification/verify_theorem_steps.py`)

18 checks, 18 passing, no floating point: flow generator identity; $2z\_j/(z\_j-z\_k) = 1 - i \cdot \cot$; the $(N-1)$ cancellation; two-body solution and collision time; $d/dx \cot(x/2) = -\tfrac{1}{2}\csc^2$; $f(0) = 0$ and $f' = \tfrac{1}{4}(2\tan(x/2) - x)$; nonnegative Taylor coefficients of $\tan t - t$ ($\tfrac{1}{3}$, 2/15, 17/315, 62/2835, 1382/155925, $\cdots$); **the sign lemma as a z3 decision problem (unsat for the negation)**; ACUE pigeonhole; $\Sigma\tfrac{1}{2}\csc^2(\pi k/N) = (N^2-1)/6$ at $N = 4,6,8,12,20$. Other exact checks: the signed identity on 1,000 random weights across five models; β^* with verified dual; the cellwise identity $a + G = Gp$.

7.2 Lean — written, NOT compiled

`research/riemann-rmt/lean/DepthComparison.lean` (also packaged as `riemann-impostors/lean/RiemannImpostors/DepthComparison.lean` with `lakefile.toml`, toolchain pin `leanprover/lean4:v4.33.0-rc2` — matching the `anthropics/zeta-23-lean` clone at `/workspace/anthropics/zeta-23-lean`). Declarations: `cot`, `hasDerivAt_cot`, `cot_strictAntiOn`, `background_sign`, `self_le_tan`, `neg_log_cos_ge` (proofs written), `two_body_solution` and `depth_ge` (**sorry**: scalar autonomous ODE uniqueness; Grönwall-type comparison). No toolchain could be installed: egress 403 for `elan.lean-lang.org` and for GitHub *release* downloads (`git ls-remote` of public repos works; releases do not; building Lean + Mathlib from source is not feasible in-session). **If Astra has a toolchain: `lake update && lake build`, then discharge the two sorries.** Also unformalised but small: the five-line `mult_two_pair` edit in zeta-23-lean giving $N\_d \geqslant 0.8362503 \cdot N + p$.

7.3 Repository index (branch `claude/riemann-zeta-random-matrix-udxp3f`, PR #11)

`research/riemann-rmt/`:

- Papers: `impostors_paper.md` (canonical, 10 sections), `stopping_times_paper.md`, `depth_scaling_theorem.md`, `signed_sieve_nogo.md`, `newman_depth_note.md`, `H2_H3_record_announcement.md`, `prime_gap_survey.md`, `round3_synthesis.md`, `final_verified_paper.md`, `rmt_zeta_survey.md`, `rmt_zeta_popular.md`, `tao_ah_notes.pdf`, `codex_handoff.md`, `README.md`.
- Context files given to agents: `joint_context_v2.md`, `prime_gap_context.md`, `signed_context.md` (and `lambda_context.md` in the scratchpad).

- Engines/certificates: `p9_mk_engine.py`, `p9_certify_hp.py`, `p9_exact_cert_k*.json`, `p9_tuple_k*.npy`, `p9_g_k*.npz`, `p9_scan.py`, `p9_tuples.py`, `verify_codex.py`.
- Theorem checks: `fab_*.py` (+ `*_results.json`), `sgn1_t1_exact.{py,json}`, `beta_*.py`, `tomo_fiber.py`, `d6_allN_family.py`.
- `riemann-impostors/`: standalone package (README = labeled results summary; paper/, lean/, counterexamples/, verification/, certificates/, data/ with `dyn1_results_N3..10.npz`; `PUBLISHING.md` — repo creation returned 403 from the GitHub App integration, so publish via `git subtree split`).
- `handoff/`: this document (md + pdf).

Session scratchpad (`/tmp/claude-0/…/scratchpad`, ~1,150 files) holds everything else: `dyn2_data_N*.npz` (CUE, N = 2..256), `beta_data_b{1,4}_N*.npz`, `monsep_*`, `fus_*`, `brg_*`, `dir*_`, `p1_–p10_*`, `sgn*`, `tomo_*`, `dyn1_*`, agent logs. It survived two container restarts but is not guaranteed to outlive the session; anything essential has been copied into the repo.

7.4 Key constants (verified this programme)

δ_MT = 3/2 − (1/√2)cot(1/√2) = 0.672500703679 · PairCeiling 0.6818287 · ceiling candidate 15/22 · Φ_3 (MT) = −0.0117753128 · N_d/N ⩾ 0.8362503 · deficiency ledger 0.3275 · M₋ payoffs 0.6796896 / 0.6844924 · δ_free → ½ + 2/π^2 = 0.70264237 · doubles density 1/4 − 1/π^2 = 0.148679 · M_2 = 1.385933, M_3 = 1.646440, M_4 = 1.845401, M_5 = 2.007080 (Q_5 exact above), $M_{5\ 4}$ > 4.00238, $M_{4\ 9}$ ⩾ 3.891257590916, M_{49,1/35} ⩾ 3.930490592 (float 3.959325169) · β^* = 2.0973249209 · $\pi^2/8$ = 1.2337005501 · s* = 1.419640342, u* = 1.812942145, ρ_∞ = 1.150717118 · separated-defect ρ = 1.19120 ($N^2(-\Lambda)$ = 1.46946) · ρ_max → 1.6094 · 72π = 226.19 · (72π ln2)^{2/3} = 29.08 · N* ≈ 165 · LR: 1/2 ⩽ μ ⩽ 0.606894 ⟺ μ_Λ ⩽ 1.8177 · clock stiffness (N^2−1)/6 · fibre dims 0,0,2,10,80,403,1804.

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8. Open problems, ranked, with the exact statement and its price

1. **The background bound [the linchpin].** Prove: for CβE (β fixed), with probability → 1 the background stiffness of the first-colliding pair satisfies S ⩽ A · N^2 (measured median 0.120 · N^2, max 0.79 · N^2). This single rigidity statement upgrades to theorems: 8N^{8/3}(−Λ) ⟹ G^2 (β=2), −Λ ≍ N^{−2−2/(β+1)} for all finite β, and Theorem C(ii). It is a standard-shaped estimate (Σ_k d_k^{−2} near the extremal pair) and is the best-value item on this list.
2. **ρ_∞ = O(1) for ACUE at all N** (lattice upper bound). Deterministic; proven only for N ⩽ 10 by enumeration (ρ ∈ [1.049, 1.610]); the minimising family's ρ decreases toward ≈ 1.03 – 1.05.
3. **Exact constants:** s* = 1.419640342 as a theorem (infinite clock + one dislocation under the Coulomb flow; Calogero – Moser framing); the separated-defect ρ = 1.19120⋯; ρ_∞ of the minimising family. Targets: closed forms in theta/Bessel/trigonometric data.
4. **The marked-depth law as a theorem**, Dτ[uu*] = ($\varkappa\delta$/4)u*K_ab u + (δ^2/8)$\varkappa'$ (u), with $\varkappa'$ characterised; on the null cone c_a q_a = c_b q_b the leading term should be λ_j″ = 2q_j Σ q_k/(λ_j − λ_k).
5. **Transversality theorem:** τ| *{F_m} generically non-constant; marks added until ∩ ker DΛ*{u_j} ∩ ker DM = {0}.
6. **The M₋ lemma** on the zeta side: ‖ ($c^{-1}\hat{A}$)₋ ‖ ⩽ M₋ uniform in T; worth +0.007..0.012 on 0.6725.
7. **The 15/22 ceiling identity** and *d = 3(N−3)** kill-degree law (exact-arithmetic conjectures).
8. **(E_θ)** for any θ > 1/2 (cheapest: 7/12) — the one statement between the price list and H_1 ⩽ 114/130.
9. **k = 49 ε-door:** push the Galerkin engine to d = 20 – 22 at ε ∈ {1/40..1/30}, warm-start from `p2_vec_hp_k49_d18_e35.npy`; if the float plateaus < 3.98, try the vanishing-marginal variant; non-separable dual

certificates are the only known way to *close* the door above $\varepsilon = 0.0068$.

10. **SHELL-M49:** compute M_{4} $_{9}$ on Maynard-III's shell polytope with the ε-trick layered; needs 8 – 13% of the full-shell surplus; undecided in print.
11. **Tuple k = 35,265 below 396,504** (best 397,352; add iterated merging + LP repair).
12. **Interference code / phase-rigidity conjecture:** among phase-twisted Slater states u ∈ U(1)^{2N}, vanishing of all balanced Schur minors below ⌊N²/4⌋ forces u_c = αβ^c up to gauge/dihedral symmetry (would make the det-character impostor the canonical extremiser of its whole class). Computationally attackable via circulant Toeplitz minors / Plücker relations.
13. **FUSION-ACUE, corrected:** find the charge bookkeeping under which KW fusion matches ACUE triple pairings (fails at $N = 3$ as stated).
14. **Function-field Newman universality** (§ 9.4).
15. **Formalisation:** discharge the two Lean sorries; formalise the records' certificate chain; make the five-line `mult_two_pair` edit.

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9. Wide-open research programme for GPT6 Astra: which famous conjecture, and by what route

This section answers the second half of the handoff request. It is opinionated and it is honest about odds. The ordering is by (probability of a genuine theorem) × (size of the theorem), not by glamour. The human's stated ambition is a historic result, not another small one; the way to earn that is to pick the target whose *remaining gap* is a statement of a shape people already know how to prove.

9.1 The best target: refute the Alternative Hypothesis dynamically

Claim to aim for. *Under a hypothesis strictly weaker than GUE, the zeros of ζ cannot have asymptotic hard core 1/2.* Equivalently, a proof that the Alternative Hypothesis is false — the statement Tao, Lagarias – Rodgers and others have posed as the obstruction to Montgomery.

Why this is now the right target. The static route needs pair-correlation information at Fourier bandwidth > 1, which is exactly what no one can prove. The dynamic route needs only that *the zeros are, locally, more fragile than a lattice under the de Bruijn – Newman flow*. Concretely (Theorem C(i) + the LR bridge, both **[P]**): AH ⟹ every window's local depth satisfies $\liminf N^2(-\Lambda) \geqslant \pi^2/8 = 1.2337$. So:

> **Level B target.** For ζ, with $N = (\log T)/2\pi$ zeros in a window at height T under the *true* H_t flow, prove $\liminf (\log T)^2 \cdot D\_T < \pi^2/8$ — or merely $D\_T = o((\log T)^{-2})$ — from averaged information (pair-correlation at bandwidth ⩽ 1, plus variance/energy inputs of Rodgers – Tao type).

Level A (the full N^{−8/3} CUE law) is *not* needed; Level C ($\liminf(\log T)^2 D\_T = 0$ along a subsequence) may already exclude rigid subclasses.

What Rodgers – Tao already give you. Their proof of $\Lambda\_{dBN} \geqslant 0$ runs exactly this kind of argument in reverse: assume the zeros are over-stable ($\Lambda < 0$), push the flow, obtain a local clock equilibrium, contradict known fluctuations. The needed extension is *quantitative and local*: the same energy method in a window, with the over-stability hypothesis replaced by "hard core 1/2" and the contradiction replaced by a fluctuation input weaker than GUE. The finite results say what the right normalisation is (the clock stiffness $(N^2-1)/6$; the two-body constant 1/8; the singular endpoint at $\beta = \infty$ where the background is leading-order).

Steps, in order.

1. **Truncation theorem.** One may not cut a window of zeros and call it a polynomial. Use the Polymath15 machinery for H_t (effective approximations to the flow) to define D_T rigorously and prove that the window's first collision time is controlled by the finite depth of the local configuration up to o(1) — the same localisation lemma as Theorem B, now for ξ.
2. **Hard-core $\Longrightarrow$ depth (already proved at finite N).** Transfer Theorem A to the H_t flow (the cotangent interaction becomes the Rodgers – Tao kernel; the sign argument survives since it only uses monotonicity of the pair interaction).
3. **The fluctuation input.** Show that near-collisions at scale $o(1/N)$ occur along a positive proportion of windows using *only* bandwidth-$\leqslant 1$ pair correlation plus an energy/variance bound. Note Montgomery's own theorem already gives a positive proportion of gaps < 0.68 mean spacing (unconditionally, via bandwidth-1 data); the question is whether such inputs plus the flow give gaps *below 1/2* often enough to break the floor. This is where new mathematics is required, and it is a concrete analytic-number-theory question, not a random-matrix metaphor.
4. Assemble: (2) + (3) $\Longrightarrow$ $\liminf N^2 D_T < \pi^2/8 \Longrightarrow$ AH false.

Risk. Step 3 may be equivalent in strength to the static statement one is trying to avoid. The finite theory suggests not — the depth is a *first-passage* quantity and Theorem C(ii) needed only the gap tail plus a bounded background — but this has to be settled on the zeta side. Either way the outcome is a theorem: either AH is refuted, or a precise equivalence "dynamic AH-refutation $\Longleftrightarrow$ bandwidth-$(1+\eta)$ pair correlation" is proved, which would itself be new.

9.2 Second target: the Lagarias—Rodgers conjecture μ = 1/2

Fully within random-matrix/point-process theory, no arithmetic input. **[P]** already: $\mu \leqslant \sqrt{(2\mu\_\Lambda)/\pi}$. So one needs a *depth upper bound* for every bandwidth-one sine mimicker: prove that any such process has $\liminf N^2(-\Lambda) \leqslant M$ with $M < 1.8177$ to beat the published bound, and $M \leqslant \pi^2/8 \cdot \rho\_lat$ to reach 1/2. Route: transport the Palm row-sum-square / multiwindow packing-body certificates (LR docs, Prop 5.1 – 5.2) from the row sum to the depth. The depth is smooth in the configuration (gradient computed in § 4.9), has an explicit critical-pair derivative law, and — unlike the row sum — has no known local obstruction of the $\{-h,0,h\}$ type. A win here is a clean, publishable theorem in its own right and directly feeds 9.1.

9.3 Third target: a Dyson—Montgomery statement one can actually prove

The honest position: the full pair-correlation conjecture is not reachable by these tools. What *is* plausibly reachable, and would count as a partial Dyson – Montgomery result, is a **dynamic universality theorem**:

> *Any point process with (a) bandwidth-one sine pair correlation, (b) a repulsion exponent β at short range, and (c) bounded background stiffness, has finite depth law $-\Lambda \asymp N^{-2-2/(\beta+1)}$.*

This is Theorems A + B plus the extreme-gap input, stated abstractly; the CβE case reduces to Feng – Wei, and the ACUE case is the $\beta = \infty$ endpoint. It says the Newman depth is a *fingerprint of the repulsion exponent* — a random-matrix theorem with a clear statement, four confirmed data points and one missing lemma (open problem 1). Expected effort: weeks, not years.

9.4 Fourth target: function-field Newman-depth universality (Katz–Sarnak)

Function-field Newman constants exist in the literature (Andrade – Chang – Miller-type explicit formulas; e.g. Λ_{D_p} = log(|a_p|/2√p) at genus 1 is exactly our N = 2 law in arithmetic clothing). Define, for a family of L-functions over F_q with Frobenius classes Θ_C ∈ G, the depth of det(1 − uΘ_C); Λ is a highly nonlinear class function on G. **Theorem to prove:** as q → ∞ at fixed rank N, Law(Λ_C) → Law(Λ_Haar(G)) for USp/SO/U families with big monodromy (Katz – Sarnak equidistribution pushes through a *singular* statistic because Λ is continuous off the discriminant locus, whose Haar measure is zero; the delicate part is the near-collision set). Then N → ∞ gives the universality exponents per symmetry class, with the ±1 hard edge in Sp/SO possibly changing the minimal-gap law. Our monodromy-separator computation (§ 3.5) shows the det-sector is already dark at q^{−1/2}; the U(N) family machinery (`monsep_core.py`) is reusable. This is the most *reliable* big theorem on the list, and the one best suited to a geometric collaborator.

9.5 Prime gaps: from 246 (or from the reported 186) downward — what is and is not possible

Consistency check on "186". H(40) = 186 exactly, so a proof of $H_1 \leqslant 186$ is a proof of DHL(40, 2). Under Bombieri – Vinogradov (θ = 1/2) that needs a Maynard – Tao variant > 4 at k = 40. The classical bound gives pure $M_{40} \leqslant$ (40/39)log 40 = **3.78347**, so the pure problem is *impossible*; an ε-variant needs (1+ε) · 3.78347 > 4, i.e. **ε > 0.05723** — far beyond anything Polymath ever certified (their ε-trick at k = 50 used ε ≈ 1/25 and moved M by ≈ 0.02); alternatively a level of distribution θ with 2/θ < M_{40}-variant, i.e. **θ > 0.529** at least (with pure M_{40} ≈ 3.75 one needs θ > 0.533), which is beyond Maynard III's 11/21 = 0.5238 and would need residue-uniform input of (E_θ) type. So *if* 186 is real, its proof contains either a new equidistribution theorem at level ≳ 0.53 uniform in CRT residues, or a variational enlargement beyond every known variant. Astra should first identify which, since that determines everything below. (If the number was actually 246, ignore this paragraph.)

From wherever the record stands, the levers, priced:

- **k = 49 ε-door → 240:** open, undecided; float 3.9593 vs 4; degree-22 Galerkin + ε-scan (open problem 9). Cheapest possible improvement of 246.
- **k = 47 → 226:** pays 20; needs ≈ 0.06 more in M or a level δ ≈ 0.006.
- **(E_{7/12}) → 114, (E_{4/7}) → 130, (E_{3/5}) → 94, (E_{5/8}) → 80:** the well-factorable route, blocked only by the tuple-residue estimate (E_θ). This is the route with the largest payoff and the clearest missing statement; the structurally cheapest is 7/12 where λ can literally be Iwaniec's λ^±.
- **EH ⟹ 12, EH + OMC_8 ⟹ 8, EH + quartic block ⟹ 6**; parity floor 6 absolute.
- **What cannot work** (walls 1 – 5): post-processing, matrix decodes, PSD/copositive cones, signed weights at face-value debt, zero-density inputs, GRH.
- **m ⩾ 2 records are ours and are far from optimal**: Galerkin/ε refinement would shave 3 – 10% off k*; H-R tuple repair at k = 5.6 · 10^7; the Deligne cap-normalisation (verify verbatim → $H_2 \leqslant$ 145,226).

9.6 A formalisation programme in the FLT style

We could not read Anthropic's post on formalising Fermat's Last Theorem; Astra should, and should import its agent architecture. What in this programme is *ready* to formalise, in order of value per effort: (1) Theorem A (two sorries away); (2) the five-line `mult_two_pair` edit giving N_d ⩾ 0.8362503 · N + p in `zeta-23-lean` — a strictly stronger published constant, in an existing verified codebase; (3) the records' hypothesis chain (Maynard's theorem is not in Mathlib, but the *certificate* — ball-arithmetic bounds on I and J for an explicit F — is a finite verification and the tuples' admissibility is decidable); (4) the signed no-gain identity (trivial to formalise, useful as a lemma); (5) P(clock) = 2^{1-N} and $\delta_{min} = \pi/N$.

9.7 Speculative but structured: the "Newmanised parity barrier"

ACUE proved once that a static information barrier is not a dynamic one. The twin-prime parity barrier is also a static barrier (sieve-accessible observables cannot separate a and b). Look for a flow S_t on arithmetic sequences — the Dobner flow $a_n \mapsto a_n e^{(t/4)\log^2 n}$ is the canonical candidate, being the dBN flow on the Dirichlet side — under which sieve observables stay matched but a positivity/stability hitting time differs. No theorem is known; the problem is now well-posed enough to search.

9.8 What not to spend the first month on

Larger N simulations of the depth (the exponents are settled); five-Fermi-sea adversaries (L = 2 is optimal); decoupling on the bandwidth-one gap; pure-support k = 49; higher traces for Guth – Maynard by deflation alone; any "new phase" from signed weights; window engineering on the zeta side.

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10. Closing note to Astra

The programme's single deepest fact is that the invisibility hierarchy of Tao's impostors and the stability spectrum of the zero dynamics at the clock are the same operator: impostors hide exactly where the flow acts hardest. Everything else — the depth laws, the separation, the LR bridge, the $\pi^2/8$ threshold — is that fact read in different coordinates. The route to a historic theorem runs through § 9.1, and its one genuinely new ingredient (step 3) is an analytic statement about near-collisions of zeros that current technique might reach. Prove open problem 1 first; it costs little and upgrades three numerical laws to theorems. Then go for AH.

Everything here is reproducible from the repository. Where we were wrong, we have said so; where we were uncertain, the tag says so. Good hunting.